

Caracterización integral de jets b

ALL VAL · actualización 6 de octubre de 2026

46 modelos únicos · 67.551 jets · 32 variables raw · punto de trabajo: score de al menos 0,8.

Este documento conserva las 24 caracterizaciones completas de VQC8/32/16/12 y RF32; añade nueve modelos únicos de QRF12/RFF36 y trece de la evaluación boosted. RF12 y VQC12 que vuelven a aparecer en las nuevas exportaciones se deduplican.

Misma cohorte congelada VAL: 67.551 jets (6.675 b, 8.366 c, 21.011 g, 31.499 uds). Scores alineados por clave física; las 32 variables raw mantienen sus unidades. No se abrió TEST, TRAIN ni ROOT, ni se reajustaron modelos.

Para cada modelo: una página de perfiles 3D y una tabla compacta con media $\pm$ desviación estándar poblacional (ddof=0) de las 32 variables en TP/FN/FP/TN. También se calcularon y verificaron N, mediana, p05/p95, mínimos, máximos y fracción de ceros.

Perfiles comparables: cero separado más diez cuantiles globales positivos de $d0/dz$ ponderados por pT; altura = pT medio (GeV), color = conteo absoluto con LogNorm entre 1 y 67.551. Todas las páginas comparten bins, cámara y techo Z de 45 GeV. Son descripciones condicionadas por la predicción, no evidencia causal.

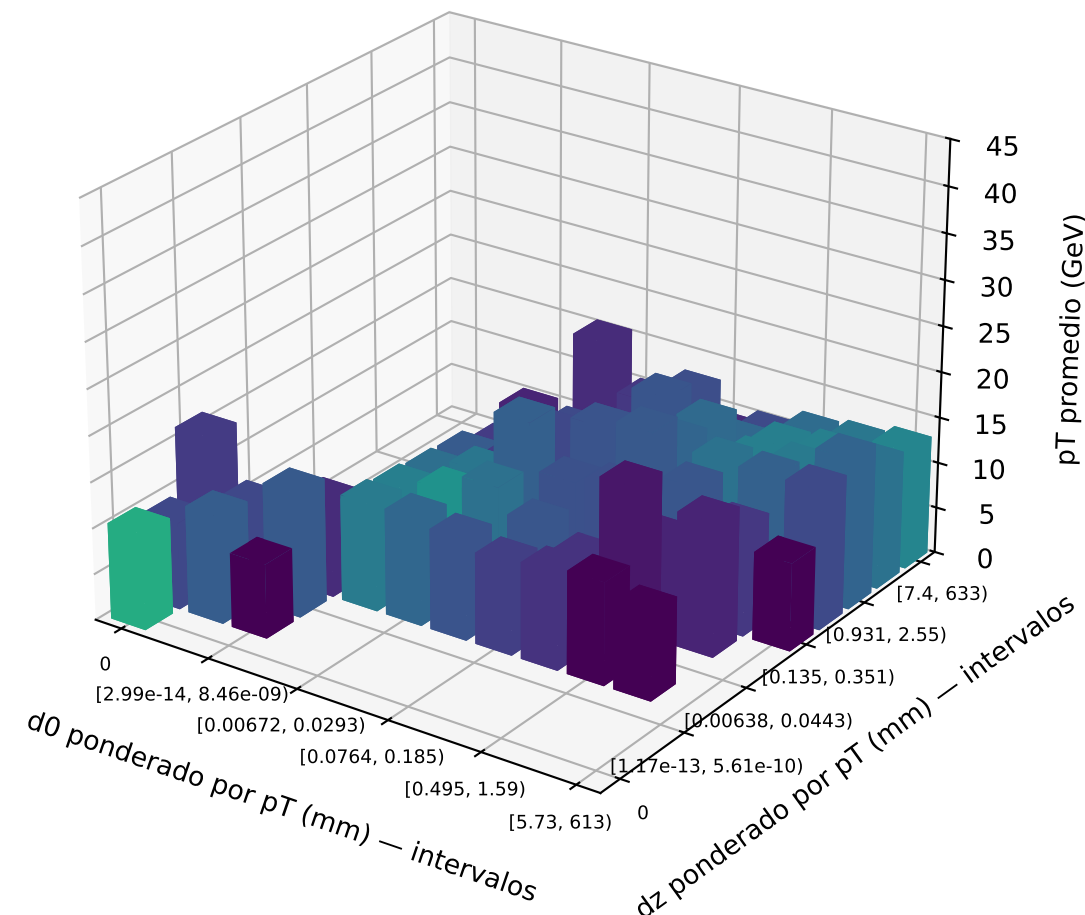
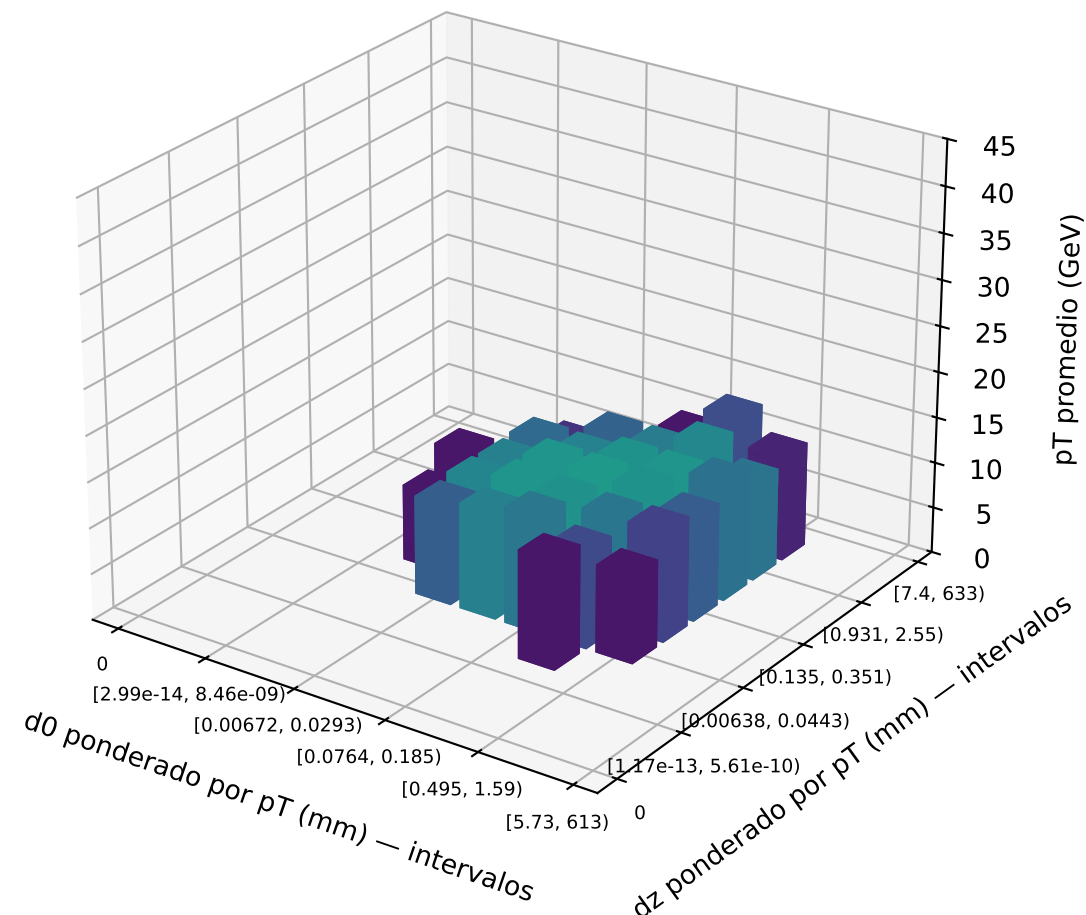
Resultados de ranking global publicados por los productores: AP HGB12 0,5561 frente a RF12 0,5433; ensembles QRF36-HGB 0,5275, QRF48-HGB 0,5512 y RFF48-HGB 0,5603. La mejora no es uniforme: HGB12 y RFF48-HGB encabezan esos controles; añadir ramas/mapas no mejora todas las familias. AP/AUC son métricas de ranking, no métricas al umbral 0,8.

La rama cuántica lateral propuesta tras el pooling de HMPNet es una hipótesis de diseño, no un resultado aquí. HMPNet/híbrido aún no verificados no se presentan como completados. RF32 es una referencia local existente, no un ajuste nuevo.

Perfil de pT medio en intervalos de d0/dz — VQC8 · ensemble

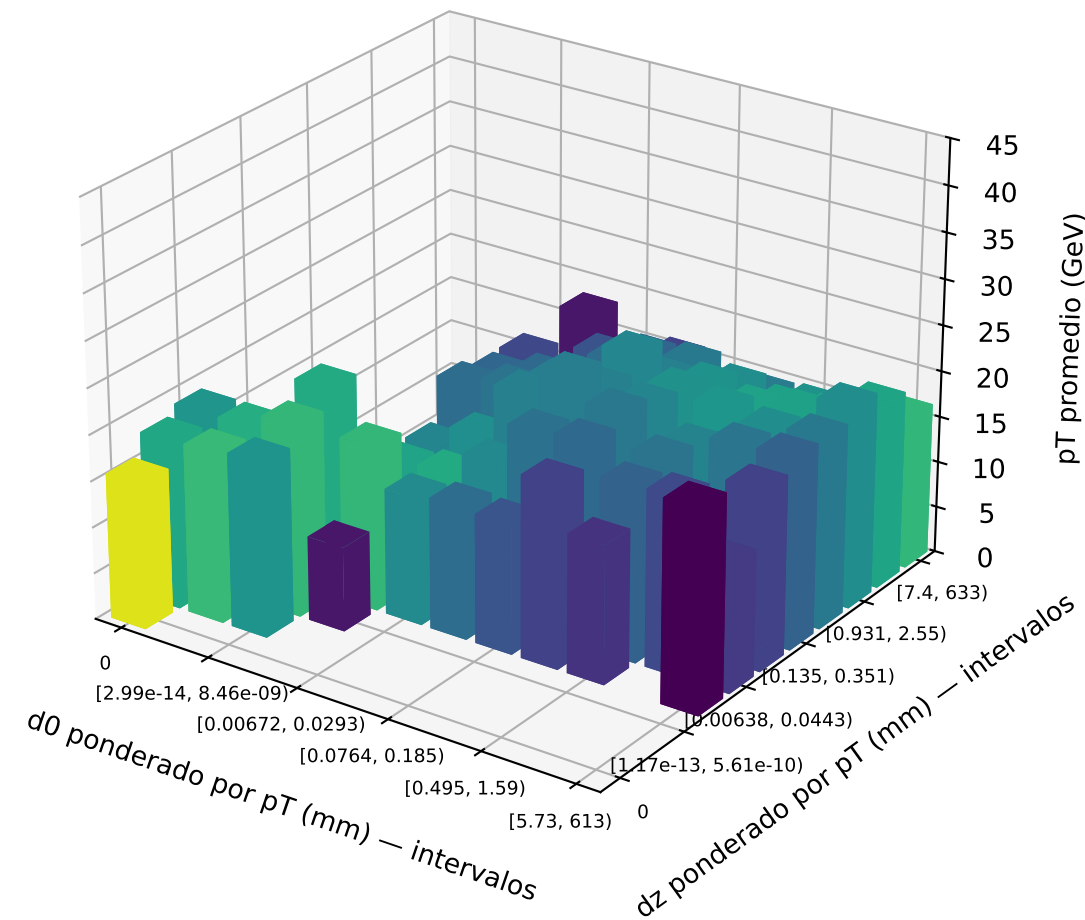
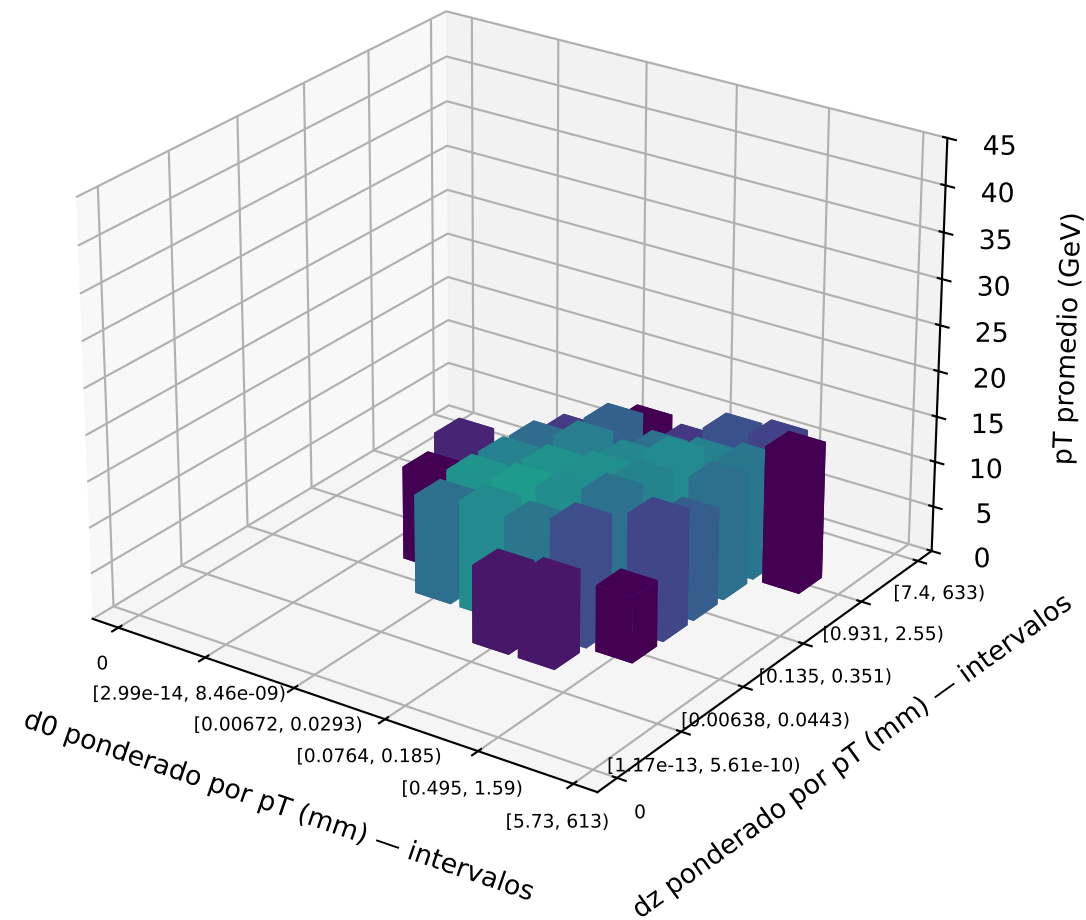
TP — N=3,513

FN — N=3,162



FP — N=3,175

TN — N=57,701



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · ensemble

Variable	Unidad	TP N=3,513	FN N=3,162	FP N=3,175	TN N=57,701
jet_pt	GeV	12.23 ± 6.325	12.3 ± 6.966	12.1 ± 5.807	16.65 ± 11.85
jet_eta	1	0.001792 ± 1.398	0.003268 ± 1.434	-0.007309 ± 1.457	0.003244 ± 1.498
jet_abs_eta	1	1.205 ± 0.7081	1.227 ± 0.742	1.258 ± 0.7354	1.276 ± 0.7838
jet_mass	GeV	2.513 ± 1.374	2.549 ± 1.698	2.392 ± 1.196	2.877 ± 1.894
jet_ncharged	count	2.757 ± 1.537	2.681 ± 2.017	2.423 ± 1.304	2.597 ± 1.958
jet_nneutral	count	1.857 ± 1.6	2.046 ± 1.636	1.858 ± 1.428	2.397 ± 1.799
n_tracks_cone	count	2.785 ± 1.521	2.724 ± 2.022	2.447 ± 1.288	2.626 ± 1.958
track_pt_sum	GeV	5.783 ± 4.472	5.017 ± 5.273	5.126 ± 3.83	6.844 ± 8.217
track_pt_sum_over_jet_pt	1	0.4747 ± 0.2237	0.3895 ± 0.2697	0.4256 ± 0.225	0.3717 ± 0.2684
avg_track_pt	GeV	2.247 ± 1.413	1.607 ± 1.245	2.135 ± 1.222	2.167 ± 2.091
std_track_pt	GeV	0.7893 ± 0.8669	0.678 ± 0.8643	0.6658 ± 0.7858	1.023 ± 1.627
n_tracks_below_avg_pt	count	1.426 ± 1.141	1.452 ± 1.378	1.177 ± 0.9873	1.389 ± 1.369
n_tracks_above_avg_pt	count	1.359 ± 0.6416	1.272 ± 0.8699	1.27 ± 0.5317	1.237 ± 0.8139
max_pt_ratio	1	0.2845 ± 0.161	0.2129 ± 0.1605	0.2565 ± 0.1347	0.2097 ± 0.1559
min_pt_ratio	1	0.1361 ± 0.1466	0.08613 ± 0.1078	0.1344 ± 0.1145	0.08389 ± 0.09749
max_dr_ratio	1	0.1604 ± 0.1462	0.1074 ± 0.1133	0.157 ± 0.1171	0.1026 ± 0.1068
min_dr_ratio	1	0.2465 ± 0.1775	0.1779 ± 0.1637	0.2241 ± 0.1442	0.1789 ± 0.1549
dr_max_pt	1	0.1414 ± 0.08528	0.133 ± 0.09934	0.1498 ± 0.08548	0.1259 ± 0.0992
dr_min_pt	1	0.2023 ± 0.1004	0.1906 ± 0.1195	0.2037 ± 0.09471	0.19 ± 0.1178
dr_max_dr	1	0.2397 ± 0.09657	0.2285 ± 0.1232	0.2328 ± 0.08953	0.221 ± 0.1213
dr_min_dr	1	0.1123 ± 0.07407	0.1037 ± 0.08457	0.1262 ± 0.07955	0.1031 ± 0.08826
pt_difference	GeV	1.851 ± 2.116	1.66 ± 2.261	1.532 ± 1.829	2.512 ± 4.111
r50	1	0.1476 ± 0.08264	0.1409 ± 0.09718	0.1562 ± 0.08261	0.1325 ± 0.09758
r95	1	0.2371 ± 0.09604	0.2259 ± 0.1223	0.2307 ± 0.08954	0.2146 ± 0.1199
mean_abs_d0 *	mm	0.1552 ± 0.158	2.484 ± 11.44	0.135 ± 0.176	1.257 ± 7.925
max_abs_d0 *	mm	0.2678 ± 0.2677	6.774 ± 21.94	0.2453 ± 0.3015	3.332 ± 16.67
std_abs_d0 *	mm	0.08614 ± 0.1083	2.612 ± 8.494	0.08619 ± 0.1326	1.282 ± 6.7
mean_abs_dz *	mm	0.3828 ± 0.445	3.679 ± 17.11	0.3392 ± 0.4748	1.674 ± 9.721
max_abs_dz *	mm	0.6529 ± 0.7371	9.59 ± 28.09	0.5968 ± 0.7767	4.387 ± 23.14
std_abs_dz *	mm	0.2079 ± 0.2962	3.642 ± 10.08	0.2052 ± 0.3356	1.7 ± 9.543
pt_weighted_abs_d0 *	mm	0.1637 ± 0.1653	2.332 ± 11.33	0.146 ± 0.1879	1.197 ± 7.937
pt_weighted_abs_dz *	mm	0.4048 ± 0.4725	3.146 ± 16.88	0.3682 ± 0.5189	1.373 ± 8.416

* Variable utilizada como entrada del modelo.

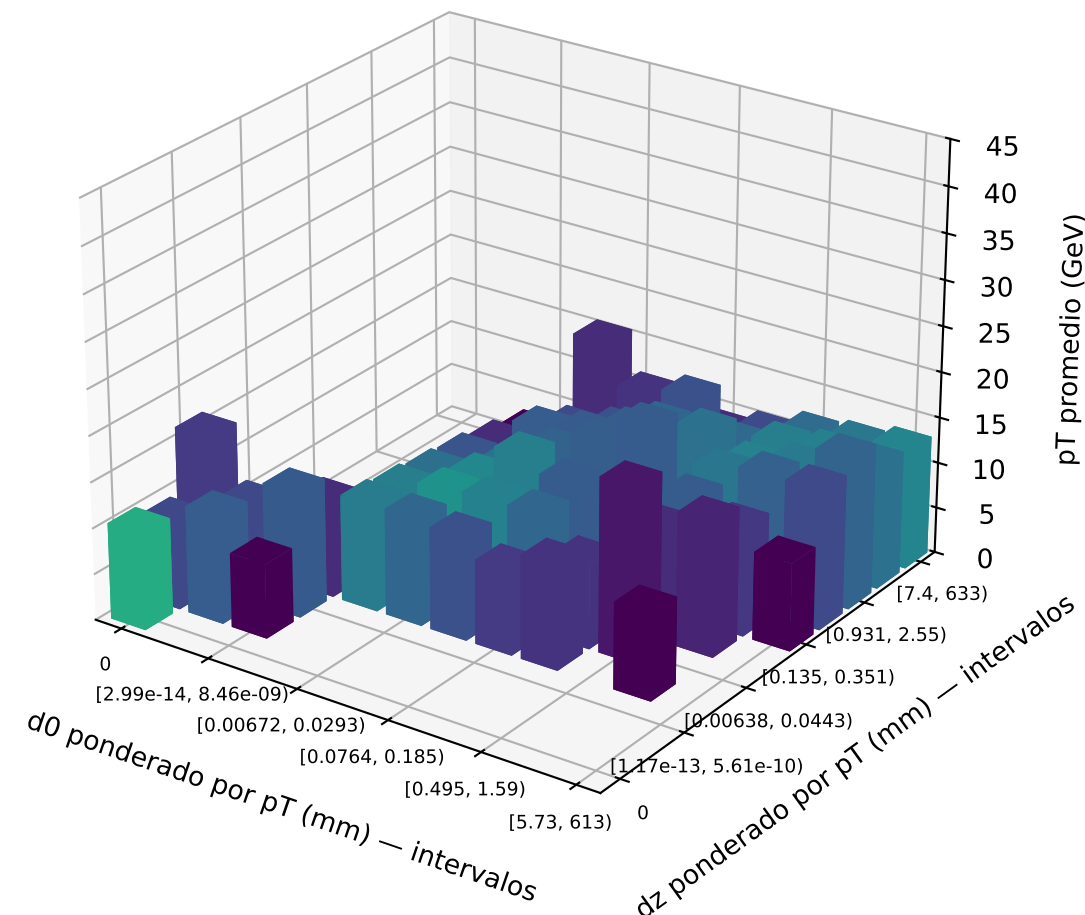
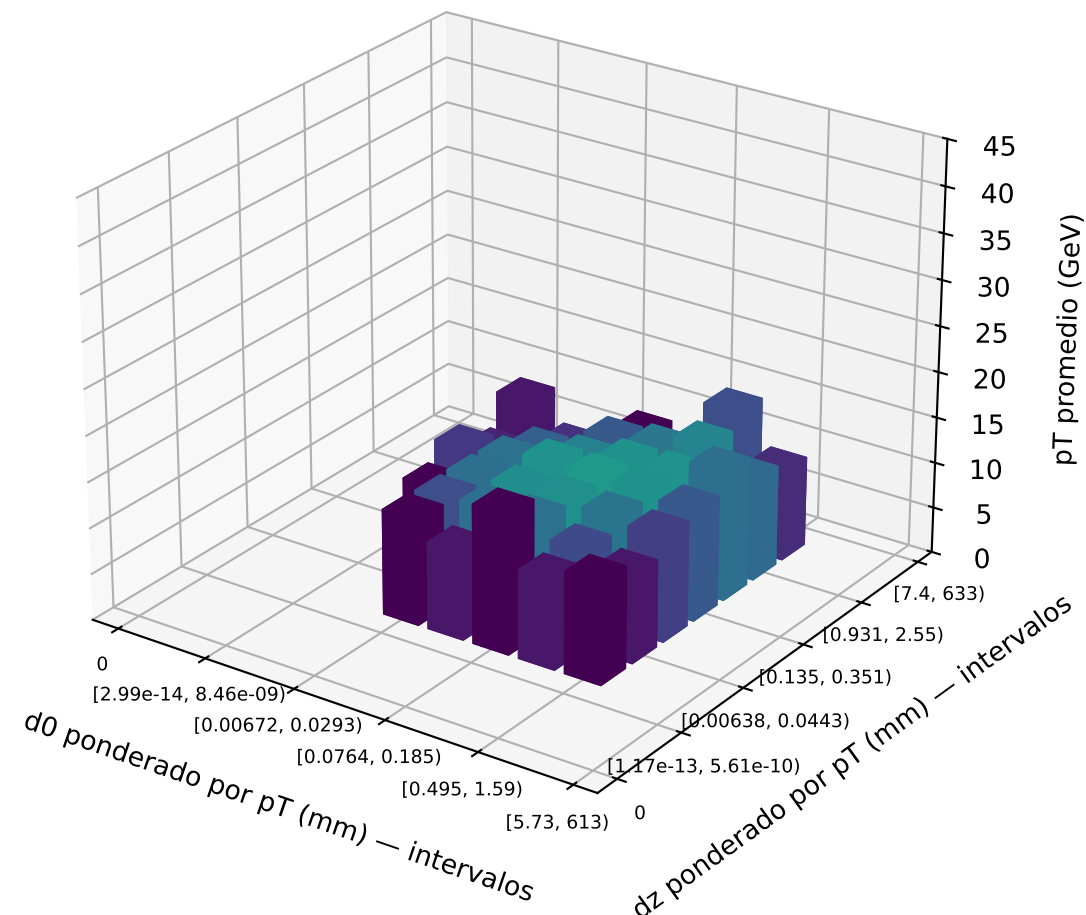
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RF8

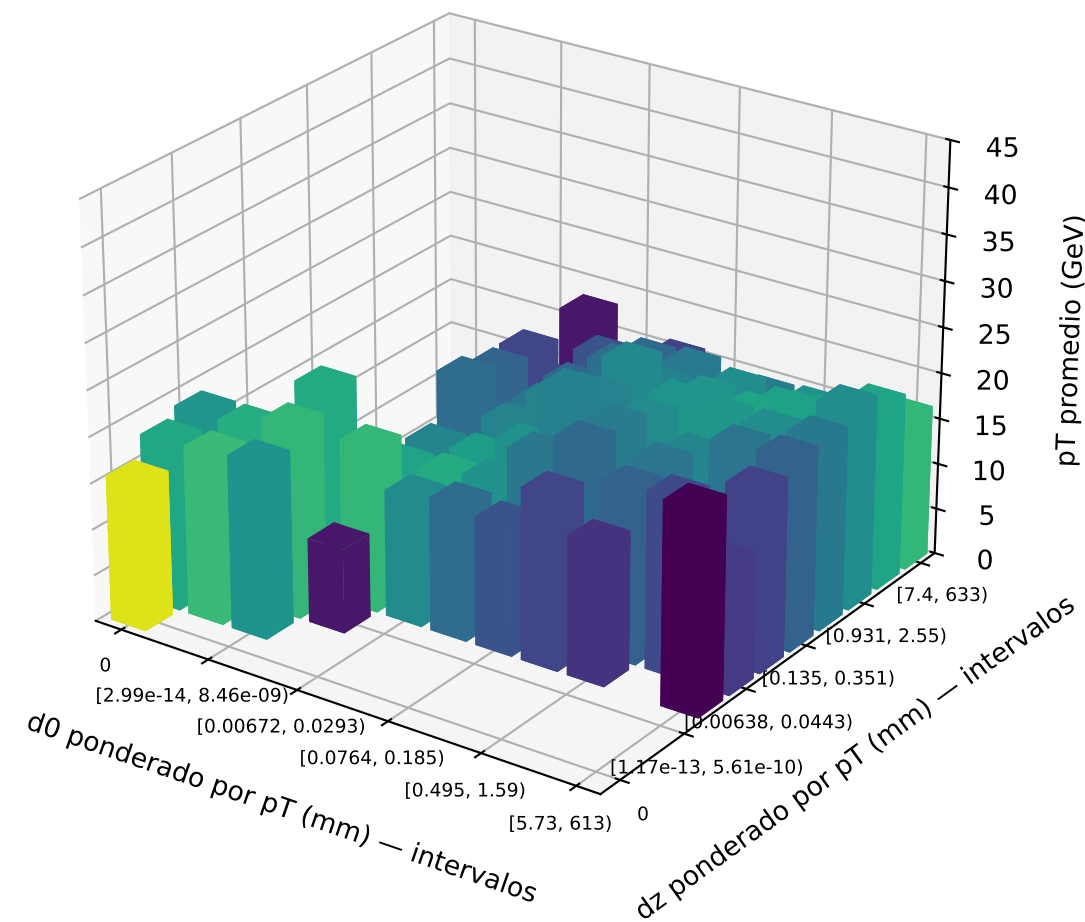
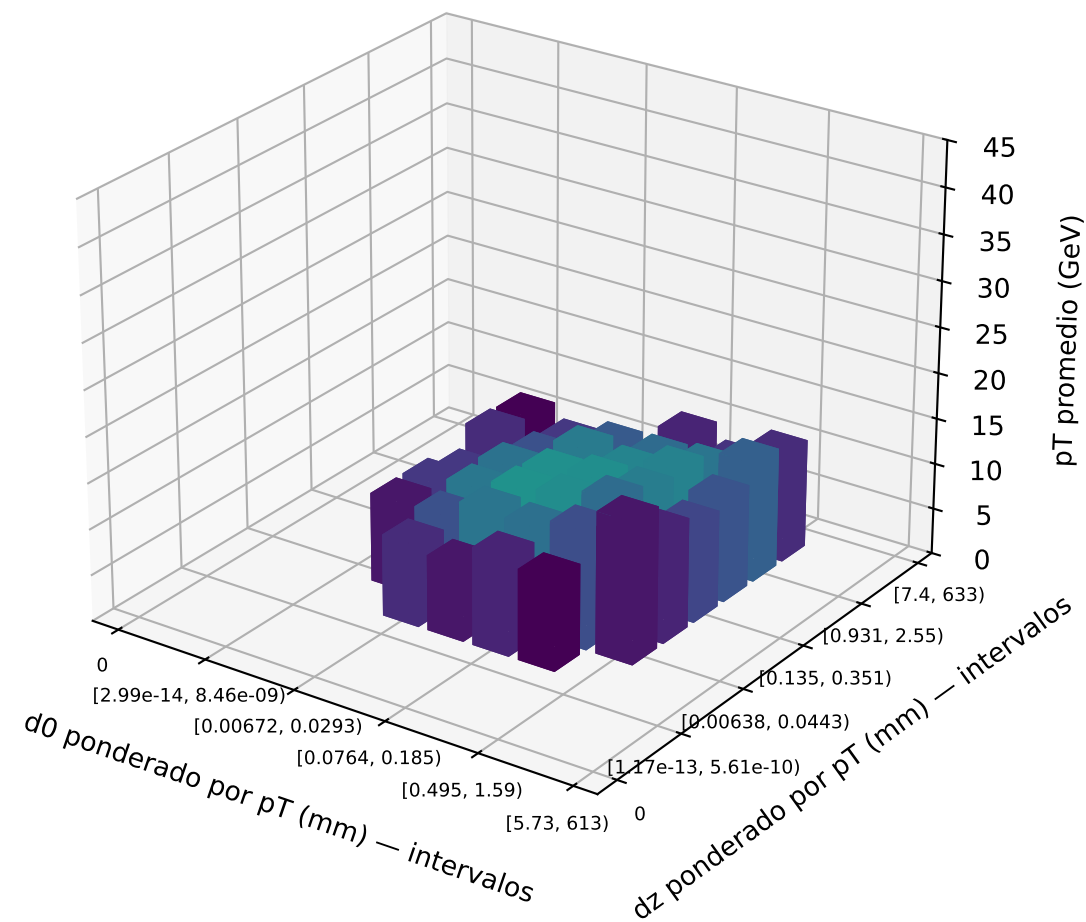
TP — N=2,987

FN — N=3,688



FP — N=2,112

TN — N=58,764



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF8

Variable	Unidad	TP N=2,987	FN N=3,688	FP N=2,112	TN N=58,764
jet_pt	GeV	12.24 ± 6.241	12.28 ± 6.94	11.9 ± 5.522	16.58 ± 11.78
jet_eta	1	0.01025 ± 1.403	-0.003789 ± 1.424	0.01912 ± 1.483	0.002103 ± 1.496
jet_abs_eta	1	1.211 ± 0.7098	1.219 ± 0.736	1.284 ± 0.742	1.275 ± 0.7827
jet_mass	GeV	2.509 ± 1.35	2.547 ± 1.672	2.286 ± 1.112	2.872 ± 1.885
jet_ncharged	count	2.776 ± 1.538	2.676 ± 1.954	2.295 ± 1.248	2.598 ± 1.949
jet_nneutral	count	1.845 ± 1.586	2.03 ± 1.643	1.835 ± 1.387	2.388 ± 1.796
n_tracks_cone	count	2.798 ± 1.523	2.722 ± 1.957	2.313 ± 1.234	2.627 ± 1.949
track_pt_sum	GeV	5.862 ± 4.583	5.062 ± 5.084	4.955 ± 3.686	6.819 ± 8.163
track_pt_sum_over_jet_pt	1	0.4794 ± 0.2243	0.3978 ± 0.2638	0.419 ± 0.2287	0.3729 ± 0.2677
avg_track_pt	GeV	2.269 ± 1.43	1.681 ± 1.267	2.175 ± 1.231	2.165 ± 2.078
std_track_pt	GeV	0.788 ± 0.8746	0.695 ± 0.8594	0.6446 ± 0.761	1.017 ± 1.617
n_tracks_below_avg_pt	count	1.431 ± 1.14	1.444 ± 1.347	1.085 ± 0.9667	1.389 ± 1.363
n_tracks_above_avg_pt	count	1.367 ± 0.6474	1.278 ± 0.8374	1.229 ± 0.4907	1.239 ± 0.8107
max_pt_ratio	1	0.285 ± 0.1601	0.2228 ± 0.1631	0.2607 ± 0.135	0.2104 ± 0.1556
min_pt_ratio	1	0.1374 ± 0.1475	0.09213 ± 0.114	0.141 ± 0.1156	0.08456 ± 0.09788
max_dr_ratio	1	0.1623 ± 0.1473	0.1134 ± 0.1183	0.1622 ± 0.1178	0.1034 ± 0.1071
min_dr_ratio	1	0.2464 ± 0.1765	0.1877 ± 0.1683	0.2294 ± 0.1435	0.1795 ± 0.1548
dr_max_pt	1	0.1403 ± 0.08417	0.135 ± 0.09833	0.1489 ± 0.08477	0.1263 ± 0.09905
dr_min_pt	1	0.2011 ± 0.1002	0.1932 ± 0.1173	0.2012 ± 0.09407	0.1904 ± 0.1174
dr_max_dr	1	0.2388 ± 0.09644	0.2308 ± 0.1199	0.2267 ± 0.08966	0.2214 ± 0.1208
dr_min_dr	1	0.1113 ± 0.07327	0.1057 ± 0.08384	0.1287 ± 0.08117	0.1034 ± 0.08808
pt_difference	GeV	1.852 ± 2.15	1.686 ± 2.215	1.471 ± 1.76	2.496 ± 4.084
r50	1	0.147 ± 0.08168	0.1423 ± 0.09597	0.154 ± 0.08171	0.133 ± 0.09742
r95	1	0.2362 ± 0.09595	0.2282 ± 0.1191	0.2247 ± 0.0898	0.2151 ± 0.1194
mean_abs_d0 *	mm	0.1638 ± 0.1542	2.145 ± 10.62	0.1285 ± 0.1401	1.237 ± 7.854
max_abs_d0 *	mm	0.2771 ± 0.2871	5.839 ± 20.44	0.2083 ± 0.2123	3.277 ± 16.53
std_abs_d0 *	mm	0.0867 ± 0.1148	2.251 ± 7.915	0.06464 ± 0.08748	1.261 ± 6.641
mean_abs_dz *	mm	0.4011 ± 0.4659	3.194 ± 15.89	0.3367 ± 0.4563	1.65 ± 9.634
max_abs_dz *	mm	0.666 ± 0.7531	8.305 ± 26.2	0.5212 ± 0.6278	4.321 ± 22.93
std_abs_dz *	mm	0.2031 ± 0.2772	3.156 ± 9.411	0.1546 ± 0.2395	1.675 ± 9.459
pt_weighted_abs_d0 *	mm	0.1693 ± 0.1516	2.018 ± 10.52	0.133 ± 0.1361	1.179 ± 7.866
pt_weighted_abs_dz *	mm	0.4164 ± 0.4755	2.745 ± 15.66	0.3486 ± 0.4483	1.355 ± 8.341

* Variable utilizada como entrada del modelo.

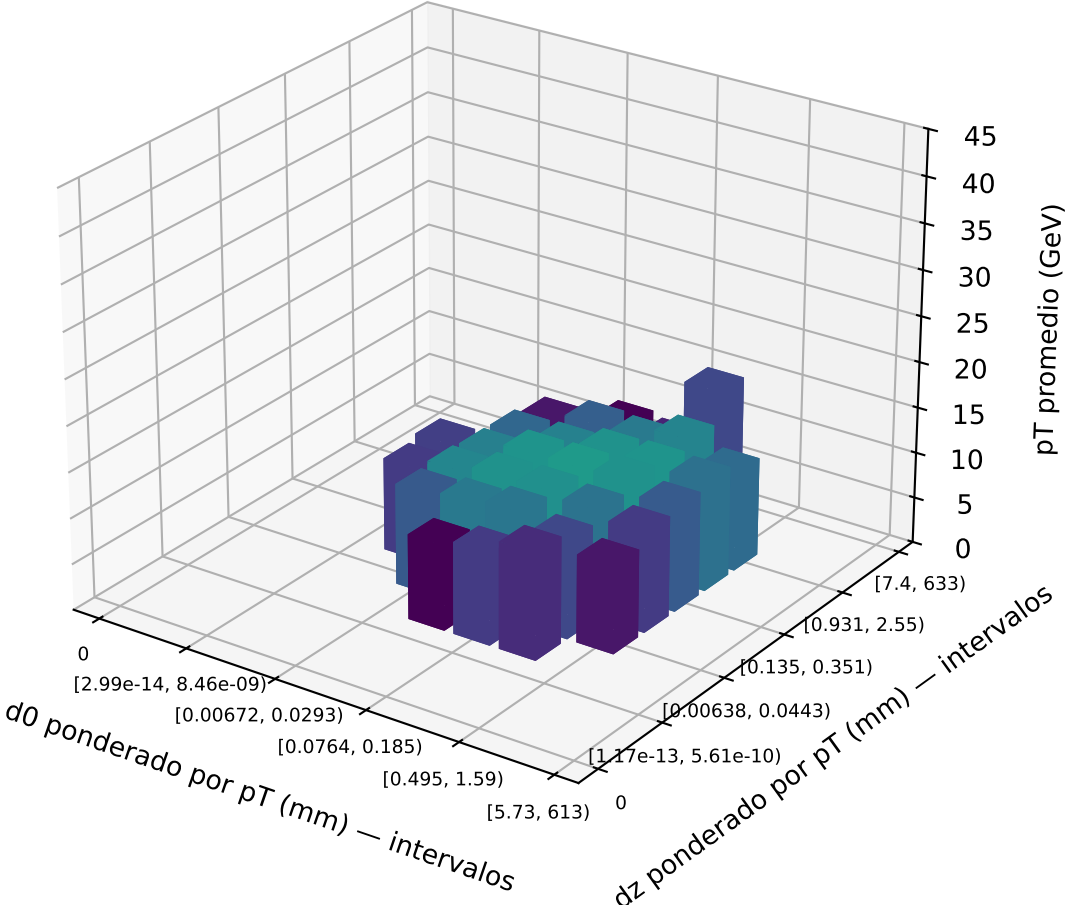
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

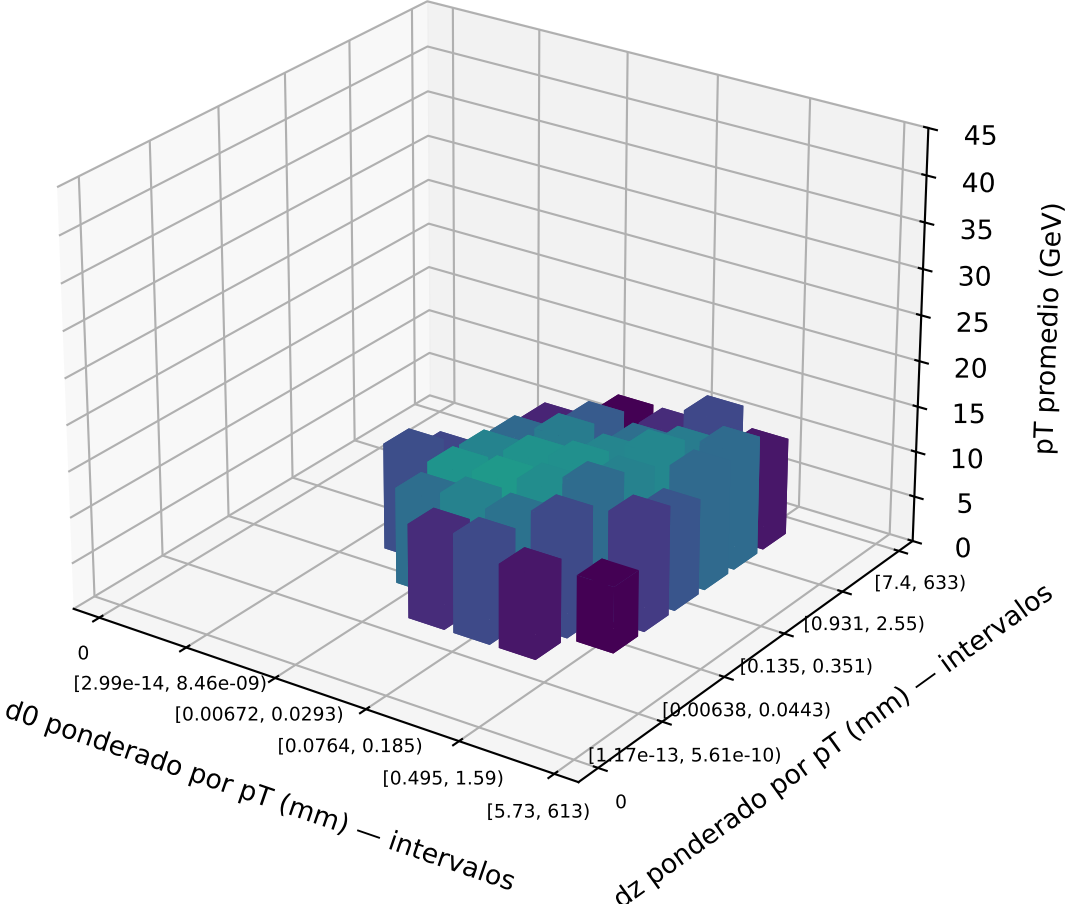
Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 42

TP — N=3,339

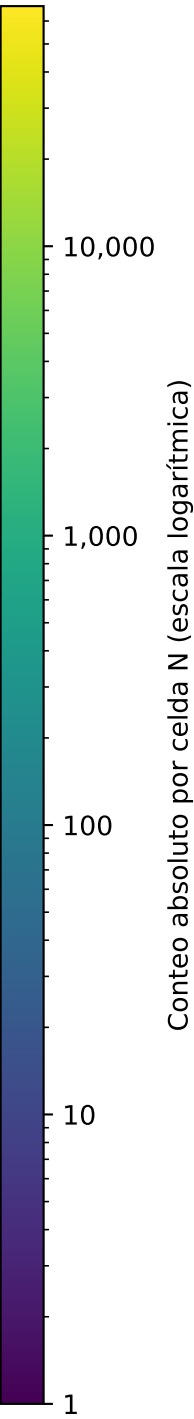
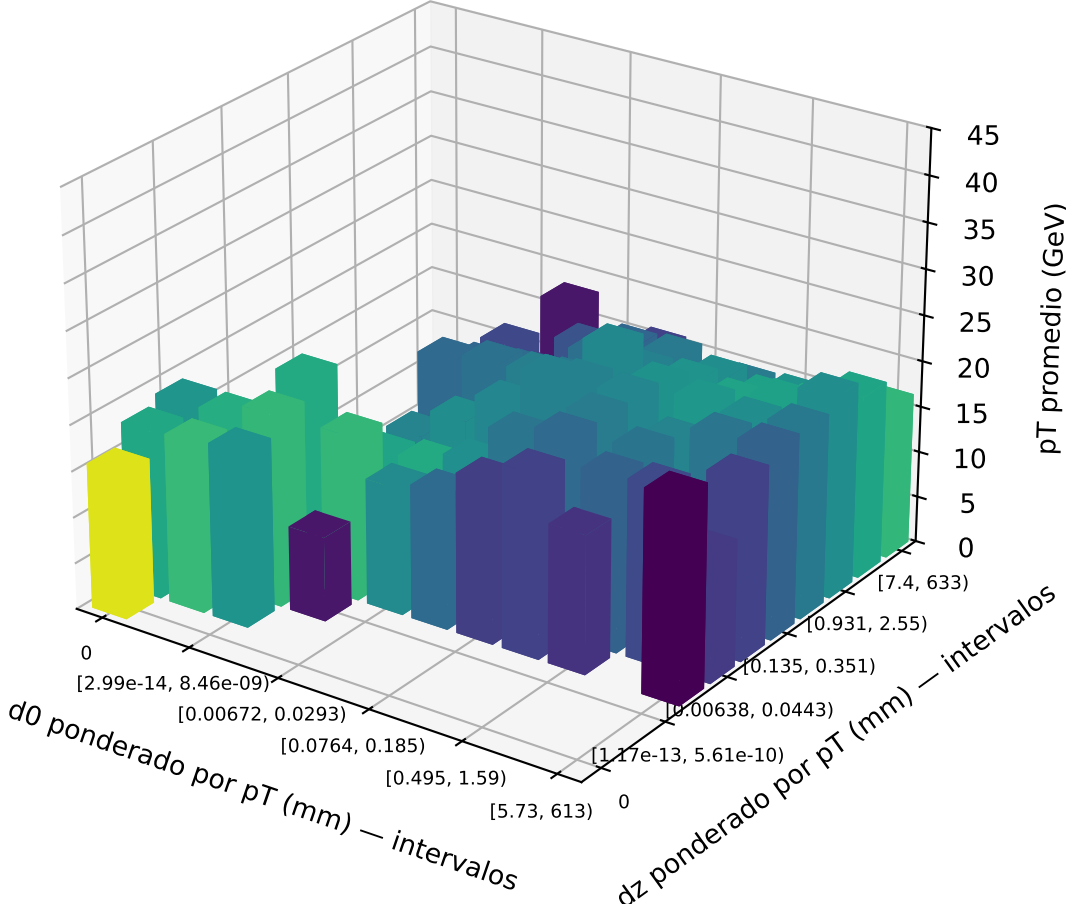
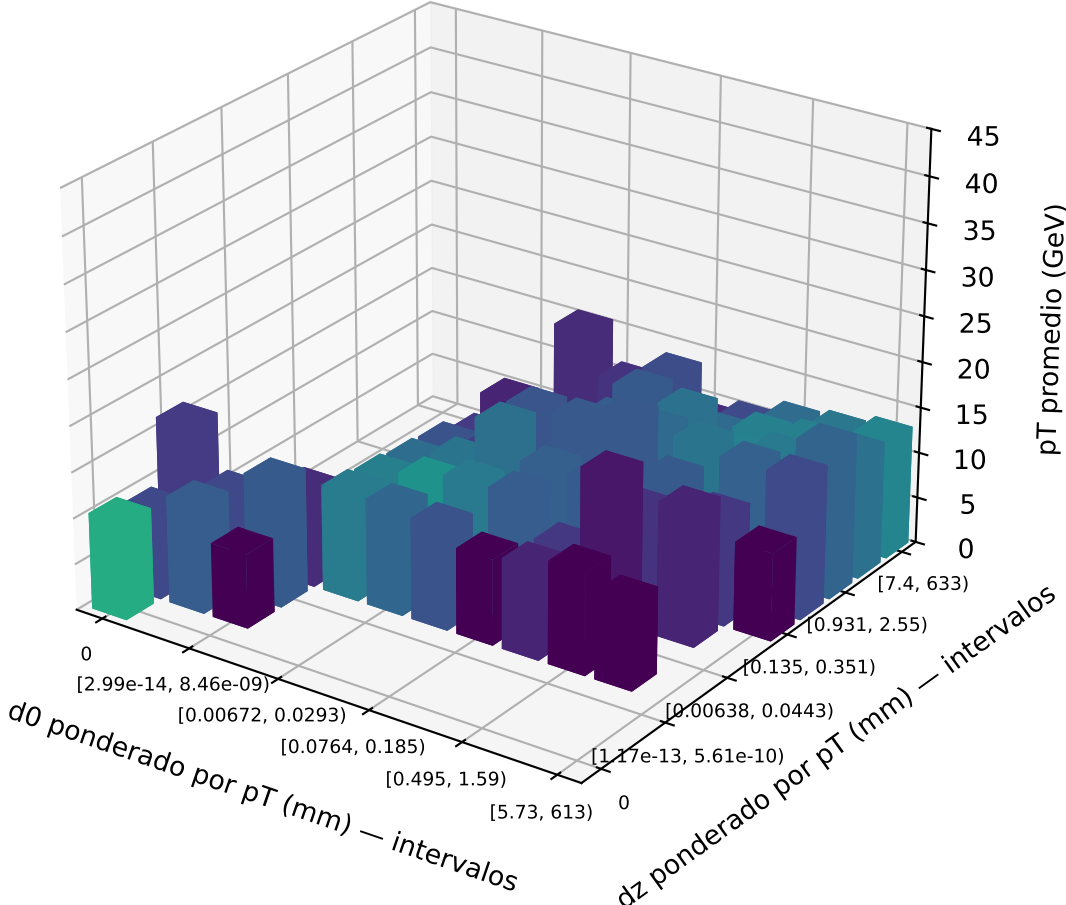
FN — N=3,336



FP — N=2,886



TN — N=57,990



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 42

Variable	Unidad	TP N=3,339	FN N=3,336	FP N=2,886	TN N=57,990
jet_pt	GeV	12.11 ± 6.142	12.42 ± 7.093	11.85 ± 5.49	16.64 ± 11.83
jet_eta	1	0.01025 ± 1.399	-0.005271 ± 1.43	-0.006923 ± 1.486	0.003172 ± 1.496
jet_abs_eta	1	1.207 ± 0.7077	1.223 ± 0.7407	1.29 ± 0.7362	1.275 ± 0.7835
jet_mass	GeV	2.475 ± 1.344	2.585 ± 1.706	2.305 ± 1.118	2.879 ± 1.892
jet_ncharged	count	2.697 ± 1.504	2.744 ± 2.02	2.292 ± 1.227	2.602 ± 1.957
jet_nneutral	count	1.844 ± 1.588	2.05 ± 1.645	1.859 ± 1.409	2.395 ± 1.799
n_tracks_cone	count	2.721 ± 1.487	2.792 ± 2.024	2.316 ± 1.209	2.631 ± 1.957
track_pt_sum	GeV	5.694 ± 4.352	5.146 ± 5.348	4.914 ± 3.658	6.846 ± 8.204
track_pt_sum_over_jet_pt	1	0.472 ± 0.2239	0.3966 ± 0.2688	0.418 ± 0.2255	0.3724 ± 0.2683
avg_track_pt	GeV	2.272 ± 1.437	1.616 ± 1.222	2.17 ± 1.251	2.166 ± 2.087
std_track_pt	GeV	0.7711 ± 0.8611	0.7021 ± 0.8724	0.6347 ± 0.7759	1.023 ± 1.624
n_tracks_below_avg_pt	count	1.378 ± 1.121	1.499 ± 1.38	1.084 ± 0.9368	1.393 ± 1.368
n_tracks_above_avg_pt	count	1.343 ± 0.6202	1.293 ± 0.8764	1.232 ± 0.4953	1.239 ± 0.8139
max_pt_ratio	1	0.2855 ± 0.1612	0.2157 ± 0.1607	0.2598 ± 0.1376	0.2098 ± 0.1556
min_pt_ratio	1	0.1403 ± 0.1491	0.08449 ± 0.1052	0.1423 ± 0.1205	0.08375 ± 0.09707
max_dr_ratio	1	0.1646 ± 0.1485	0.106 ± 0.1108	0.1644 ± 0.1225	0.1025 ± 0.1065
min_dr_ratio	1	0.2485 ± 0.1774	0.1795 ± 0.1644	0.2286 ± 0.1466	0.1789 ± 0.1547
dr_max_pt	1	0.1411 ± 0.08529	0.1337 ± 0.09868	0.1503 ± 0.08499	0.126 ± 0.09917
dr_min_pt	1	0.2008 ± 0.1005	0.1926 ± 0.1186	0.2006 ± 0.09458	0.1902 ± 0.1177
dr_max_dr	1	0.2367 ± 0.09697	0.2321 ± 0.1218	0.2276 ± 0.09001	0.2213 ± 0.1212
dr_min_dr	1	0.1129 ± 0.07475	0.1035 ± 0.08341	0.1283 ± 0.08031	0.1031 ± 0.08817
pt_difference	GeV	1.8 ± 2.096	1.721 ± 2.275	1.445 ± 1.793	2.511 ± 4.103
r50	1	0.1472 ± 0.08284	0.1415 ± 0.09634	0.1556 ± 0.08248	0.1327 ± 0.09754
r95	1	0.2342 ± 0.0964	0.2294 ± 0.121	0.2256 ± 0.09016	0.215 ± 0.1197
mean_abs_d0 *	mm	0.1497 ± 0.146	2.368 ± 11.14	0.122 ± 0.1586	1.252 ± 7.905
max_abs_d0 *	mm	0.254 ± 0.2454	6.448 ± 21.4	0.2073 ± 0.2492	3.318 ± 16.63
std_abs_d0 *	mm	0.08071 ± 0.09995	2.485 ± 8.287	0.06915 ± 0.1077	1.277 ± 6.684
mean_abs_dz *	mm	0.3666 ± 0.4245	3.523 ± 16.68	0.3045 ± 0.4139	1.669 ± 9.697
max_abs_dz *	mm	0.6079 ± 0.6617	9.169 ± 27.4	0.5152 ± 0.6636	4.372 ± 23.08
std_abs_dz *	mm	0.1908 ± 0.2629	3.48 ± 9.839	0.1743 ± 0.2915	1.695 ± 9.52
pt_weighted_abs_d0 *	mm	0.1572 ± 0.1504	2.225 ± 11.04	0.1308 ± 0.1672	1.193 ± 7.917
pt_weighted_abs_dz *	mm	0.3904 ± 0.4558	3.017 ± 16.44	0.332 ± 0.462	1.37 ± 8.395

* Variable utilizada como entrada del modelo.

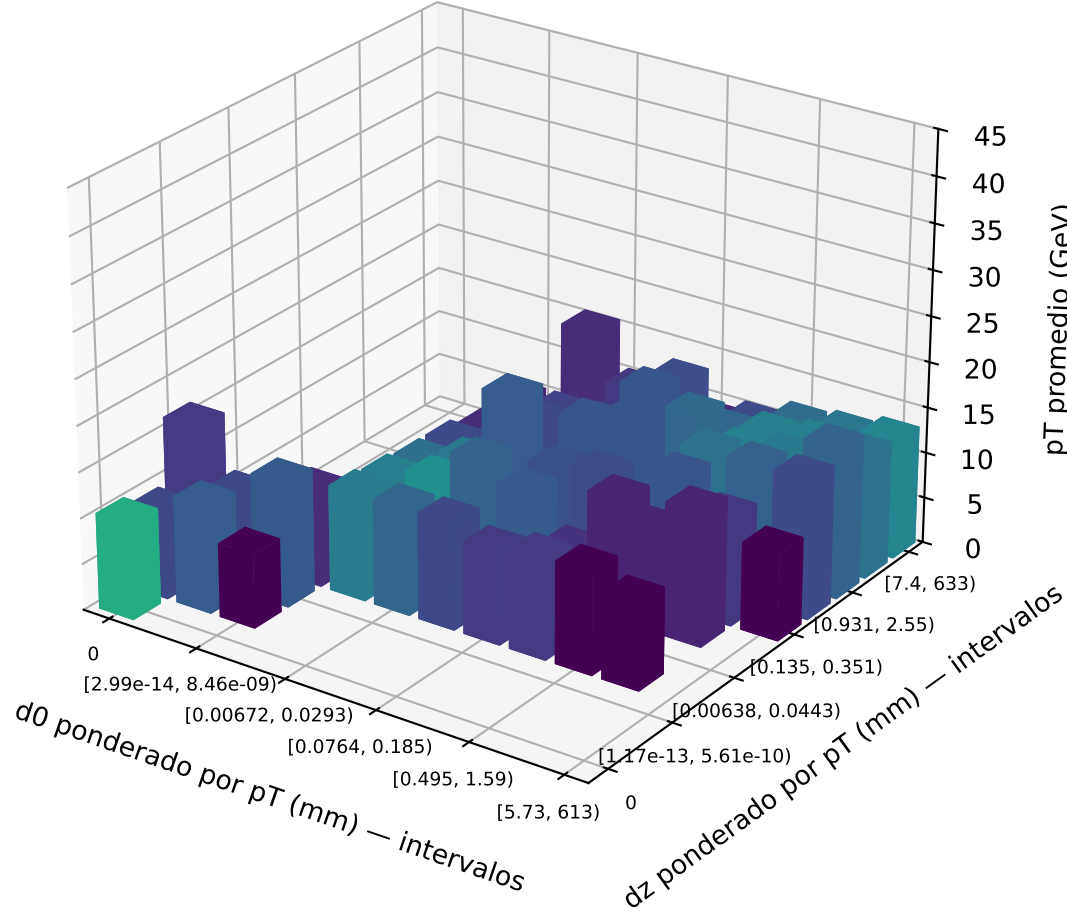
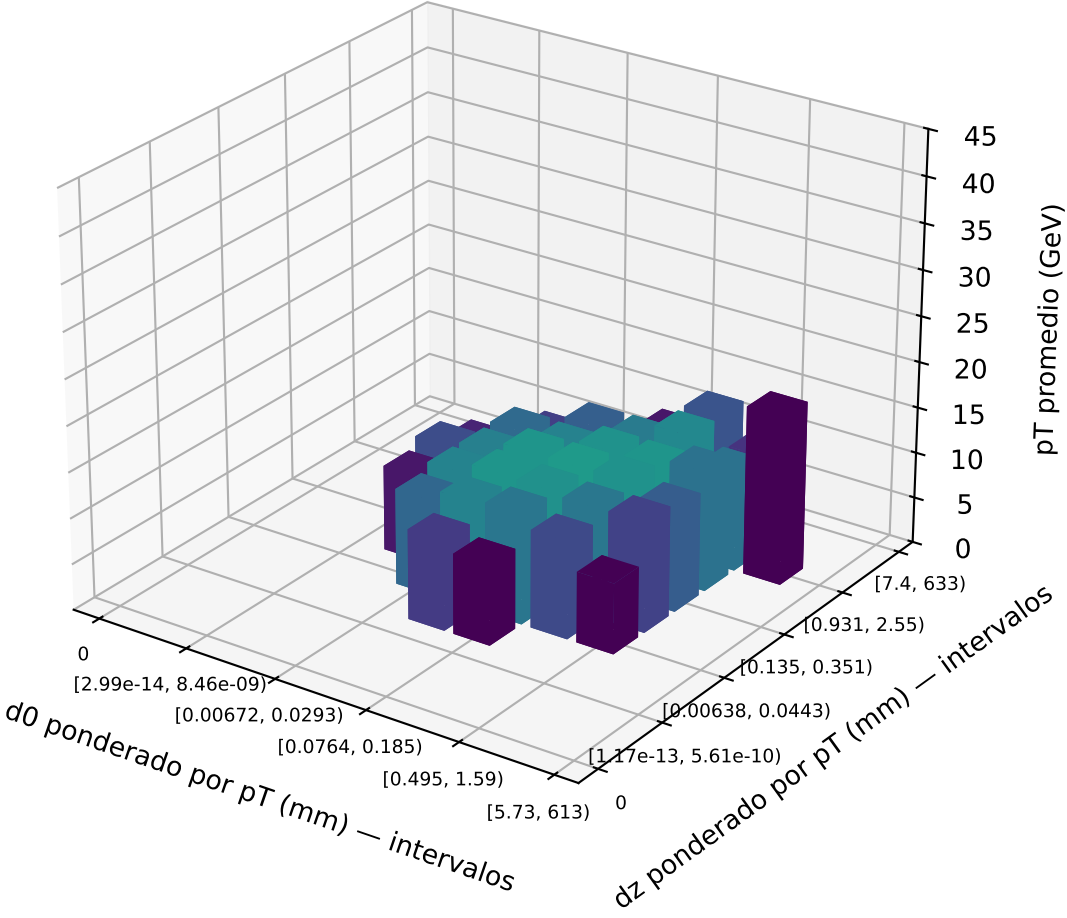
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Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 73

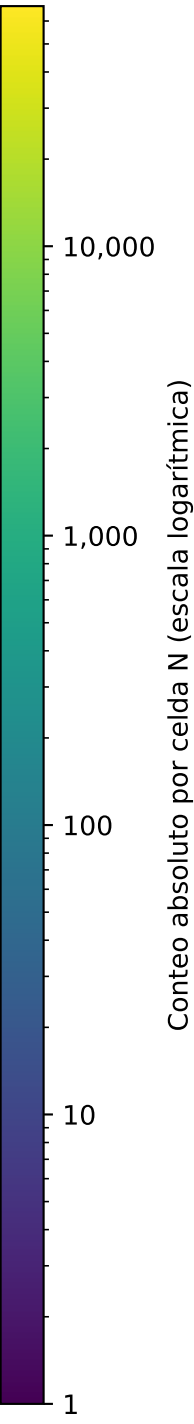
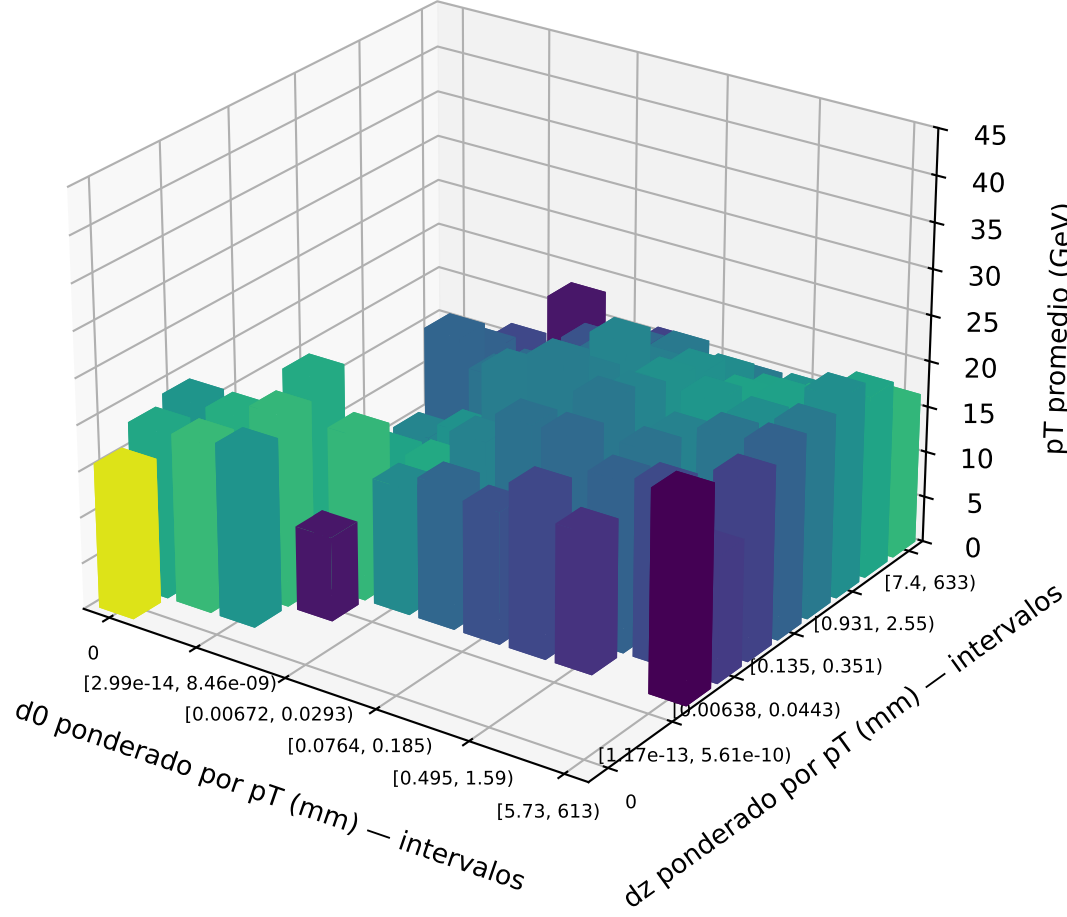
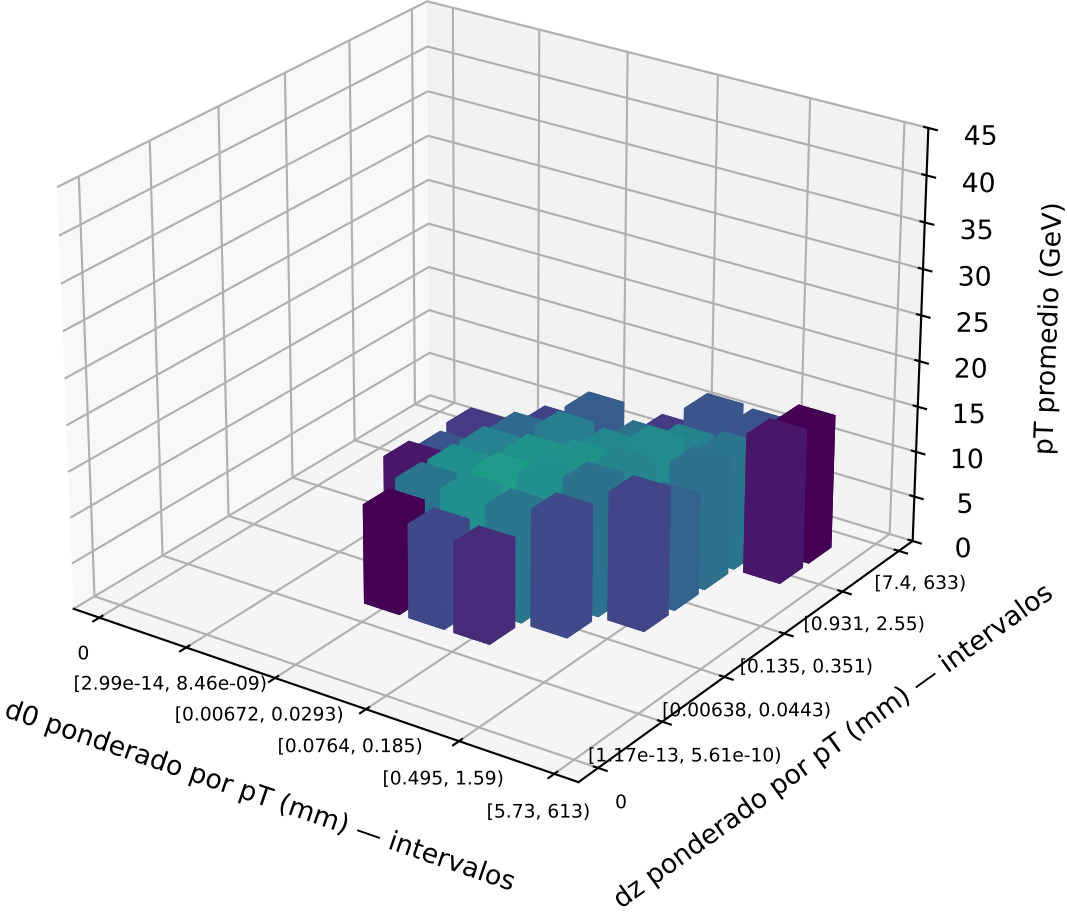
TP — N=3,562

FN — N=3,113



FP — N=3,294

TN — N=57,582



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 73

Variable	Unidad	TP N=3,562	FN N=3,113	FP N=3,294	TN N=57,582
jet_pt	GeV	12.21 ± 6.285	12.32 ± 7.017	12.21 ± 5.965	16.66 ± 11.85
jet_eta	1	-0.000559 ± 1.391	0.005981 ± 1.441	-0.000785 ± 1.44	0.002892 ± 1.499
jet_abs_eta	1	1.198 ± 0.7076	1.235 ± 0.7427	1.239 ± 0.7344	1.278 ± 0.7839
jet_mass	GeV	2.512 ± 1.367	2.55 ± 1.709	2.427 ± 1.228	2.876 ± 1.894
jet_ncharged	count	2.763 ± 1.546	2.672 ± 2.015	2.468 ± 1.346	2.595 ± 1.957
jet_nneutral	count	1.847 ± 1.599	2.061 ± 1.637	1.84 ± 1.421	2.4 ± 1.8
n_tracks_cone	count	2.793 ± 1.535	2.713 ± 2.016	2.49 ± 1.332	2.624 ± 1.958
track_pt_sum	GeV	5.781 ± 4.456	5.006 ± 5.299	5.25 ± 3.999	6.84 ± 8.22
track_pt_sum_over_jet_pt	1	0.4755 ± 0.2237	0.3873 ± 0.2699	0.4298 ± 0.2257	0.3714 ± 0.2684
avg_track_pt	GeV	2.243 ± 1.4	1.602 ± 1.258	2.149 ± 1.227	2.167 ± 2.092
std_track_pt	GeV	0.7886 ± 0.8653	0.6771 ± 0.8661	0.6803 ± 0.8035	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.431 ± 1.148	1.446 ± 1.375	1.206 ± 1.017	1.388 ± 1.369
n_tracks_above_avg_pt	count	1.362 ± 0.6482	1.267 ± 0.8669	1.284 ± 0.5455	1.236 ± 0.8138
max_pt_ratio	1	0.285 ± 0.1614	0.2113 ± 0.1595	0.2576 ± 0.135	0.2095 ± 0.1559
min_pt_ratio	1	0.1362 ± 0.147	0.08517 ± 0.1061	0.1337 ± 0.1151	0.08382 ± 0.0974
max_dr_ratio	1	0.1605 ± 0.1466	0.1064 ± 0.1119	0.1564 ± 0.1176	0.1025 ± 0.1068
min_dr_ratio	1	0.2469 ± 0.178	0.1764 ± 0.1624	0.2246 ± 0.1447	0.1788 ± 0.1548
dr_max_pt	1	0.1416 ± 0.08534	0.1326 ± 0.09945	0.1486 ± 0.08535	0.1259 ± 0.09924
dr_min_pt	1	0.2022 ± 0.1007	0.1904 ± 0.1195	0.2034 ± 0.09506	0.19 ± 0.1178
dr_max_dr	1	0.2402 ± 0.09672	0.2278 ± 0.1233	0.2331 ± 0.0899	0.2209 ± 0.1213
dr_min_dr	1	0.1123 ± 0.07399	0.1035 ± 0.0848	0.125 ± 0.07947	0.1031 ± 0.08829
pt_difference	GeV	1.85 ± 2.11	1.658 ± 2.269	1.576 ± 1.892	2.511 ± 4.113
r50	1	0.1477 ± 0.0826	0.1406 ± 0.09742	0.1554 ± 0.0826	0.1325 ± 0.09761
r95	1	0.2376 ± 0.09615	0.2252 ± 0.1225	0.2306 ± 0.08985	0.2146 ± 0.1199
mean_abs_d0 *	mm	0.1532 ± 0.1593	2.523 ± 11.52	0.1352 ± 0.1796	1.259 ± 7.933
max_abs_d0 *	mm	0.2672 ± 0.2749	6.877 ± 22.09	0.2561 ± 0.3228	3.338 ± 16.69
std_abs_d0 *	mm	0.08688 ± 0.1099	2.651 ± 8.555	0.09236 ± 0.1404	1.284 ± 6.707
mean_abs_dz *	mm	0.3838 ± 0.4624	3.73 ± 17.24	0.3503 ± 0.5199	1.676 ± 9.73
max_abs_dz *	mm	0.6615 ± 0.7883	9.721 ± 28.29	0.6382 ± 0.8944	4.392 ± 23.16
std_abs_dz *	mm	0.2108 ± 0.3114	3.692 ± 10.15	0.2255 ± 0.3922	1.702 ± 9.553
pt_weighted_abs_d0 *	mm	0.162 ± 0.1665	2.368 ± 11.41	0.1484 ± 0.195	1.199 ± 7.945
pt_weighted_abs_dz *	mm	0.4048 ± 0.487	3.189 ± 17.01	0.3824 ± 0.5699	1.374 ± 8.424

* Variable utilizada como entrada del modelo.

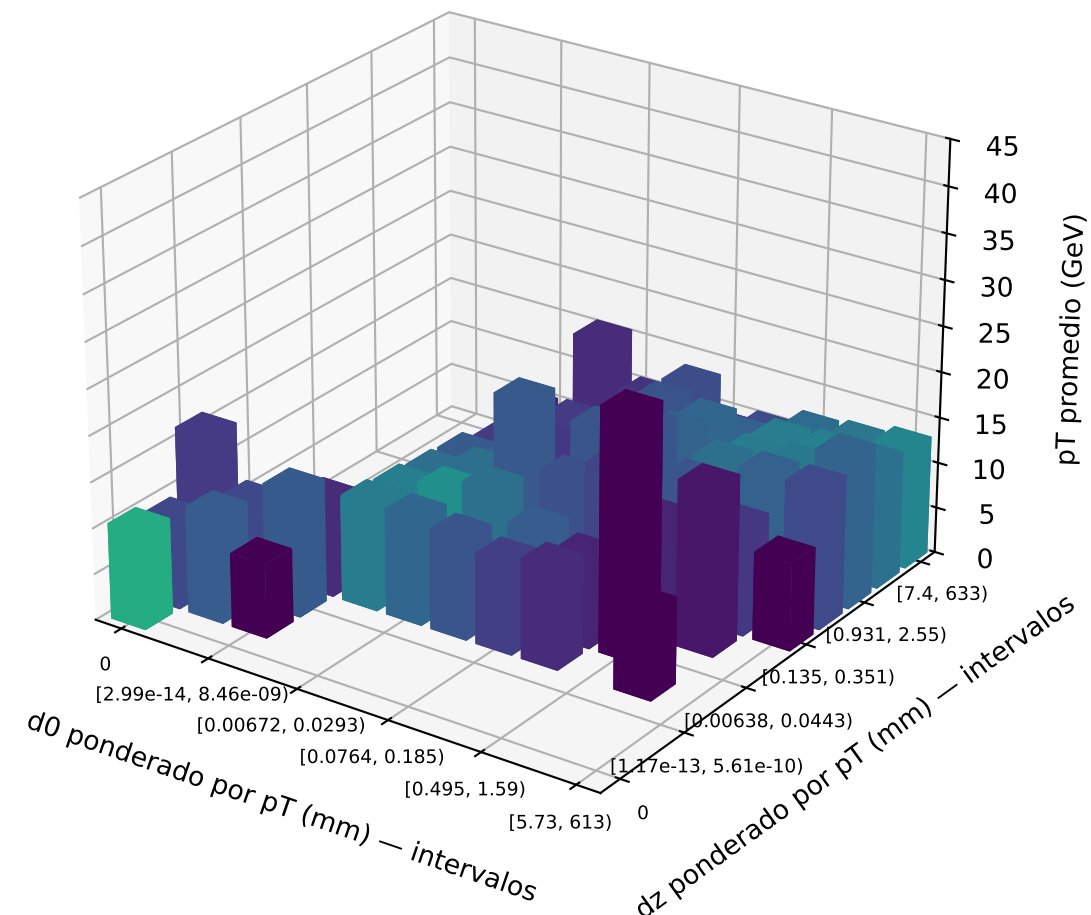
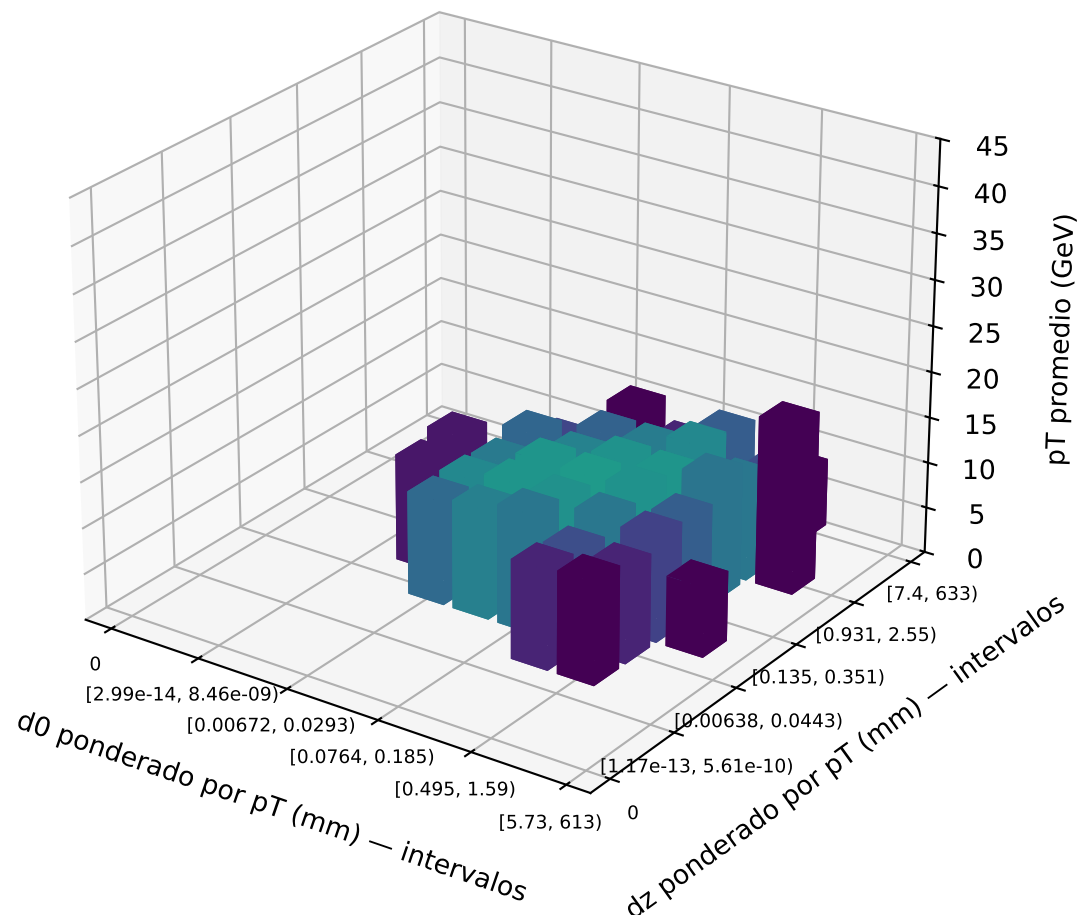
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 101

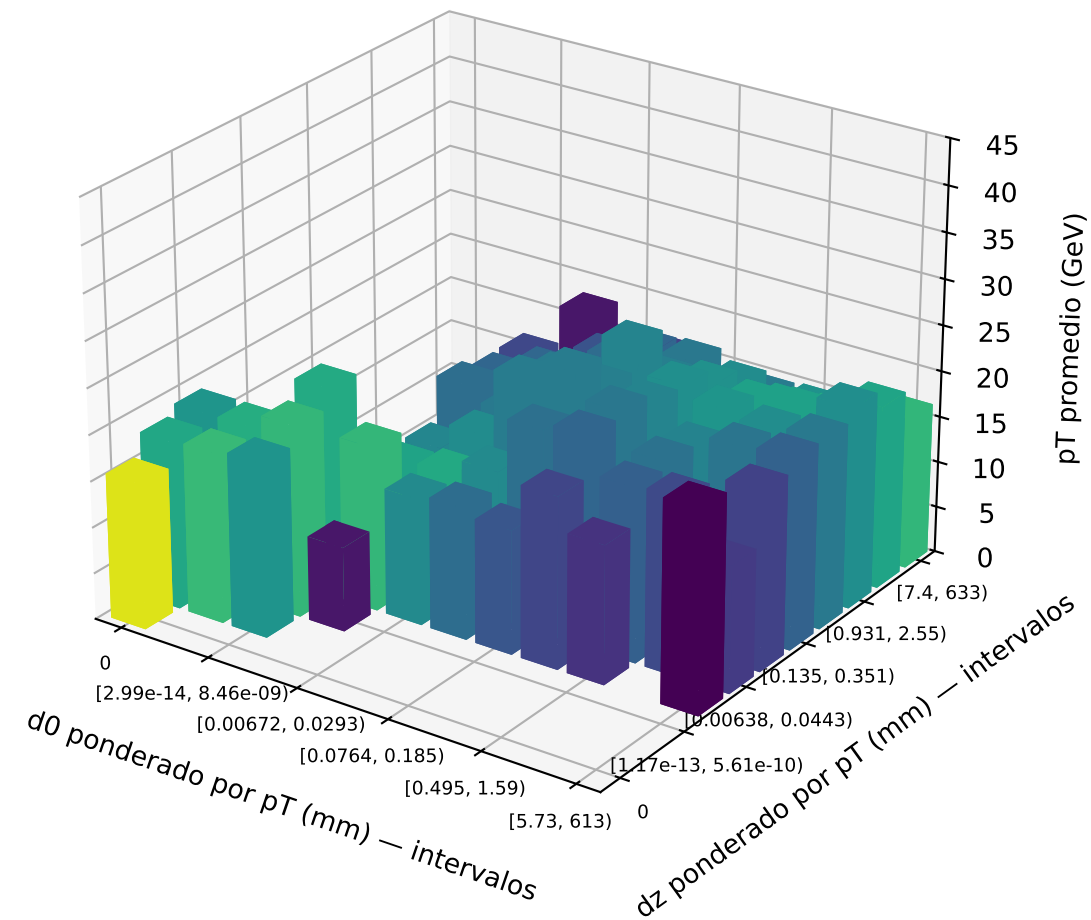
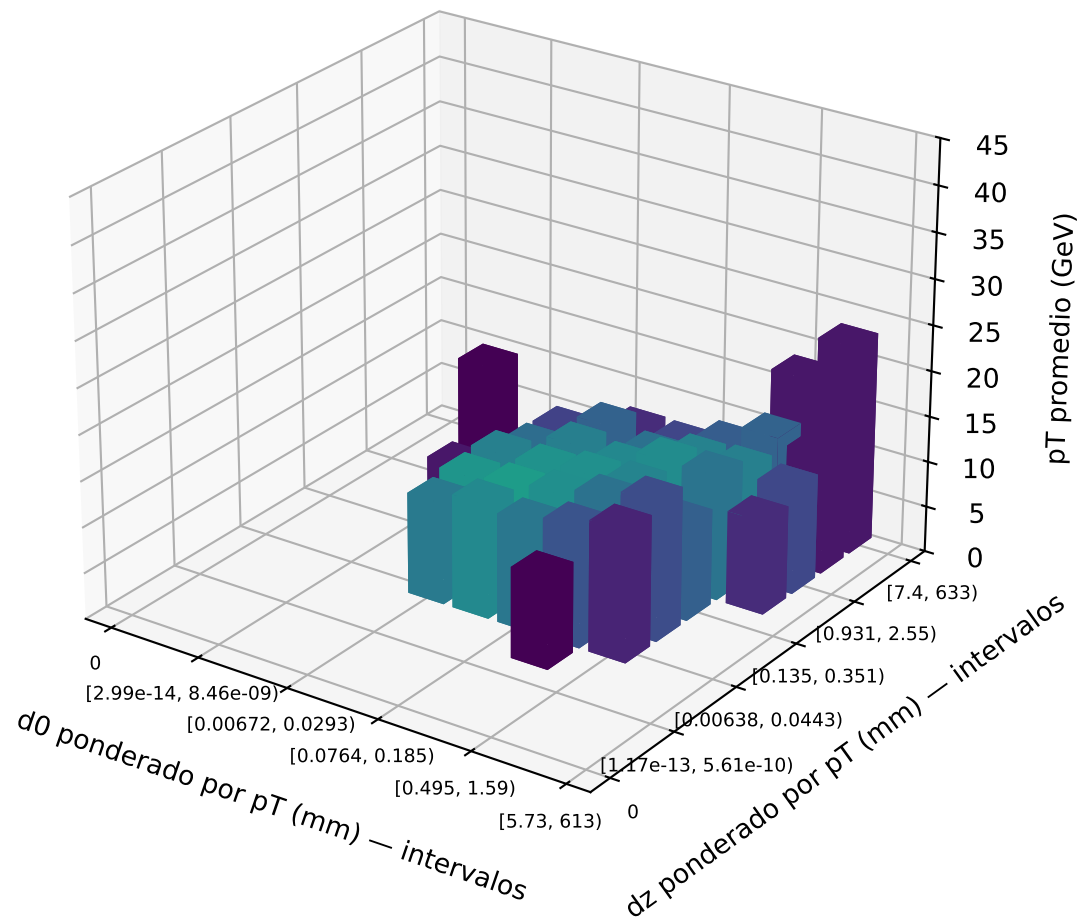
TP — N=3,614

FN — N=3,061



FP — N=3,443

TN — N=57,433



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 101

Variable	Unidad	TP N=3,614	FN N=3,061	FP N=3,443	TN N=57,433
jet_pt	GeV	12.28 ± 6.338	12.24 ± 6.972	12.24 ± 5.929	16.67 ± 11.86
jet_eta	1	-0.001635 ± 1.399	0.007363 ± 1.433	0.01688 ± 1.448	0.001843 ± 1.499
jet_abs_eta	1	1.208 ± 0.7063	1.224 ± 0.7451	1.248 ± 0.7339	1.277 ± 0.7841
jet_mass	GeV	2.547 ± 1.38	2.51 ± 1.702	2.454 ± 1.23	2.876 ± 1.896
jet_ncharged	count	2.812 ± 1.545	2.613 ± 2.019	2.502 ± 1.322	2.593 ± 1.96
jet_nneutral	count	1.856 ± 1.596	2.054 ± 1.642	1.846 ± 1.431	2.401 ± 1.8
n_tracks_cone	count	2.842 ± 1.528	2.654 ± 2.025	2.528 ± 1.3	2.622 ± 1.961
track_pt_sum	GeV	5.858 ± 4.53	4.903 ± 5.222	5.249 ± 3.942	6.844 ± 8.23
track_pt_sum_over_jet_pt	1	0.4782 ± 0.2232	0.3825 ± 0.2697	0.4311 ± 0.2264	0.3711 ± 0.2685
avg_track_pt	GeV	2.215 ± 1.372	1.624 ± 1.305	2.111 ± 1.276	2.169 ± 2.092
std_track_pt	GeV	0.8078 ± 0.8757	0.6526 ± 0.85	0.6878 ± 0.7941	1.023 ± 1.63
n_tracks_below_avg_pt	count	1.472 ± 1.147	1.398 ± 1.377	1.241 ± 0.9986	1.386 ± 1.37
n_tracks_above_avg_pt	count	1.37 ± 0.6486	1.256 ± 0.8686	1.288 ± 0.543	1.235 ± 0.8145
max_pt_ratio	1	0.2828 ± 0.1589	0.2126 ± 0.1633	0.2543 ± 0.1335	0.2096 ± 0.156
min_pt_ratio	1	0.131 ± 0.1413	0.09042 ± 0.1164	0.1282 ± 0.1094	0.08402 ± 0.09787
max_dr_ratio	1	0.156 ± 0.1415	0.1108 ± 0.1207	0.1516 ± 0.1131	0.1027 ± 0.1071
min_dr_ratio	1	0.2435 ± 0.1756	0.1792 ± 0.1666	0.2203 ± 0.143	0.1789 ± 0.155
dr_max_pt	1	0.1419 ± 0.08509	0.132 ± 0.09988	0.1508 ± 0.08567	0.1257 ± 0.09922
dr_min_pt	1	0.2041 ± 0.09996	0.188 ± 0.1202	0.2069 ± 0.09483	0.1898 ± 0.1178
dr_max_dr	1	0.2432 ± 0.09514	0.2241 ± 0.1247	0.2377 ± 0.08841	0.2206 ± 0.1214
dr_min_dr	1	0.1123 ± 0.07351	0.1033 ± 0.08545	0.126 ± 0.07948	0.103 ± 0.08829
pt_difference	GeV	1.9 ± 2.145	1.596 ± 2.226	1.593 ± 1.867	2.512 ± 4.118
r50	1	0.1484 ± 0.08229	0.1397 ± 0.09788	0.1579 ± 0.08267	0.1323 ± 0.0976
r95	1	0.2405 ± 0.09467	0.2215 ± 0.1238	0.2353 ± 0.08855	0.2143 ± 0.12
mean_abs_d0 *	mm	0.1641 ± 0.1798	2.551 ± 11.62	0.1634 ± 0.25	1.261 ± 7.943
max_abs_d0 *	mm	0.2846 ± 0.2984	6.969 ± 22.27	0.294 ± 0.385	3.343 ± 16.71
std_abs_d0 *	mm	0.09134 ± 0.1181	2.689 ± 8.622	0.1009 ± 0.157	1.287 ± 6.715
mean_abs_dz *	mm	0.4125 ± 0.5419	3.753 ± 17.39	0.4421 ± 1.916	1.674 ± 9.733
max_abs_dz *	mm	0.7044 ± 0.8433	9.824 ± 28.51	0.7565 ± 2.081	4.395 ± 23.19
std_abs_dz *	mm	0.2228 ± 0.3208	3.737 ± 10.23	0.2469 ± 0.4058	1.705 ± 9.565
pt_weighted_abs_d0 *	mm	0.1716 ± 0.1847	2.394 ± 11.51	0.1749 ± 0.2622	1.2 ± 7.955
pt_weighted_abs_dz *	mm	0.4327 ± 0.5614	3.203 ± 17.15	0.4691 ± 1.913	1.371 ± 8.424

* Variable utilizada como entrada del modelo.

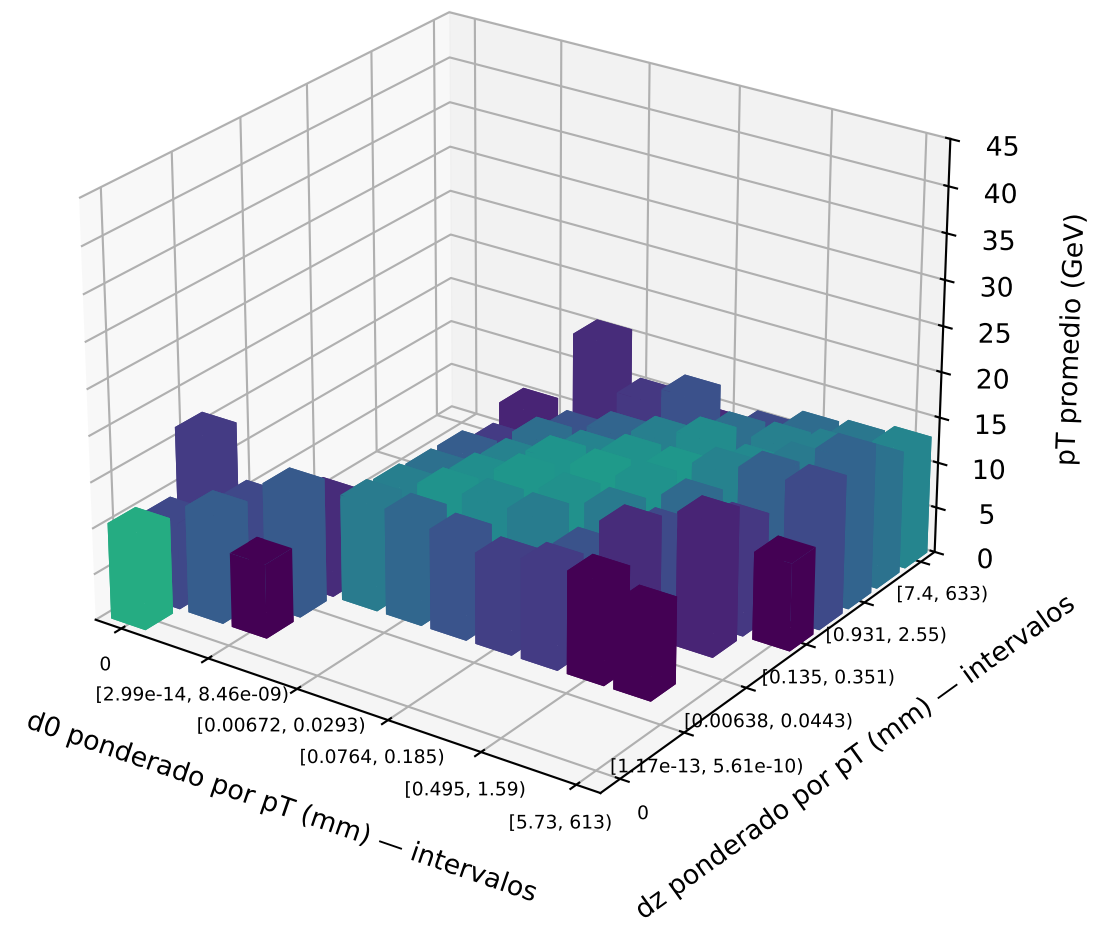
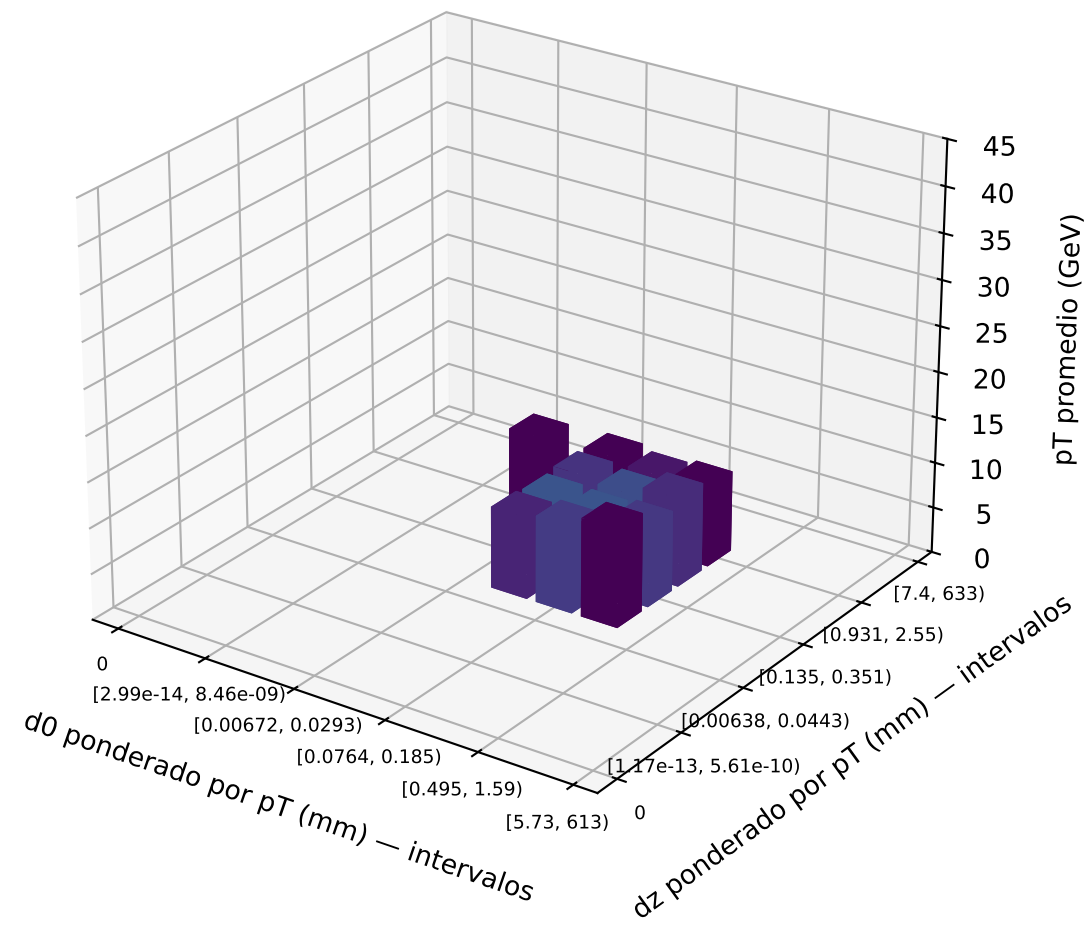
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · ensemble

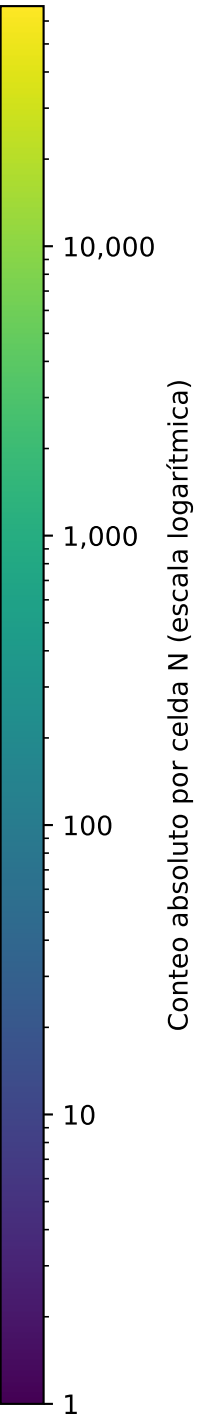
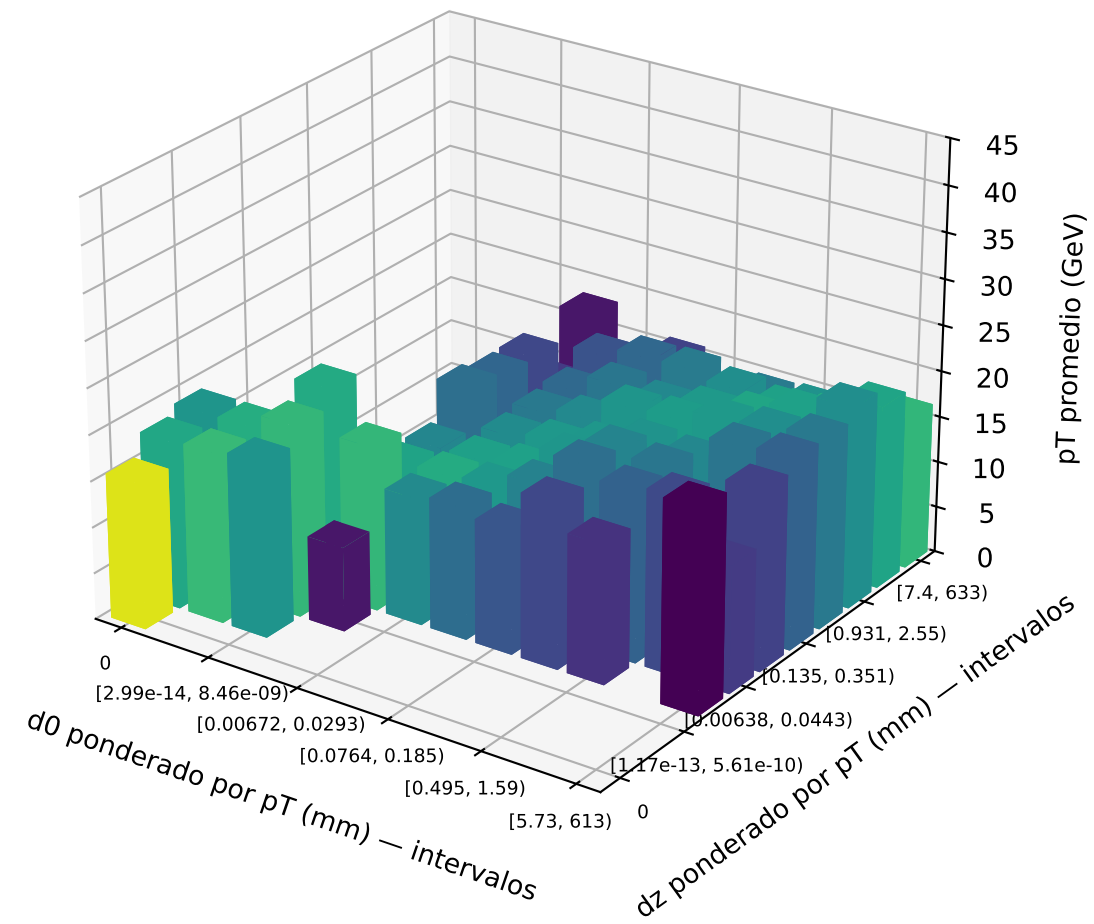
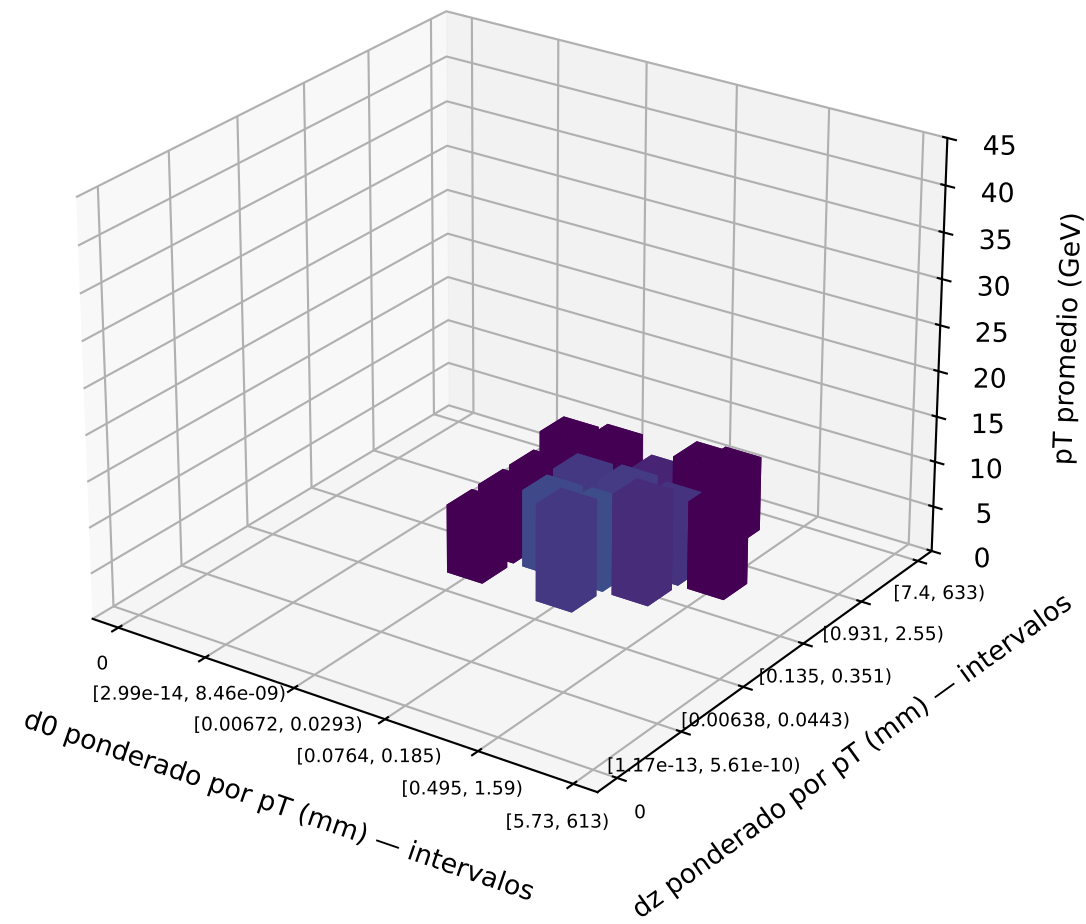
TP — N=80

FN — N=6,595



FP — N=63

TN — N=60,813



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · ensemble

Variable	Unidad	TP N=80	FN N=6,595	FP N=63	TN N=60,813
jet_pt *	GeV	8.9 ± 1.541	12.3 ± 6.664	9.016 ± 1.886	16.42 ± 11.66
jet_eta *	1	0.02875 ± 1.119	0.002173 ± 1.418	-0.1885 ± 1.431	0.002891 ± 1.496
jet_abs_eta *	1	0.9278 ± 0.6267	1.219 ± 0.7248	1.303 ± 0.6194	1.275 ± 0.7815
jet_mass *	GeV	1.851 ± 0.6772	2.538 ± 1.542	2.149 ± 0.7905	2.852 ± 1.868
jet_ncharged *	count	1.962 ± 0.8866	2.73 ± 1.787	2 ± 0.9759	2.588 ± 1.93
jet_nneutral *	count	1.562 ± 0.7219	1.952 ± 1.627	1.524 ± 0.7095	2.37 ± 1.787
n_tracks_cone *	count	1.95 ± 0.8646	2.766 ± 1.782	1.984 ± 0.9511	2.617 ± 1.93
track_pt_sum *	GeV	3.761 ± 1.304	5.44 ± 4.907	3.482 ± 1.281	6.757 ± 8.06
track_pt_sum_over_jet_pt *	1	0.4227 ± 0.1334	0.4345 ± 0.2513	0.3891 ± 0.1395	0.3745 ± 0.2667
avg_track_pt *	GeV	2.147 ± 0.9364	1.942 ± 1.378	1.908 ± 0.7414	2.166 ± 2.056
std_track_pt *	GeV	0.2693 ± 0.2781	0.7423 ± 0.8706	0.2955 ± 0.3562	1.005 ± 1.597
n_tracks_below_avg_pt *	count	0.875 ± 0.7479	1.445 ± 1.262	0.873 ± 0.7662	1.379 ± 1.353
n_tracks_above_avg_pt *	count	1.075 ± 0.2634	1.321 ± 0.7631	1.111 ± 0.3143	1.238 ± 0.802
max_pt_ratio *	1	0.283 ± 0.1116	0.2502 ± 0.1652	0.2515 ± 0.09436	0.2121 ± 0.1552
min_pt_ratio *	1	0.2192 ± 0.1389	0.1111 ± 0.1314	0.1817 ± 0.1073	0.08642 ± 0.09903
max_dr_ratio *	1	0.2341 ± 0.1337	0.1341 ± 0.1338	0.1985 ± 0.1028	0.1053 ± 0.108
min_dr_ratio *	1	0.269 ± 0.12	0.2133 ± 0.1749	0.2341 ± 0.1049	0.1812 ± 0.1547
dr_max_pt *	1	0.1345 ± 0.06329	0.1374 ± 0.0926	0.1506 ± 0.06423	0.1271 ± 0.0987
dr_min_pt *	1	0.1745 ± 0.06859	0.197 ± 0.1104	0.1952 ± 0.07683	0.1907 ± 0.1168
dr_max_dr *	1	0.1878 ± 0.07379	0.235 ± 0.1104	0.2138 ± 0.07061	0.2216 ± 0.1199
dr_min_dr *	1	0.1285 ± 0.06107	0.1079 ± 0.0795	0.136 ± 0.06543	0.1043 ± 0.08799
pt_difference *	GeV	0.5919 ± 0.6306	1.775 ± 2.196	0.6399 ± 0.7496	2.462 ± 4.032
r50 *	1	0.1429 ± 0.06298	0.1444 ± 0.09016	0.1575 ± 0.06083	0.1337 ± 0.09703
r95 *	1	0.1878 ± 0.07379	0.2323 ± 0.1097	0.2138 ± 0.07061	0.2155 ± 0.1186
mean_abs_d0 *	mm	0.1191 ± 0.07604	1.272 ± 8.004	0.1255 ± 0.09311	1.2 ± 7.723
max_abs_d0 *	mm	0.1797 ± 0.133	3.388 ± 15.54	0.2361 ± 0.282	3.174 ± 16.25
std_abs_d0 *	mm	0.04999 ± 0.05975	1.298 ± 6.016	0.08339 ± 0.1412	1.221 ± 6.532
mean_abs_dz *	mm	0.3412 ± 0.2811	1.964 ± 11.97	0.4868 ± 0.5319	1.606 ± 9.474
max_abs_dz *	mm	0.551 ± 0.6172	4.939 ± 19.96	0.8873 ± 1.21	4.193 ± 22.55
std_abs_dz *	mm	0.1777 ± 0.2886	1.855 ± 7.191	0.3151 ± 0.604	1.624 ± 9.302
pt_weighted_abs_d0 *	mm	0.1208 ± 0.07972	1.204 ± 7.92	0.1211 ± 0.08851	1.143 ± 7.734
pt_weighted_abs_dz *	mm	0.3393 ± 0.2836	1.72 ± 11.77	0.4557 ± 0.4588	1.321 ± 8.202

* Variable utilizada como entrada del modelo.

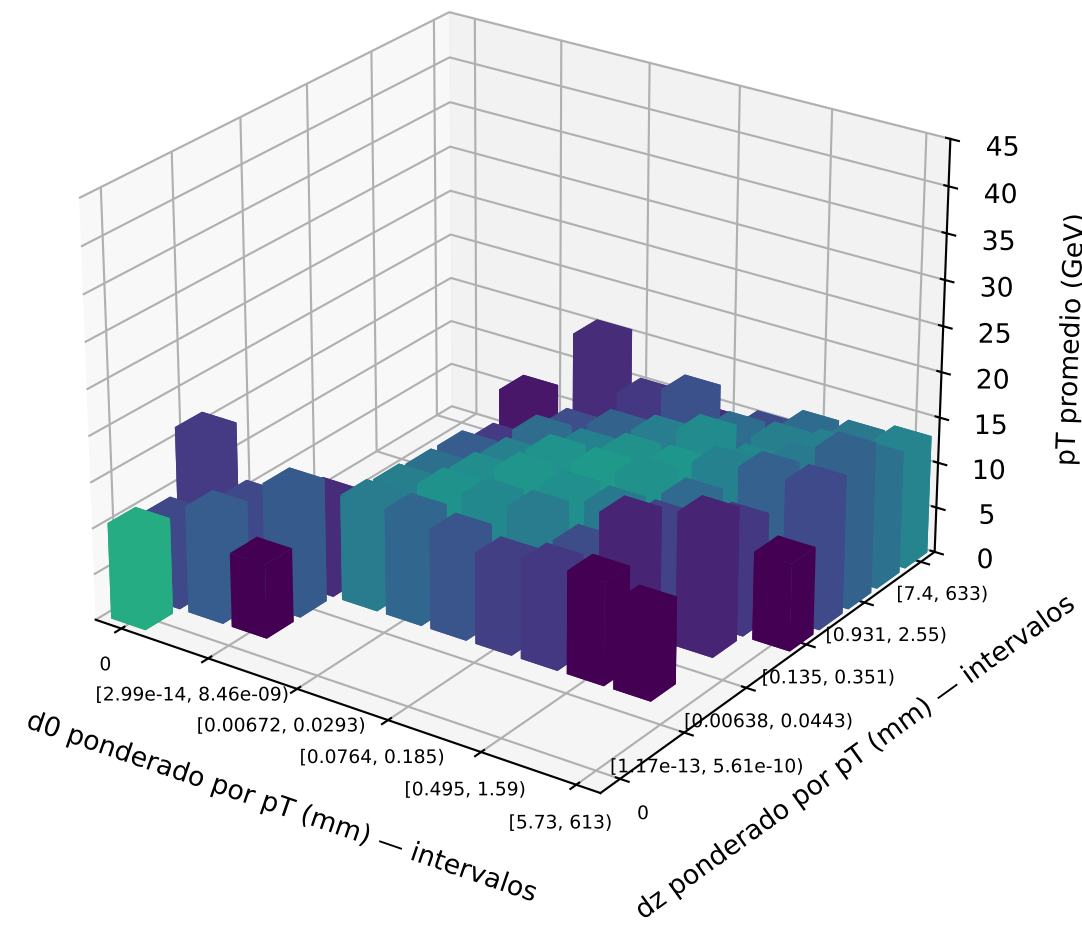
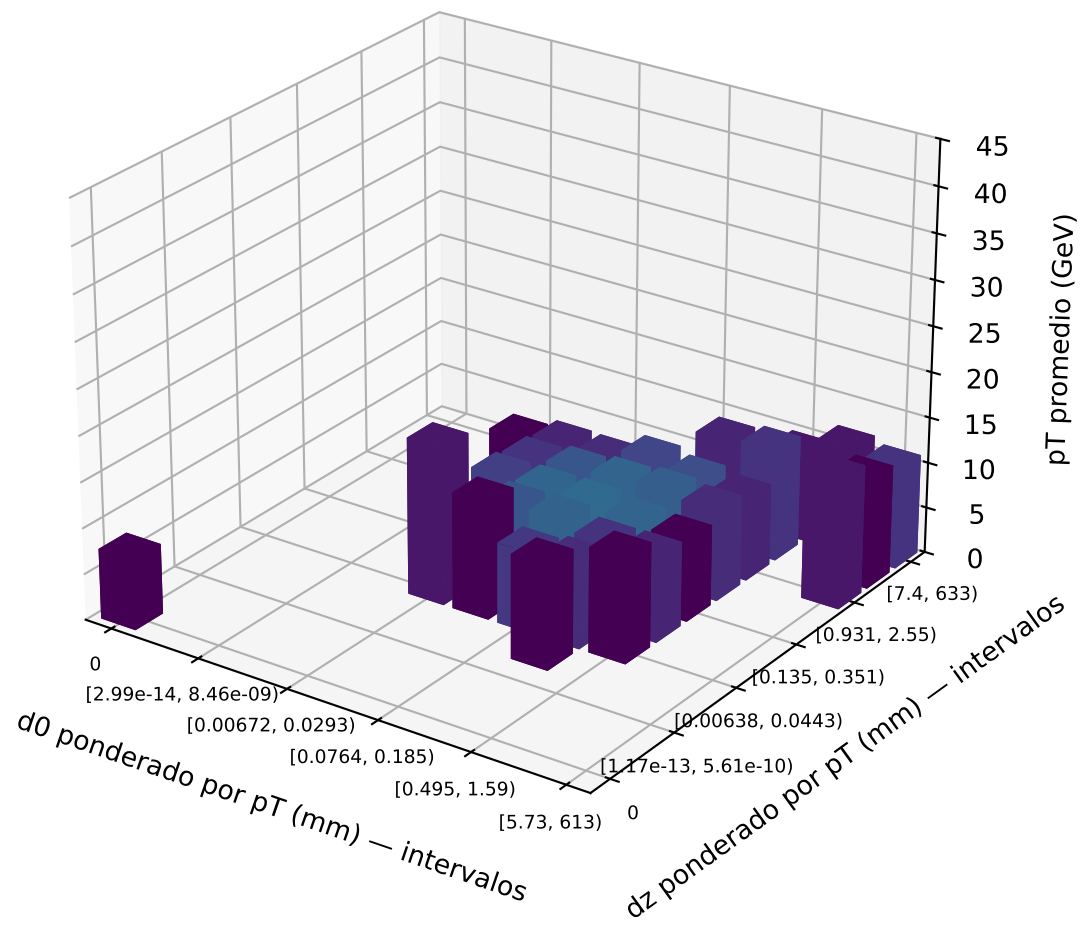
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 42

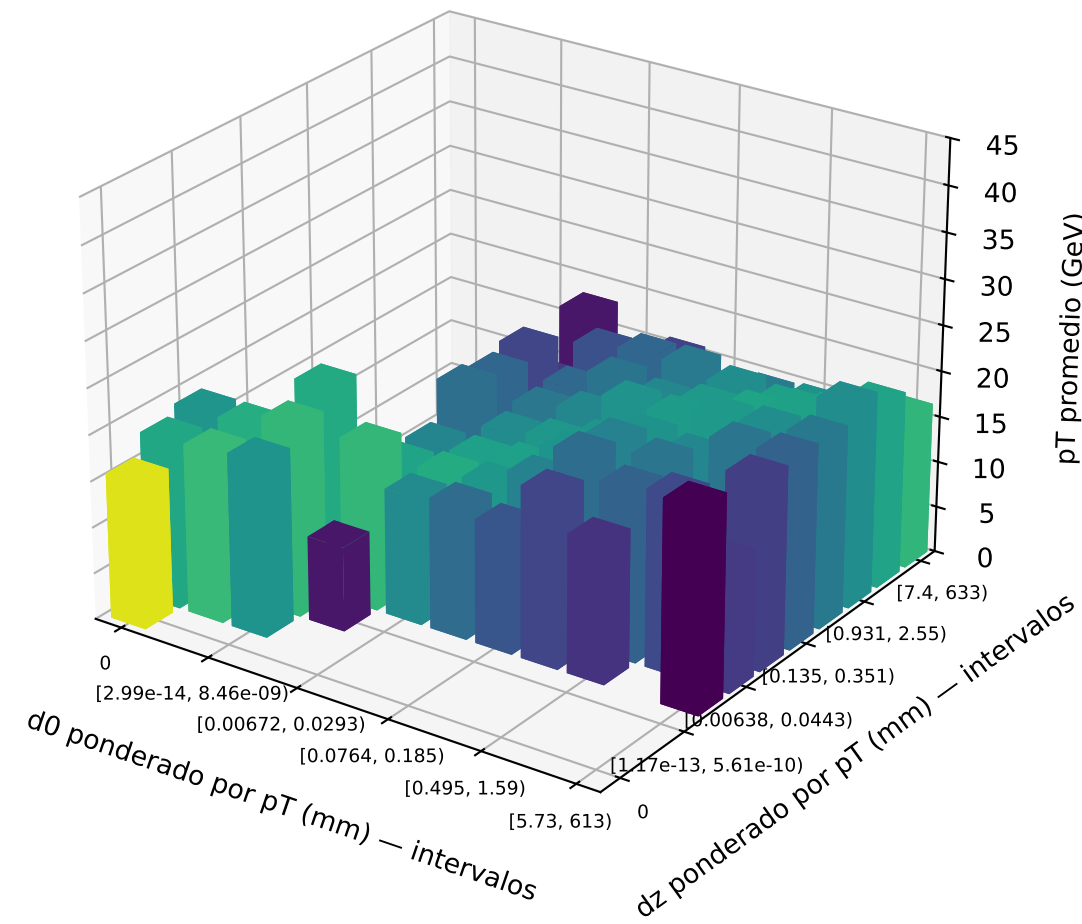
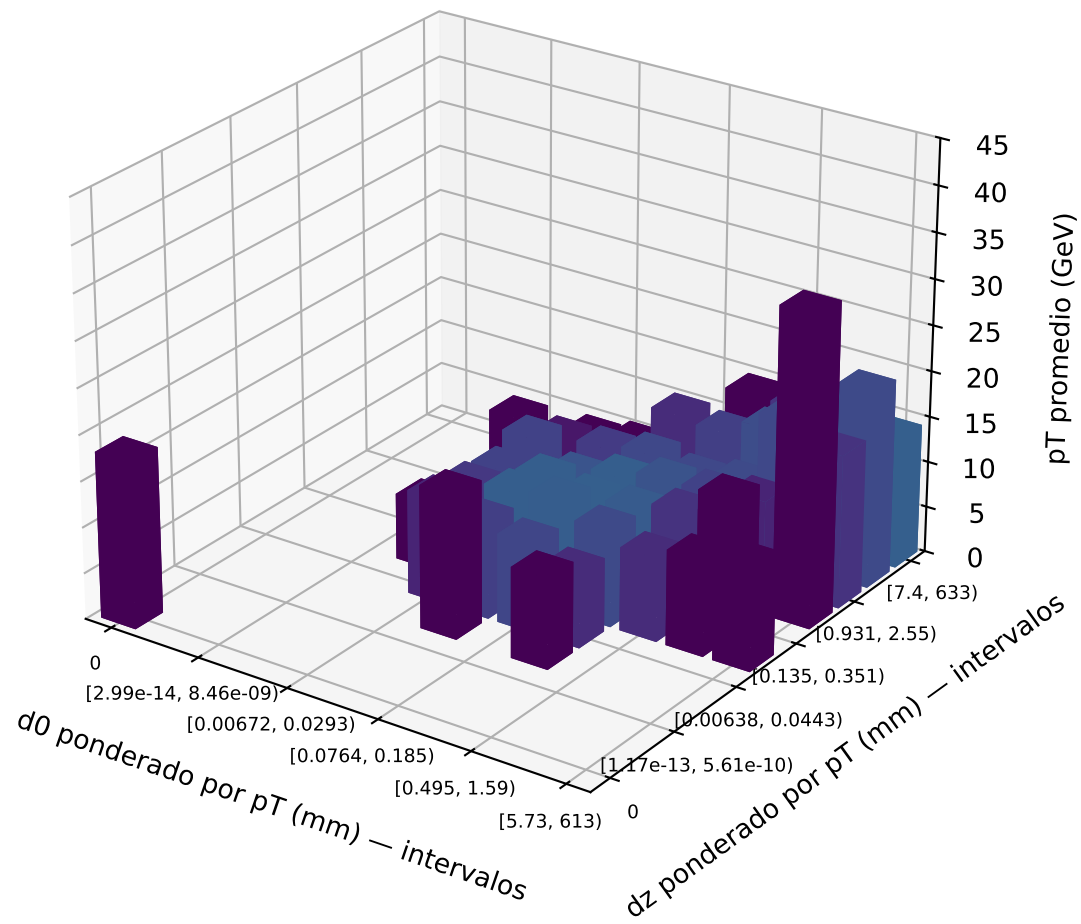
TP — N=349

FN — N=6,326



FP — N=435

TN — N=60,441



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 42

Variable	Unidad	TP N=349	FN N=6,326	FP N=435	TN N=60,441
jet_pt *	GeV	9.913 ± 2.296	12.39 ± 6.772	11.75 ± 4.869	16.45 ± 11.68
jet_eta *	1	-0.07897 ± 1.391	0.006985 ± 1.416	0.01054 ± 1.453	0.002637 ± 1.496
jet_abs_eta *	1	1.19 ± 0.7251	1.217 ± 0.7243	1.254 ± 0.733	1.276 ± 0.7817
jet_mass *	GeV	2.013 ± 1.041	2.559 ± 1.554	2.637 ± 1.254	2.853 ± 1.871
jet_ncharged *	count	2.464 ± 1.342	2.735 ± 1.801	3.021 ± 1.595	2.585 ± 1.931
jet_nneutral *	count	1.573 ± 1.119	1.968 ± 1.641	1.736 ± 1.232	2.374 ± 1.789
n_tracks_cone *	count	2.481 ± 1.362	2.771 ± 1.795	3.057 ± 1.626	2.613 ± 1.931
track_pt_sum *	GeV	4.756 ± 2.061	5.456 ± 4.99	5.671 ± 3.721	6.762 ± 8.079
track_pt_sum_over_jet_pt *	1	0.4853 ± 0.1941	0.4315 ± 0.2526	0.4799 ± 0.2047	0.3738 ± 0.2669
avg_track_pt *	GeV	2.341 ± 1.54	1.922 ± 1.361	2.037 ± 1.115	2.167 ± 2.06
std_track_pt *	GeV	0.5339 ± 0.5834	0.7478 ± 0.8791	0.7149 ± 0.7426	1.006 ± 1.601
n_tracks_below_avg_pt *	count	1.192 ± 1.024	1.452 ± 1.269	1.589 ± 1.178	1.377 ± 1.354
n_tracks_above_avg_pt *	count	1.289 ± 0.5347	1.319 ± 0.77	1.469 ± 0.6615	1.237 ± 0.8024
max_pt_ratio *	1	0.3167 ± 0.1814	0.247 ± 0.163	0.2684 ± 0.1348	0.2117 ± 0.1553
min_pt_ratio *	1	0.196 ± 0.207	0.1078 ± 0.125	0.1283 ± 0.138	0.08622 ± 0.09869
max_dr_ratio *	1	0.2162 ± 0.2027	0.1308 ± 0.128	0.151 ± 0.1391	0.1051 ± 0.1077
min_dr_ratio *	1	0.2898 ± 0.1959	0.2098 ± 0.1723	0.2252 ± 0.1465	0.1809 ± 0.1547
dr_max_pt *	1	0.1221 ± 0.07832	0.1382 ± 0.09294	0.1379 ± 0.07668	0.127 ± 0.09881
dr_min_pt *	1	0.1826 ± 0.09934	0.1975 ± 0.1105	0.2148 ± 0.1005	0.1906 ± 0.1168
dr_max_dr *	1	0.2088 ± 0.104	0.2358 ± 0.1103	0.2492 ± 0.09326	0.2214 ± 0.12
dr_min_dr *	1	0.1041 ± 0.06952	0.1084 ± 0.07983	0.1098 ± 0.0665	0.1043 ± 0.08811
pt_difference *	GeV	1.221 ± 1.278	1.79 ± 2.223	1.742 ± 1.916	2.466 ± 4.041
r50 *	1	0.1321 ± 0.07869	0.1451 ± 0.09041	0.1486 ± 0.0722	0.1337 ± 0.09715
r95 *	1	0.2084 ± 0.1035	0.2331 ± 0.1096	0.2463 ± 0.09207	0.2153 ± 0.1187
mean_abs_d0 *	mm	0.9043 ± 5.837	1.278 ± 8.058	3.366 ± 12.78	1.183 ± 7.669
max_abs_d0 *	mm	1.984 ± 11.78	3.425 ± 15.62	10.37 ± 33.85	3.119 ± 16.04
std_abs_d0 *	mm	0.8516 ± 5.339	1.306 ± 6.014	4.255 ± 14.01	1.198 ± 6.438
mean_abs_dz *	mm	1.054 ± 3.569	1.993 ± 12.19	3.64 ± 9.674	1.59 ± 9.466
max_abs_dz *	mm	2.767 ± 10.52	5.003 ± 20.23	11.77 ± 26.86	4.135 ± 22.5
std_abs_dz *	mm	1.111 ± 4.344	1.874 ± 7.271	4.937 ± 11.58	1.599 ± 9.275
pt_weighted_abs_d0 *	mm	0.9683 ± 6.482	1.203 ± 7.943	3.336 ± 12.56	1.127 ± 7.682
pt_weighted_abs_dz *	mm	1.017 ± 3.751	1.741 ± 11.99	2.942 ± 7.865	1.309 ± 8.199

* Variable utilizada como entrada del modelo.

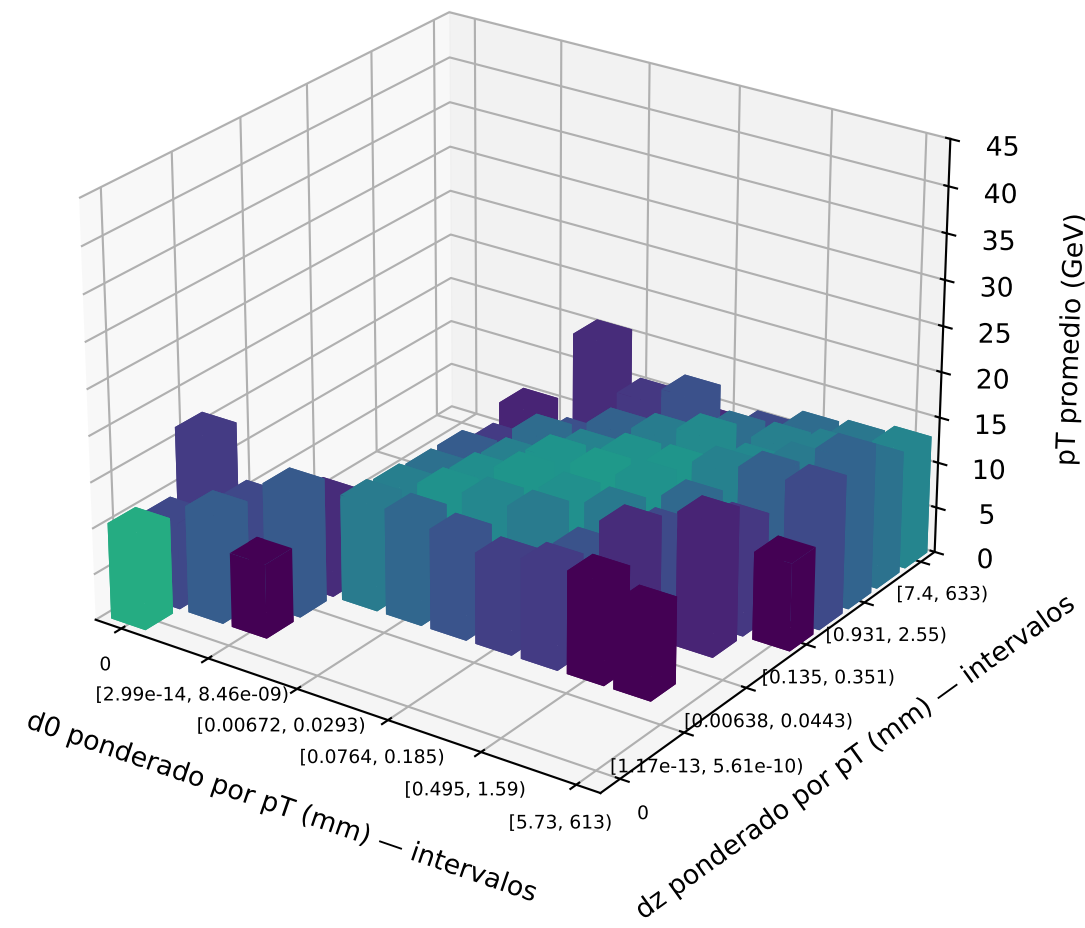
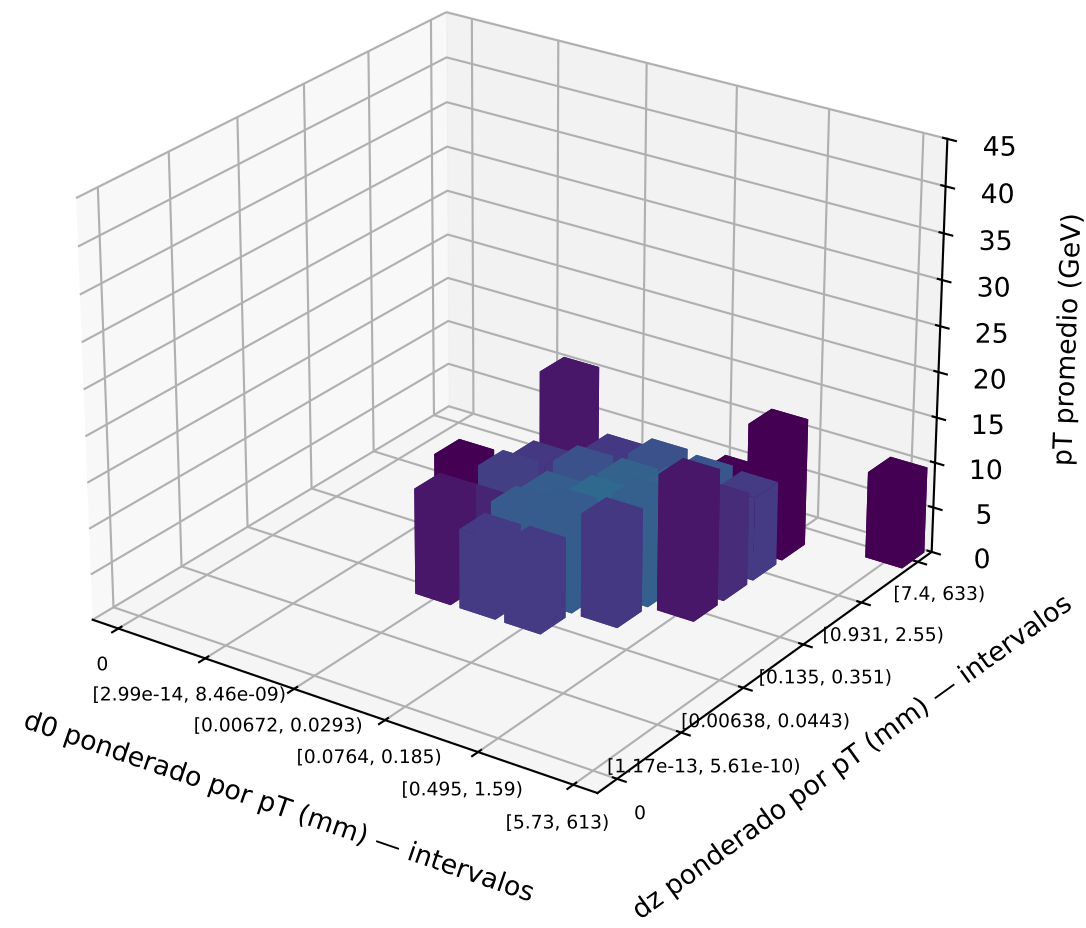
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 73

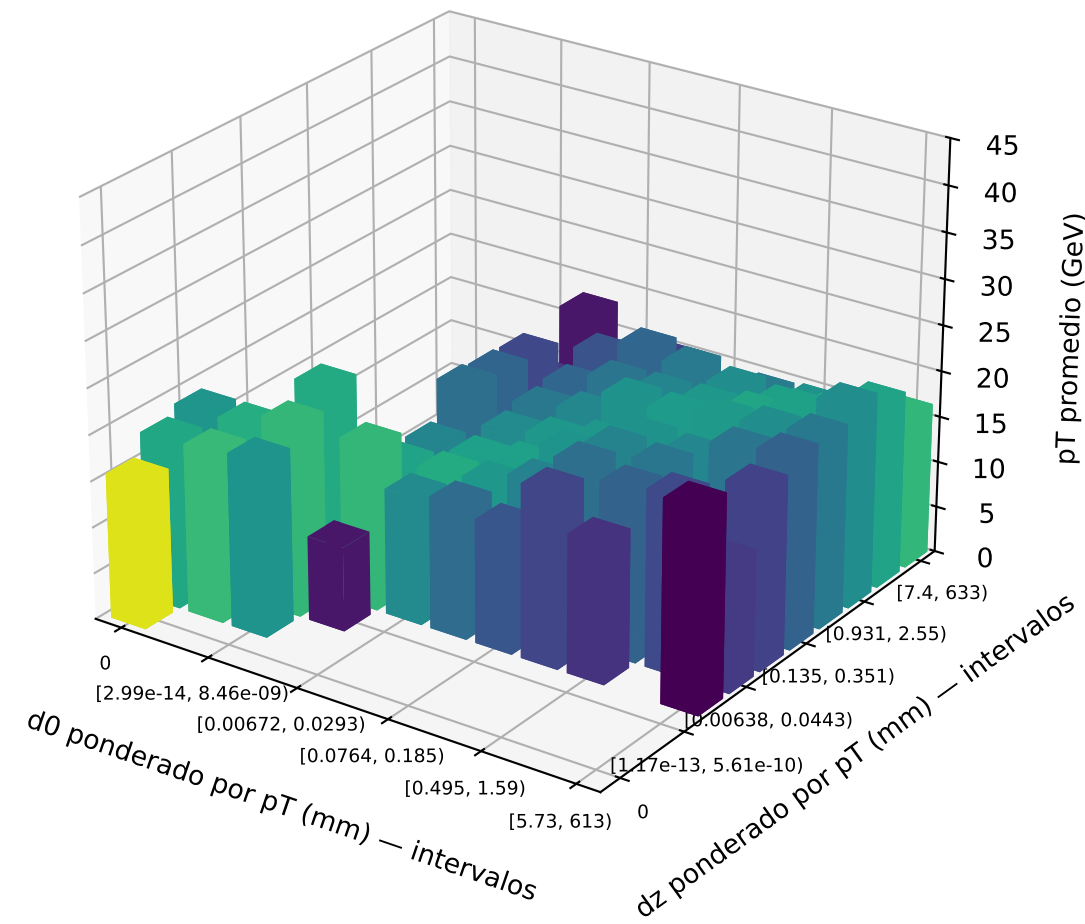
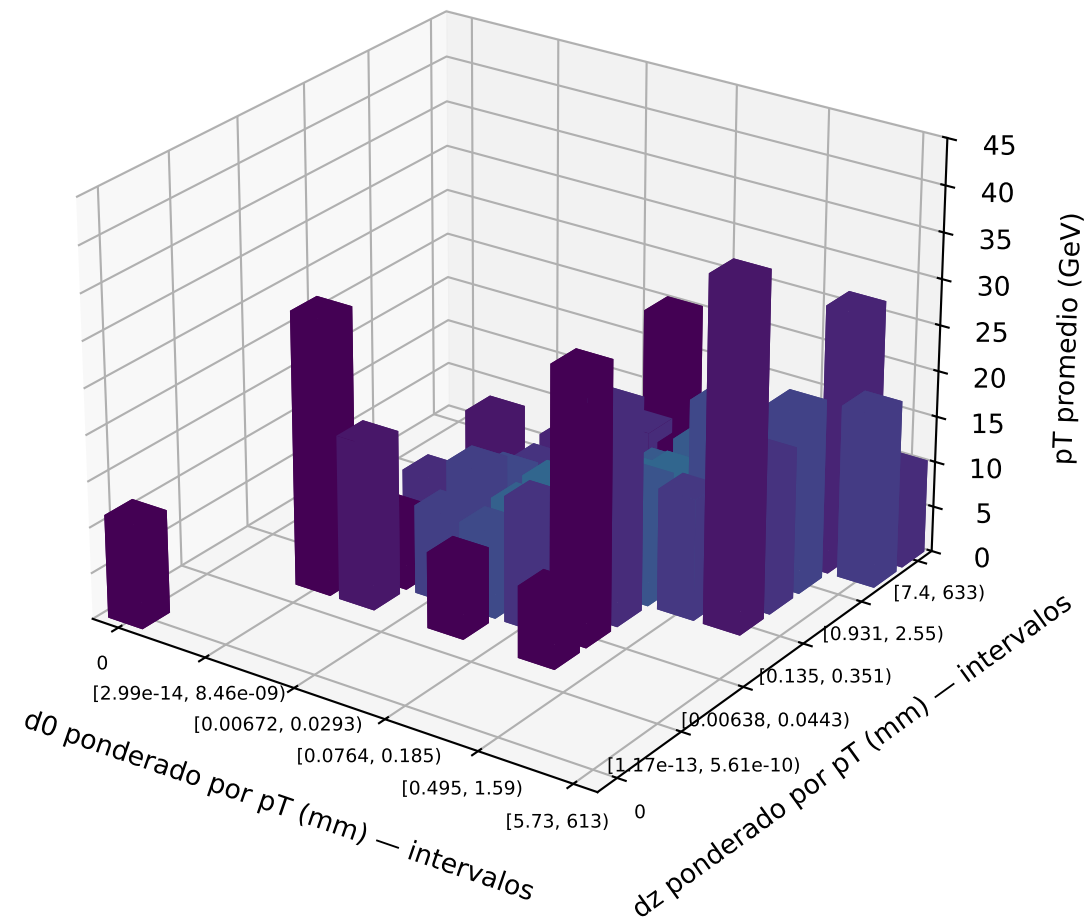
TP — N=324

FN — N=6,351



FP — N=425

TN — N=60,451



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 73

Variable	Unidad	TP N=324	FN N=6,351	FP N=425	TN N=60,451
jet_pt *	GeV	10.03 ± 2.808	12.38 ± 6.754	11.99 ± 7.591	16.45 ± 11.67
jet_eta *	1	-0.04922 ± 1.401	0.005129 ± 1.415	-0.1387 ± 1.516	0.003688 ± 1.496
jet_abs_eta *	1	1.198 ± 0.7279	1.216 ± 0.7242	1.323 ± 0.7541	1.275 ± 0.7815
jet_mass *	GeV	2.183 ± 0.8589	2.548 ± 1.561	2.591 ± 1.565	2.854 ± 1.869
jet_ncharged *	count	2.438 ± 1.237	2.735 ± 1.803	2.739 ± 1.642	2.587 ± 1.931
jet_nneutral *	count	1.565 ± 1.191	1.966 ± 1.637	1.807 ± 1.447	2.373 ± 1.788
n_tracks_cone *	count	2.448 ± 1.26	2.772 ± 1.797	2.746 ± 1.638	2.615 ± 1.932
track_pt_sum *	GeV	4.662 ± 2.691	5.458 ± 4.966	5.563 ± 4.77	6.762 ± 8.074
track_pt_sum_over_jet_pt *	1	0.4598 ± 0.2076	0.433 ± 0.2521	0.4538 ± 0.2028	0.374 ± 0.2669
avg_track_pt *	GeV	2.008 ± 0.9372	1.941 ± 1.392	2.04 ± 0.9765	2.167 ± 2.06
std_track_pt *	GeV	0.6472 ± 0.6449	0.7412 ± 0.877	0.6756 ± 0.832	1.006 ± 1.6
n_tracks_below_avg_pt *	count	1.222 ± 0.9876	1.449 ± 1.27	1.409 ± 1.22	1.378 ± 1.353
n_tracks_above_avg_pt *	count	1.225 ± 0.4732	1.322 ± 0.771	1.336 ± 0.6076	1.238 ± 0.8029
max_pt_ratio *	1	0.2841 ± 0.1313	0.2489 ± 0.166	0.2623 ± 0.1156	0.2118 ± 0.1554
min_pt_ratio *	1	0.1415 ± 0.09935	0.1109 ± 0.1333	0.1372 ± 0.1044	0.08617 ± 0.09896
max_dr_ratio *	1	0.1551 ± 0.09677	0.1343 ± 0.1358	0.1551 ± 0.1033	0.1051 ± 0.108
min_dr_ratio *	1	0.2636 ± 0.1417	0.2115 ± 0.1756	0.2298 ± 0.1285	0.1809 ± 0.1548
dr_max_pt *	1	0.1419 ± 0.07037	0.1372 ± 0.09328	0.1503 ± 0.06857	0.127 ± 0.09884
dr_min_pt *	1	0.223 ± 0.07804	0.1954 ± 0.1112	0.21 ± 0.08244	0.1906 ± 0.1169
dr_max_dr *	1	0.2433 ± 0.07494	0.2339 ± 0.1116	0.2414 ± 0.077	0.2215 ± 0.1201
dr_min_dr *	1	0.1249 ± 0.06576	0.1073 ± 0.07987	0.1257 ± 0.06438	0.1042 ± 0.0881
pt_difference *	GeV	1.451 ± 1.444	1.776 ± 2.218	1.626 ± 2.164	2.466 ± 4.04
r50 *	1	0.1469 ± 0.06617	0.1443 ± 0.09093	0.1569 ± 0.06549	0.1336 ± 0.09716
r95 *	1	0.2422 ± 0.07436	0.2313 ± 0.1109	0.2402 ± 0.07667	0.2153 ± 0.1188
mean_abs_d0 *	mm	0.263 ± 1.632	1.309 ± 8.146	0.8379 ± 3.557	1.201 ± 7.741
max_abs_d0 *	mm	0.7265 ± 6.546	3.484 ± 15.75	2.998 ± 13.54	3.172 ± 16.26
std_abs_d0 *	mm	0.2841 ± 2.831	1.333 ± 6.094	1.232 ± 5.697	1.22 ± 6.534
mean_abs_dz *	mm	0.6436 ± 2.291	2.011 ± 12.18	1.146 ± 3.001	1.608 ± 9.499
max_abs_dz *	mm	1.484 ± 9.038	5.06 ± 20.23	3.513 ± 10.84	4.194 ± 22.6
std_abs_dz *	mm	0.545 ± 3.908	1.9 ± 7.27	1.382 ± 4.457	1.624 ± 9.323
pt_weighted_abs_d0 *	mm	0.2077 ± 0.703	1.241 ± 8.067	0.9241 ± 4.734	1.144 ± 7.747
pt_weighted_abs_dz *	mm	0.5364 ± 1.059	1.763 ± 11.99	1.119 ± 3.513	1.322 ± 8.221

* Variable utilizada como entrada del modelo.

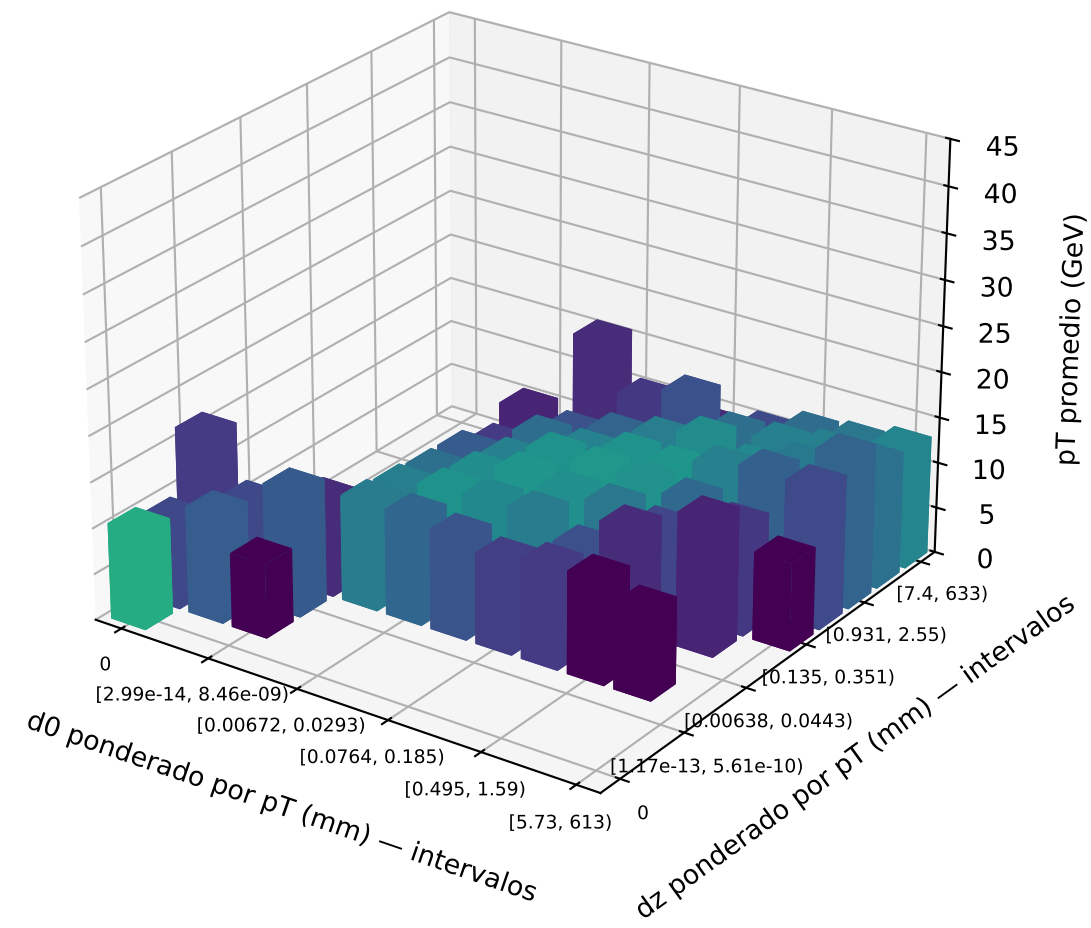
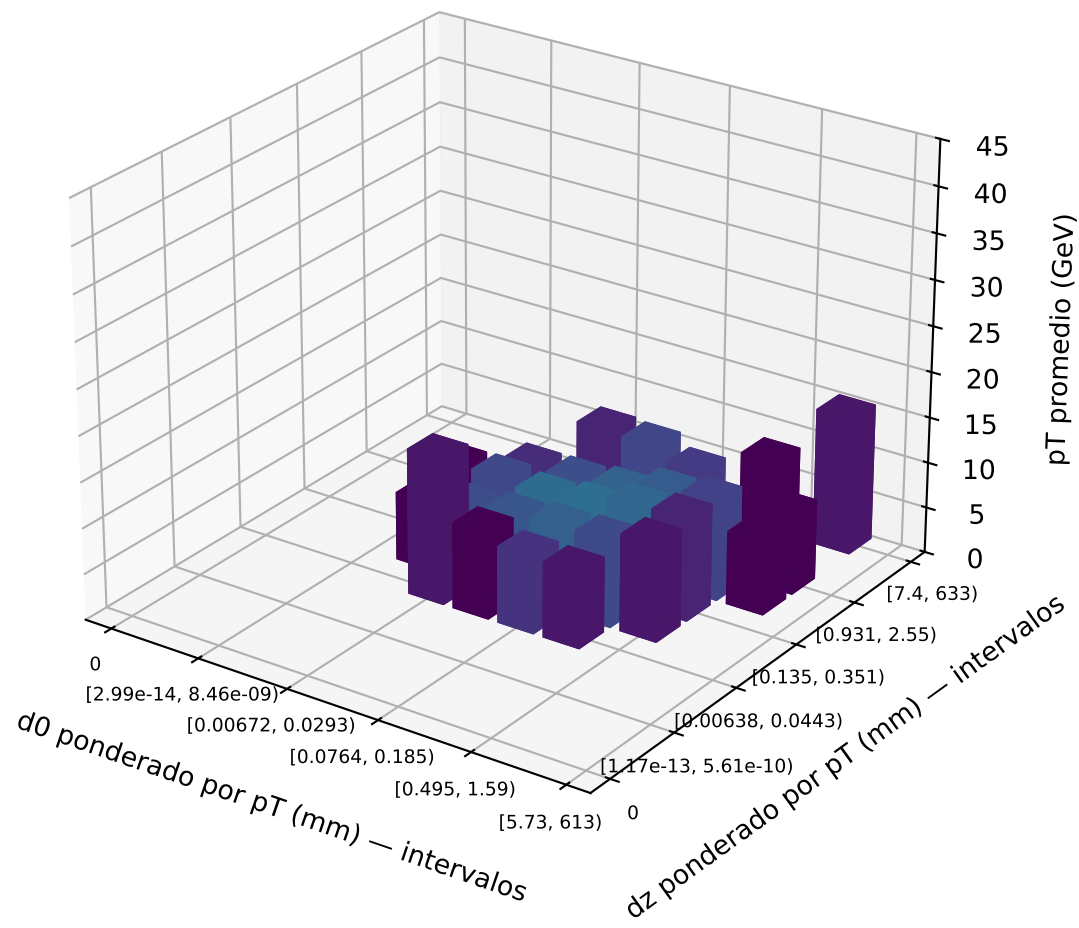
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 101

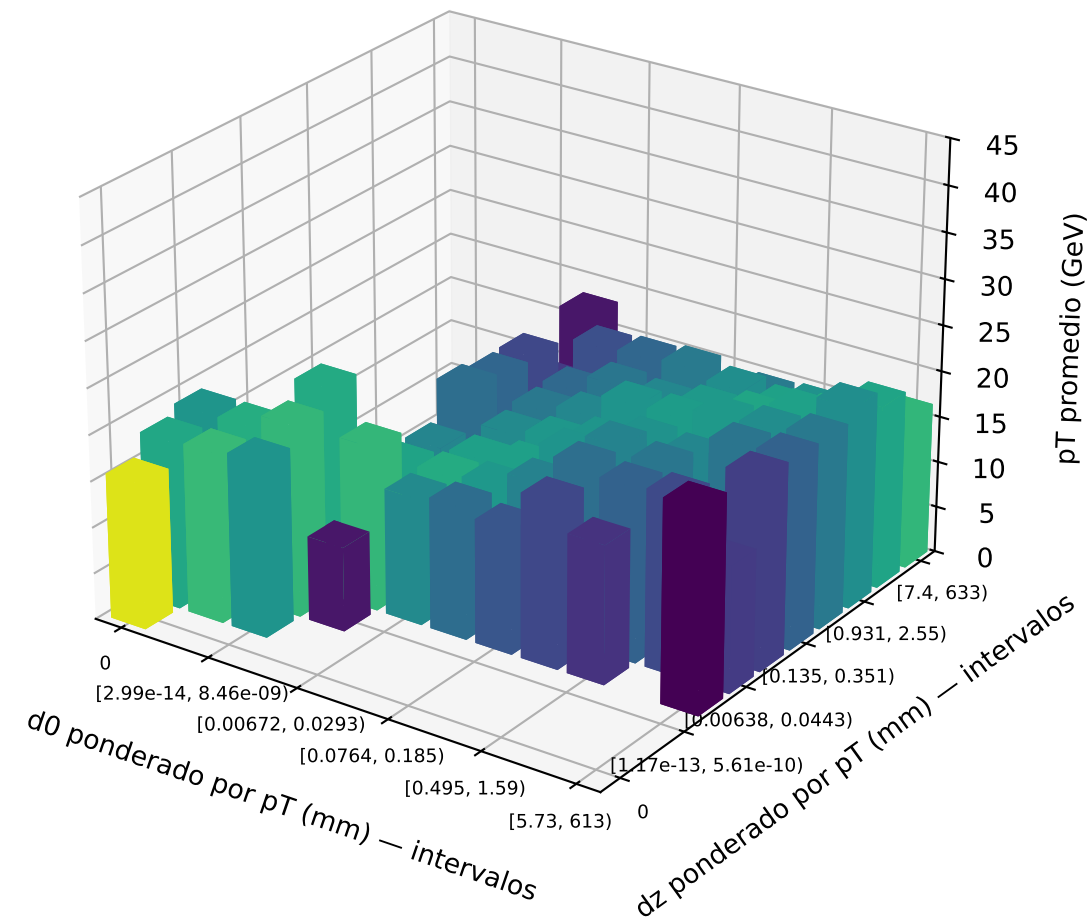
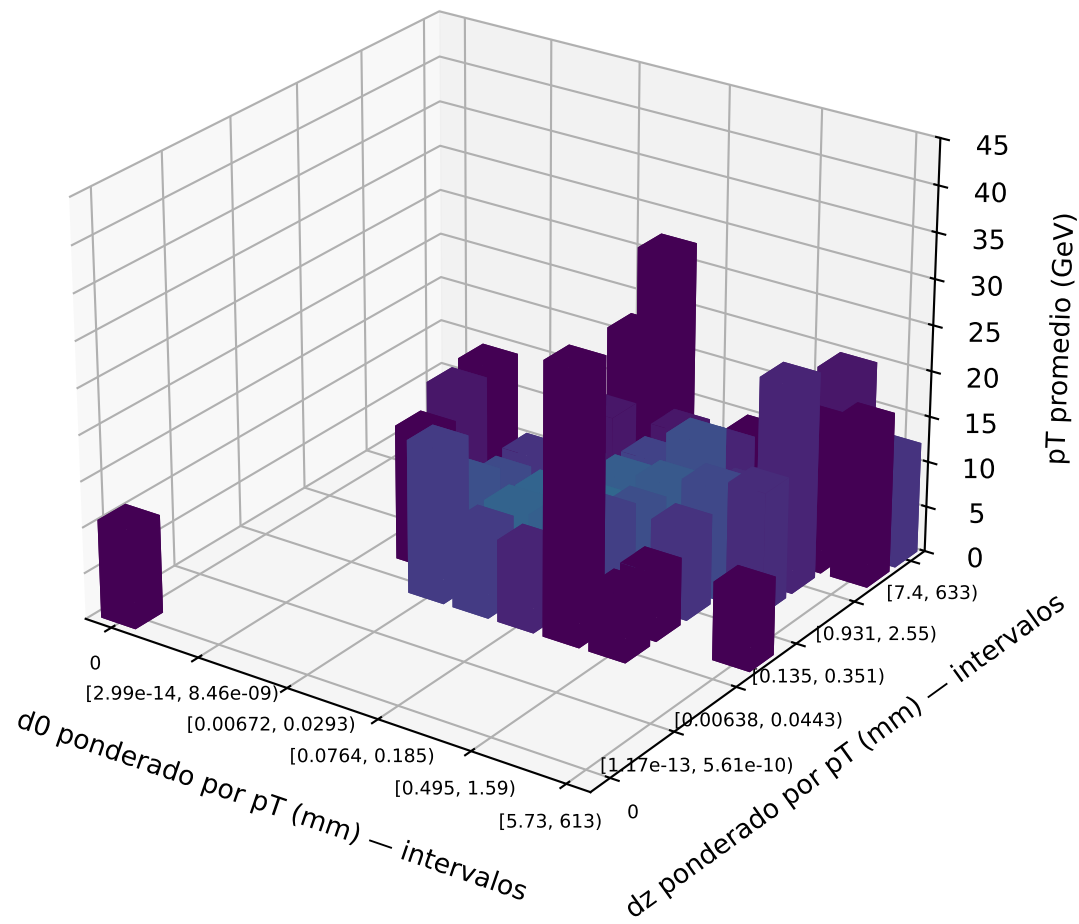
TP — N=350

FN — N=6,325



FP — N=356

TN — N=60,520



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 101

Variable	Unidad	TP N=350	FN N=6,325	FP N=356	TN N=60,520
jet_pt *	GeV	9.643 ± 2.776	12.41 ± 6.757	10.71 ± 4.418	16.45 ± 11.67
jet_eta *	1	0.1605 ± 1.301	-0.00625 ± 1.42	0.09337 ± 1.414	0.00216 ± 1.496
jet_abs_eta *	1	1.112 ± 0.6949	1.221 ± 0.7256	1.217 ± 0.727	1.276 ± 0.7816
jet_mass *	GeV	1.913 ± 1.048	2.564 ± 1.552	2.27 ± 1.229	2.855 ± 1.87
jet_ncharged *	count	2.143 ± 1.165	2.753 ± 1.804	2.374 ± 1.363	2.589 ± 1.932
jet_nneutral *	count	1.414 ± 1.153	1.976 ± 1.637	1.545 ± 1.099	2.374 ± 1.788
n_tracks_cone *	count	2.109 ± 1.144	2.792 ± 1.798	2.376 ± 1.353	2.618 ± 1.932
track_pt_sum *	GeV	4.536 ± 2.211	5.469 ± 4.984	4.927 ± 3.193	6.765 ± 8.075
track_pt_sum_over_jet_pt *	1	0.4737 ± 0.189	0.4322 ± 0.253	0.4594 ± 0.1957	0.374 ± 0.2669
avg_track_pt *	GeV	2.534 ± 1.483	1.911 ± 1.36	2.293 ± 1.198	2.165 ± 2.059
std_track_pt *	GeV	0.4045 ± 0.4765	0.755 ± 0.8804	0.496 ± 0.5555	1.007 ± 1.6
n_tracks_below_avg_pt *	count	0.96 ± 0.9493	1.465 ± 1.268	1.163 ± 1.074	1.379 ± 1.354
n_tracks_above_avg_pt *	count	1.149 ± 0.379	1.327 ± 0.7741	1.213 ± 0.4851	1.238 ± 0.8032
max_pt_ratio *	1	0.3279 ± 0.1663	0.2463 ± 0.1636	0.291 ± 0.1397	0.2117 ± 0.1552
min_pt_ratio *	1	0.2374 ± 0.201	0.1055 ± 0.1235	0.1879 ± 0.1564	0.08593 ± 0.09834
max_dr_ratio *	1	0.2516 ± 0.195	0.1288 ± 0.127	0.2055 ± 0.1498	0.1048 ± 0.1075
min_dr_ratio *	1	0.3086 ± 0.179	0.2088 ± 0.1727	0.2736 ± 0.1525	0.1807 ± 0.1545
dr_max_pt *	1	0.1217 ± 0.07971	0.1383 ± 0.09287	0.1306 ± 0.07427	0.1271 ± 0.0988
dr_min_pt *	1	0.1685 ± 0.09677	0.1983 ± 0.1105	0.1858 ± 0.09144	0.1908 ± 0.1169
dr_max_dr *	1	0.1869 ± 0.1002	0.237 ± 0.11	0.2128 ± 0.09444	0.2216 ± 0.12
dr_min_dr *	1	0.1097 ± 0.07498	0.1081 ± 0.07957	0.1139 ± 0.06867	0.1043 ± 0.08807
pt_difference *	GeV	0.9203 ± 1.111	1.807 ± 2.223	1.146 ± 1.328	2.468 ± 4.04
r50 *	1	0.1298 ± 0.07738	0.1452 ± 0.09046	0.1396 ± 0.07342	0.1337 ± 0.09712
r95 *	1	0.1869 ± 0.1002	0.2343 ± 0.1094	0.2119 ± 0.09422	0.2155 ± 0.1187
mean_abs_d0 *	mm	0.2017 ± 0.4021	1.317 ± 8.17	0.986 ± 6.227	1.2 ± 7.727
max_abs_d0 *	mm	0.3772 ± 1.124	3.514 ± 15.85	2.013 ± 10.91	3.178 ± 16.27
std_abs_d0 *	mm	0.1262 ± 0.4579	1.347 ± 6.137	0.8631 ± 4.734	1.222 ± 6.538
mean_abs_dz *	mm	0.4162 ± 1.118	2.029 ± 12.21	1.383 ± 5.559	1.606 ± 9.487
max_abs_dz *	mm	0.795 ± 3.272	5.113 ± 20.35	3.14 ± 12.13	4.195 ± 22.59
std_abs_dz *	mm	0.2736 ± 1.426	1.921 ± 7.328	1.328 ± 5.574	1.624 ± 9.315
pt_weighted_abs_d0 *	mm	0.1903 ± 0.3415	1.246 ± 8.084	0.9838 ± 6.466	1.143 ± 7.737
pt_weighted_abs_dz *	mm	0.4642 ± 1.818	1.772 ± 12.01	1.256 ± 5.211	1.321 ± 8.212

* Variable utilizada como entrada del modelo.

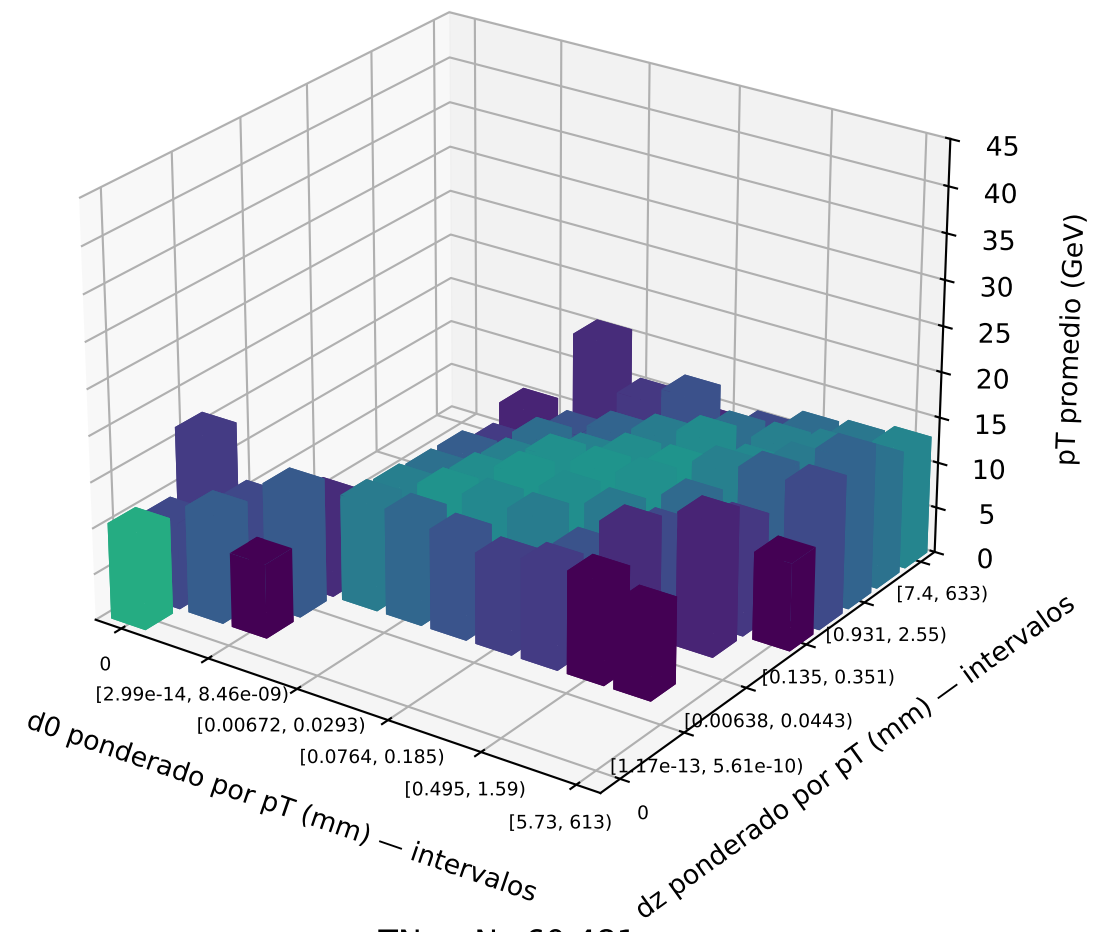
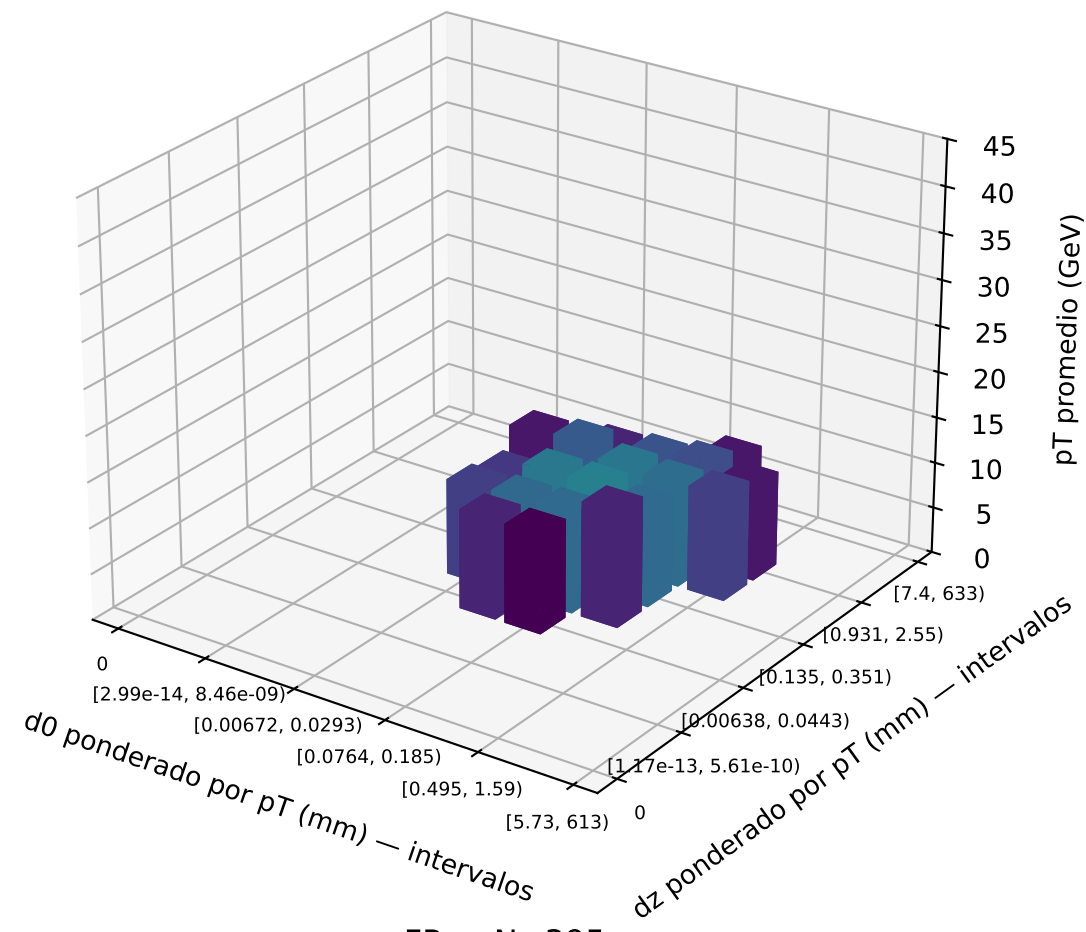
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · ensemble

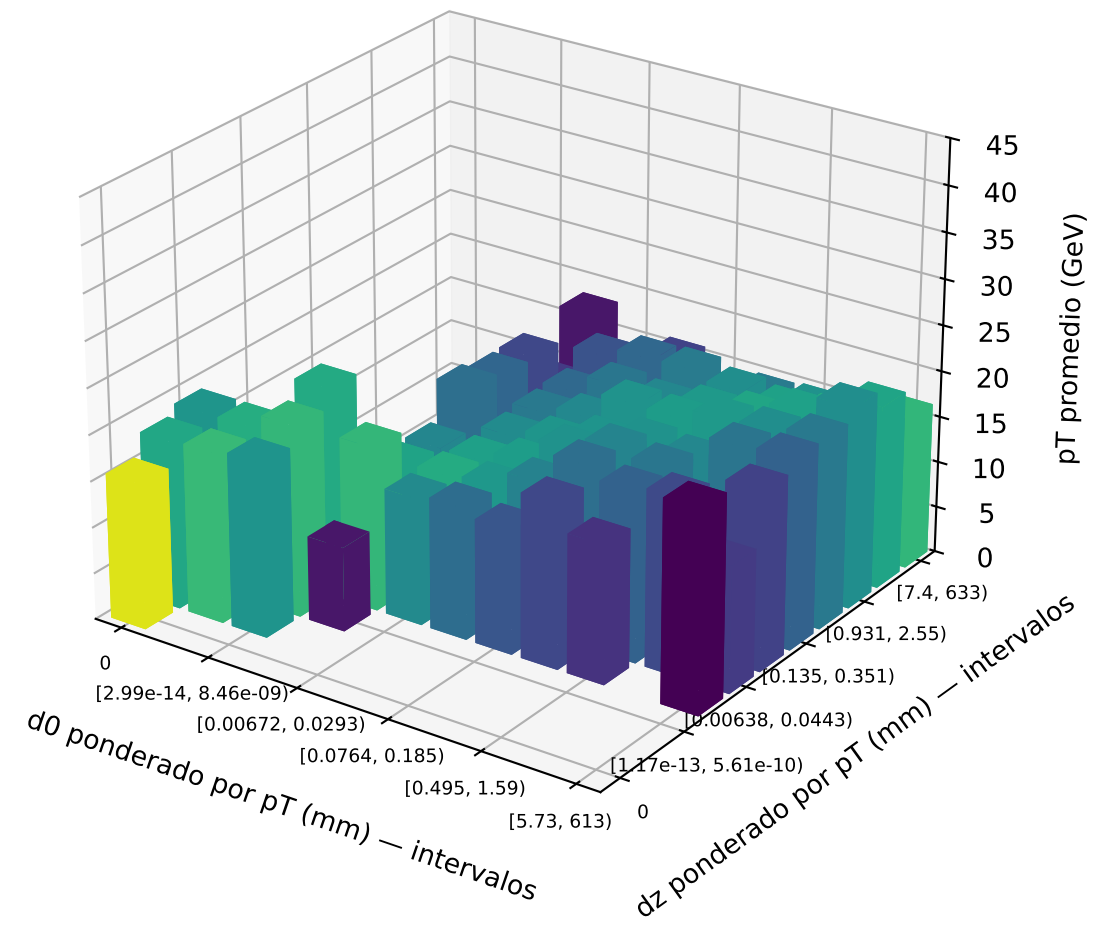
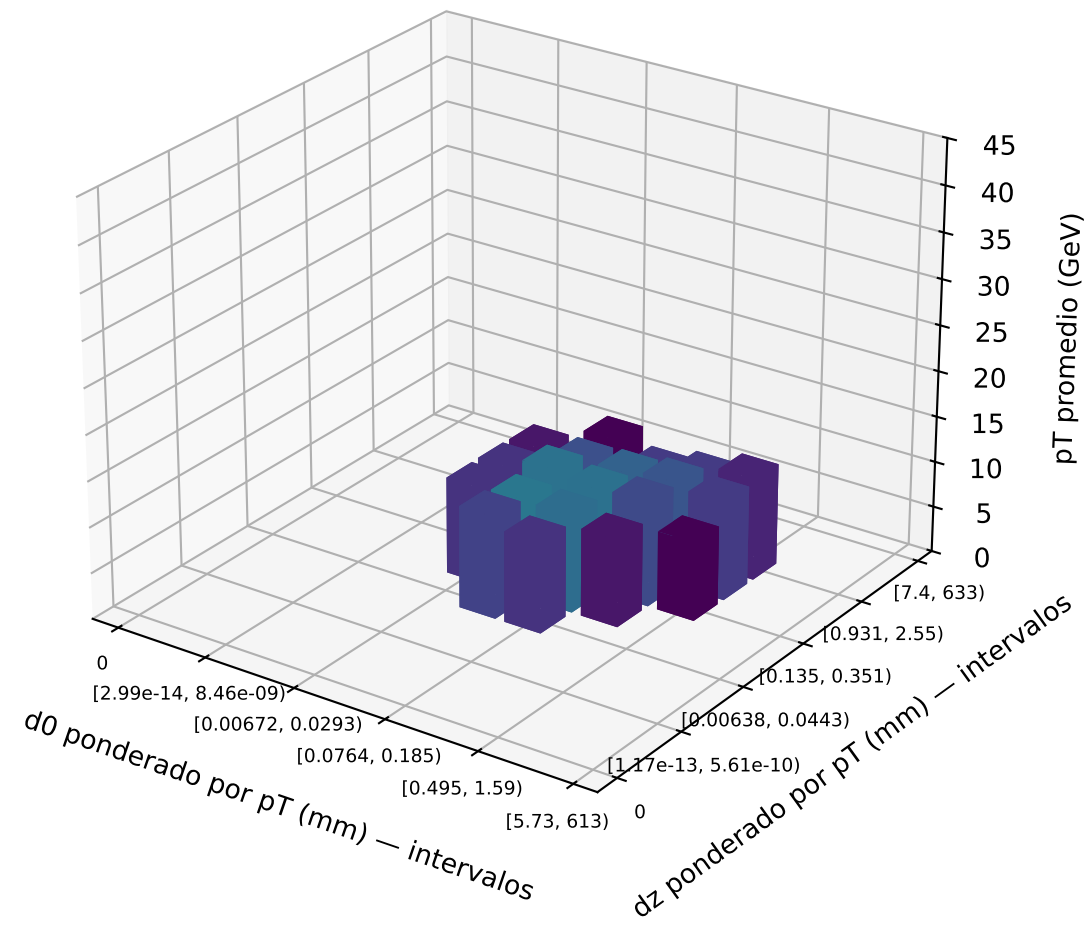
TP — N=562

FN — N=6,113



FP — N=395

TN — N=60,481



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · ensemble

Variable	Unidad	TP N=562	FN N=6,113	FP N=395	TN N=60,481
jet_pt *	GeV	11.29 ± 3.018	12.35 ± 6.867	10.95 ± 2.716	16.45 ± 11.68
jet_eta *	1	0.1264 ± 1.395	-0.008904 ± 1.416	0.2734 ± 1.347	0.000925 ± 1.497
jet_abs_eta *	1	1.266 ± 0.6003	1.211 ± 0.7346	1.237 ± 0.601	1.276 ± 0.7824
jet_mass *	GeV	2.483 ± 0.8399	2.534 ± 1.585	2.311 ± 0.7828	2.855 ± 1.871
jet_ncharged	count	3.319 ± 1.147	2.666 ± 1.818	2.886 ± 1.05	2.586 ± 1.934
jet_nneutral	count	1.6 ± 1.154	1.979 ± 1.653	1.453 ± 1.009	2.375 ± 1.788
n_tracks_cone	count	3.368 ± 1.131	2.7 ± 1.813	2.942 ± 1.023	2.614 ± 1.934
track_pt_sum *	GeV	6.095 ± 2.286	5.358 ± 5.051	5.692 ± 1.961	6.761 ± 8.081
track_pt_sum_over_jet_pt *	1	0.5417 ± 0.1515	0.4245 ± 0.2551	0.5243 ± 0.1447	0.3735 ± 0.267
avg_track_pt *	GeV	1.876 ± 0.6023	1.95 ± 1.424	2.046 ± 0.6941	2.167 ± 2.061
std_track_pt	GeV	0.8775 ± 0.5582	0.7237 ± 0.8894	0.7834 ± 0.5459	1.006 ± 1.601
n_tracks_below_avg_pt	count	1.865 ± 0.9449	1.399 ± 1.277	1.587 ± 0.8445	1.377 ± 1.355
n_tracks_above_avg_pt	count	1.504 ± 0.6032	1.301 ± 0.7701	1.354 ± 0.5656	1.238 ± 0.803
max_pt_ratio	1	0.2761 ± 0.0991	0.2483 ± 0.1693	0.2829 ± 0.09322	0.2117 ± 0.1554
min_pt_ratio	1	0.08893 ± 0.05729	0.1146 ± 0.1367	0.1157 ± 0.07421	0.08633 ± 0.0992
max_dr_ratio	1	0.1232 ± 0.0749	0.1364 ± 0.1384	0.1529 ± 0.08922	0.1051 ± 0.1081
min_dr_ratio	1	0.2136 ± 0.1253	0.214 ± 0.1783	0.224 ± 0.1191	0.181 ± 0.1548
dr_max_pt	1	0.1396 ± 0.06974	0.1372 ± 0.0941	0.1352 ± 0.07285	0.1271 ± 0.09882
dr_min_pt	1	0.2182 ± 0.09312	0.1947 ± 0.1112	0.2034 ± 0.09768	0.1907 ± 0.1168
dr_max_dr	1	0.2681 ± 0.06904	0.2313 ± 0.1126	0.2454 ± 0.07984	0.2214 ± 0.1201
dr_min_dr	1	0.1011 ± 0.05637	0.1088 ± 0.08109	0.09874 ± 0.05622	0.1044 ± 0.08814
pt_difference *	GeV	2.104 ± 1.32	1.729 ± 2.248	1.829 ± 1.296	2.465 ± 4.042
r50	1	0.1481 ± 0.06437	0.1441 ± 0.09187	0.1472 ± 0.06743	0.1337 ± 0.09716
r95	1	0.2649 ± 0.06846	0.2288 ± 0.112	0.2421 ± 0.07985	0.2153 ± 0.1187
mean_abs_d0 *	mm	0.1354 ± 0.1104	1.362 ± 8.307	0.1138 ± 0.1438	1.206 ± 7.744
max_abs_d0 *	mm	0.2689 ± 0.2289	3.633 ± 16.11	0.2638 ± 0.4147	3.19 ± 16.3
std_abs_d0 *	mm	0.09849 ± 0.09135	1.391 ± 6.239	0.1082 ± 0.1799	1.227 ± 6.549
mean_abs_dz *	mm	0.3598 ± 0.3558	2.09 ± 12.42	0.2614 ± 0.2726	1.613 ± 9.499
max_abs_dz *	mm	0.706 ± 0.6515	5.271 ± 20.7	0.6031 ± 0.726	4.213 ± 22.61
std_abs_dz *	mm	0.2579 ± 0.252	1.979 ± 7.454	0.2507 ± 0.3238	1.631 ± 9.327
pt_weighted_abs_d0 *	mm	0.1419 ± 0.1174	1.287 ± 8.22	0.1189 ± 0.1347	1.149 ± 7.755
pt_weighted_abs_dz *	mm	0.3794 ± 0.3801	1.825 ± 12.22	0.2787 ± 0.3025	1.327 ± 8.224

* Variable utilizada como entrada del modelo.

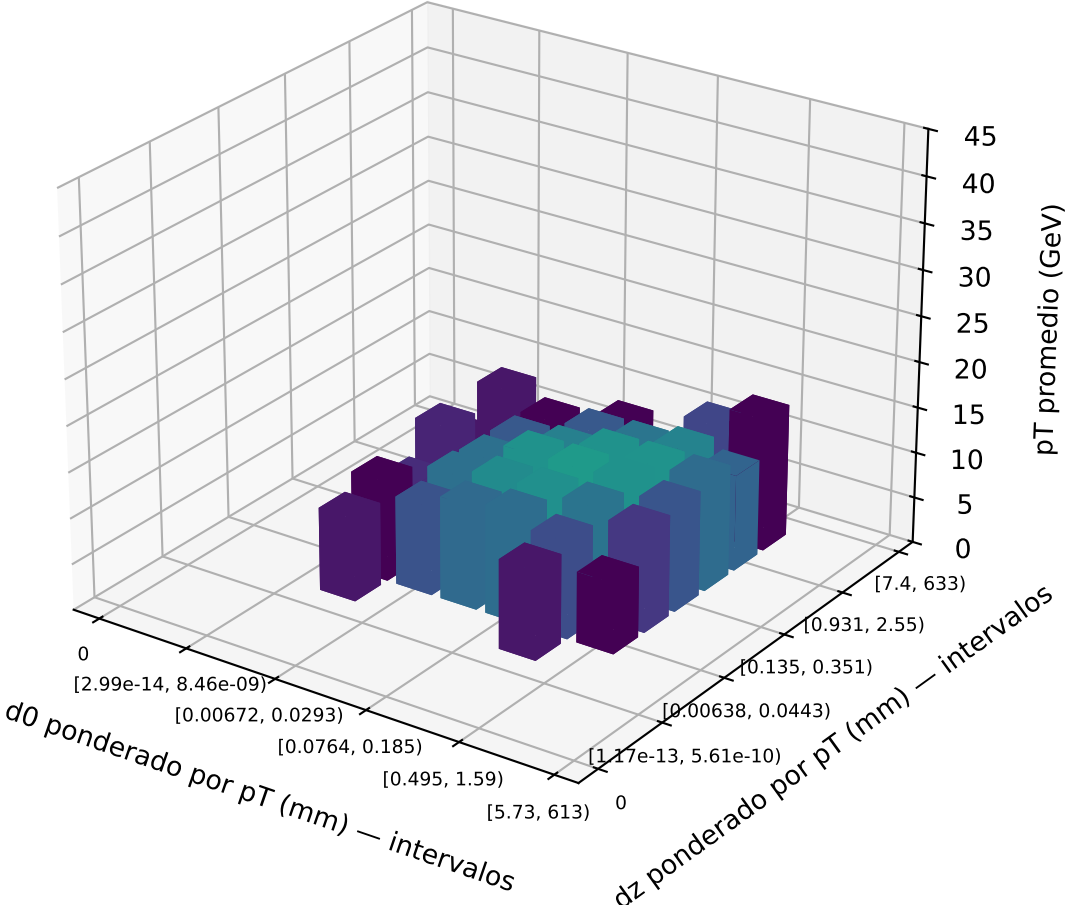
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

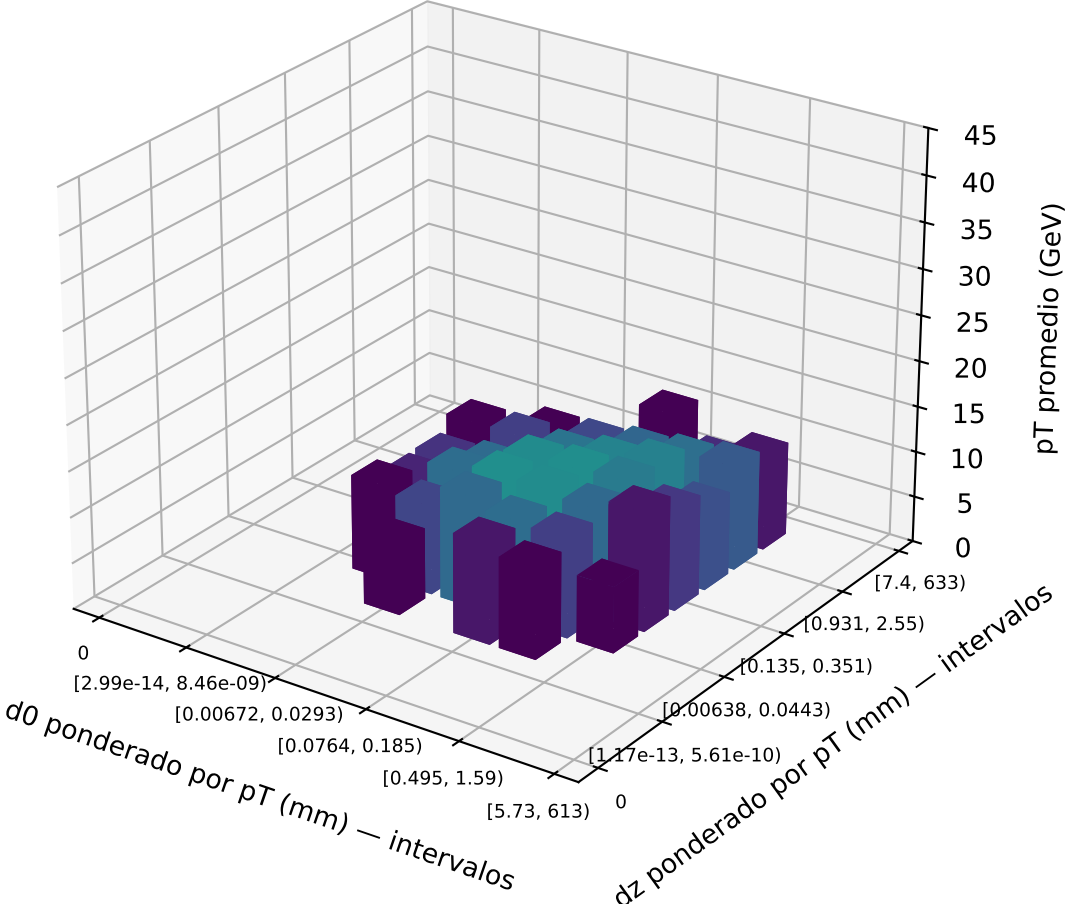
Perfil de pT medio en intervalos de d0/dz — RF16

TP — N=2,819

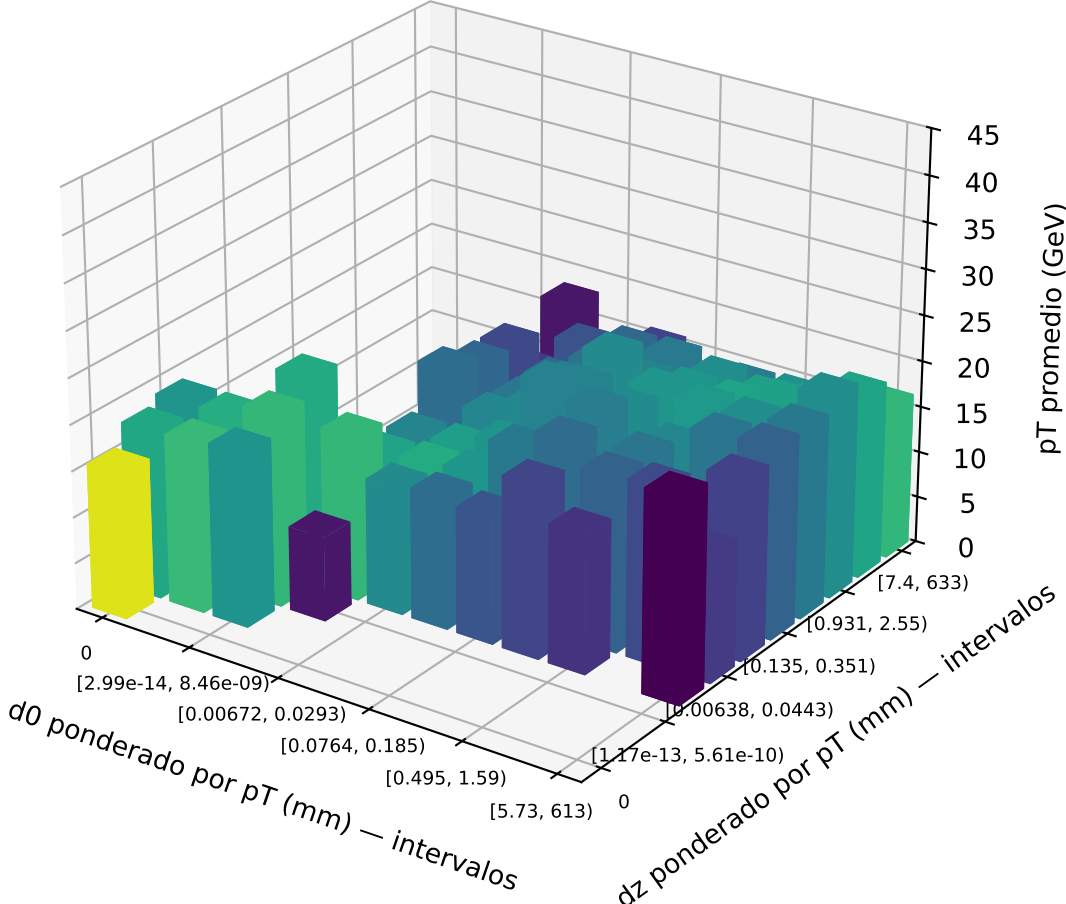
FN — N=3,856



FP — N=1,769



TN — N=59,107



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF16

Variable	Unidad	TP N=2,819	FN N=3,856	FP N=1,769	TN N=59,107
jet_pt *	GeV	11.92 ± 5.068	12.51 ± 7.571	11.7 ± 4.989	16.56 ± 11.76
jet_eta *	1	0.003177 ± 1.349	0.00199 ± 1.461	0.02252 ± 1.355	0.0021 ± 1.5
jet_abs_eta *	1	1.161 ± 0.6869	1.255 ± 0.7481	1.168 ± 0.6876	1.279 ± 0.7837
jet_mass *	GeV	2.46 ± 1.314	2.581 ± 1.678	2.296 ± 1.12	2.868 ± 1.882
jet_ncharged	count	2.783 ± 1.528	2.675 ± 1.944	2.4 ± 1.284	2.593 ± 1.945
jet_nneutral	count	1.737 ± 1.524	2.1 ± 1.671	1.679 ± 1.313	2.39 ± 1.794
n_tracks_cone	count	2.807 ± 1.516	2.719 ± 1.943	2.417 ± 1.268	2.622 ± 1.946
track_pt_sum *	GeV	5.847 ± 3.52	5.107 ± 5.655	5.196 ± 3.524	6.801 ± 8.149
track_pt_sum_over_jet_pt *	1	0.4953 ± 0.2184	0.3898 ± 0.2623	0.4464 ± 0.2242	0.3724 ± 0.2675
avg_track_pt *	GeV	2.339 ± 1.472	1.656 ± 1.219	2.238 ± 1.236	2.164 ± 2.074
std_track_pt	GeV	0.7818 ± 0.809	0.7036 ± 0.9064	0.6719 ± 0.7267	1.014 ± 1.614
n_tracks_below_avg_pt	count	1.437 ± 1.14	1.439 ± 1.339	1.162 ± 0.9911	1.385 ± 1.361
n_tracks_above_avg_pt	count	1.37 ± 0.6445	1.28 ± 0.8317	1.256 ± 0.5124	1.238 ± 0.8088
max_pt_ratio	1	0.2982 ± 0.1662	0.2158 ± 0.1546	0.2741 ± 0.1385	0.2103 ± 0.1553
min_pt_ratio	1	0.1473 ± 0.1624	0.08692 ± 0.09672	0.1454 ± 0.1256	0.08476 ± 0.09764
max_dr_ratio	1	0.1716 ± 0.1605	0.1087 ± 0.1035	0.1676 ± 0.1267	0.1036 ± 0.1069
min_dr_ratio	1	0.2599 ± 0.1847	0.1805 ± 0.1585	0.2415 ± 0.1506	0.1794 ± 0.1544
dr_max_pt	1	0.1362 ± 0.0837	0.1383 ± 0.0981	0.1454 ± 0.08257	0.1266 ± 0.09906
dr_min_pt	1	0.1988 ± 0.1016	0.1952 ± 0.1157	0.1998 ± 0.09415	0.1905 ± 0.1173
dr_max_dr	1	0.2361 ± 0.09922	0.2332 ± 0.1174	0.2276 ± 0.08975	0.2214 ± 0.1207
dr_min_dr	1	0.1078 ± 0.07205	0.1084 ± 0.08426	0.1231 ± 0.07788	0.1038 ± 0.0882
pt_difference *	GeV	1.828 ± 1.862	1.711 ± 2.397	1.541 ± 1.691	2.488 ± 4.076
r50	1	0.1429 ± 0.08151	0.1455 ± 0.09553	0.151 ± 0.07925	0.1332 ± 0.09743
r95	1	0.2335 ± 0.09876	0.2306 ± 0.1166	0.2255 ± 0.08984	0.2152 ± 0.1193
mean_abs_d0 *	mm	0.1608 ± 0.1406	2.061 ± 10.4	0.1292 ± 0.1278	1.231 ± 7.832
max_abs_d0 *	mm	0.2705 ± 0.2311	5.601 ± 20.02	0.212 ± 0.1937	3.259 ± 16.48
std_abs_d0 *	mm	0.08402 ± 0.09225	2.159 ± 7.753	0.06593 ± 0.08172	1.254 ± 6.622
mean_abs_dz *	mm	0.3863 ± 0.4291	3.083 ± 15.55	0.3146 ± 0.3957	1.643 ± 9.607
max_abs_dz *	mm	0.6368 ± 0.6515	7.993 ± 25.66	0.4992 ± 0.5478	4.3 ± 22.87
std_abs_dz *	mm	0.1931 ± 0.2434	3.034 ± 9.222	0.1505 ± 0.2129	1.666 ± 9.432
pt_weighted_abs_d0 *	mm	0.1673 ± 0.1421	1.939 ± 10.29	0.135 ± 0.1278	1.172 ± 7.843
pt_weighted_abs_dz *	mm	0.4056 ± 0.4478	2.652 ± 15.32	0.3295 ± 0.4002	1.35 ± 8.317

* Variable utilizada como entrada del modelo.

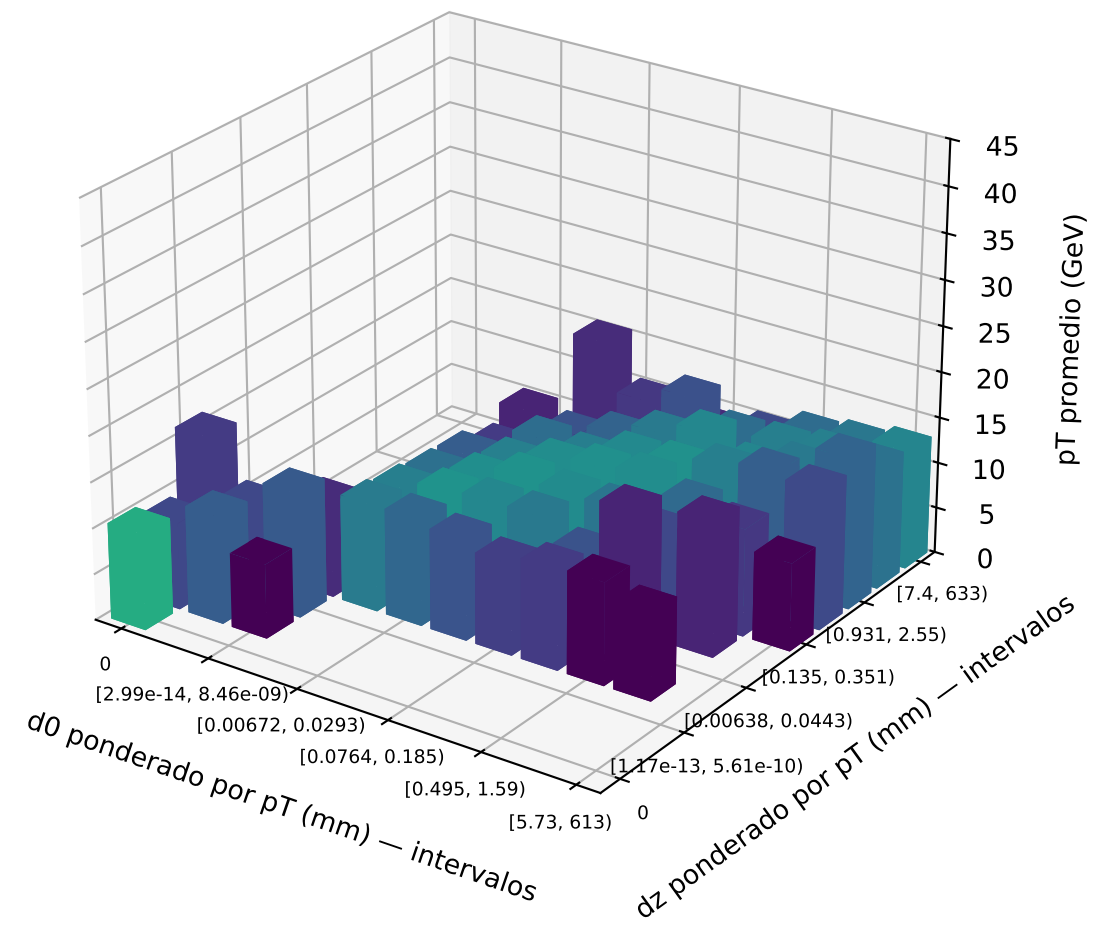
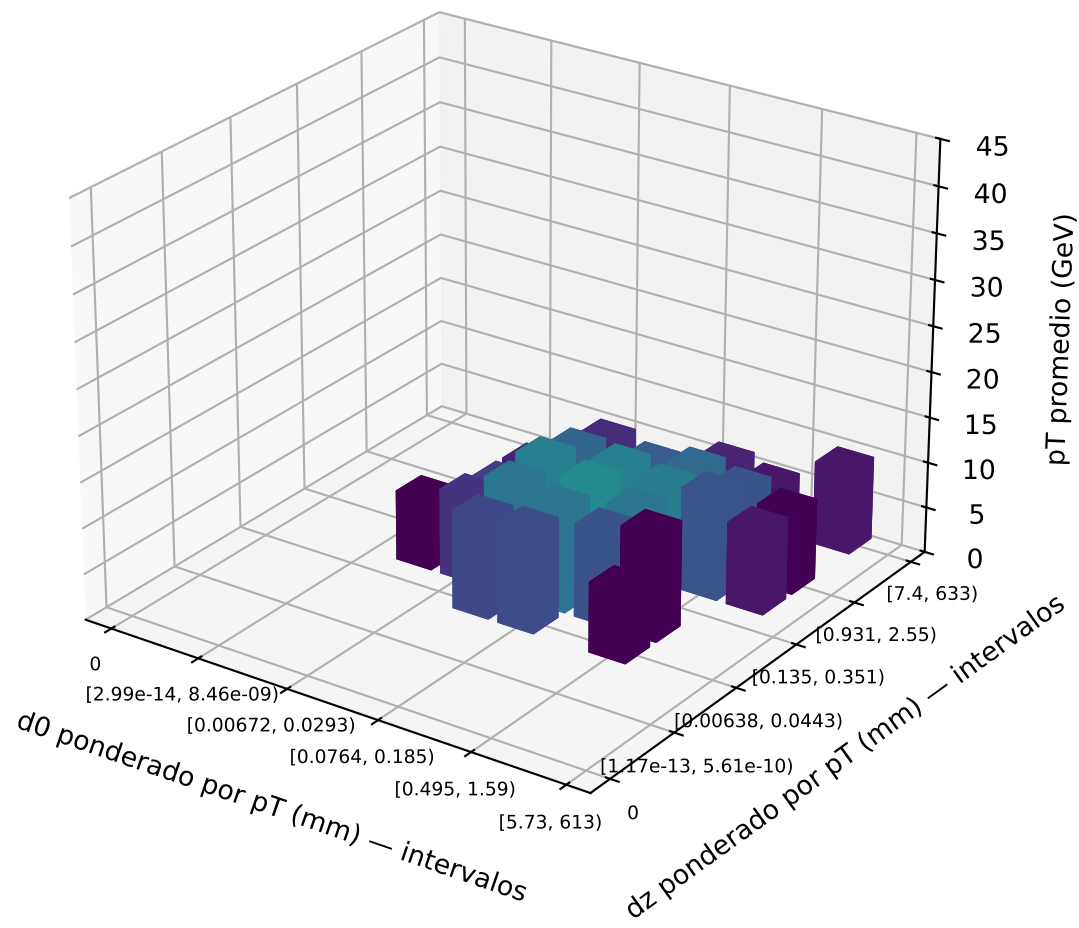
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 42

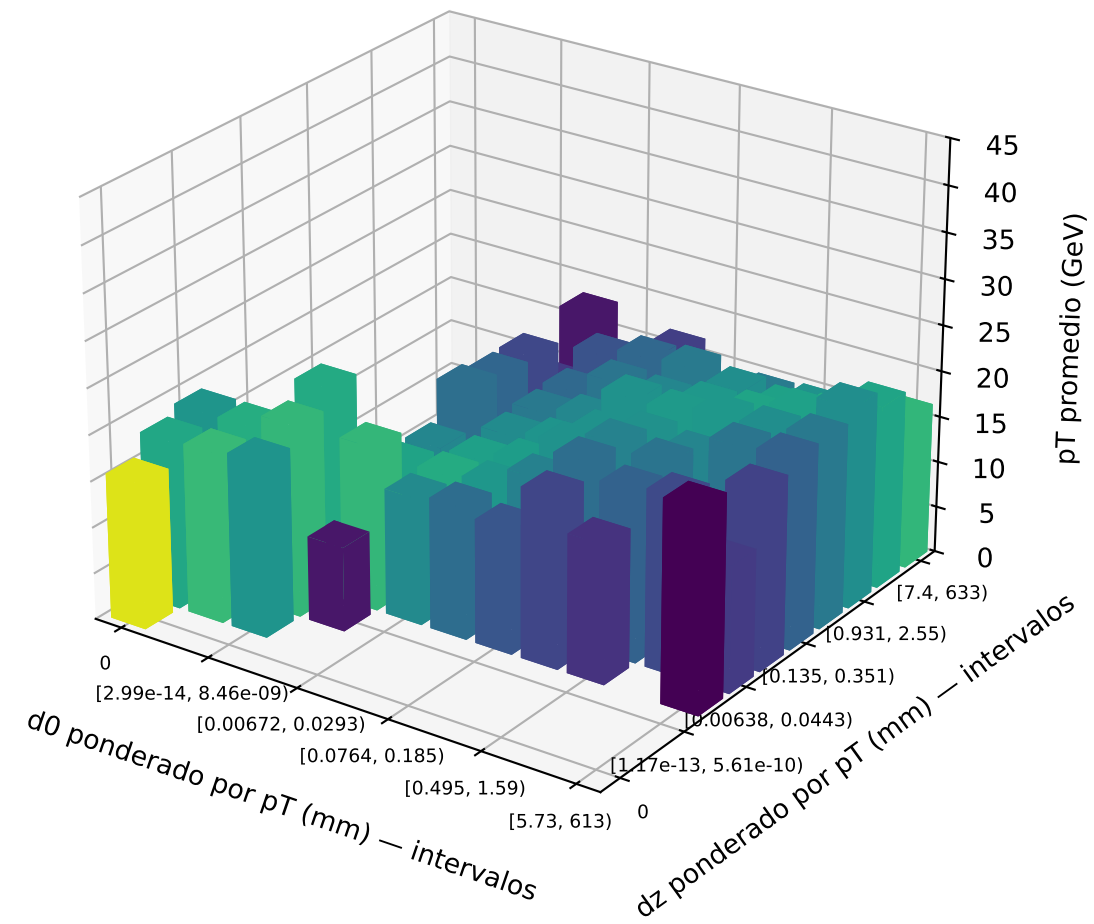
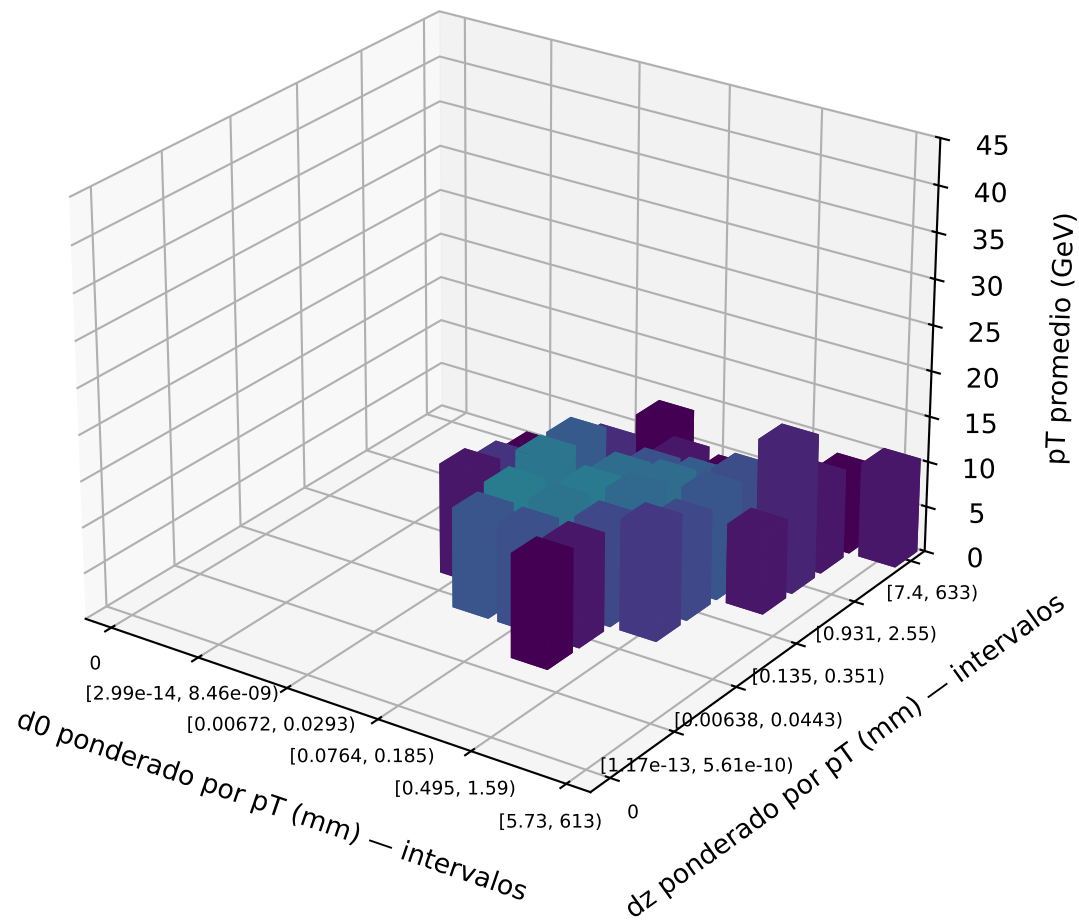
TP — N=952

FN — N=5,723



FP — N=663

TN — N=60,213



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 42

Variable	Unidad	TP N=952	FN N=5,723	FP N=663	TN N=60,213
jet_pt *	GeV	11.33 ± 3.336	12.42 ± 7.025	11.29 ± 4.017	16.47 ± 11.7
jet_eta *	1	0.07876 ± 1.386	-0.0102 ± 1.419	0.3035 ± 1.333	-0.000619 ± 1.497
jet_abs_eta *	1	1.241 ± 0.6226	1.211 ± 0.7399	1.203 ± 0.6496	1.276 ± 0.7826
jet_mass *	GeV	2.603 ± 1.032	2.518 ± 1.605	2.602 ± 1.05	2.854 ± 1.874
jet_ncharged	count	3.38 ± 1.316	2.611 ± 1.824	3.192 ± 1.254	2.581 ± 1.935
jet_nneutral	count	1.616 ± 1.287	2.002 ± 1.663	1.558 ± 1.154	2.378 ± 1.79
n_tracks_cone	count	3.424 ± 1.276	2.645 ± 1.822	3.232 ± 1.213	2.61 ± 1.935
track_pt_sum *	GeV	6.157 ± 2.561	5.297 ± 5.159	5.892 ± 3.217	6.764 ± 8.093
track_pt_sum_over_jet_pt *	1	0.5463 ± 0.1698	0.4157 ± 0.2565	0.521 ± 0.1756	0.3729 ± 0.267
avg_track_pt *	GeV	1.926 ± 0.9102	1.947 ± 1.436	1.887 ± 0.8376	2.169 ± 2.064
std_track_pt	GeV	0.8882 ± 0.6678	0.7114 ± 0.8939	0.7906 ± 0.6609	1.006 ± 1.603
n_tracks_below_avg_pt	count	1.897 ± 0.9957	1.362 ± 1.281	1.738 ± 0.9144	1.374 ± 1.356
n_tracks_above_avg_pt	count	1.527 ± 0.6808	1.283 ± 0.7663	1.495 ± 0.6588	1.235 ± 0.8027
max_pt_ratio	1	0.2823 ± 0.1248	0.2453 ± 0.1699	0.2644 ± 0.1001	0.2116 ± 0.1556
min_pt_ratio	1	0.09482 ± 0.08899	0.1153 ± 0.1377	0.09961 ± 0.06776	0.08638 ± 0.09937
max_dr_ratio	1	0.128 ± 0.1016	0.1365 ± 0.1389	0.1317 ± 0.0819	0.1051 ± 0.1083
min_dr_ratio	1	0.2244 ± 0.1473	0.2123 ± 0.1786	0.213 ± 0.1203	0.1809 ± 0.155
dr_max_pt	1	0.1386 ± 0.07229	0.1372 ± 0.09522	0.143 ± 0.07752	0.1269 ± 0.09887
dr_min_pt	1	0.2176 ± 0.09638	0.1933 ± 0.1117	0.2159 ± 0.09243	0.1905 ± 0.1169
dr_max_dr	1	0.2679 ± 0.07736	0.2288 ± 0.1137	0.2624 ± 0.07616	0.2211 ± 0.1202
dr_min_dr	1	0.09988 ± 0.05565	0.1096 ± 0.08254	0.1052 ± 0.06076	0.1043 ± 0.08823
pt_difference *	GeV	2.121 ± 1.528	1.7 ± 2.273	1.891 ± 1.665	2.467 ± 4.048
r50	1	0.1485 ± 0.06911	0.1437 ± 0.09287	0.1561 ± 0.06985	0.1335 ± 0.09723
r95	1	0.264 ± 0.07773	0.2264 ± 0.113	0.26 ± 0.07701	0.215 ± 0.1188
mean_abs_d0 *	mm	0.172 ± 0.2732	1.439 ± 8.58	0.2516 ± 0.6609	1.209 ± 7.761
max_abs_d0 *	mm	0.3553 ± 0.5395	3.848 ± 16.63	0.6767 ± 1.705	3.198 ± 16.33
std_abs_d0 *	mm	0.1317 ± 0.2087	1.474 ± 6.439	0.2844 ± 0.7657	1.23 ± 6.563
mean_abs_dz *	mm	0.4265 ± 1.029	2.197 ± 12.83	0.5224 ± 2.217	1.616 ± 9.517
max_abs_dz *	mm	0.8321 ± 1.233	5.561 ± 21.35	1.222 ± 3.94	4.222 ± 22.66
std_abs_dz *	mm	0.2968 ± 0.3263	2.09 ± 7.691	0.5077 ± 1.747	1.635 ± 9.346
pt_weighted_abs_d0 *	mm	0.1815 ± 0.2694	1.359 ± 8.491	0.2329 ± 0.5743	1.152 ± 7.772
pt_weighted_abs_dz *	mm	0.4542 ± 1.039	1.911 ± 12.62	0.5307 ± 2.128	1.329 ± 8.239

* Variable utilizada como entrada del modelo.

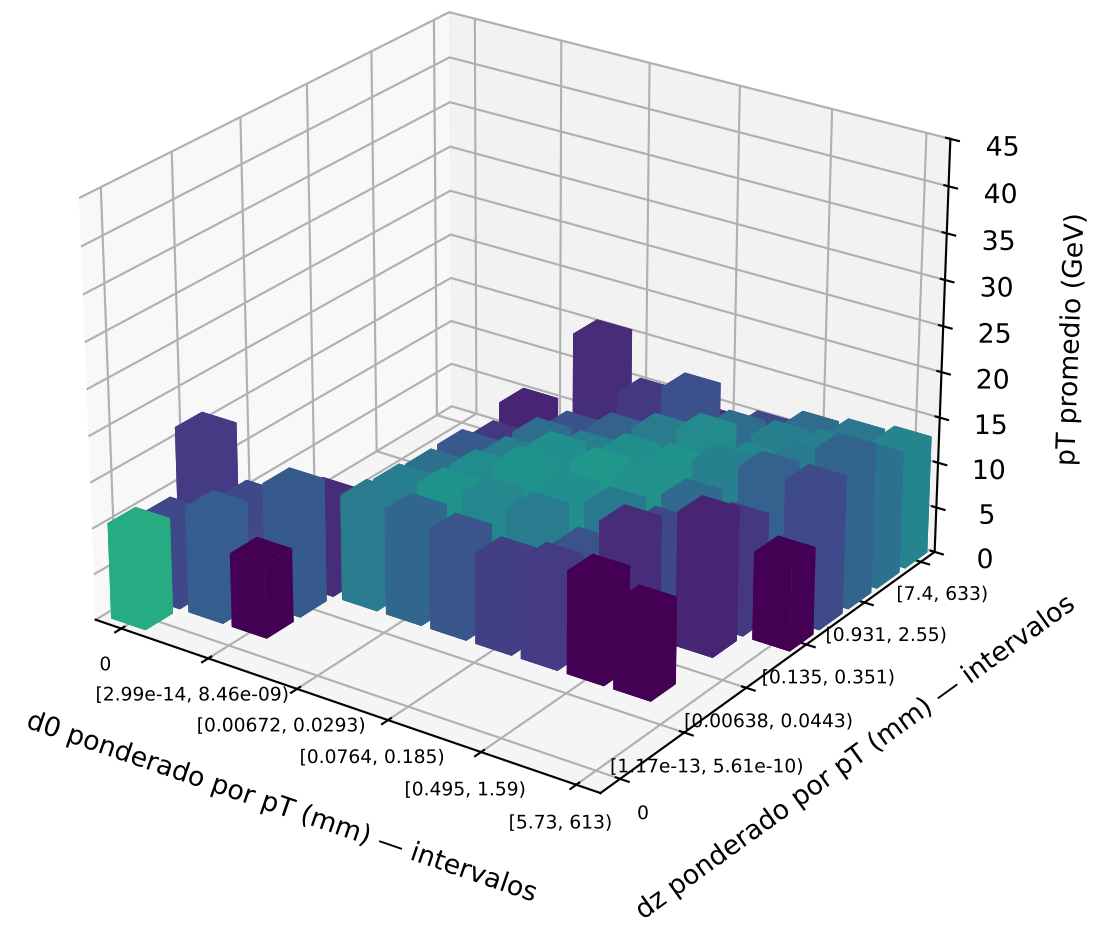
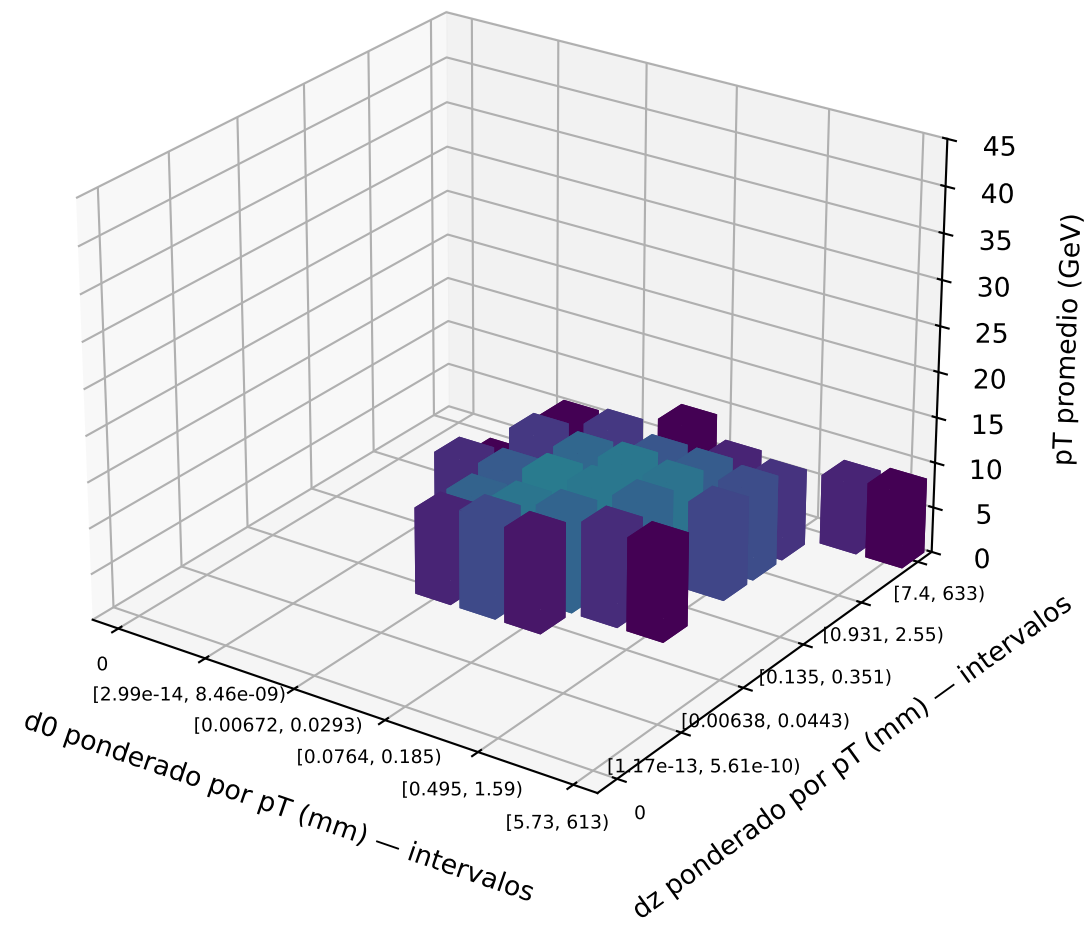
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 73

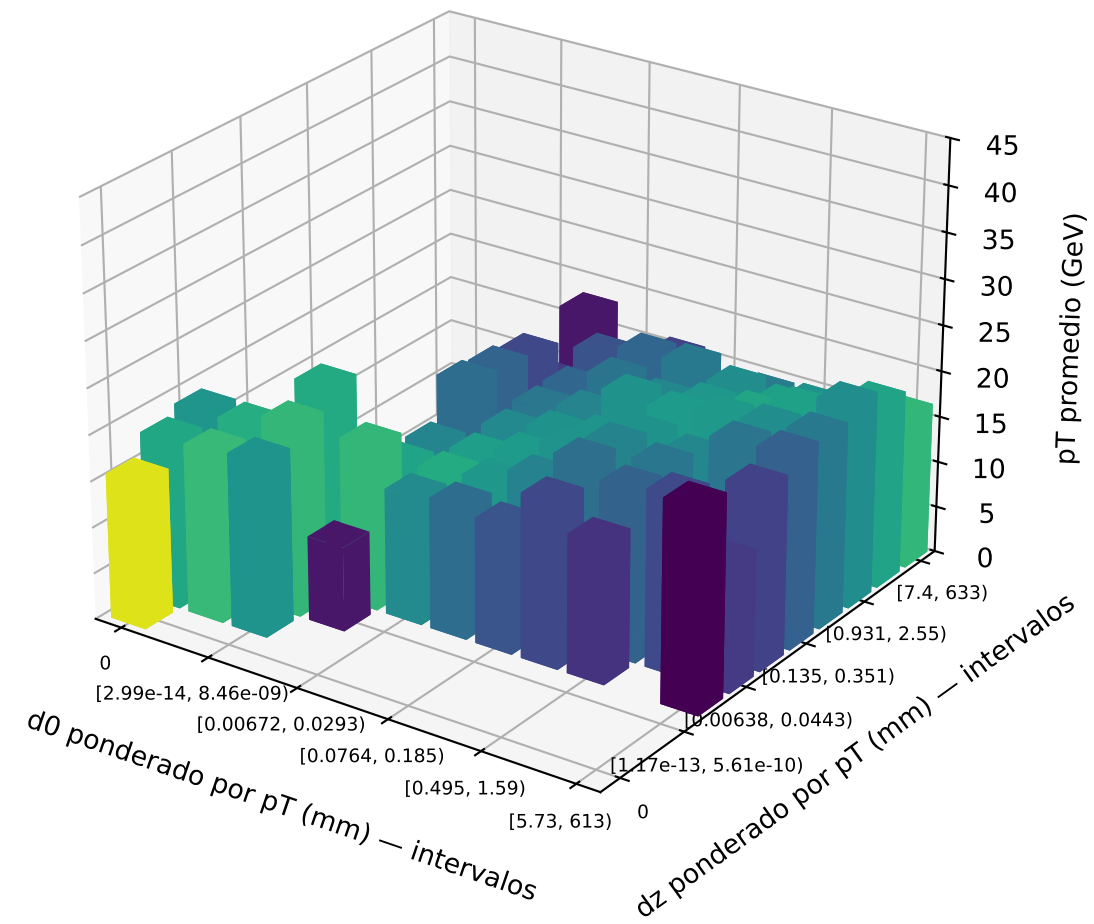
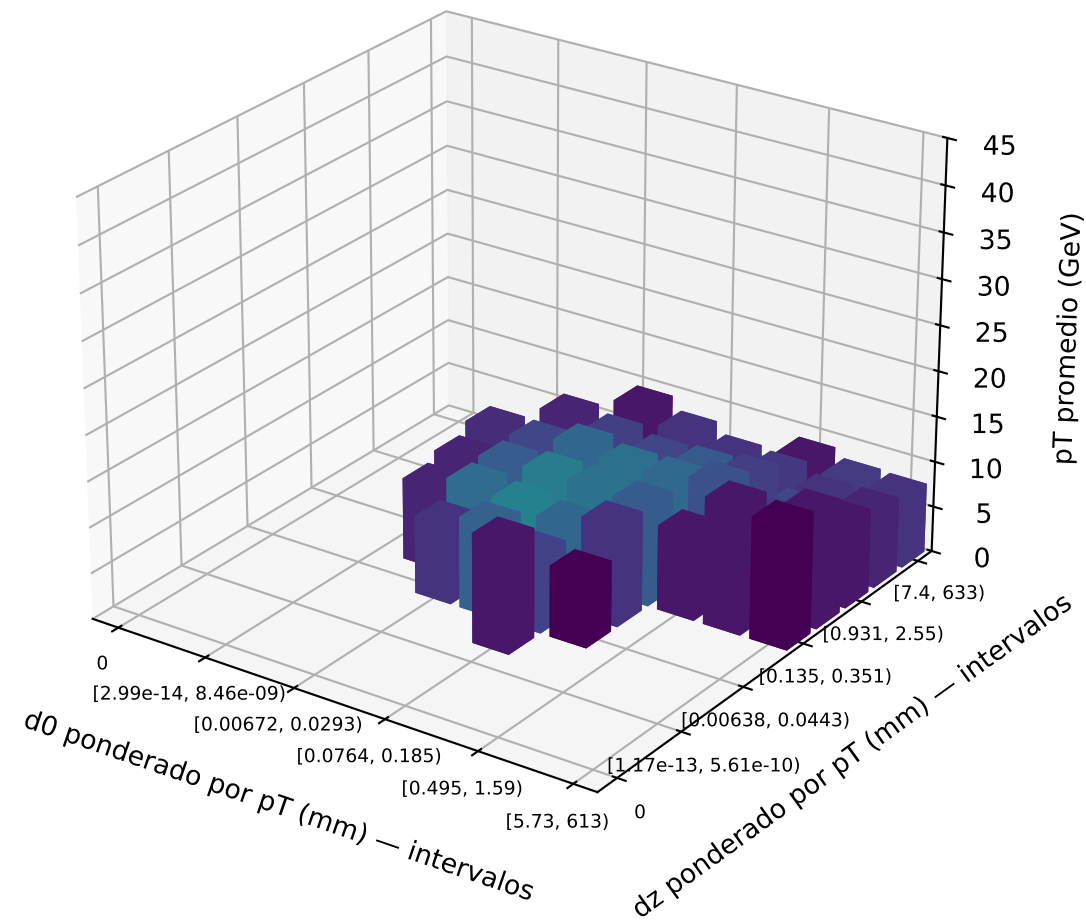
TP — N=692

FN — N=5,983



FP — N=774

TN — N=60,102



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 73

Variable	Unidad	TP N=692	FN N=5,983	FP N=774	TN N=60,102
jet_pt *	GeV	10.79 ± 3.263	12.43 ± 6.901	10.62 ± 3.488	16.49 ± 11.7
jet_eta *	1	0.1162 ± 1.541	-0.01066 ± 1.399	0.05706 ± 1.53	0.001993 ± 1.495
jet_abs_eta *	1	1.396 ± 0.6637	1.194 ± 0.7282	1.37 ± 0.6819	1.274 ± 0.7825
jet_mass *	GeV	2.15 ± 0.9066	2.574 ± 1.587	2.123 ± 0.8569	2.861 ± 1.875
jet_ncharged	count	2.679 ± 1.24	2.726 ± 1.833	2.545 ± 1.105	2.588 ± 1.938
jet_nneutral	count	1.825 ± 1.404	1.961 ± 1.643	1.658 ± 1.272	2.378 ± 1.79
n_tracks_cone	count	2.759 ± 1.21	2.756 ± 1.83	2.628 ± 1.106	2.616 ± 1.938
track_pt_sum *	GeV	5.207 ± 2.68	5.444 ± 5.076	5.004 ± 2.494	6.777 ± 8.101
track_pt_sum_over_jet_pt *	1	0.483 ± 0.1934	0.4287 ± 0.2554	0.4771 ± 0.1917	0.3732 ± 0.2672
avg_track_pt *	GeV	2.011 ± 0.9817	1.936 ± 1.412	2.044 ± 1.071	2.167 ± 2.064
std_track_pt	GeV	0.7691 ± 0.6177	0.7329 ± 0.8918	0.7228 ± 0.6689	1.008 ± 1.604
n_tracks_below_avg_pt	count	1.447 ± 0.9504	1.437 ± 1.29	1.348 ± 0.8727	1.379 ± 1.358
n_tracks_above_avg_pt	count	1.312 ± 0.5646	1.318 ± 0.779	1.28 ± 0.5309	1.238 ± 0.8046
max_pt_ratio	1	0.2872 ± 0.1353	0.2464 ± 0.1673	0.283 ± 0.1223	0.2112 ± 0.1554
min_pt_ratio	1	0.1242 ± 0.1246	0.111 ± 0.1328	0.1295 ± 0.111	0.08597 ± 0.0988
max_dr_ratio	1	0.1514 ± 0.1272	0.1334 ± 0.135	0.1599 ± 0.1148	0.1047 ± 0.1078
min_dr_ratio	1	0.2413 ± 0.1572	0.2108 ± 0.1761	0.2413 ± 0.1424	0.1805 ± 0.1547
dr_max_pt	1	0.1358 ± 0.07874	0.1376 ± 0.09374	0.1375 ± 0.07668	0.127 ± 0.09892
dr_min_pt	1	0.2084 ± 0.0991	0.1954 ± 0.1111	0.2035 ± 0.09534	0.1906 ± 0.117
dr_max_dr	1	0.2428 ± 0.08936	0.2334 ± 0.1122	0.2365 ± 0.08847	0.2214 ± 0.1202
dr_min_dr	1	0.1076 ± 0.06853	0.1083 ± 0.08049	0.1093 ± 0.0647	0.1043 ± 0.08823
pt_difference *	GeV	1.772 ± 1.432	1.759 ± 2.259	1.636 ± 1.48	2.471 ± 4.052
r50	1	0.1423 ± 0.07629	0.1446 ± 0.09132	0.1439 ± 0.07379	0.1336 ± 0.09725
r95	1	0.2405 ± 0.08835	0.2308 ± 0.1116	0.2351 ± 0.08835	0.2152 ± 0.1188
mean_abs_d0 *	mm	0.1847 ± 0.5566	1.383 ± 8.394	0.4968 ± 1.726	1.208 ± 7.766
max_abs_d0 *	mm	0.3738 ± 1.506	3.694 ± 16.27	1.138 ± 3.912	3.197 ± 16.34
std_abs_d0 *	mm	0.1389 ± 0.6357	1.415 ± 6.301	0.5054 ± 1.843	1.229 ± 6.567
mean_abs_dz *	mm	0.5727 ± 2.092	2.103 ± 12.54	1.538 ± 8.939	1.605 ± 9.476
max_abs_dz *	mm	1.251 ± 6.287	5.307 ± 20.81	3.85 ± 21.6	4.194 ± 22.55
std_abs_dz *	mm	0.482 ± 2.901	1.991 ± 7.471	1.745 ± 10.39	1.621 ± 9.282
pt_weighted_abs_d0 *	mm	0.1715 ± 0.3701	1.309 ± 8.307	0.4044 ± 1.485	1.152 ± 7.778
pt_weighted_abs_dz *	mm	0.52 ± 1.189	1.84 ± 12.35	0.9099 ± 4.471	1.326 ± 8.234

* Variable utilizada como entrada del modelo.

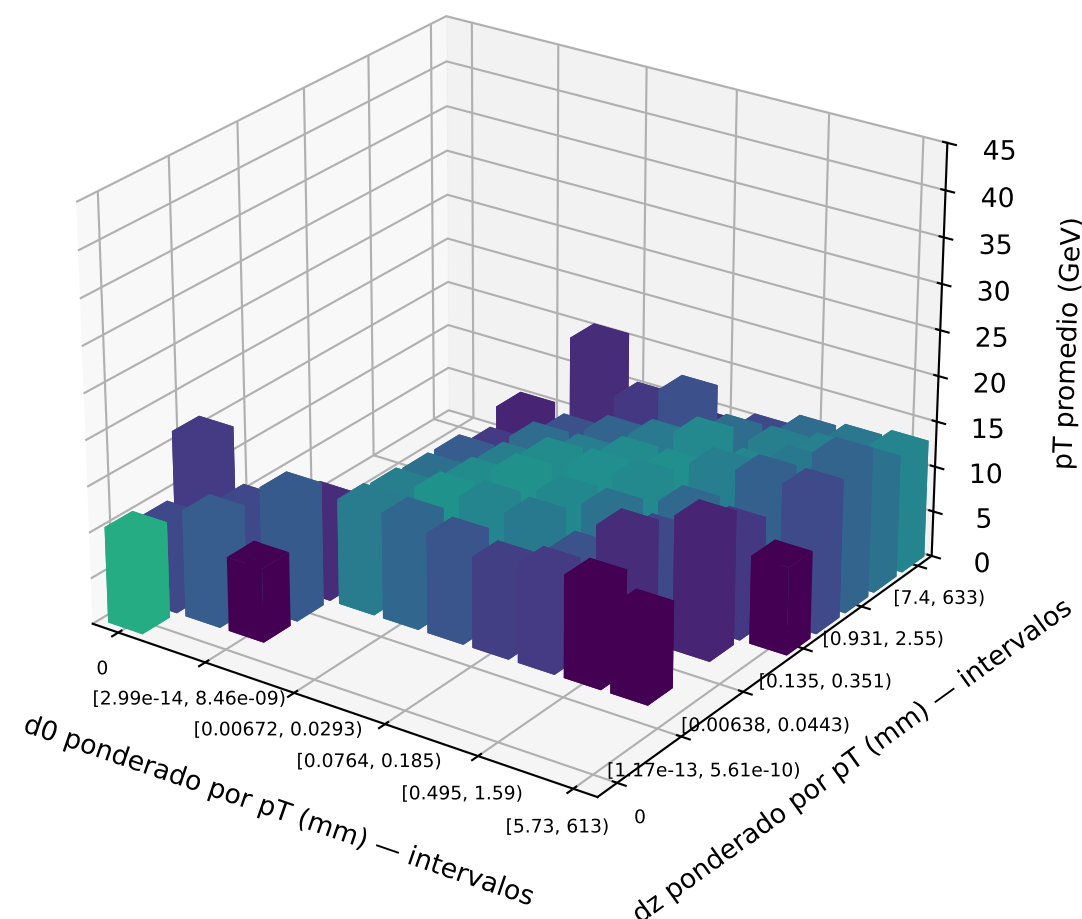
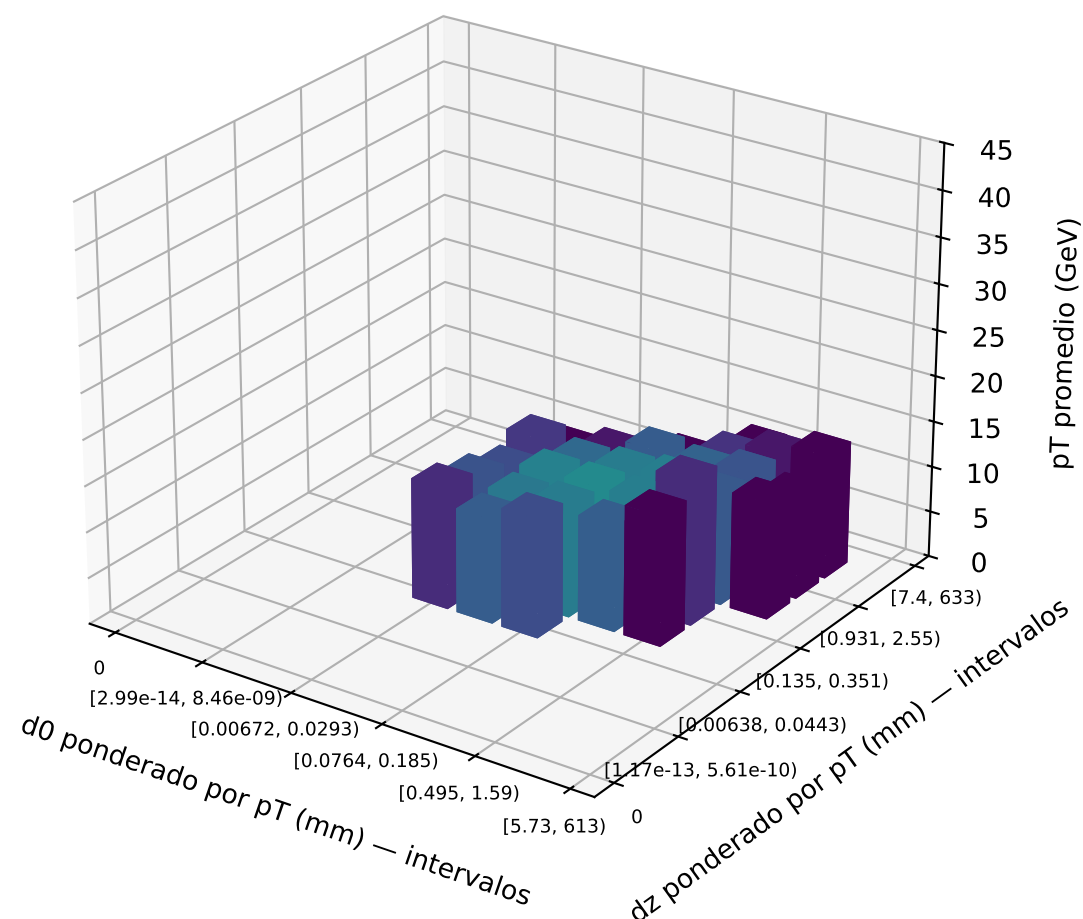
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 101

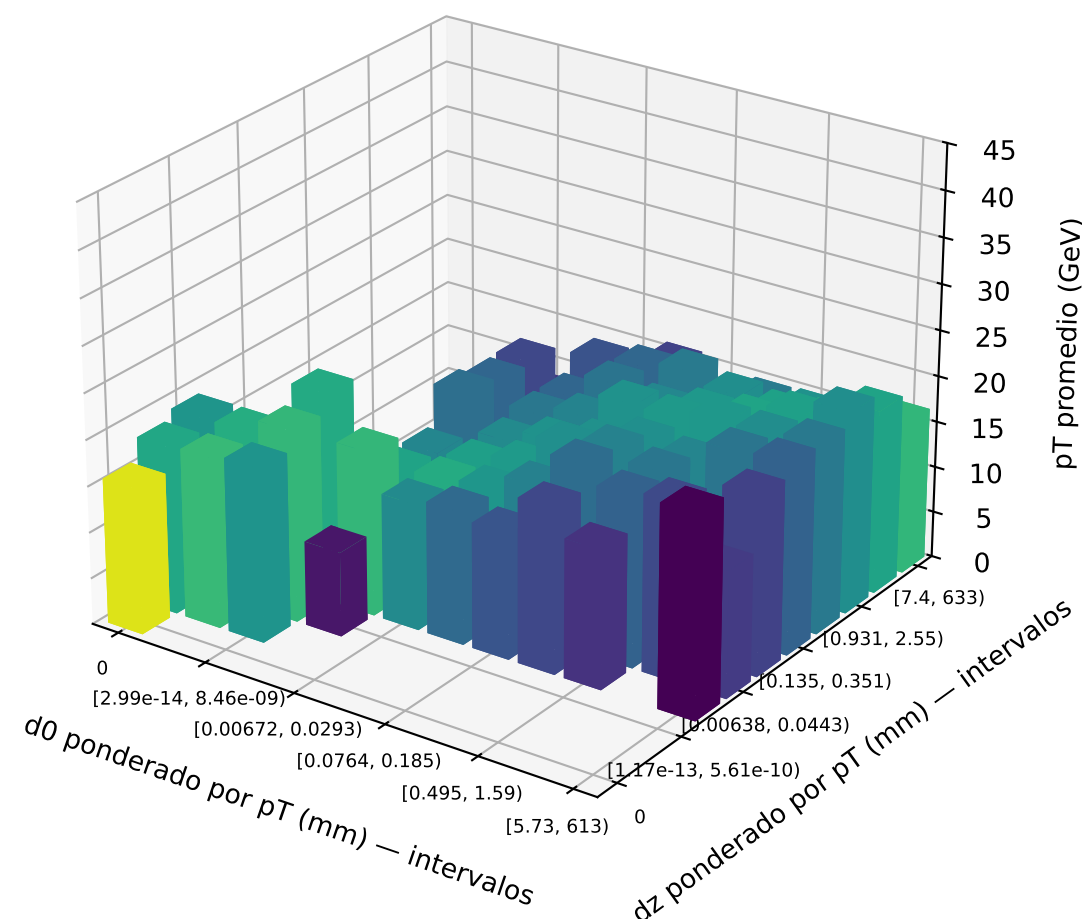
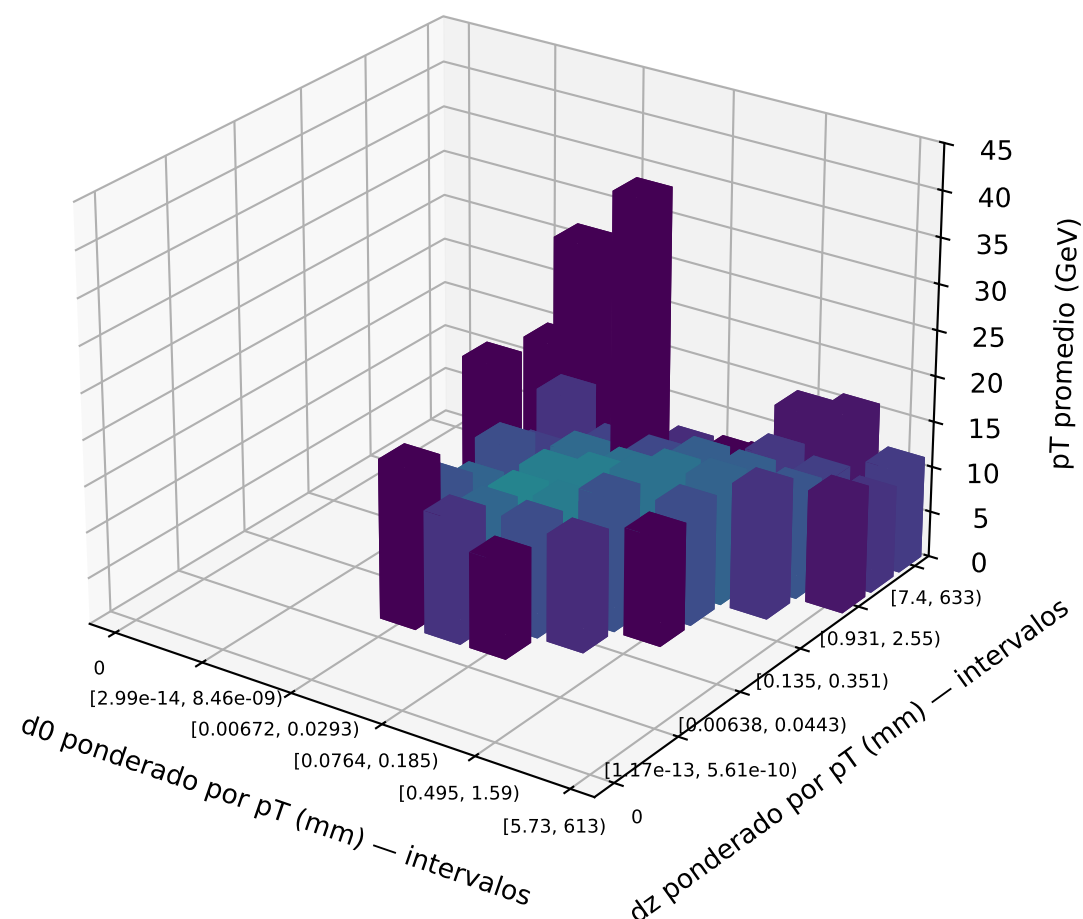
TP — N=1,096

FN — N=5,579



FP — N=1,039

TN — N=59,837



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 101

Variable	Unidad	TP N=1,096	FN N=5,579	FP N=1,039	TN N=59,837
jet_pt *	GeV	11.88 ± 3.863	12.34 ± 7.052	12.38 ± 4.394	16.48 ± 11.73
jet_eta *	1	0.01727 ± 1.263	-0.000412 ± 1.443	0.05285 ± 1.288	0.001822 ± 1.499
jet_abs_eta *	1	1.096 ± 0.6274	1.239 ± 0.7397	1.118 ± 0.6427	1.278 ± 0.7832
jet_mass *	GeV	2.617 ± 1.053	2.513 ± 1.614	2.689 ± 1.042	2.855 ± 1.878
jet_ncharged	count	3.222 ± 1.422	2.622 ± 1.827	3.163 ± 1.443	2.578 ± 1.935
jet_nneutral	count	1.625 ± 1.252	2.01 ± 1.676	1.659 ± 1.185	2.382 ± 1.792
n_tracks_cone	count	3.255 ± 1.408	2.658 ± 1.824	3.202 ± 1.427	2.606 ± 1.936
track_pt_sum *	GeV	6.216 ± 2.742	5.263 ± 5.186	6.212 ± 3.007	6.763 ± 8.116
track_pt_sum_over_jet_pt *	1	0.5263 ± 0.1728	0.4163 ± 0.2589	0.5051 ± 0.1723	0.3722 ± 0.2674
avg_track_pt *	GeV	2.049 ± 0.8451	1.923 ± 1.454	2.046 ± 0.74	2.168 ± 2.07
std_track_pt	GeV	0.8298 ± 0.6397	0.7183 ± 0.9044	0.8065 ± 0.6272	1.008 ± 1.608
n_tracks_below_avg_pt	count	1.779 ± 1.081	1.371 ± 1.28	1.746 ± 1.109	1.372 ± 1.356
n_tracks_above_avg_pt	count	1.476 ± 0.6502	1.287 ± 0.7754	1.456 ± 0.6311	1.234 ± 0.8038
max_pt_ratio	1	0.2782 ± 0.1156	0.2452 ± 0.1722	0.2643 ± 0.09797	0.2112 ± 0.1558
min_pt_ratio	1	0.1091 ± 0.09656	0.1131 ± 0.1379	0.1074 ± 0.08065	0.08616 ± 0.09934
max_dr_ratio	1	0.1392 ± 0.1016	0.1345 ± 0.1398	0.136 ± 0.08915	0.1049 ± 0.1083
min_dr_ratio	1	0.2272 ± 0.1395	0.2114 ± 0.1805	0.2147 ± 0.1175	0.1807 ± 0.1552
dr_max_pt	1	0.1405 ± 0.07393	0.1368 ± 0.09548	0.1411 ± 0.07776	0.1269 ± 0.09898
dr_min_pt	1	0.2118 ± 0.09718	0.1938 ± 0.1121	0.216 ± 0.09535	0.1903 ± 0.117
dr_max_dr	1	0.2602 ± 0.08239	0.2293 ± 0.1141	0.2592 ± 0.08224	0.2209 ± 0.1203
dr_min_dr	1	0.1048 ± 0.0599	0.1089 ± 0.0826	0.1065 ± 0.06392	0.1043 ± 0.08833
pt_difference *	GeV	1.992 ± 1.508	1.715 ± 2.295	1.951 ± 1.552	2.469 ± 4.059
r50	1	0.1492 ± 0.06992	0.1435 ± 0.09328	0.1524 ± 0.07291	0.1334 ± 0.09733
r95	1	0.2571 ± 0.08189	0.2268 ± 0.1134	0.2558 ± 0.08283	0.2148 ± 0.1189
mean_abs_d0 *	mm	0.1728 ± 0.2067	1.472 ± 8.687	0.4898 ± 3.128	1.211 ± 7.775
max_abs_d0 *	mm	0.3893 ± 0.8267	3.931 ± 16.83	1.413 ± 9.111	3.201 ± 16.34
std_abs_d0 *	mm	0.1408 ± 0.3213	1.507 ± 6.517	0.6112 ± 4.262	1.23 ± 6.56
mean_abs_dz *	mm	0.3905 ± 0.5804	2.249 ± 12.99	0.623 ± 1.65	1.621 ± 9.547
max_abs_dz *	mm	0.7954 ± 1.42	5.69 ± 21.61	1.654 ± 4.225	4.233 ± 22.73
std_abs_dz *	mm	0.2858 ± 0.6152	2.139 ± 7.78	0.6958 ± 1.912	1.639 ± 9.374
pt_weighted_abs_d0 *	mm	0.1765 ± 0.1999	1.39 ± 8.597	0.4691 ± 2.532	1.154 ± 7.79
pt_weighted_abs_dz *	mm	0.3928 ± 0.5542	1.96 ± 12.78	0.6024 ± 1.729	1.333 ± 8.265

* Variable utilizada como entrada del modelo.

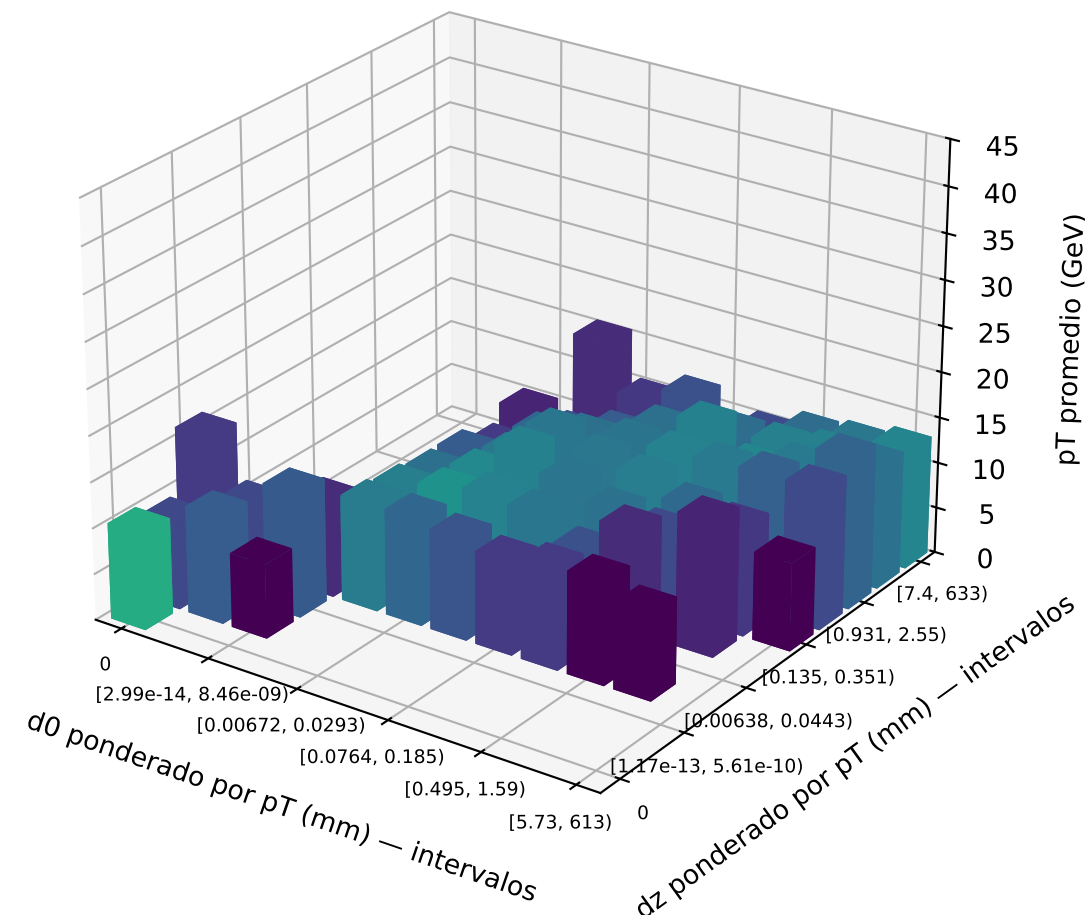
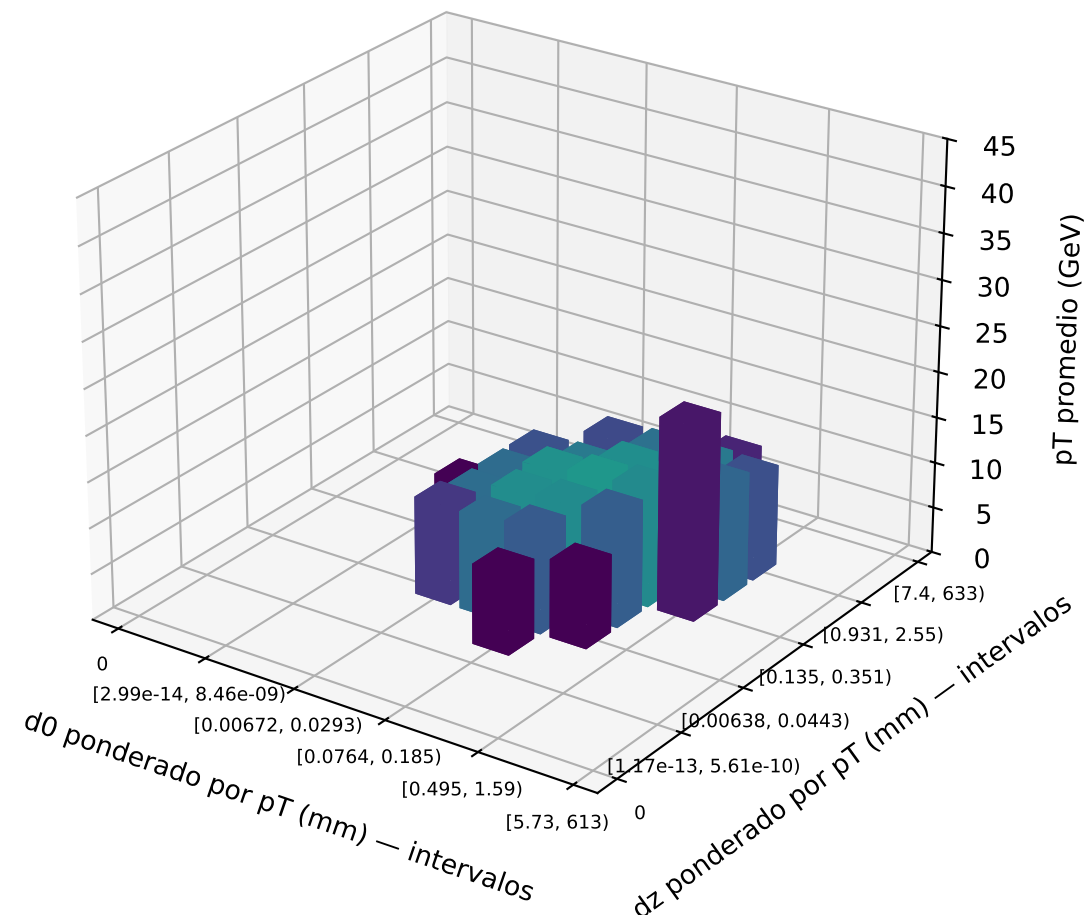
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC12 · ensemble

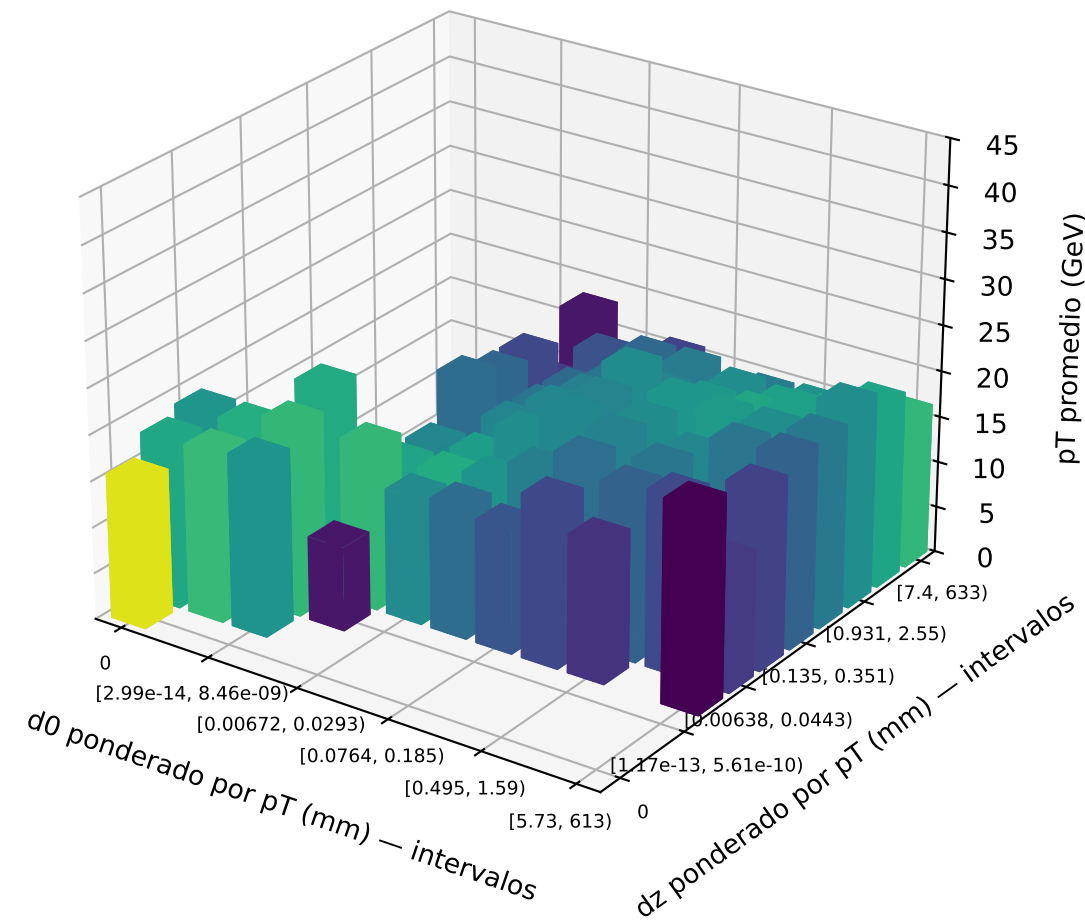
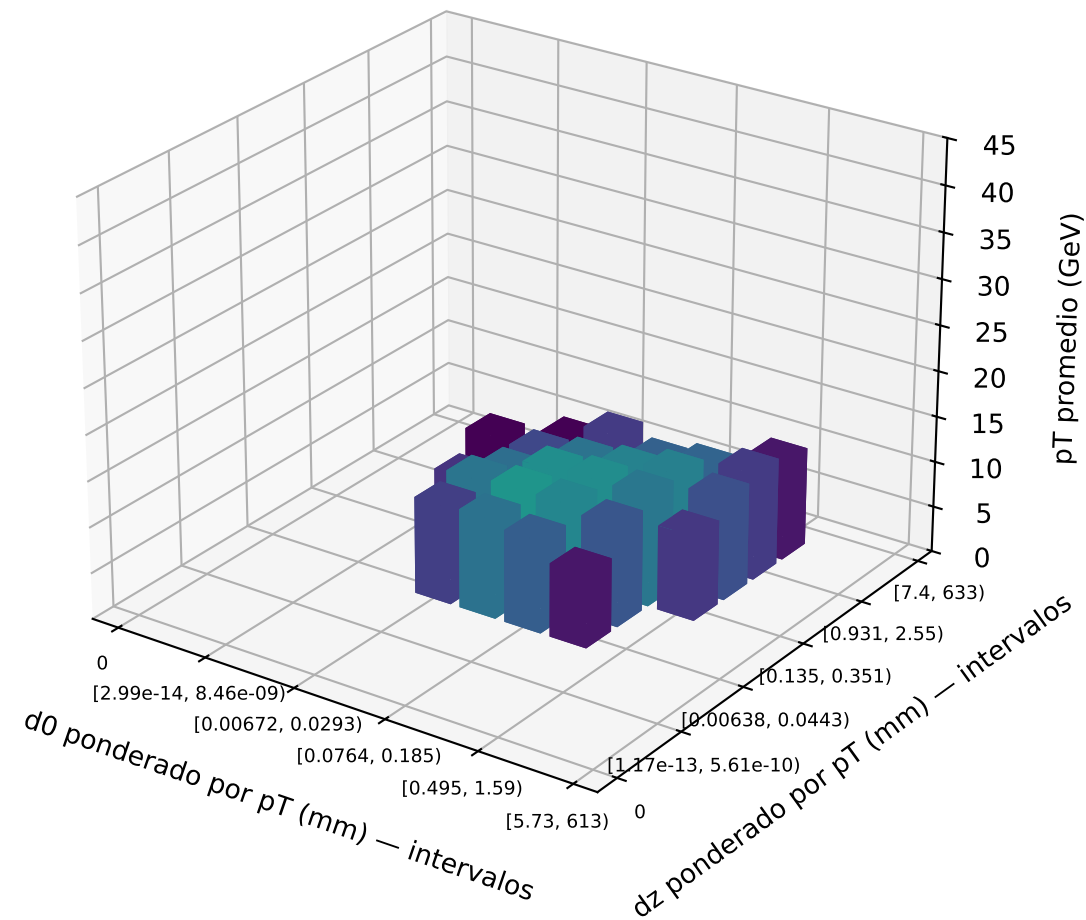
TP — N=2,233

FN — N=4,442



FP — N=1,733

TN — N=59,143



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC12 · ensemble

Variable	Unidad	TP N=2,233	FN N=4,442	FP N=1,733	TN N=59,143
jet_pt *	GeV	11.68 ± 4.203	12.55 ± 7.553	11.52 ± 4.43	16.56 ± 11.77
jet_eta	1	-0.02116 ± 1.344	0.01438 ± 1.449	0.02729 ± 1.376	0.001973 ± 1.499
jet_abs_eta *	1	1.175 ± 0.6523	1.236 ± 0.7572	1.202 ± 0.671	1.278 ± 0.7842
jet_mass	GeV	2.507 ± 1.159	2.542 ± 1.694	2.418 ± 1.11	2.864 ± 1.883
jet_ncharged	count	2.973 ± 1.39	2.594 ± 1.936	2.684 ± 1.265	2.585 ± 1.946
jet_nneutral	count	1.581 ± 1.347	2.131 ± 1.712	1.556 ± 1.226	2.393 ± 1.794
n_tracks_cone	count	3.003 ± 1.373	2.632 ± 1.936	2.707 ± 1.255	2.614 ± 1.946
track_pt_sum *	GeV	6.12 ± 3.211	5.068 ± 5.502	5.692 ± 3.207	6.785 ± 8.153
track_pt_sum_over_jet_pt	1	0.5281 ± 0.1967	0.3872 ± 0.2607	0.4972 ± 0.2043	0.3709 ± 0.2674
avg_track_pt *	GeV	2.23 ± 1.182	1.8 ± 1.439	2.215 ± 0.9857	2.164 ± 2.078
std_track_pt	GeV	0.872 ± 0.766	0.6686 ± 0.9066	0.7438 ± 0.6848	1.012 ± 1.615
n_tracks_below_avg_pt	count	1.606 ± 1.052	1.354 ± 1.343	1.381 ± 0.9718	1.378 ± 1.362
n_tracks_above_avg_pt	count	1.397 ± 0.6258	1.278 ± 0.8157	1.326 ± 0.5638	1.236 ± 0.8075
max_pt_ratio	1	0.3006 ± 0.1446	0.2255 ± 0.1684	0.2837 ± 0.118	0.21 ± 0.1557
min_pt_ratio	1	0.1268 ± 0.1295	0.1052 ± 0.1327	0.1363 ± 0.1074	0.08506 ± 0.09845
max_dr_ratio	1	0.1553 ± 0.1319	0.1252 ± 0.1344	0.1624 ± 0.1101	0.1038 ± 0.1076
min_dr_ratio	1	0.2546 ± 0.1674	0.1936 ± 0.1744	0.2455 ± 0.1356	0.1794 ± 0.1548
dr_max_pt	1	0.1359 ± 0.08053	0.1382 ± 0.09767	0.1385 ± 0.07605	0.1268 ± 0.09924
dr_min_pt	1	0.2049 ± 0.09815	0.1926 ± 0.1153	0.2025 ± 0.09399	0.1904 ± 0.1173
dr_max_dr	1	0.2466 ± 0.09098	0.2283 ± 0.1181	0.2369 ± 0.08724	0.2211 ± 0.1207
dr_min_dr	1	0.1035 ± 0.06496	0.1106 ± 0.08556	0.1122 ± 0.06685	0.1041 ± 0.08851
pt_difference	GeV	2.038 ± 1.771	1.621 ± 2.357	1.729 ± 1.618	2.482 ± 4.077
r50	1	0.1431 ± 0.07748	0.145 ± 0.09551	0.147 ± 0.07289	0.1334 ± 0.09759
r95	1	0.2439 ± 0.09092	0.2257 ± 0.1172	0.2346 ± 0.08735	0.2149 ± 0.1193
mean_abs_d0 *	mm	0.1405 ± 0.1216	1.821 ± 9.705	0.111 ± 0.1168	1.231 ± 7.829
max_abs_d0 *	mm	0.2659 ± 0.2527	4.9 ± 18.74	0.2386 ± 0.2909	3.257 ± 16.47
std_abs_d0 *	mm	0.09407 ± 0.1045	1.88 ± 7.259	0.09347 ± 0.1324	1.253 ± 6.621
mean_abs_dz *	mm	0.3533 ± 0.3571	2.744 ± 14.52	0.2753 ± 0.3025	1.643 ± 9.604
max_abs_dz *	mm	0.6451 ± 0.6678	7.019 ± 24.04	0.5531 ± 0.6465	4.296 ± 22.86
std_abs_dz *	mm	0.2219 ± 0.2731	2.645 ± 8.65	0.2076 ± 0.2936	1.664 ± 9.429
pt_weighted_abs_d0 *	mm	0.1477 ± 0.1256	1.715 ± 9.608	0.1146 ± 0.1133	1.172 ± 7.841
pt_weighted_abs_dz *	mm	0.3697 ± 0.3735	2.373 ± 14.3	0.2851 ± 0.316	1.351 ± 8.315

* Variable utilizada como entrada del modelo.

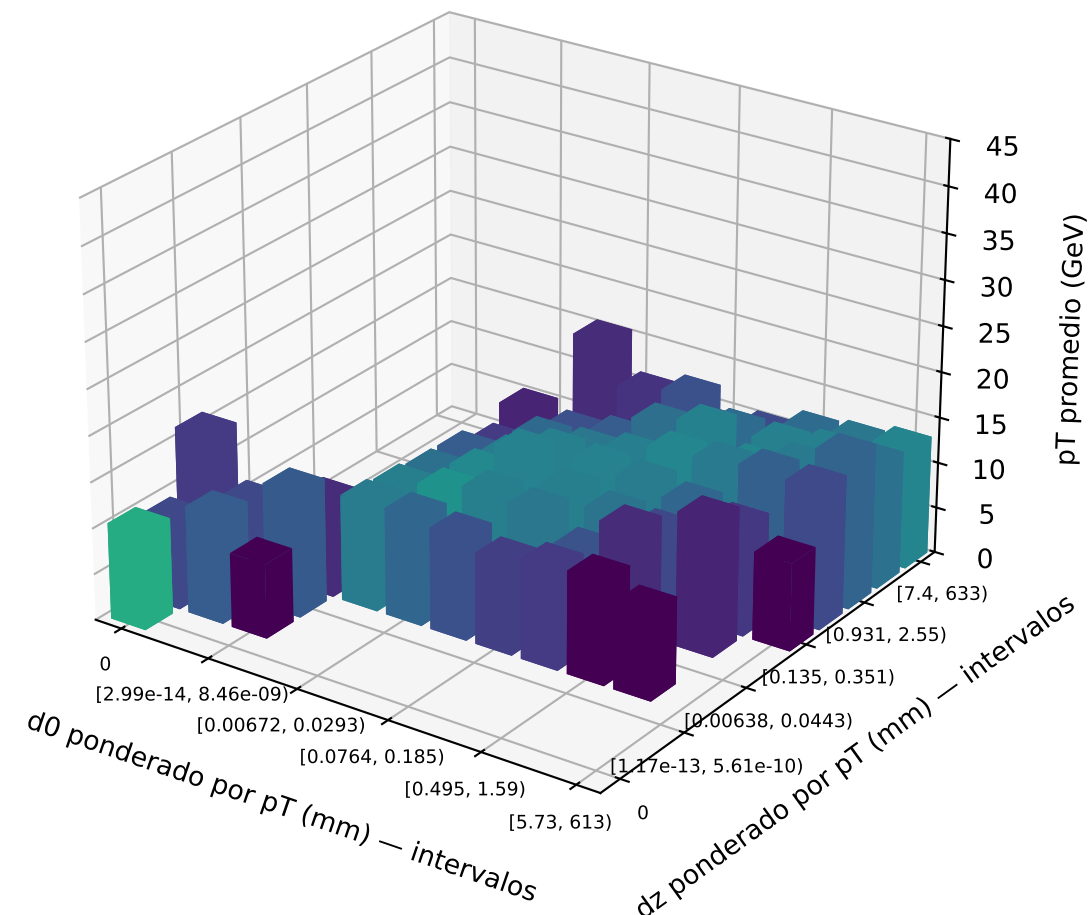
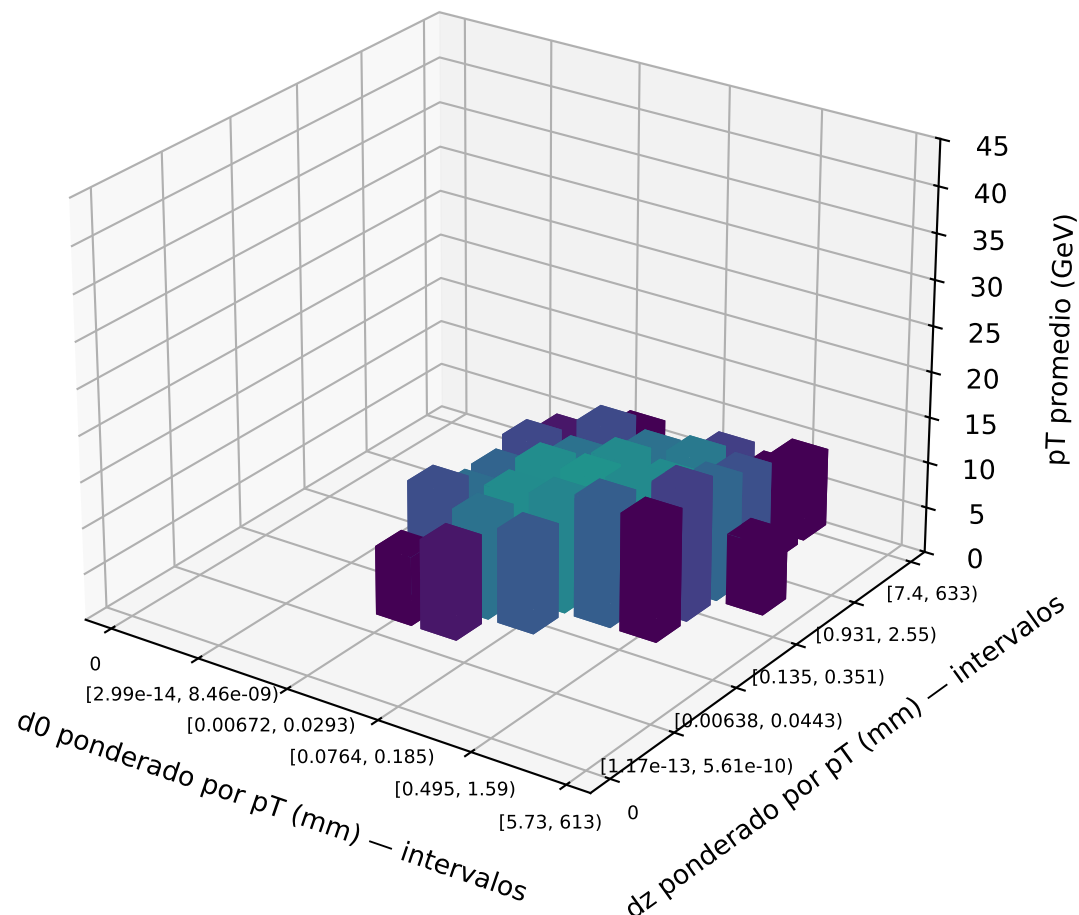
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC12 · semilla 42

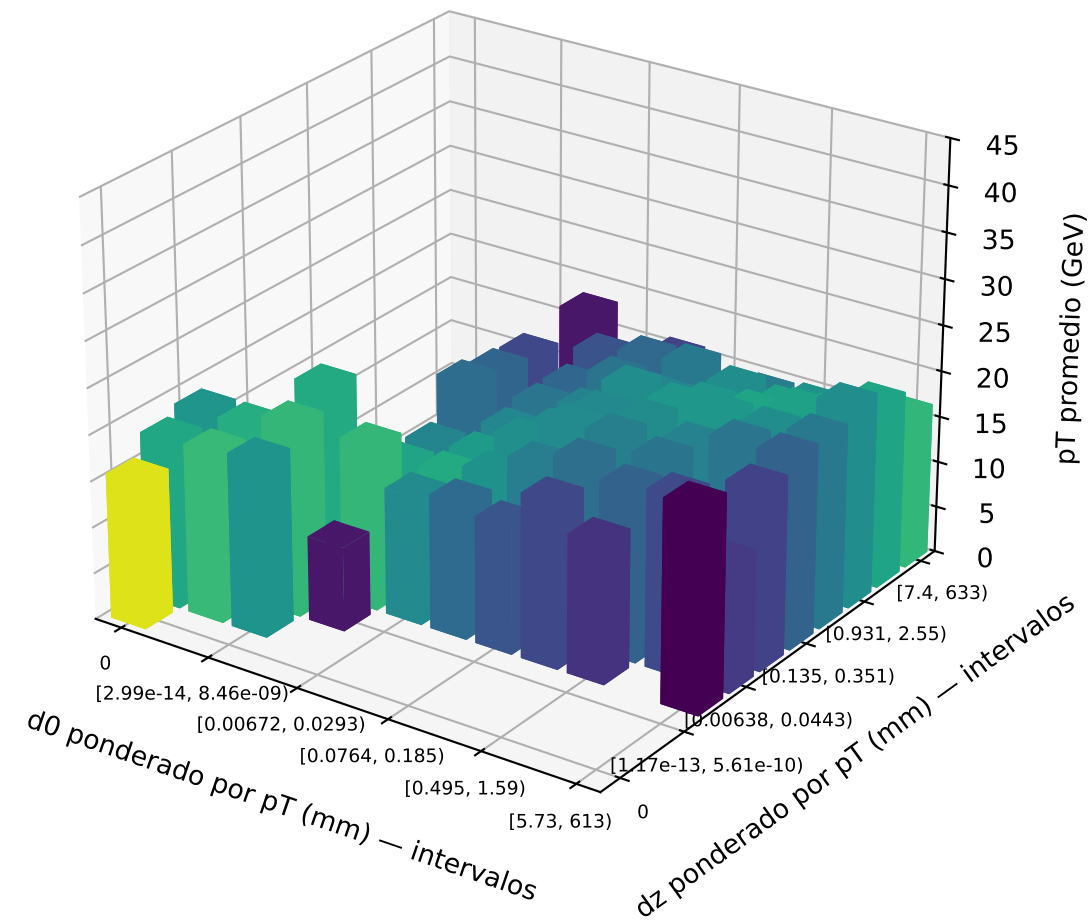
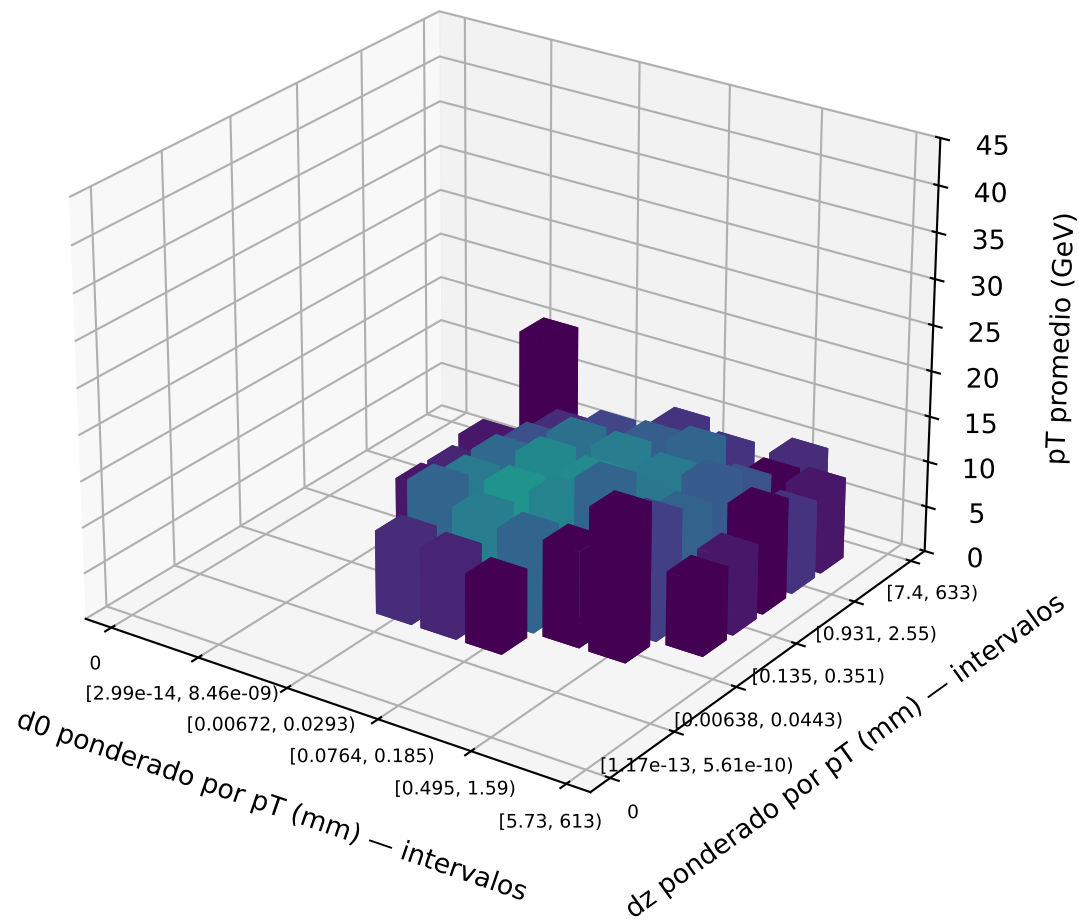
TP — N=2,021

FN — N=4,654



FP — N=1,728

TN — N=59,148



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC12 · semilla 42

Variable	Unidad	TP N=2,021	FN N=4,654	FP N=1,728	TN N=59,148
jet_pt *	GeV	11.87 ± 4.386	12.43 ± 7.397	11.97 ± 4.947	16.54 ± 11.77
jet_eta	1	-0.05589 ± 1.298	0.02785 ± 1.462	0.05024 ± 1.267	0.001304 ± 1.502
jet_abs_eta *	1	1.142 ± 0.6208	1.247 ± 0.7629	1.085 ± 0.6578	1.281 ± 0.7839
jet_mass	GeV	2.555 ± 1.19	2.519 ± 1.664	2.551 ± 1.221	2.861 ± 1.882
jet_ncharged	count	3.012 ± 1.438	2.594 ± 1.897	2.793 ± 1.364	2.582 ± 1.943
jet_nneutral	count	1.644 ± 1.376	2.078 ± 1.699	1.63 ± 1.274	2.391 ± 1.794
n_tracks_cone	count	3.05 ± 1.432	2.628 ± 1.892	2.826 ± 1.367	2.61 ± 1.943
track_pt_sum *	GeV	5.915 ± 2.987	5.205 ± 5.493	5.499 ± 3.213	6.791 ± 8.152
track_pt_sum_over_jet_pt	1	0.5106 ± 0.204	0.4012 ± 0.2609	0.4713 ± 0.21	0.3717 ± 0.2676
avg_track_pt *	GeV	2.161 ± 1.216	1.85 ± 1.427	2.08 ± 1.035	2.168 ± 2.077
std_track_pt	GeV	0.8091 ± 0.7304	0.7051 ± 0.9188	0.7071 ± 0.726	1.013 ± 1.614
n_tracks_below_avg_pt	count	1.629 ± 1.089	1.355 ± 1.317	1.453 ± 1.029	1.376 ± 1.361
n_tracks_above_avg_pt	count	1.421 ± 0.6541	1.273 ± 0.7969	1.373 ± 0.6096	1.234 ± 0.8063
max_pt_ratio	1	0.288 ± 0.1516	0.2344 ± 0.1675	0.2648 ± 0.1284	0.2106 ± 0.1556
min_pt_ratio	1	0.1257 ± 0.1369	0.1066 ± 0.1294	0.1265 ± 0.1138	0.08535 ± 0.09838
max_dr_ratio	1	0.1526 ± 0.1376	0.1277 ± 0.1321	0.1517 ± 0.1161	0.1041 ± 0.1075
min_dr_ratio	1	0.2454 ± 0.1713	0.2003 ± 0.1741	0.2272 ± 0.1436	0.1799 ± 0.1548
dr_max_pt	1	0.1403 ± 0.08245	0.1361 ± 0.09624	0.148 ± 0.08226	0.1265 ± 0.09905
dr_min_pt	1	0.2087 ± 0.09968	0.1915 ± 0.1138	0.2098 ± 0.09565	0.1902 ± 0.1172
dr_max_dr	1	0.2516 ± 0.09271	0.2269 ± 0.1161	0.248 ± 0.08888	0.2208 ± 0.1206
dr_min_dr	1	0.1071 ± 0.06791	0.1087 ± 0.08381	0.1182 ± 0.0724	0.1039 ± 0.08836
pt_difference	GeV	1.899 ± 1.662	1.7 ± 2.378	1.658 ± 1.736	2.484 ± 4.076
r50	1	0.1481 ± 0.07947	0.1428 ± 0.094	0.1567 ± 0.07786	0.1331 ± 0.09742
r95	1	0.2492 ± 0.0927	0.2242 ± 0.1151	0.2458 ± 0.0888	0.2146 ± 0.1192
mean_abs_d0 *	mm	0.1514 ± 0.1472	1.739 ± 9.489	0.1546 ± 0.251	1.229 ± 7.829
max_abs_d0 *	mm	0.2948 ± 0.3421	4.677 ± 18.34	0.3591 ± 0.6501	3.253 ± 16.47
std_abs_d0 *	mm	0.1038 ± 0.144	1.794 ± 7.102	0.1374 ± 0.2751	1.251 ± 6.62
mean_abs_dz *	mm	0.3776 ± 0.488	2.625 ± 14.19	0.3416 ± 0.8128	1.641 ± 9.603
max_abs_dz *	mm	0.6885 ± 0.8101	6.709 ± 23.53	0.7332 ± 1.333	4.29 ± 22.86
std_abs_dz *	mm	0.2309 ± 0.3015	2.531 ± 8.466	0.2576 ± 0.4657	1.662 ± 9.429
pt_weighted_abs_d0 *	mm	0.1561 ± 0.1459	1.64 ± 9.393	0.1535 ± 0.254	1.171 ± 7.841
pt_weighted_abs_dz *	mm	0.39 ± 0.4908	2.273 ± 13.97	0.34 ± 0.8136	1.349 ± 8.313

* Variable utilizada como entrada del modelo.

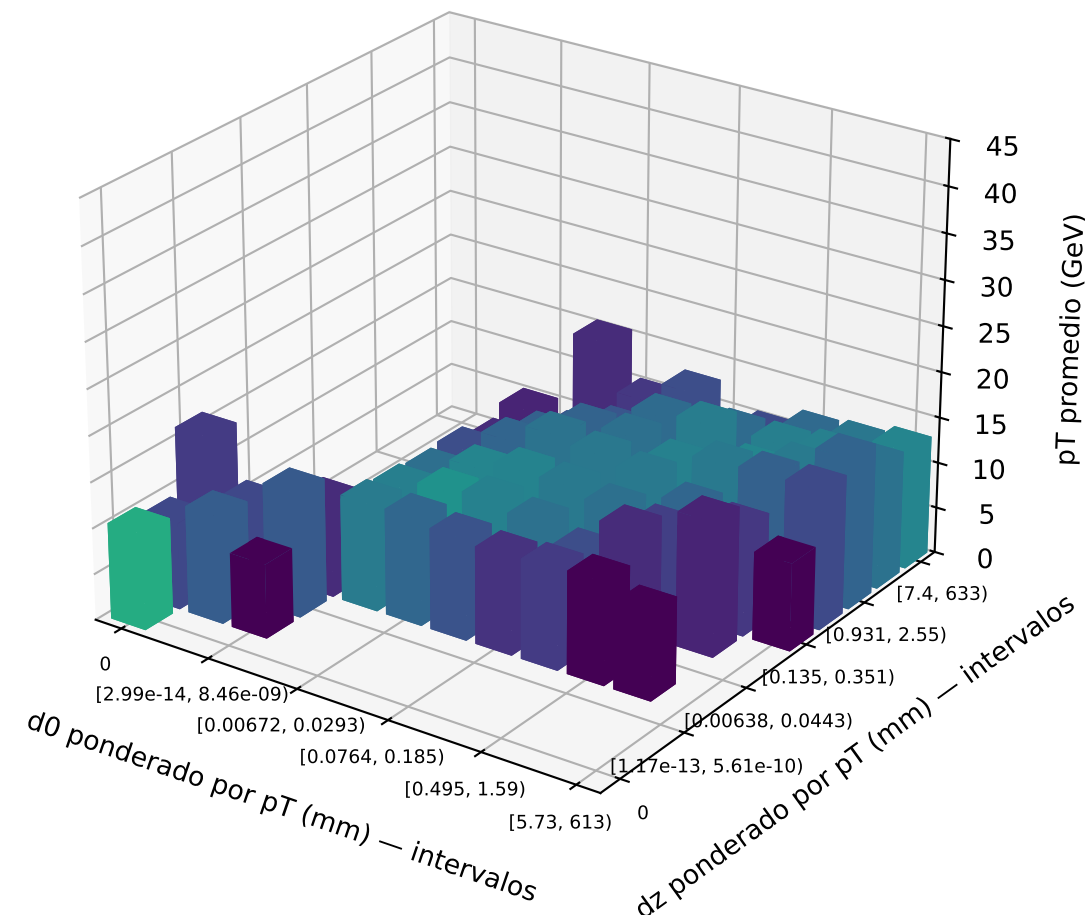
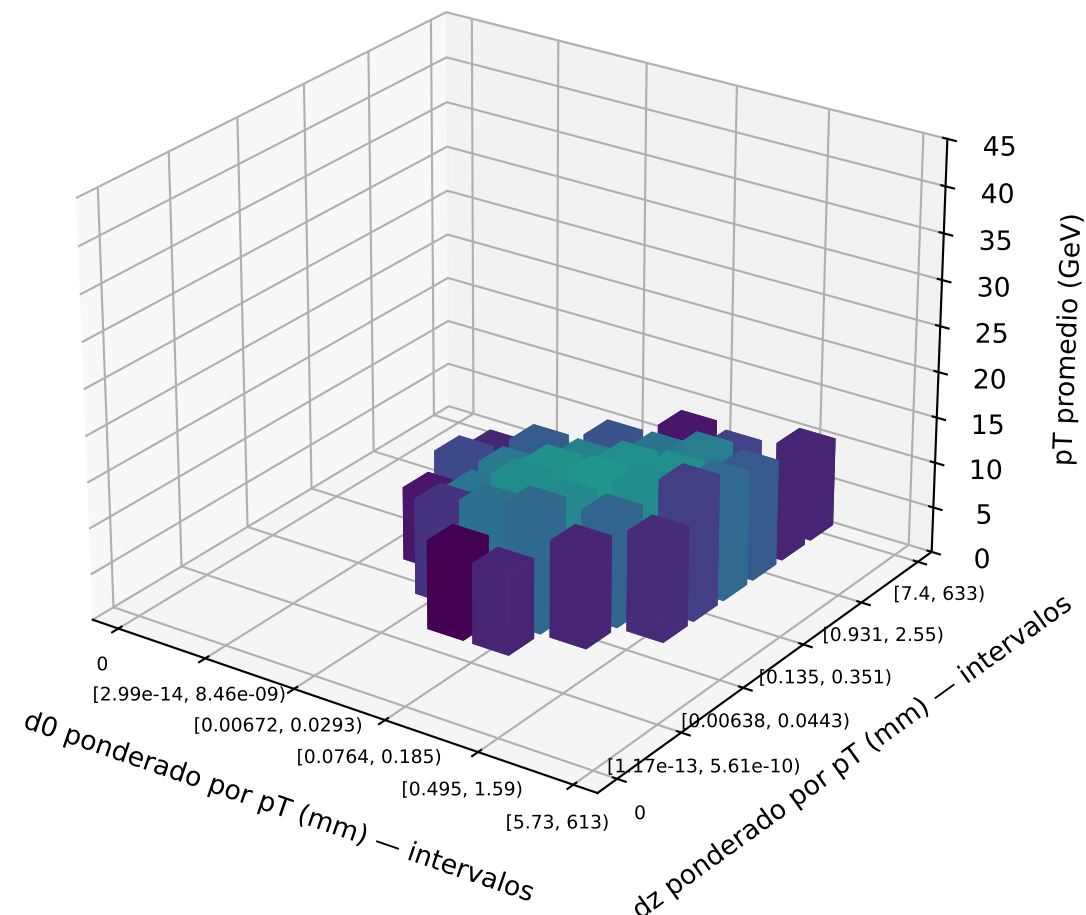
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC12 · semilla 73

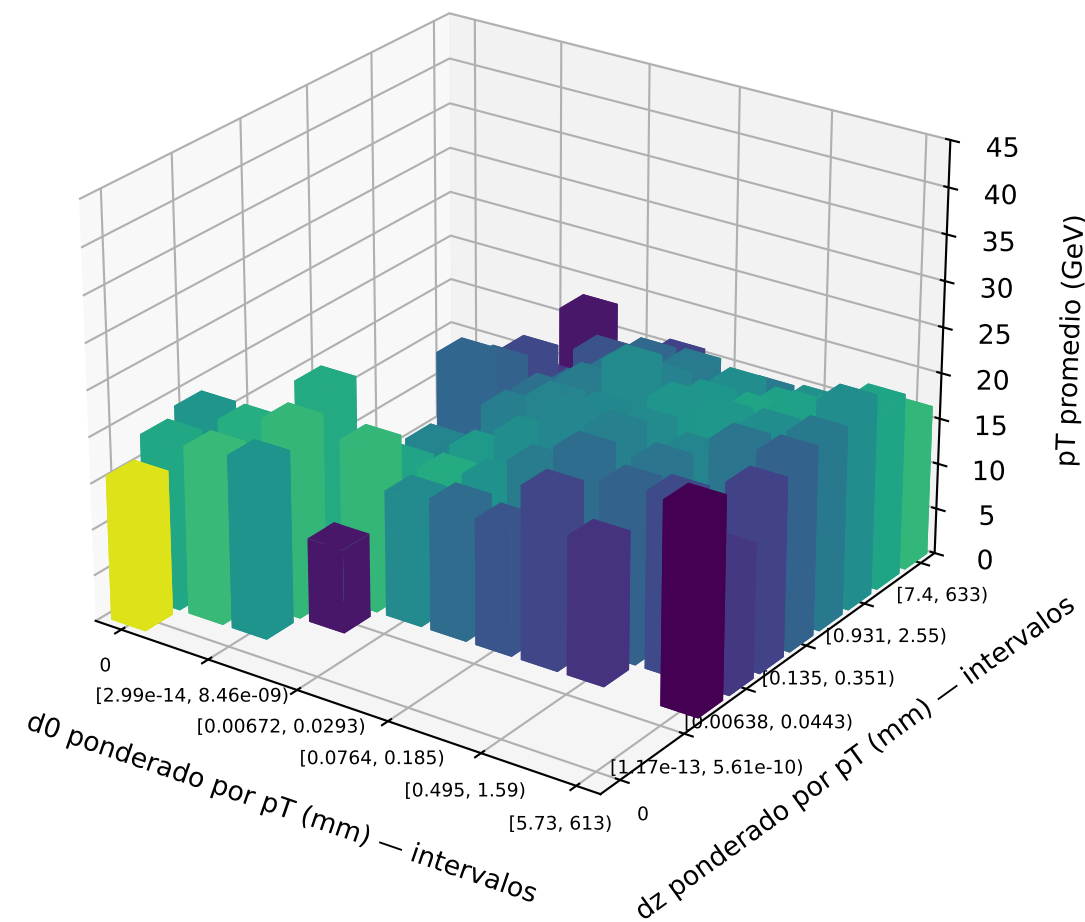
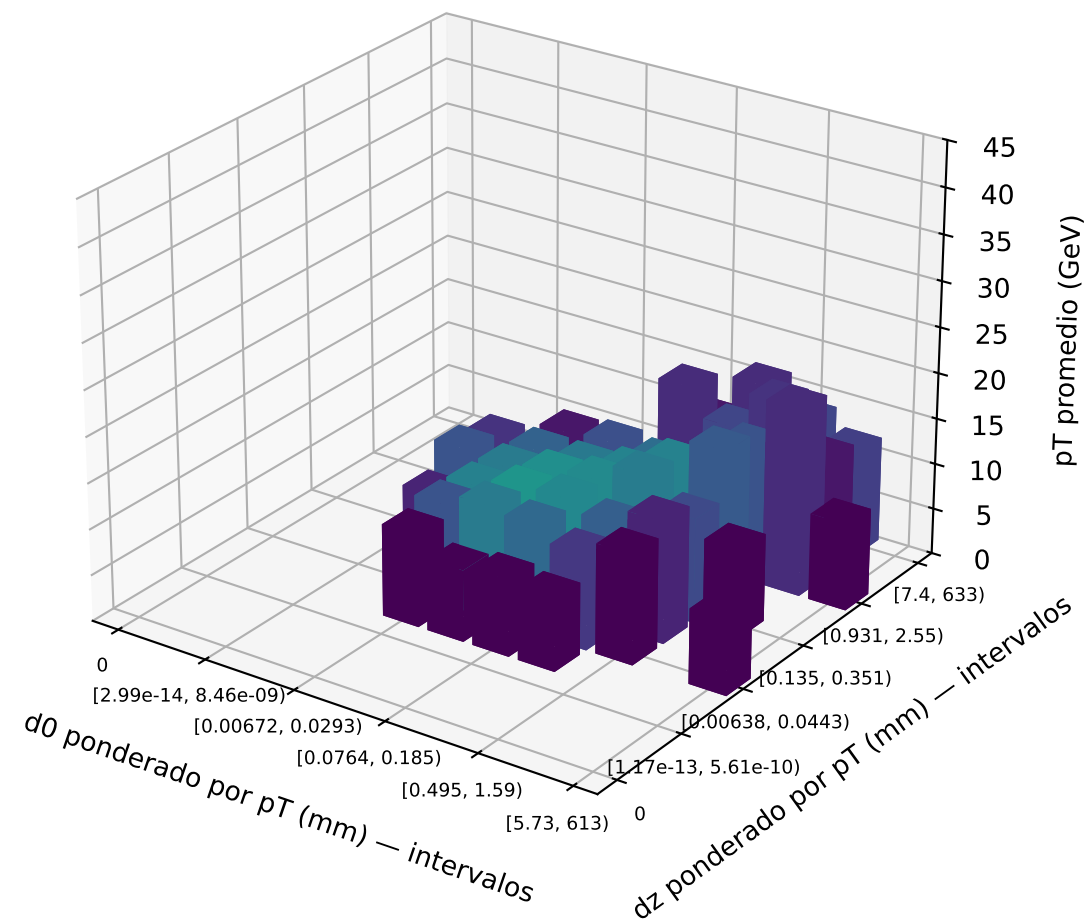
TP — N=2,487

FN — N=4,188



FP — N=2,132

TN — N=58,744



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC12 · semilla 73

Variable	Unidad	TP N=2,487	FN N=4,188	FP N=2,132	TN N=58,744
jet_pt *	GeV	11.87 ± 5.19	12.49 ± 7.352	12 ± 5.792	16.57 ± 11.78
jet_eta	1	-0.004891 ± 1.371	0.006875 ± 1.44	0.02814 ± 1.457	0.00177 ± 1.497
jet_abs_eta *	1	1.186 ± 0.6879	1.232 ± 0.7447	1.264 ± 0.7249	1.276 ± 0.7833
jet_mass	GeV	2.438 ± 1.251	2.585 ± 1.681	2.426 ± 1.219	2.867 ± 1.884
jet_ncharged	count	2.793 ± 1.466	2.678 ± 1.943	2.568 ± 1.328	2.588 ± 1.948
jet_nneutral	count	1.622 ± 1.459	2.14 ± 1.679	1.697 ± 1.405	2.394 ± 1.794
n_tracks_cone	count	2.824 ± 1.452	2.716 ± 1.942	2.593 ± 1.326	2.617 ± 1.948
track_pt_sum *	GeV	6.147 ± 3.646	4.988 ± 5.441	5.744 ± 3.864	6.791 ± 8.166
track_pt_sum_over_jet_pt	1	0.5219 ± 0.2058	0.3823 ± 0.2596	0.4841 ± 0.2124	0.3705 ± 0.2675
avg_track_pt *	GeV	2.395 ± 1.329	1.677 ± 1.329	2.302 ± 1.063	2.161 ± 2.082
std_track_pt	GeV	0.8685 ± 0.8348	0.6583 ± 0.8769	0.7516 ± 0.7777	1.013 ± 1.617
n_tracks_below_avg_pt	count	1.477 ± 1.114	1.415 ± 1.337	1.287 ± 1.015	1.381 ± 1.363
n_tracks_above_avg_pt	count	1.347 ± 0.6042	1.3 ± 0.8378	1.306 ± 0.5742	1.236 ± 0.8087
max_pt_ratio	1	0.3131 ± 0.1511	0.2135 ± 0.1612	0.2867 ± 0.125	0.2094 ± 0.1555
min_pt_ratio	1	0.144 ± 0.1454	0.09363 ± 0.1195	0.1431 ± 0.1106	0.08447 ± 0.09803
max_dr_ratio	1	0.1701 ± 0.1449	0.1146 ± 0.123	0.1691 ± 0.1134	0.1031 ± 0.1072
min_dr_ratio	1	0.2714 ± 0.1738	0.1799 ± 0.1658	0.249 ± 0.1411	0.1788 ± 0.1546
dr_max_pt	1	0.1316 ± 0.07983	0.1408 ± 0.0988	0.1368 ± 0.07672	0.1268 ± 0.09936
dr_min_pt	1	0.1984 ± 0.09902	0.1957 ± 0.116	0.1967 ± 0.09213	0.1905 ± 0.1175
dr_max_dr	1	0.2352 ± 0.09535	0.2339 ± 0.118	0.2284 ± 0.08704	0.2213 ± 0.1209
dr_min_dr	1	0.1038 ± 0.06698	0.1108 ± 0.08572	0.1128 ± 0.06842	0.104 ± 0.08859
pt_difference	GeV	2.022 ± 1.95	1.605 ± 2.303	1.749 ± 1.872	2.486 ± 4.085
r50	1	0.1374 ± 0.07751	0.1486 ± 0.09624	0.1443 ± 0.07386	0.1334 ± 0.09771
r95	1	0.2326 ± 0.09461	0.2313 ± 0.1173	0.226 ± 0.08695	0.2151 ± 0.1195
mean_abs_d0 *	mm	0.1502 ± 0.2055	1.917 ± 9.987	0.1537 ± 0.4638	1.237 ± 7.855
max_abs_d0 *	mm	0.2767 ± 0.7778	5.175 ± 19.26	0.3354 ± 1.366	3.274 ± 16.53
std_abs_d0 *	mm	0.09214 ± 0.3116	1.989 ± 7.458	0.1239 ± 0.5701	1.259 ± 6.642
mean_abs_dz *	mm	0.392 ± 0.5163	2.866 ± 14.94	0.4695 ± 1.385	1.646 ± 9.633
max_abs_dz *	mm	0.7136 ± 1.326	7.364 ± 24.7	1.065 ± 3.821	4.303 ± 22.93
std_abs_dz *	mm	0.2358 ± 0.5515	2.784 ± 8.881	0.4056 ± 1.58	1.667 ± 9.457
pt_weighted_abs_d0 *	mm	0.1533 ± 0.1472	1.807 ± 9.888	0.1532 ± 0.3533	1.178 ± 7.867
pt_weighted_abs_dz *	mm	0.4096 ± 0.5952	2.471 ± 14.71	0.5069 ± 1.614	1.35 ± 8.338

* Variable utilizada como entrada del modelo.

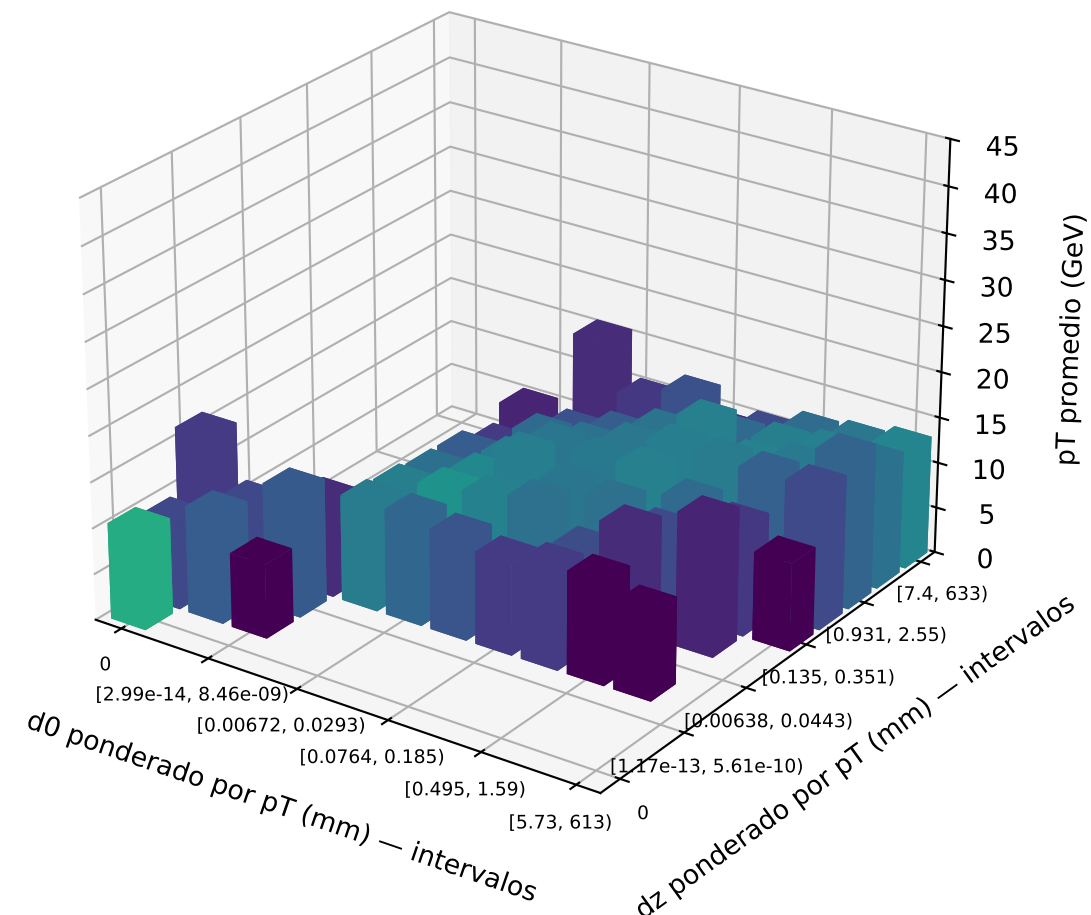
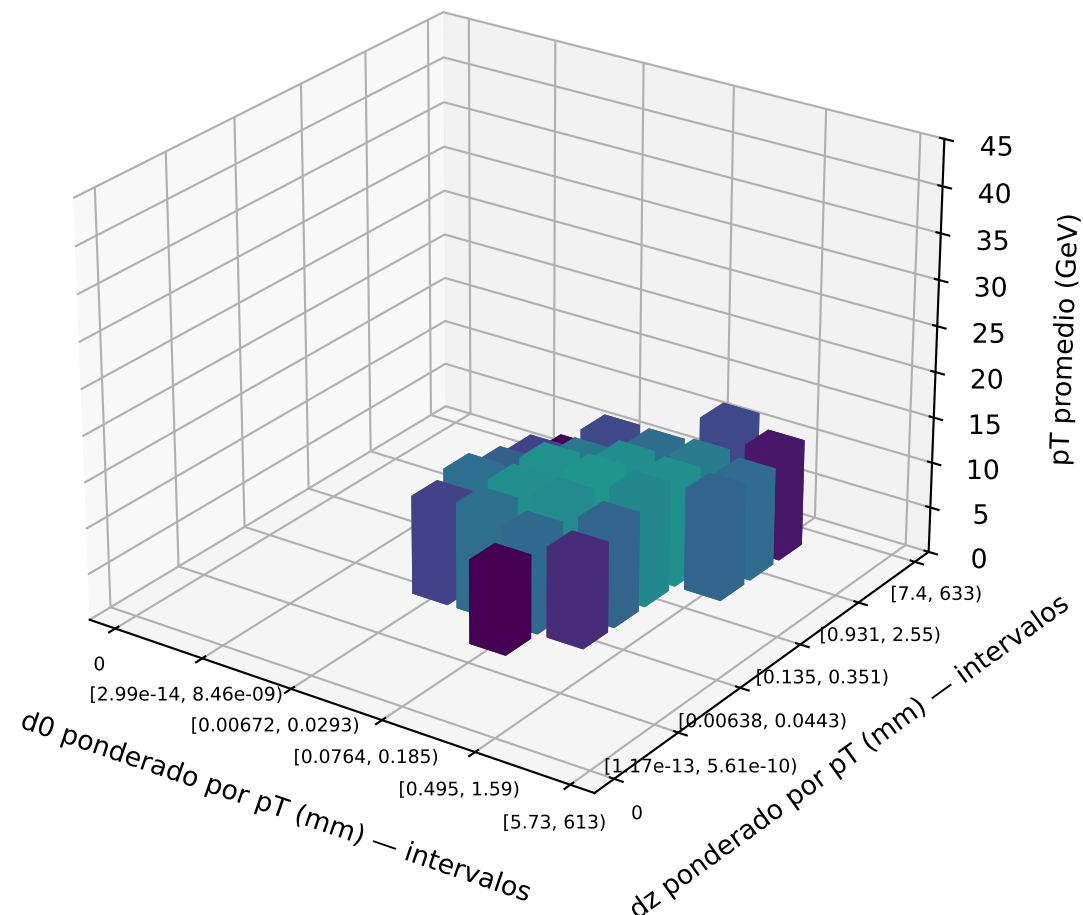
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC12 · semilla 101

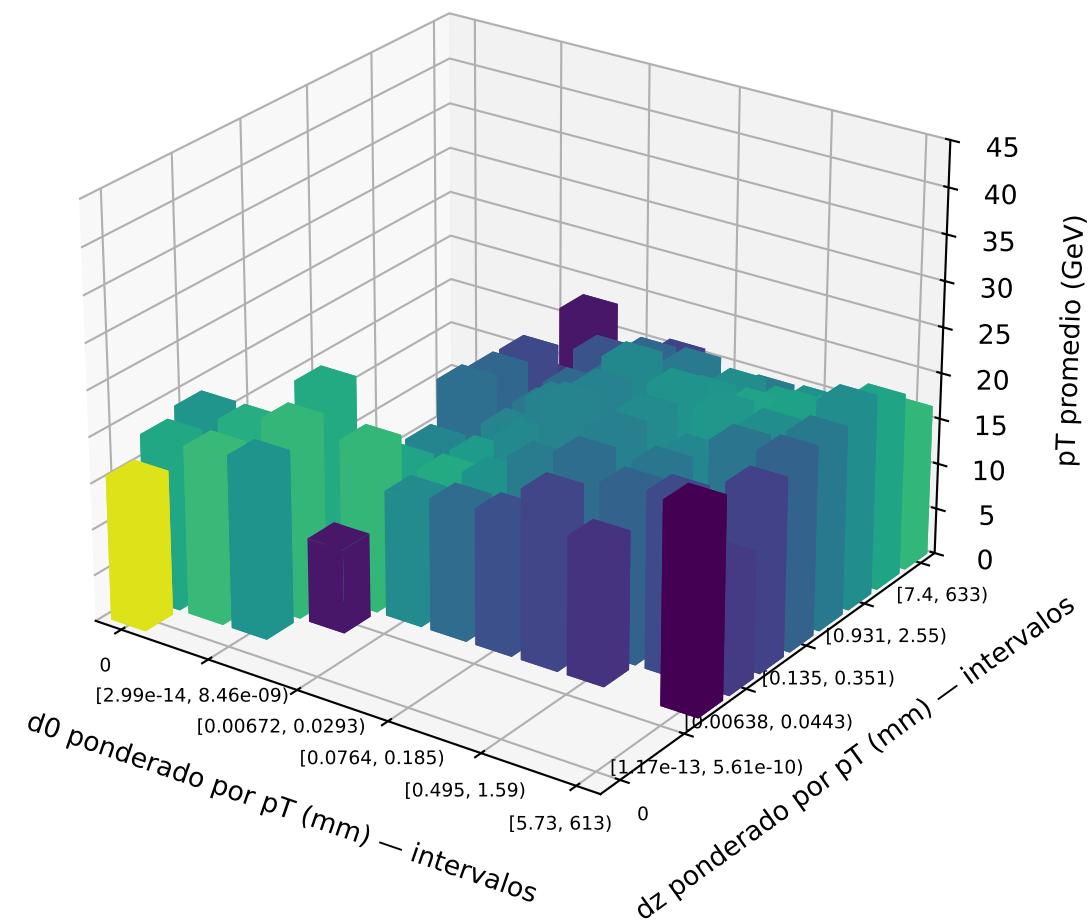
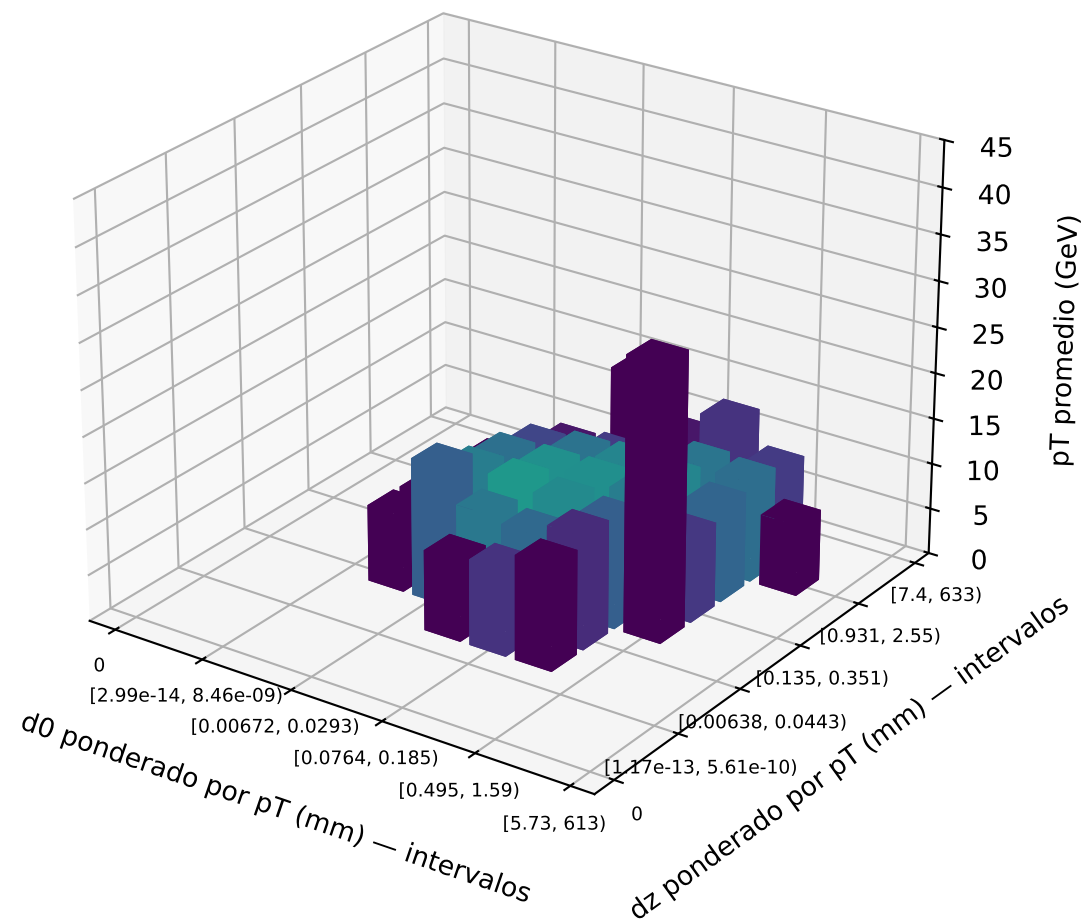
TP — N=2,284

FN — N=4,391



FP — N=2,167

TN — N=58,709



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC12 · semilla 101

Variable	Unidad	TP N=2,284	FN N=4,391	FP N=2,167	TN N=58,709
jet_pt *	GeV	11.78 ± 4.427	12.51 ± 7.522	12.04 ± 5.2	16.58 ± 11.79
jet_eta	1	-0.01074 ± 1.388	0.009376 ± 1.429	-0.004548 ± 1.416	0.002961 ± 1.499
jet_abs_eta *	1	1.213 ± 0.6743	1.216 ± 0.7492	1.234 ± 0.6944	1.277 ± 0.7843
jet_mass	GeV	2.574 ± 1.166	2.508 ± 1.697	2.614 ± 1.221	2.861 ± 1.886
jet_ncharged	count	3.064 ± 1.453	2.542 ± 1.905	2.985 ± 1.447	2.573 ± 1.944
jet_nneutral	count	1.669 ± 1.442	2.091 ± 1.688	1.649 ± 1.334	2.396 ± 1.795
n_tracks_cone	count	3.104 ± 1.426	2.575 ± 1.908	3.022 ± 1.443	2.601 ± 1.944
track_pt_sum *	GeV	6.153 ± 3.456	5.038 ± 5.441	6.091 ± 3.822	6.779 ± 8.17
track_pt_sum_over_jet_pt	1	0.5231 ± 0.2081	0.3882 ± 0.2577	0.5035 ± 0.2091	0.3698 ± 0.2673
avg_track_pt *	GeV	2.11 ± 1.126	1.858 ± 1.479	2.078 ± 1.089	2.169 ± 2.082
std_track_pt	GeV	0.893 ± 0.7877	0.6553 ± 0.8954	0.7883 ± 0.702	1.012 ± 1.619
n_tracks_below_avg_pt	count	1.673 ± 1.082	1.316 ± 1.325	1.605 ± 1.086	1.37 ± 1.361
n_tracks_above_avg_pt	count	1.431 ± 0.6571	1.259 ± 0.8015	1.418 ± 0.6484	1.232 ± 0.806
max_pt_ratio	1	0.2886 ± 0.1437	0.2309 ± 0.1713	0.268 ± 0.1153	0.2101 ± 0.1561
min_pt_ratio	1	0.1114 ± 0.1084	0.1129 ± 0.1428	0.1156 ± 0.09362	0.08545 ± 0.09912
max_dr_ratio	1	0.1404 ± 0.1139	0.1326 ± 0.1437	0.1419 ± 0.09925	0.1041 ± 0.1081
min_dr_ratio	1	0.2406 ± 0.1644	0.2002 ± 0.178	0.2264 ± 0.132	0.1796 ± 0.1552
dr_max_pt	1	0.1402 ± 0.08157	0.1359 ± 0.09738	0.141 ± 0.0783	0.1266 ± 0.09931
dr_min_pt	1	0.2123 ± 0.09697	0.1886 ± 0.1154	0.2117 ± 0.09517	0.19 ± 0.1174
dr_max_dr	1	0.2548 ± 0.08636	0.2238 ± 0.1193	0.251 ± 0.0845	0.2205 ± 0.1209
dr_min_dr	1	0.107 ± 0.06764	0.1088 ± 0.08477	0.111 ± 0.06815	0.1041 ± 0.08861
pt_difference	GeV	2.098 ± 1.82	1.585 ± 2.337	1.879 ± 1.722	2.482 ± 4.089
r50	1	0.1474 ± 0.07824	0.1428 ± 0.09534	0.1517 ± 0.07424	0.1331 ± 0.09767
r95	1	0.2517 ± 0.08628	0.2214 ± 0.1184	0.248 ± 0.0844	0.2143 ± 0.1194
mean_abs_d0 *	mm	0.1534 ± 0.1463	1.833 ± 9.761	0.1472 ± 0.174	1.237 ± 7.858
max_abs_d0 *	mm	0.3 ± 0.3261	4.936 ± 18.85	0.3635 ± 0.4922	3.274 ± 16.53
std_abs_d0 *	mm	0.1075 ± 0.1367	1.894 ± 7.299	0.1467 ± 0.2162	1.259 ± 6.644
mean_abs_dz *	mm	0.3828 ± 0.4466	2.756 ± 14.6	0.3607 ± 0.5017	1.65 ± 9.639
max_abs_dz *	mm	0.699 ± 0.7721	7.065 ± 24.18	0.8032 ± 1.112	4.314 ± 22.94
std_abs_dz *	mm	0.2358 ± 0.3086	2.666 ± 8.697	0.3125 ± 0.5075	1.671 ± 9.463
pt_weighted_abs_d0 *	mm	0.1603 ± 0.15	1.727 ± 9.663	0.1455 ± 0.1811	1.179 ± 7.869
pt_weighted_abs_dz *	mm	0.3961 ± 0.4587	2.383 ± 14.38	0.3474 ± 0.4766	1.356 ± 8.345

* Variable utilizada como entrada del modelo.

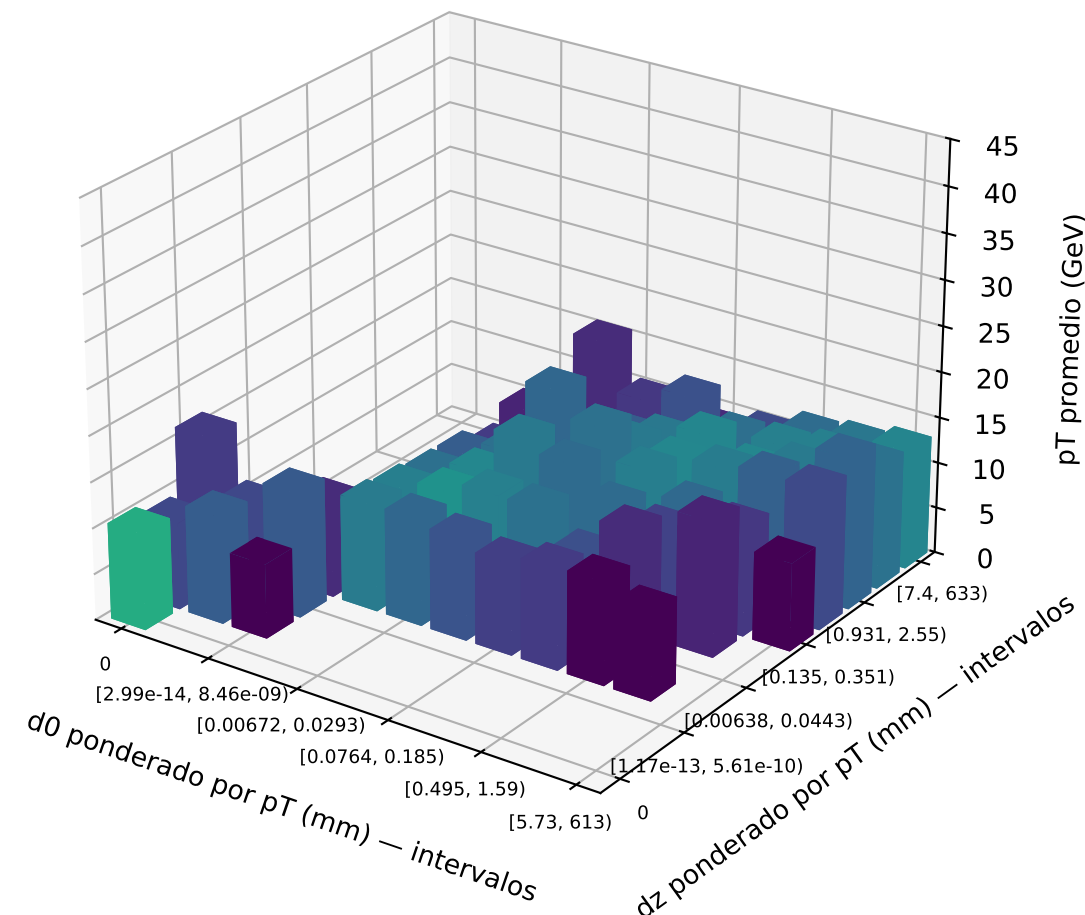
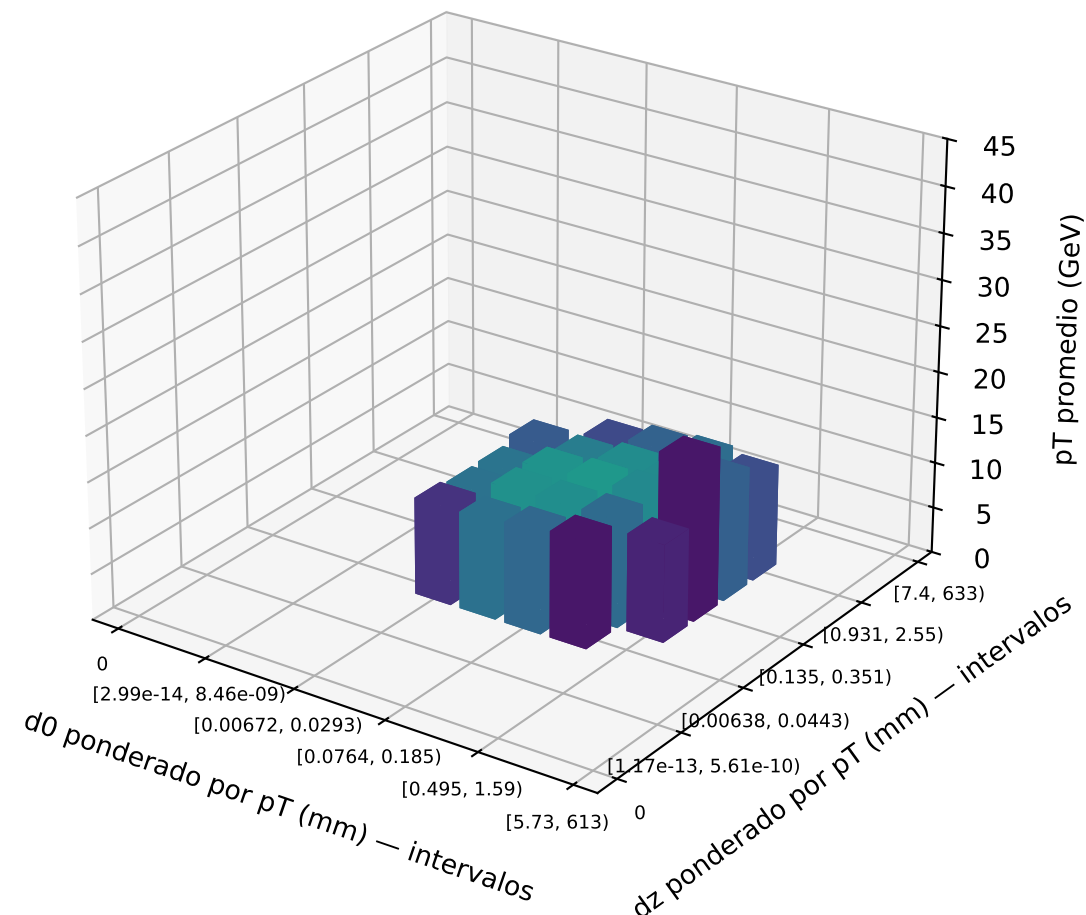
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — Alternative-VQC12 · ensemble

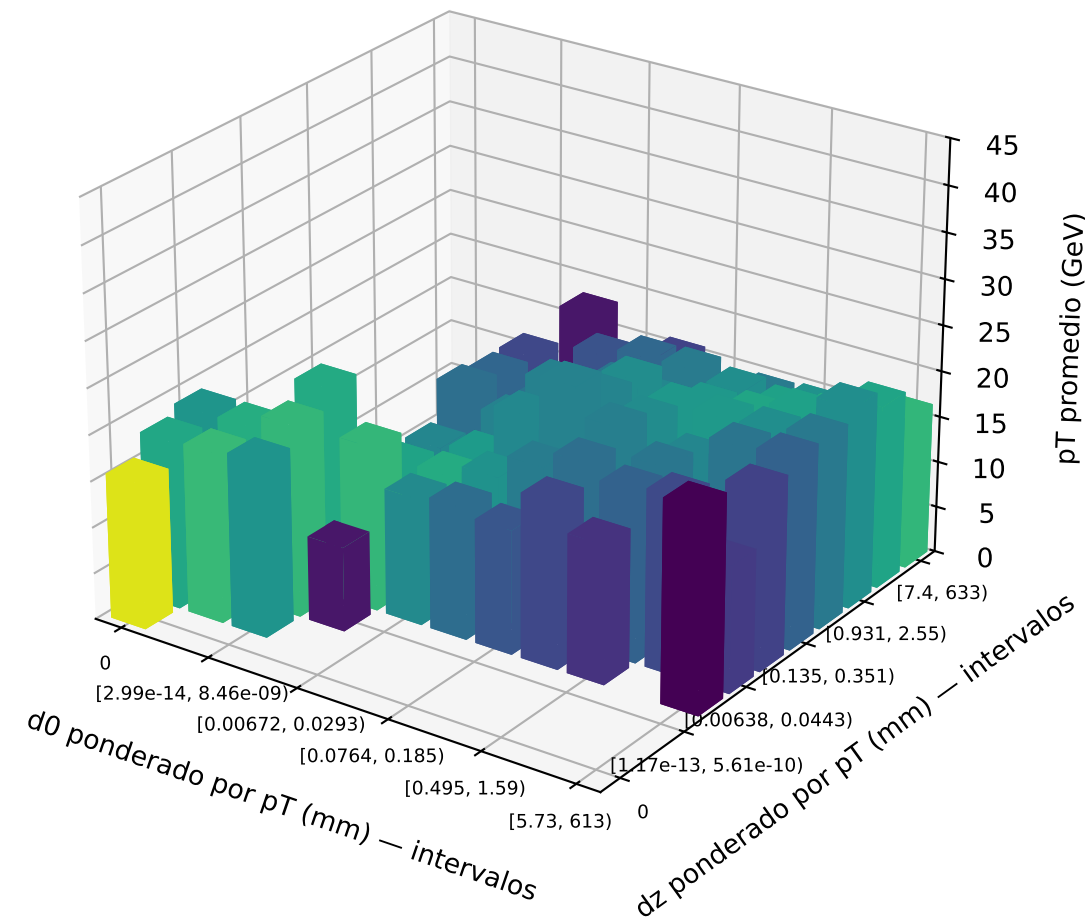
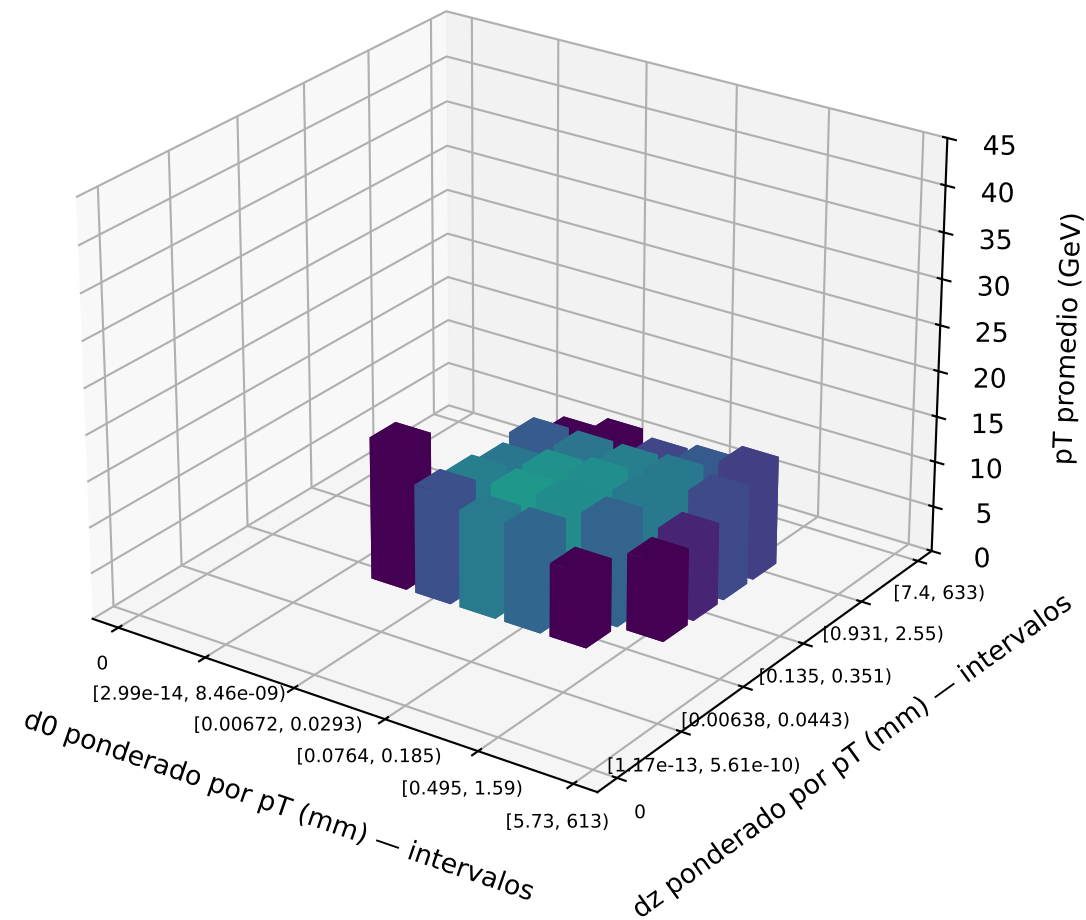
TP — N=2,328

FN — N=4,347



FP — N=1,913

TN — N=58,963



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — Alternative-VQC12 · ensemble

Variable	Unidad	TP N=2,328	FN N=4,347	FP N=1,913	TN N=58,963
jet_pt *	GeV	11.6 ± 3.942	12.62 ± 7.678	11.39 ± 3.96	16.58 ± 11.78
jet_eta	1	-0.01462 ± 1.321	0.01165 ± 1.462	0.03338 ± 1.371	0.001698 ± 1.5
jet_abs_eta *	1	1.151 ± 0.6485	1.25 ± 0.7597	1.201 ± 0.6634	1.278 ± 0.7847
jet_mass	GeV	2.516 ± 1.199	2.538 ± 1.689	2.436 ± 1.115	2.865 ± 1.885
jet_ncharged	count	2.985 ± 1.457	2.579 ± 1.917	2.758 ± 1.299	2.582 ± 1.946
jet_nneutral	count	1.649 ± 1.445	2.107 ± 1.685	1.593 ± 1.333	2.394 ± 1.793
n_tracks_cone	count	3.021 ± 1.442	2.614 ± 1.916	2.785 ± 1.283	2.611 ± 1.947
track_pt_sum *	GeV	5.869 ± 2.762	5.179 ± 5.689	5.395 ± 2.507	6.798 ± 8.17
track_pt_sum_over_jet_pt	1	0.5171 ± 0.2026	0.39 ± 0.2617	0.4891 ± 0.2039	0.3708 ± 0.2676
avg_track_pt *	GeV	2.143 ± 1.13	1.837 ± 1.477	2.077 ± 0.9108	2.169 ± 2.081
std_track_pt	GeV	0.8191 ± 0.6827	0.6925 ± 0.9488	0.7081 ± 0.6017	1.014 ± 1.617
n_tracks_below_avg_pt	count	1.611 ± 1.095	1.346 ± 1.329	1.433 ± 0.9833	1.376 ± 1.363
n_tracks_above_avg_pt	count	1.41 ± 0.6432	1.268 ± 0.8108	1.352 ± 0.5894	1.235 ± 0.8074
max_pt_ratio	1	0.2939 ± 0.1479	0.2275 ± 0.1685	0.276 ± 0.1259	0.2101 ± 0.1556
min_pt_ratio	1	0.125 ± 0.1311	0.1056 ± 0.132	0.1297 ± 0.1095	0.08512 ± 0.09841
max_dr_ratio	1	0.1531 ± 0.1326	0.1257 ± 0.1342	0.1557 ± 0.1131	0.1038 ± 0.1075
min_dr_ratio	1	0.2499 ± 0.1696	0.1948 ± 0.174	0.236 ± 0.142	0.1795 ± 0.1547
dr_max_pt	1	0.1369 ± 0.08096	0.1376 ± 0.09783	0.1415 ± 0.07707	0.1267 ± 0.09926
dr_min_pt	1	0.2052 ± 0.09926	0.1922 ± 0.1151	0.2062 ± 0.09565	0.1902 ± 0.1173
dr_max_dr	1	0.2477 ± 0.09223	0.2273 ± 0.118	0.242 ± 0.08802	0.2209 ± 0.1207
dr_min_dr	1	0.1044 ± 0.06553	0.1102 ± 0.08575	0.1129 ± 0.06732	0.104 ± 0.08855
pt_difference	GeV	1.915 ± 1.543	1.678 ± 2.461	1.637 ± 1.379	2.487 ± 4.085
r50	1	0.1446 ± 0.07796	0.1443 ± 0.09566	0.1501 ± 0.07464	0.1332 ± 0.09759
r95	1	0.2453 ± 0.09186	0.2245 ± 0.1171	0.2396 ± 0.08843	0.2147 ± 0.1193
mean_abs_d0 *	mm	0.1278 ± 0.1158	1.864 ± 9.807	0.09991 ± 0.1049	1.234 ± 7.841
max_abs_d0 *	mm	0.2467 ± 0.2526	5.012 ± 18.93	0.2203 ± 0.2671	3.266 ± 16.5
std_abs_d0 *	mm	0.08809 ± 0.1058	1.922 ± 7.332	0.08679 ± 0.1199	1.256 ± 6.63
mean_abs_dz *	mm	0.2866 ± 0.2691	2.832 ± 14.67	0.2176 ± 0.2288	1.649 ± 9.618
max_abs_dz *	mm	0.535 ± 0.5181	7.217 ± 24.27	0.4623 ± 0.5253	4.31 ± 22.9
std_abs_dz *	mm	0.1843 ± 0.2116	2.718 ± 8.731	0.178 ± 0.2375	1.669 ± 9.443
pt_weighted_abs_d0 *	mm	0.1348 ± 0.1224	1.756 ± 9.709	0.1038 ± 0.1051	1.176 ± 7.853
pt_weighted_abs_dz *	mm	0.3042 ± 0.2891	2.452 ± 14.45	0.228 ± 0.2375	1.356 ± 8.327

* Variable utilizada como entrada del modelo.

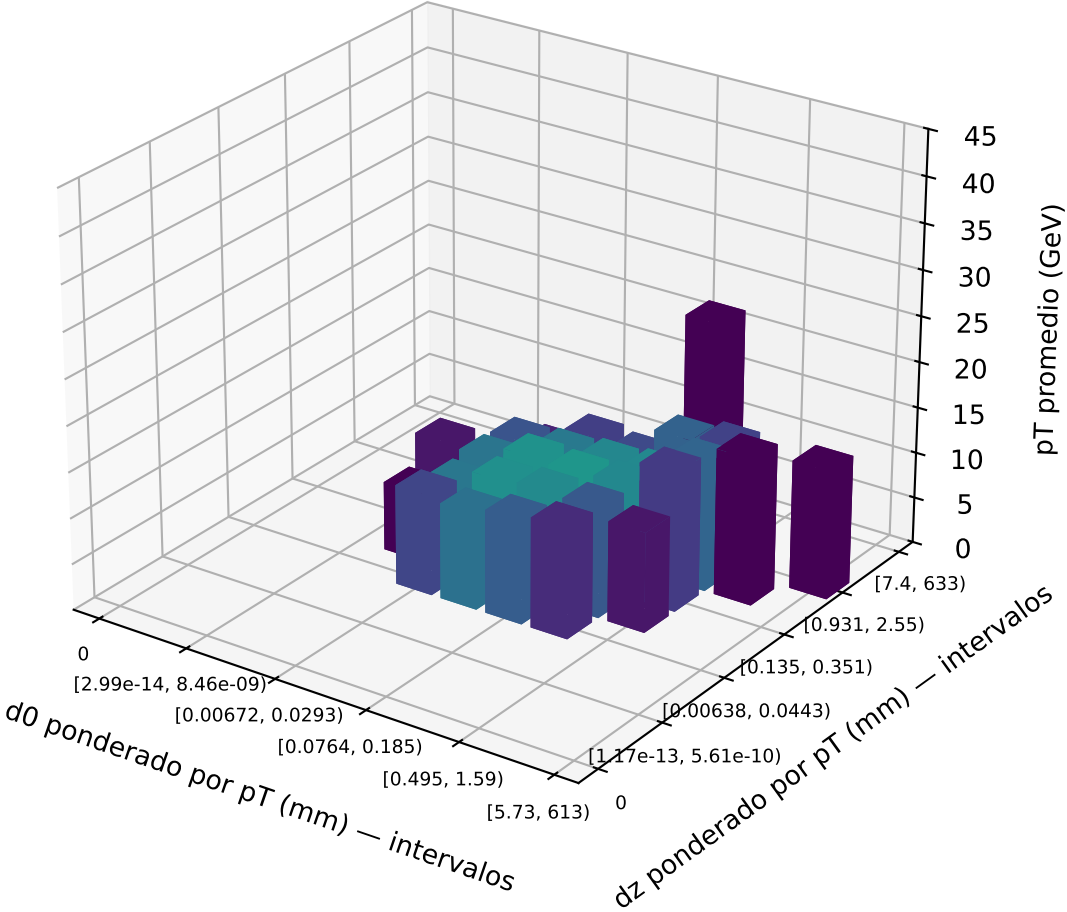
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

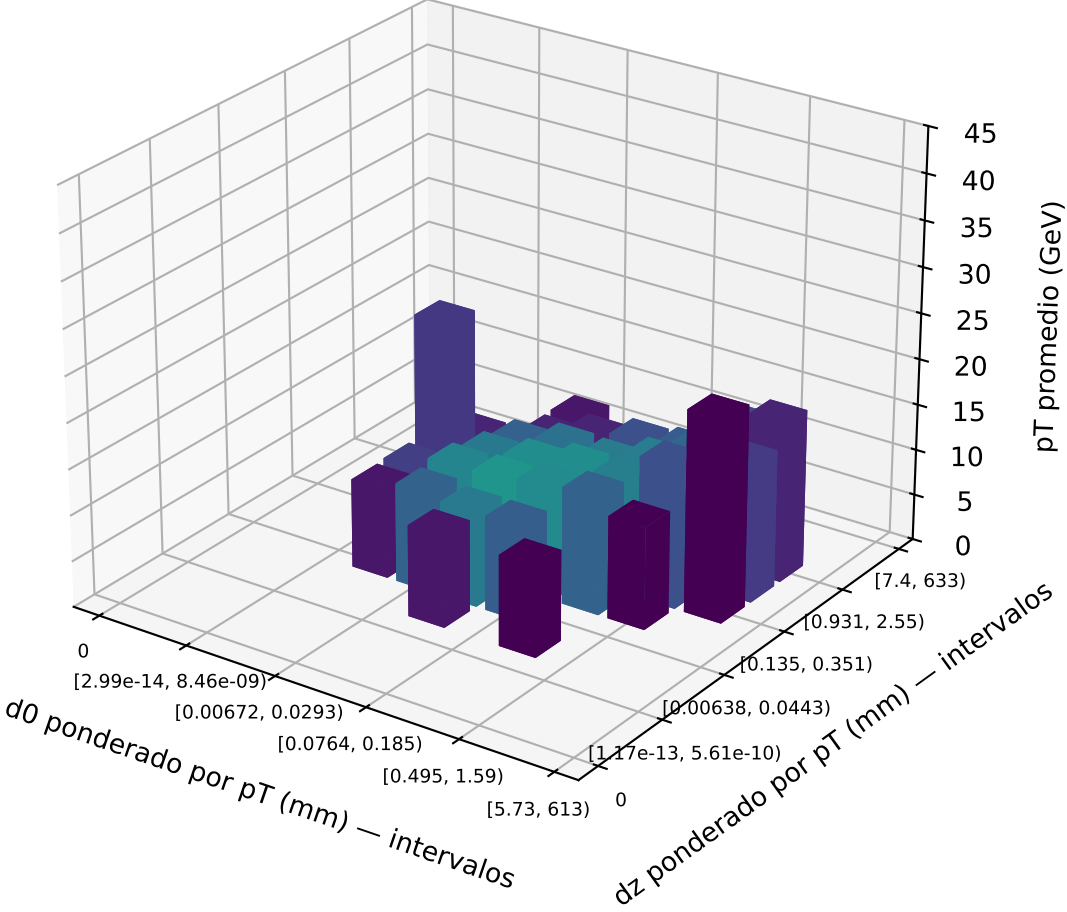
Perfil de pT medio en intervalos de d0/dz — Alternative-VQC12 · semilla 42

TP — N=2,117

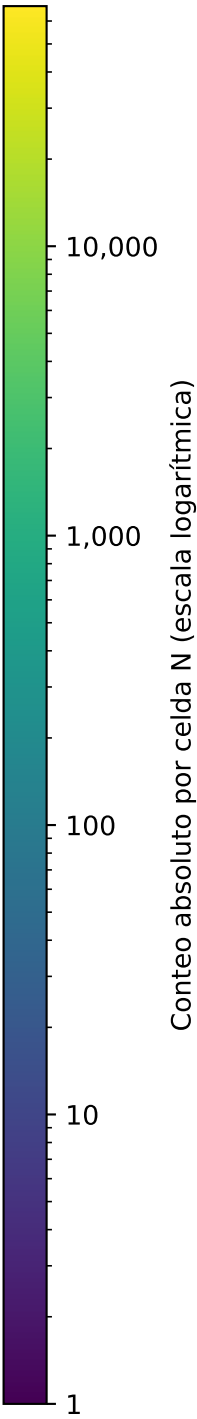
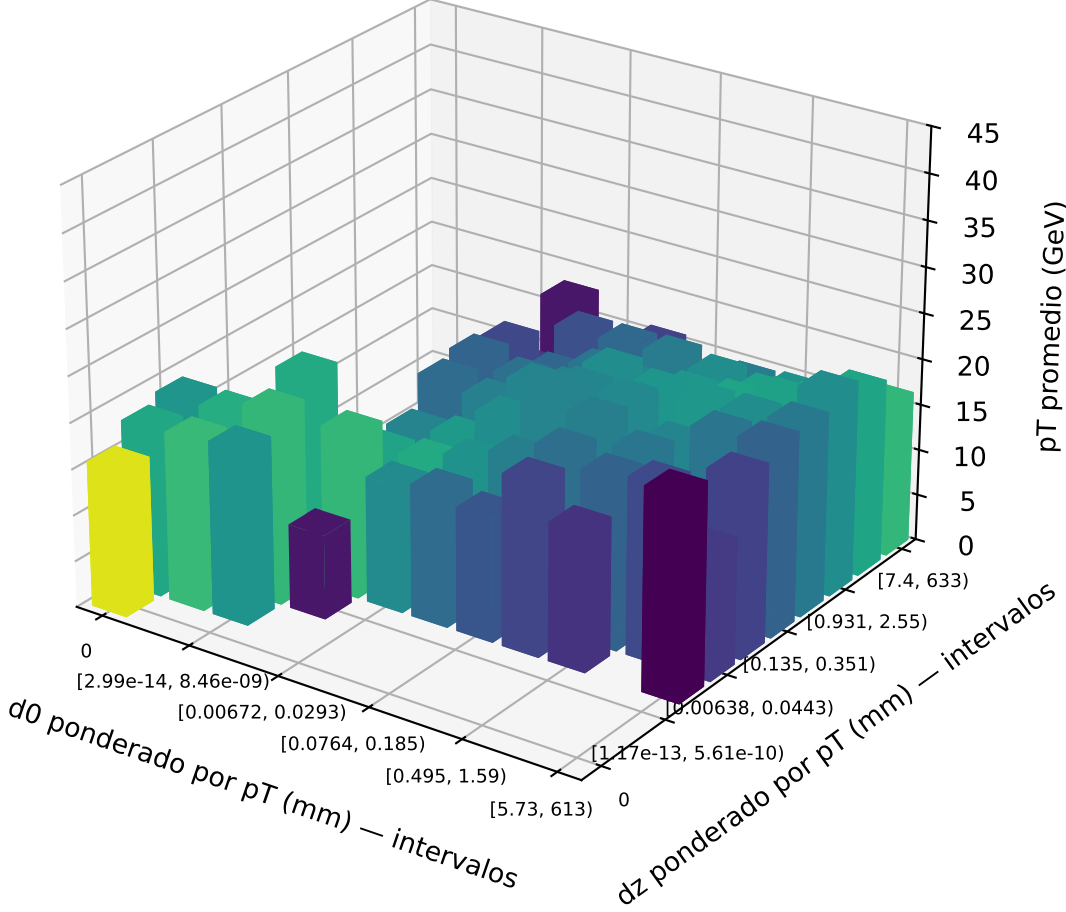
FN — N=4,558



FP — N=1,986



TN — N=58,890



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — Alternative-VQC12 · semilla 42

Variable	Unidad	TP N=2,117	FN N=4,558	FP N=1,986	TN N=58,890
jet_pt *	GeV	11.6 ± 4.446	12.57 ± 7.417	11.67 ± 4.734	16.57 ± 11.78
jet_eta	1	-0.03567 ± 1.276	0.02022 ± 1.474	0.003849 ± 1.341	0.002654 ± 1.501
jet_abs_eta *	1	1.109 ± 0.6337	1.265 ± 0.7578	1.17 ± 0.6549	1.279 ± 0.785
jet_mass	GeV	2.57 ± 1.201	2.512 ± 1.669	2.584 ± 1.21	2.861 ± 1.884
jet_ncharged	count	3.009 ± 1.44	2.587 ± 1.904	2.912 ± 1.406	2.577 ± 1.944
jet_nneutral	count	1.701 ± 1.469	2.061 ± 1.673	1.685 ± 1.363	2.392 ± 1.794
n_tracks_cone	count	3.048 ± 1.428	2.621 ± 1.901	2.943 ± 1.382	2.605 ± 1.944
track_pt_sum *	GeV	5.602 ± 2.835	5.335 ± 5.582	5.371 ± 3.103	6.801 ± 8.167
track_pt_sum_over_jet_pt	1	0.496 ± 0.2056	0.4057 ± 0.2636	0.4725 ± 0.2051	0.3712 ± 0.2678
avg_track_pt *	GeV	1.984 ± 1.023	1.925 ± 1.509	1.908 ± 0.8304	2.174 ± 2.083
std_track_pt	GeV	0.7775 ± 0.6513	0.7176 ± 0.9507	0.6941 ± 0.5859	1.015 ± 1.618
n_tracks_below_avg_pt	count	1.635 ± 1.086	1.347 ± 1.321	1.537 ± 1.015	1.373 ± 1.362
n_tracks_above_avg_pt	count	1.413 ± 0.6466	1.274 ± 0.8028	1.406 ± 0.6522	1.233 ± 0.8057
max_pt_ratio	1	0.2761 ± 0.1413	0.2388 ± 0.1732	0.2554 ± 0.1202	0.2107 ± 0.156
min_pt_ratio	1	0.1142 ± 0.1154	0.1116 ± 0.1391	0.1136 ± 0.09612	0.08561 ± 0.09905
max_dr_ratio	1	0.1423 ± 0.1181	0.132 ± 0.141	0.1395 ± 0.1004	0.1043 ± 0.1081
min_dr_ratio	1	0.2328 ± 0.1607	0.2052 ± 0.1799	0.2171 ± 0.1351	0.18 ± 0.1551
dr_max_pt	1	0.1438 ± 0.0821	0.1344 ± 0.09653	0.1482 ± 0.08115	0.1264 ± 0.09913
dr_min_pt	1	0.2107 ± 0.09799	0.1902 ± 0.1146	0.2157 ± 0.09514	0.1899 ± 0.1173
dr_max_dr	1	0.255 ± 0.08815	0.2248 ± 0.1177	0.2549 ± 0.08444	0.2205 ± 0.1207
dr_min_dr	1	0.1096 ± 0.06758	0.1076 ± 0.08423	0.1171 ± 0.07103	0.1039 ± 0.08846
pt_difference	GeV	1.824 ± 1.49	1.731 ± 2.444	1.624 ± 1.372	2.489 ± 4.087
r50	1	0.1519 ± 0.07922	0.1409 ± 0.09423	0.1584 ± 0.07758	0.1329 ± 0.09748
r95	1	0.2531 ± 0.08778	0.2219 ± 0.1168	0.2524 ± 0.08442	0.2142 ± 0.1193
mean_abs_d0 *	mm	0.1318 ± 0.1765	1.782 ± 9.584	0.1318 ± 0.2037	1.235 ± 7.846
max_abs_d0 *	mm	0.2706 ± 0.6581	4.78 ± 18.51	0.3388 ± 0.6884	3.266 ± 16.51
std_abs_d0 *	mm	0.09804 ± 0.2638	1.833 ± 7.17	0.1372 ± 0.2882	1.256 ± 6.634
mean_abs_dz *	mm	0.2748 ± 0.2621	2.72 ± 14.33	0.2575 ± 0.3666	1.65 ± 9.624
max_abs_dz *	mm	0.5167 ± 0.4745	6.916 ± 23.74	0.563 ± 0.6986	4.312 ± 22.91
std_abs_dz *	mm	0.1821 ± 0.2	2.602 ± 8.543	0.2215 ± 0.3143	1.67 ± 9.449
pt_weighted_abs_d0 *	mm	0.1399 ± 0.2578	1.679 ± 9.487	0.1354 ± 0.2428	1.176 ± 7.857
pt_weighted_abs_dz *	mm	0.2873 ± 0.2715	2.361 ± 14.11	0.2577 ± 0.3469	1.356 ± 8.332

* Variable utilizada como entrada del modelo.

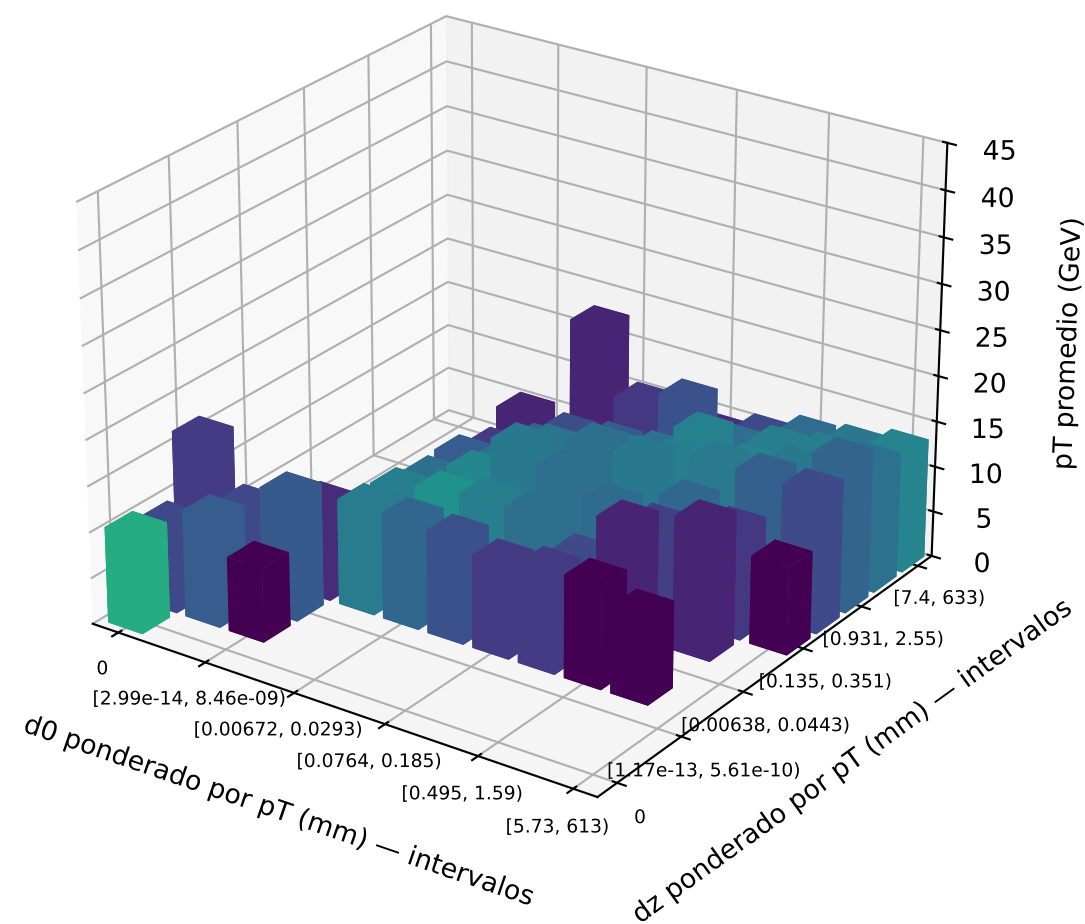
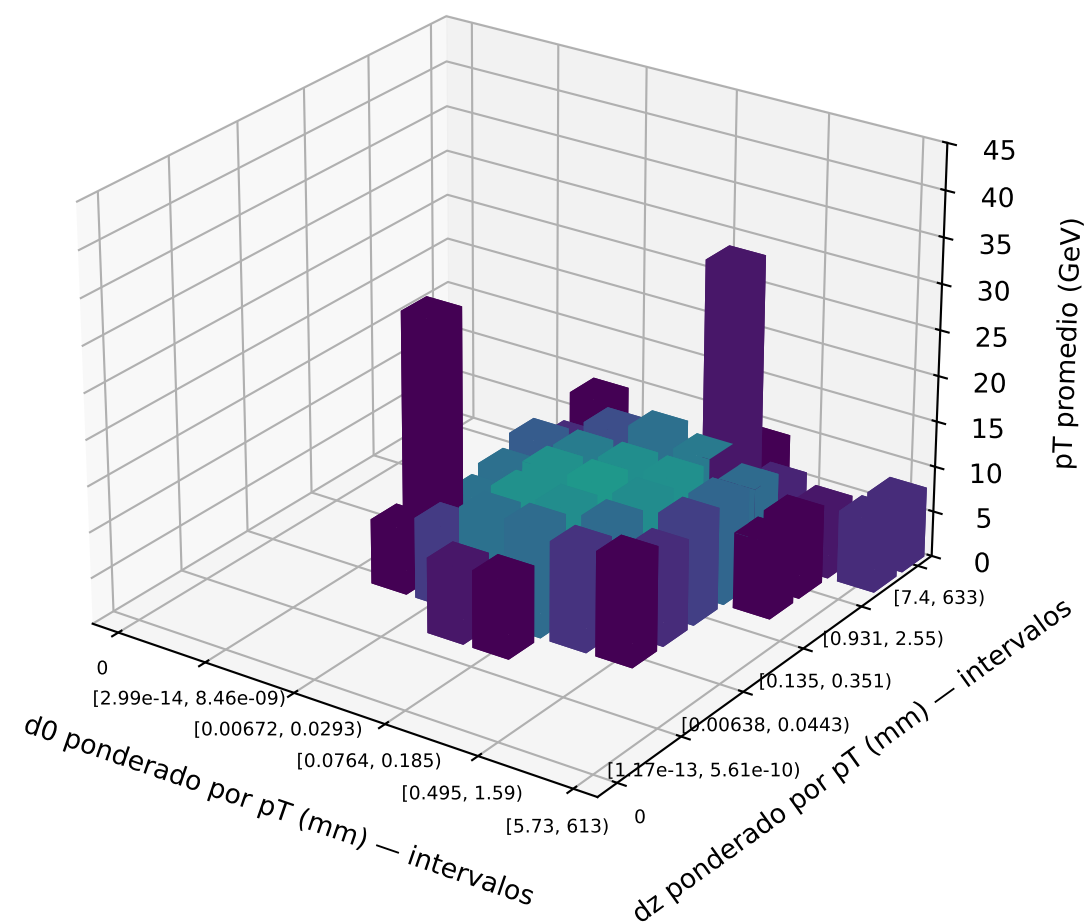
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — Alternative-VQC12 · semilla 73

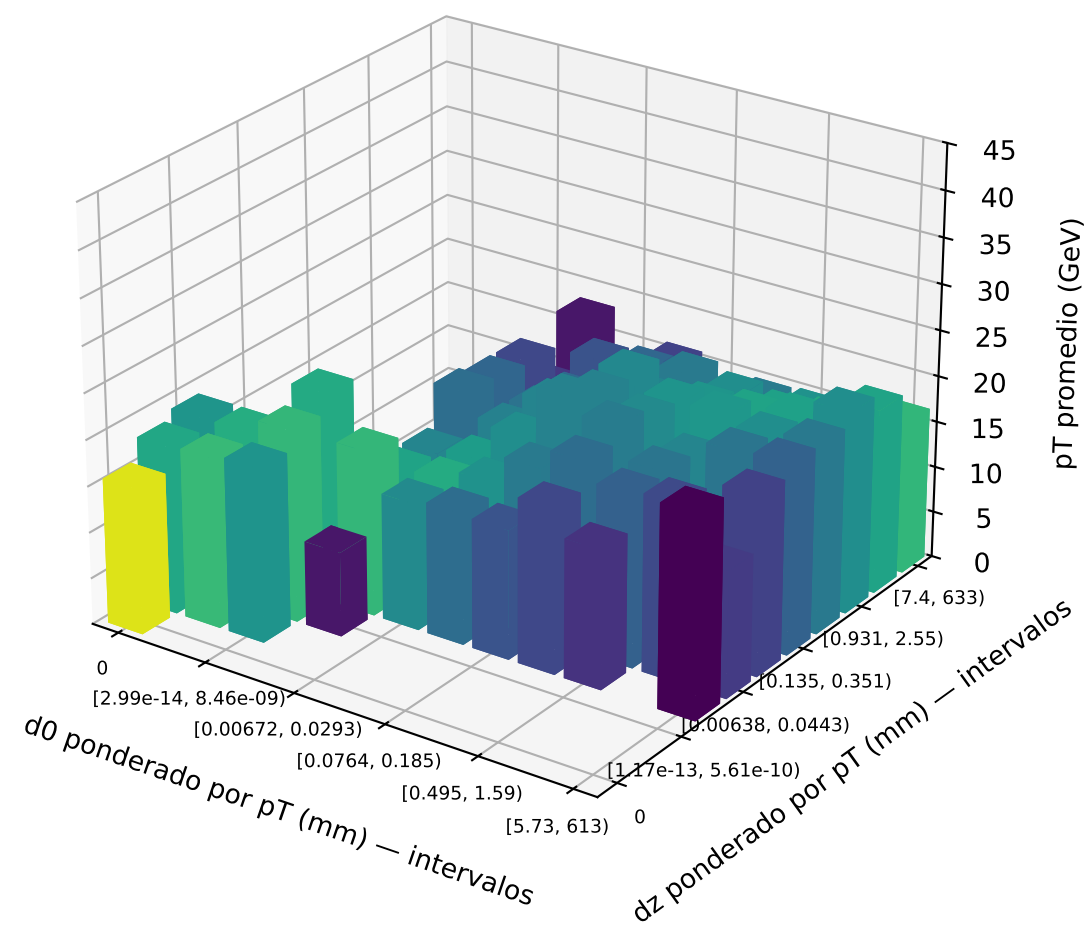
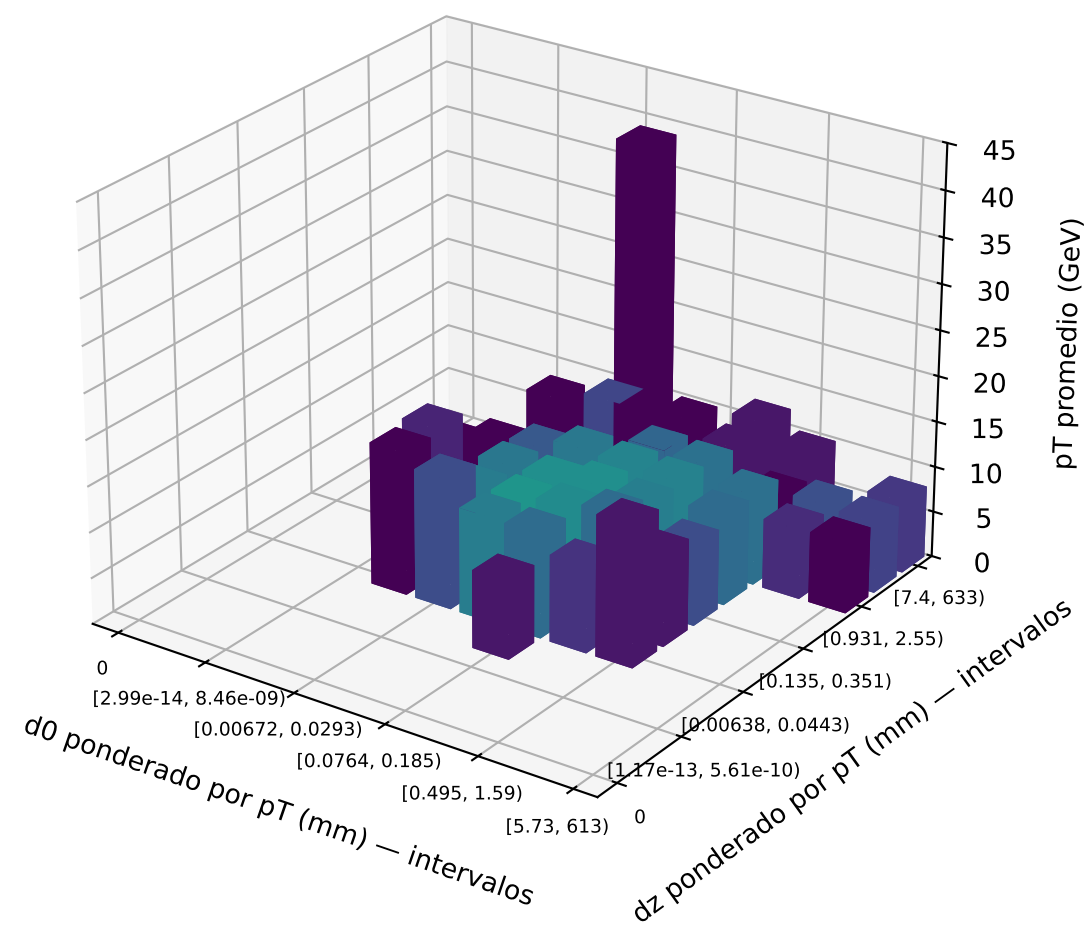
TP — N=2,532

FN — N=4,143



FP — N=2,218

TN — N=58,658



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — Alternative-VQC12 · semilla 73

Variable	Unidad	TP N=2,532	FN N=4,143	FP N=2,218	TN N=58,658
jet_pt *	GeV	11.88 ± 4.721	12.5 ± 7.562	11.58 ± 4.765	16.6 ± 11.8
jet_eta	1	-0.02326 ± 1.386	0.01823 ± 1.432	0.03924 ± 1.435	0.001311 ± 1.498
jet_abs_eta *	1	1.212 ± 0.6726	1.217 ± 0.7543	1.268 ± 0.6716	1.276 ± 0.7852
jet_mass	GeV	2.502 ± 1.287	2.547 ± 1.67	2.454 ± 1.183	2.867 ± 1.886
jet_ncharged	count	2.885 ± 1.529	2.62 ± 1.912	2.674 ± 1.381	2.584 ± 1.947
jet_nneutral	count	1.762 ± 1.521	2.06 ± 1.668	1.771 ± 1.381	2.392 ± 1.796
n_tracks_cone	count	2.922 ± 1.515	2.655 ± 1.911	2.71 ± 1.379	2.613 ± 1.947
track_pt_sum *	GeV	5.878 ± 3.45	5.14 ± 5.562	5.123 ± 2.966	6.816 ± 8.181
track_pt_sum_over_jet_pt	1	0.4977 ± 0.209	0.3956 ± 0.265	0.4535 ± 0.2059	0.3715 ± 0.2682
avg_track_pt *	GeV	2.234 ± 1.369	1.767 ± 1.346	2.042 ± 1.029	2.17 ± 2.084
std_track_pt	GeV	0.8021 ± 0.8016	0.6966 ± 0.903	0.6511 ± 0.6379	1.017 ± 1.62
n_tracks_below_avg_pt	count	1.528 ± 1.152	1.384 ± 1.316	1.361 ± 1.033	1.379 ± 1.363
n_tracks_above_avg_pt	count	1.394 ± 0.6429	1.271 ± 0.8193	1.349 ± 0.6139	1.234 ± 0.8077
max_pt_ratio	1	0.2911 ± 0.1556	0.2259 ± 0.1652	0.2613 ± 0.1247	0.2103 ± 0.1559
min_pt_ratio	1	0.1347 ± 0.1471	0.0988 ± 0.1199	0.1306 ± 0.1135	0.08486 ± 0.09811
max_dr_ratio	1	0.161 ± 0.1467	0.1195 ± 0.1235	0.1549 ± 0.1149	0.1036 ± 0.1074
min_dr_ratio	1	0.2492 ± 0.1754	0.1925 ± 0.1704	0.2251 ± 0.1391	0.1796 ± 0.155
dr_max_pt	1	0.1394 ± 0.08338	0.1362 ± 0.09733	0.1469 ± 0.08085	0.1264 ± 0.09921
dr_min_pt	1	0.2015 ± 0.1006	0.1938 ± 0.1153	0.2066 ± 0.09584	0.1901 ± 0.1174
dr_max_dr	1	0.2421 ± 0.09663	0.2297 ± 0.1173	0.241 ± 0.09037	0.2209 ± 0.1208
dr_min_dr	1	0.1076 ± 0.06997	0.1085 ± 0.08455	0.118 ± 0.07157	0.1038 ± 0.08849
pt_difference	GeV	1.884 ± 1.857	1.685 ± 2.364	1.51 ± 1.483	2.496 ± 4.091
r50	1	0.1461 ± 0.08038	0.1433 ± 0.09521	0.1551 ± 0.07731	0.133 ± 0.09757
r95	1	0.2394 ± 0.09624	0.2271 ± 0.1165	0.2389 ± 0.09065	0.2146 ± 0.1194
mean_abs_d0 *	mm	0.1907 ± 0.6058	1.911 ± 10.03	0.2797 ± 1.636	1.233 ± 7.855
max_abs_d0 *	mm	0.4028 ± 1.672	5.151 ± 19.34	0.6844 ± 3.573	3.265 ± 16.53
std_abs_d0 *	mm	0.1524 ± 0.7689	1.973 ± 7.485	0.2929 ± 1.71	1.255 ± 6.64
mean_abs_dz *	mm	0.3831 ± 0.6065	2.898 ± 15.02	0.449 ± 1.211	1.648 ± 9.641
max_abs_dz *	mm	0.7468 ± 1.458	7.416 ± 24.83	1.061 ± 2.762	4.308 ± 22.95
std_abs_dz *	mm	0.2649 ± 0.6478	2.794 ± 8.926	0.4339 ± 1.276	1.667 ± 9.465
pt_weighted_abs_d0 *	mm	0.2007 ± 0.7194	1.796 ± 9.929	0.2904 ± 1.923	1.175 ± 7.865
pt_weighted_abs_dz *	mm	0.3987 ± 0.6472	2.5 ± 14.79	0.4491 ± 1.28	1.353 ± 8.346

* Variable utilizada como entrada del modelo.

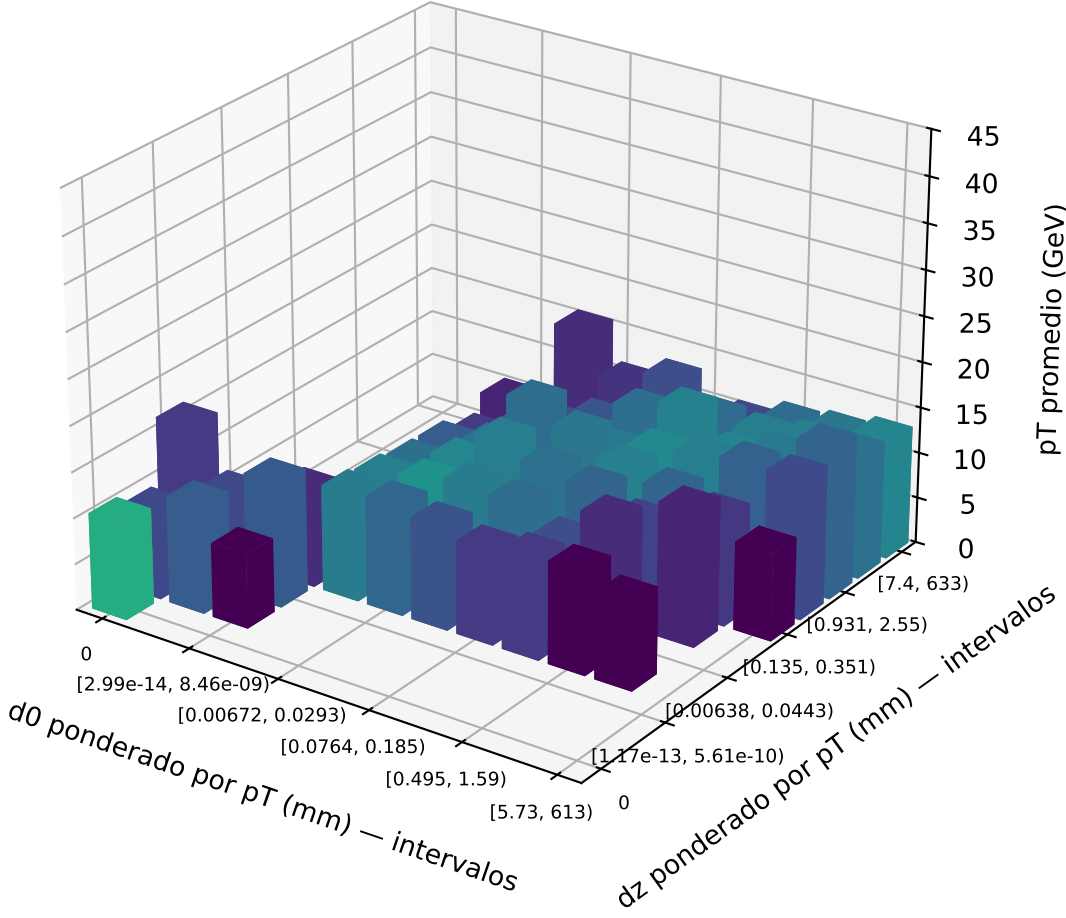
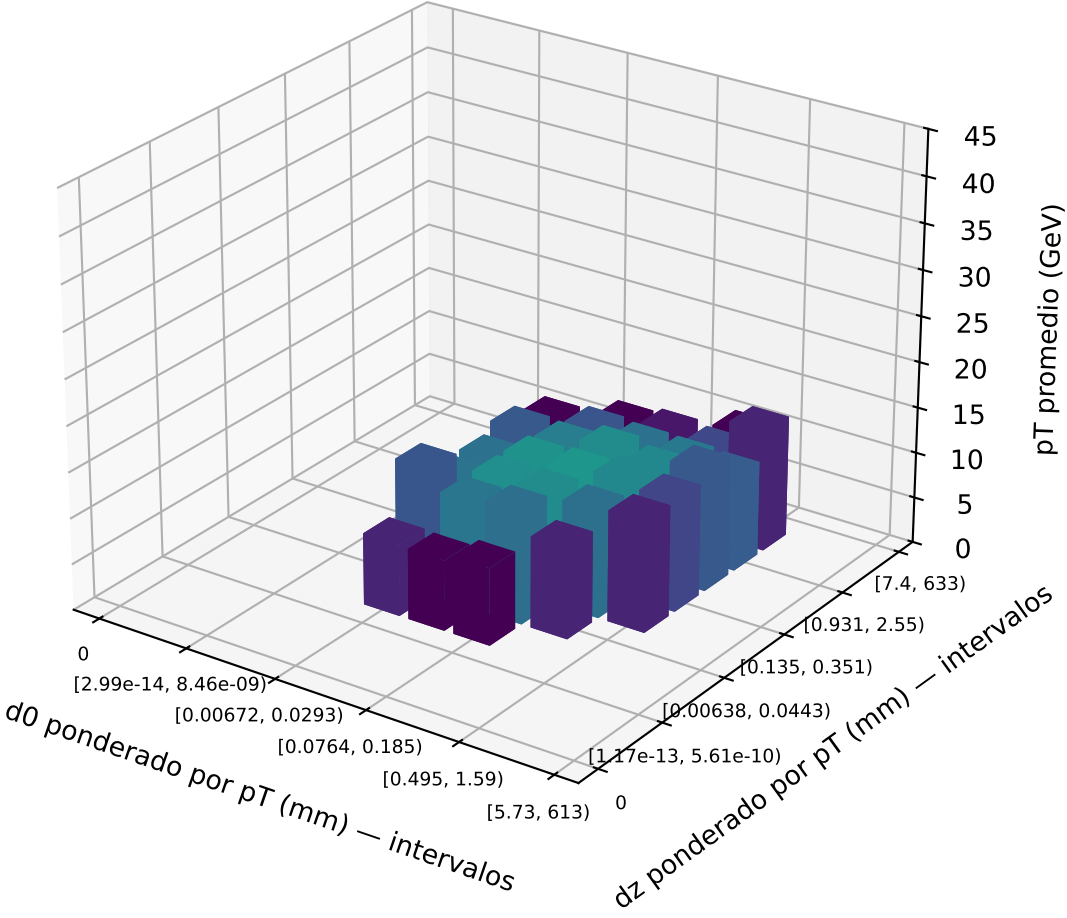
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — Alternative-VQC12 · semilla 101

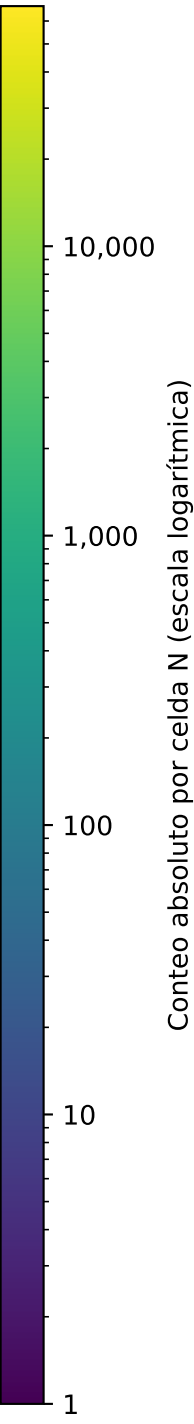
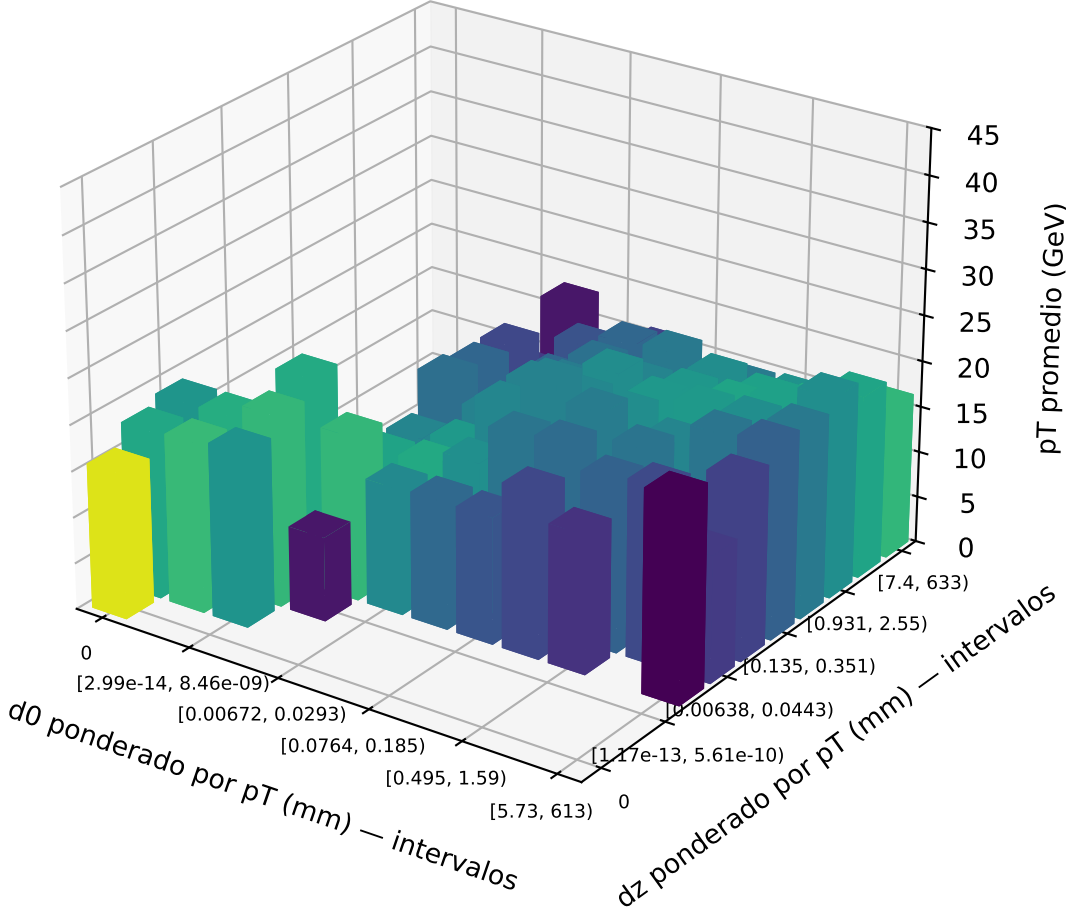
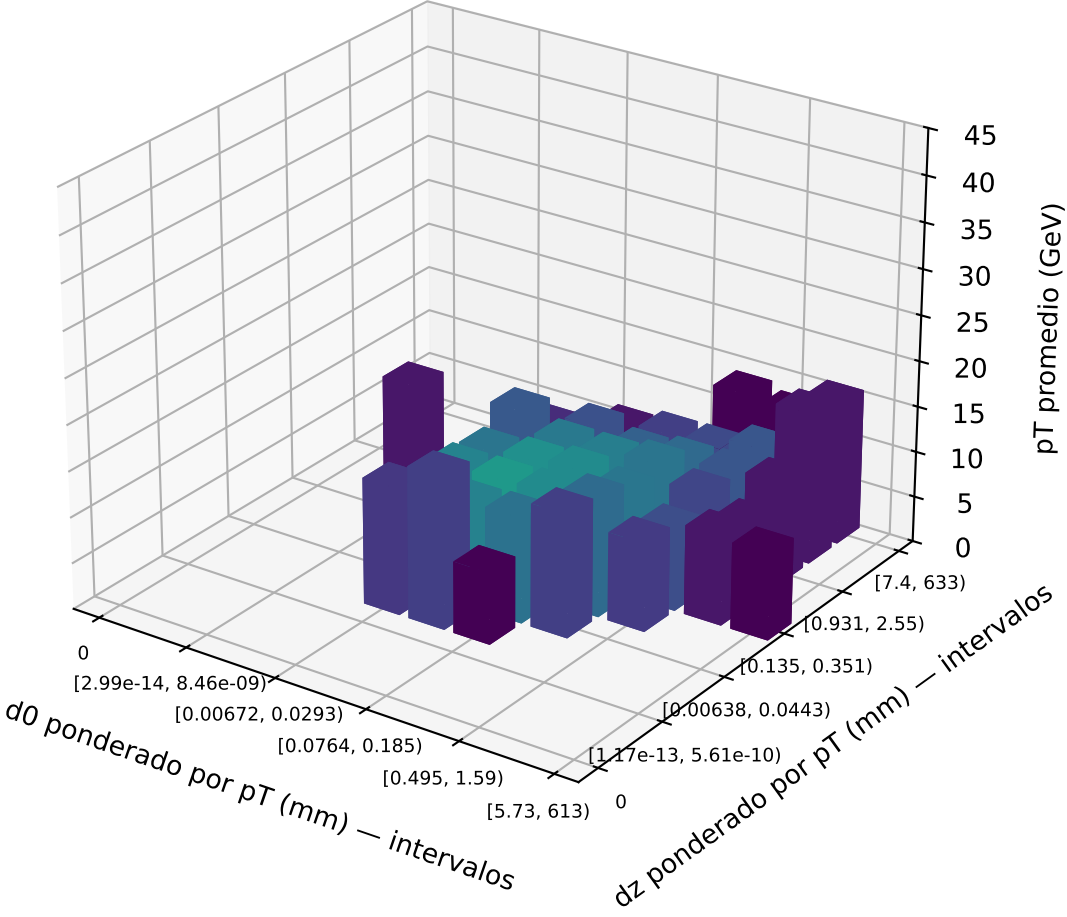
TP — N=2,408

FN — N=4,267



FP — N=2,203

TN — N=58,673



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — Alternative-VQC12 · semilla 101

Variable	Unidad	TP N=2,408	FN N=4,267	FP N=2,203	TN N=58,673
jet_pt *	GeV	11.77 ± 4.432	12.54 ± 7.59	11.65 ± 4.531	16.59 ± 11.8
jet_eta	1	-0.007772 ± 1.37	0.008283 ± 1.44	0.007169 ± 1.407	0.002525 ± 1.499
jet_abs_eta *	1	1.19 ± 0.6778	1.229 ± 0.7491	1.221 ± 0.7006	1.278 ± 0.7841
jet_mass	GeV	2.551 ± 1.236	2.519 ± 1.682	2.486 ± 1.149	2.865 ± 1.887
jet_ncharged	count	3.02 ± 1.458	2.552 ± 1.919	2.81 ± 1.33	2.579 ± 1.948
jet_nneutral	count	1.661 ± 1.468	2.109 ± 1.678	1.59 ± 1.343	2.399 ± 1.794
n_tracks_cone	count	3.056 ± 1.436	2.587 ± 1.921	2.834 ± 1.308	2.608 ± 1.949
track_pt_sum *	GeV	6.043 ± 2.954	5.068 ± 5.66	5.634 ± 2.857	6.796 ± 8.185
track_pt_sum_over_jet_pt	1	0.5259 ± 0.2014	0.3827 ± 0.2601	0.4992 ± 0.2099	0.3698 ± 0.2674
avg_track_pt *	GeV	2.164 ± 1.061	1.82 ± 1.508	2.108 ± 0.914	2.168 ± 2.086
std_track_pt	GeV	0.8529 ± 0.7042	0.671 ± 0.9409	0.7562 ± 0.6605	1.013 ± 1.62
n_tracks_below_avg_pt	count	1.637 ± 1.084	1.326 ± 1.334	1.466 ± 0.9998	1.375 ± 1.364
n_tracks_above_avg_pt	count	1.418 ± 0.6565	1.261 ± 0.8065	1.368 ± 0.5959	1.233 ± 0.808
max_pt_ratio	1	0.2974 ± 0.1454	0.2242 ± 0.1691	0.2797 ± 0.1267	0.2096 ± 0.1556
min_pt_ratio	1	0.123 ± 0.1249	0.1064 ± 0.1355	0.1258 ± 0.1027	0.08505 ± 0.09864
max_dr_ratio	1	0.1505 ± 0.1275	0.1267 ± 0.1372	0.1521 ± 0.1078	0.1037 ± 0.1077
min_dr_ratio	1	0.2529 ± 0.167	0.192 ± 0.1748	0.2382 ± 0.1434	0.1791 ± 0.1547
dr_max_pt	1	0.1345 ± 0.07968	0.139 ± 0.09868	0.1395 ± 0.07669	0.1267 ± 0.09938
dr_min_pt	1	0.207 ± 0.09851	0.1909 ± 0.1156	0.2062 ± 0.09509	0.1902 ± 0.1174
dr_max_dr	1	0.2489 ± 0.09093	0.2262 ± 0.1188	0.2429 ± 0.08692	0.2208 ± 0.1209
dr_min_dr	1	0.1028 ± 0.06424	0.1113 ± 0.08655	0.111 ± 0.06671	0.1041 ± 0.08866
pt_difference	GeV	1.997 ± 1.612	1.627 ± 2.443	1.757 ± 1.547	2.487 ± 4.092
r50	1	0.1422 ± 0.0768	0.1456 ± 0.09646	0.1483 ± 0.07393	0.1332 ± 0.09772
r95	1	0.2463 ± 0.09054	0.2236 ± 0.118	0.2403 ± 0.08709	0.2145 ± 0.1195
mean_abs_d0 *	mm	0.1348 ± 0.128	1.893 ± 9.896	0.1245 ± 0.2545	1.239 ± 7.86
max_abs_d0 *	mm	0.2621 ± 0.2698	5.092 ± 19.1	0.2683 ± 0.4852	3.28 ± 16.54
std_abs_d0 *	mm	0.09365 ± 0.1116	1.953 ± 7.397	0.1045 ± 0.219	1.261 ± 6.646
mean_abs_dz *	mm	0.3519 ± 0.4805	2.843 ± 14.8	0.324 ± 0.8738	1.652 ± 9.64
max_abs_dz *	mm	0.6371 ± 0.8443	7.285 ± 24.49	0.6508 ± 1.381	4.322 ± 22.95
std_abs_dz *	mm	0.2123 ± 0.3317	2.75 ± 8.808	0.2404 ± 0.5552	1.674 ± 9.466
pt_weighted_abs_d0 *	mm	0.1409 ± 0.1343	1.783 ± 9.797	0.1335 ± 0.3119	1.18 ± 7.871
pt_weighted_abs_dz *	mm	0.374 ± 0.5083	2.453 ± 14.58	0.3553 ± 0.9353	1.357 ± 8.346

* Variable utilizada como entrada del modelo.

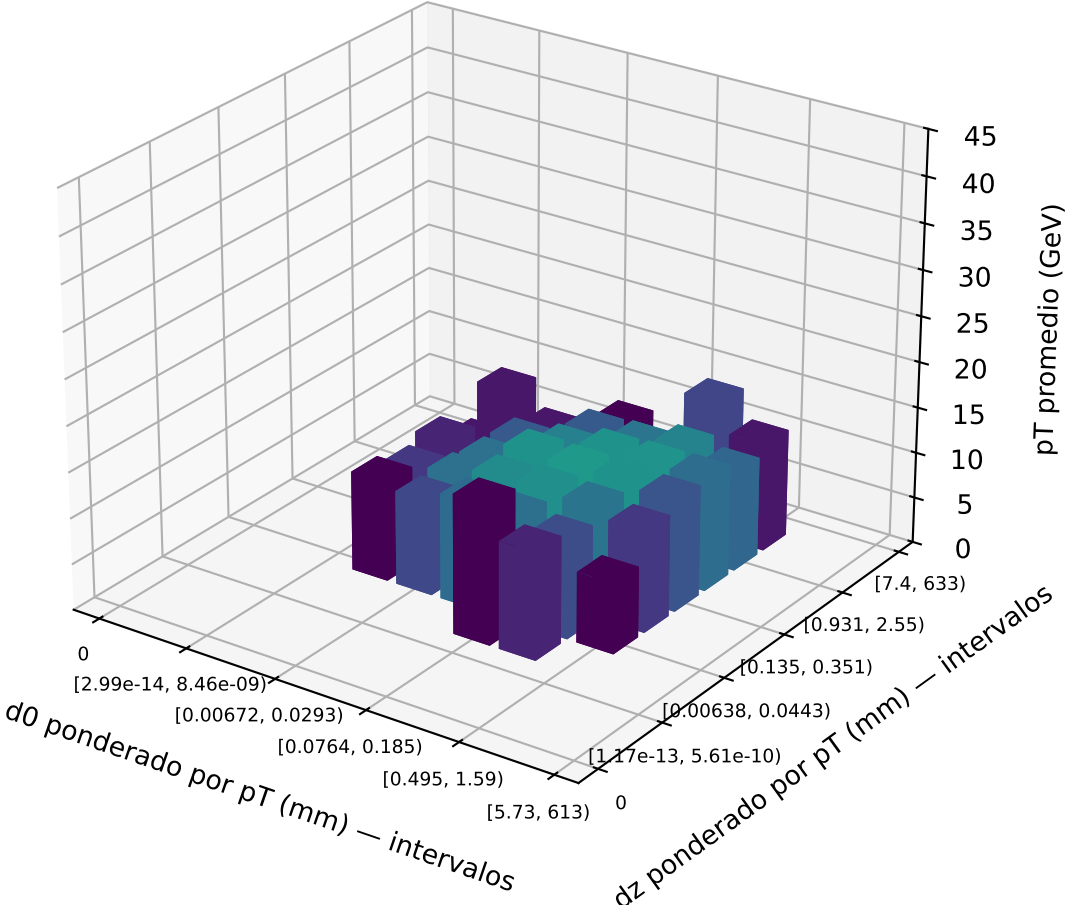
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

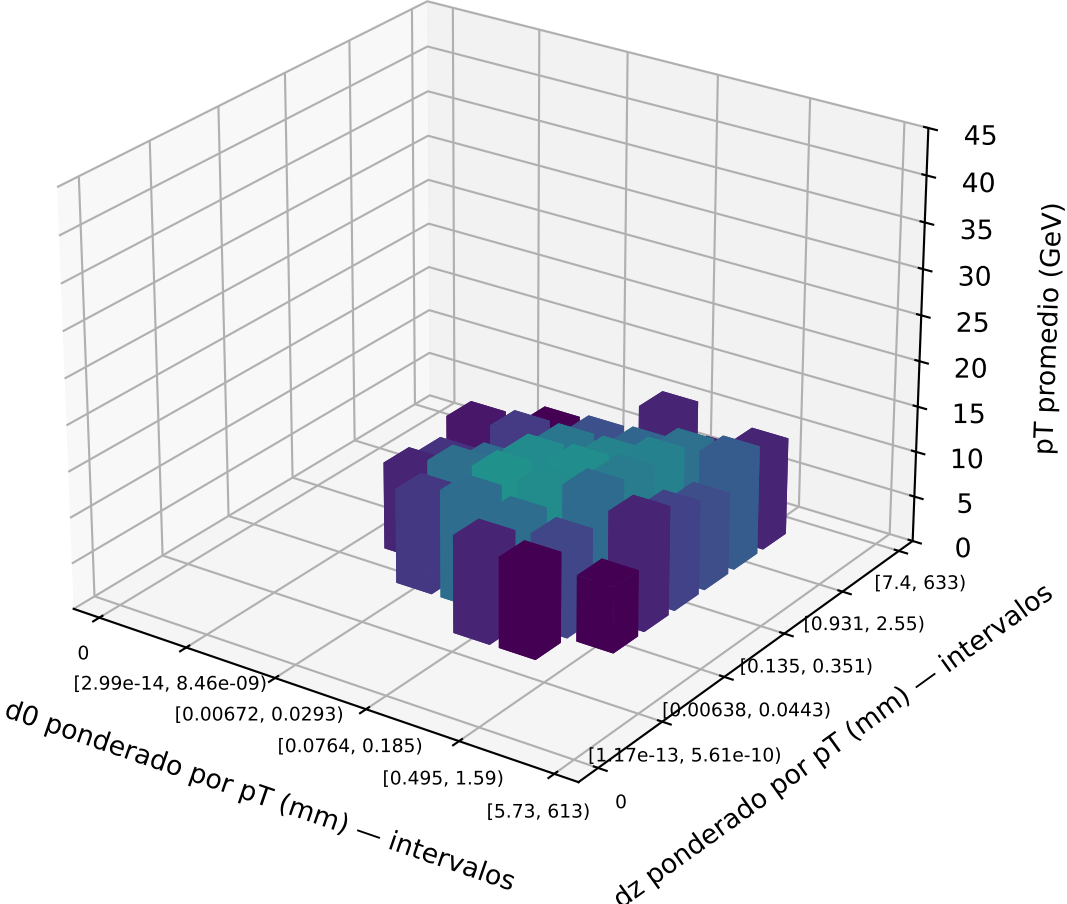
Perfil de pT medio en intervalos de d0/dz — RF12

TP — N=2,865

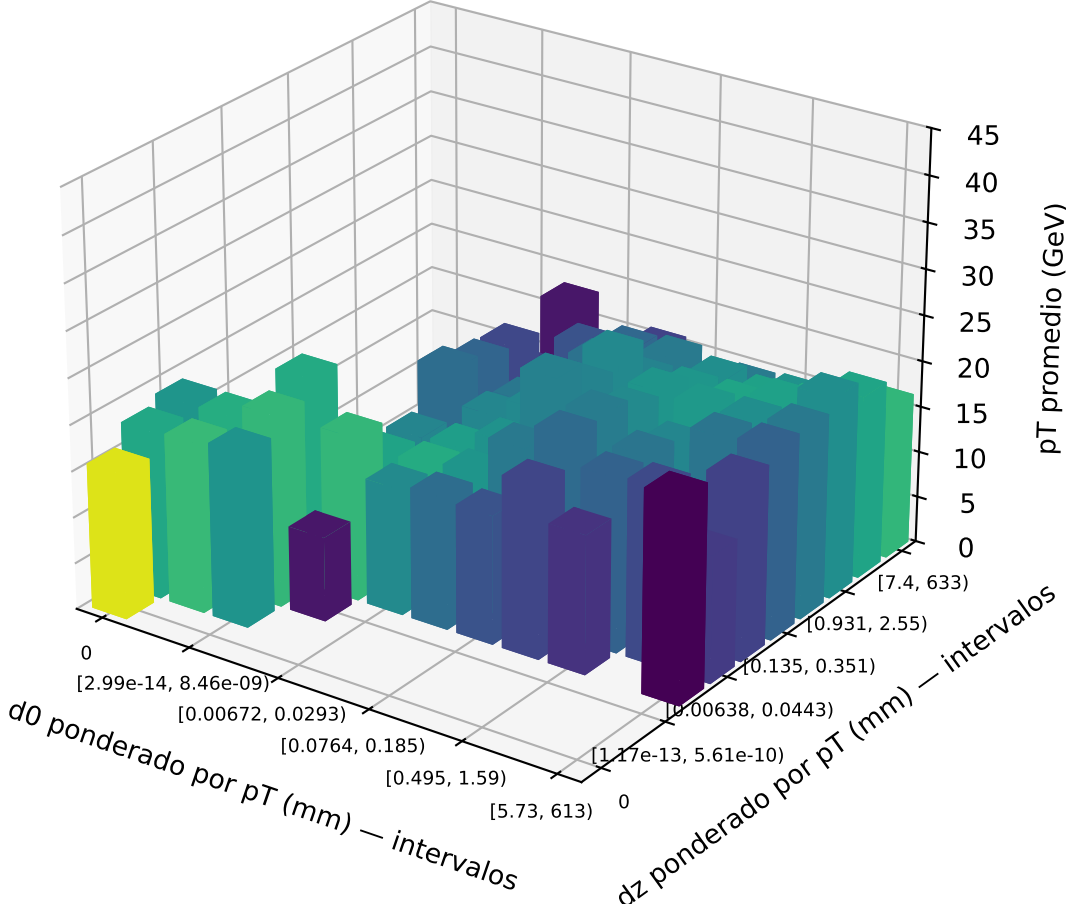
FN — N=3,810



FP — N=1,840



TN — N=59,036



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF12

Variable	Unidad	TP N=2,865	FN N=3,810	FP N=1,840	TN N=59,036
jet_pt *	GeV	12.02 ± 5.167	12.45 ± 7.55	11.79 ± 5.191	16.56 ± 11.77
jet_eta	1	-0.003428 ± 1.355	0.006943 ± 1.458	0.001546 ± 1.376	0.002729 ± 1.499
jet_abs_eta *	1	1.165 ± 0.6921	1.253 ± 0.7456	1.188 ± 0.6942	1.278 ± 0.7837
jet_mass	GeV	2.477 ± 1.31	2.57 ± 1.685	2.297 ± 1.105	2.869 ± 1.883
jet_ncharged	count	2.785 ± 1.524	2.672 ± 1.951	2.392 ± 1.276	2.594 ± 1.946
jet_nneutral	count	1.762 ± 1.534	2.086 ± 1.668	1.682 ± 1.313	2.391 ± 1.795
n_tracks_cone	count	2.807 ± 1.51	2.718 ± 1.951	2.41 ± 1.26	2.623 ± 1.946
track_pt_sum *	GeV	5.838 ± 3.586	5.106 ± 5.646	5.201 ± 3.466	6.802 ± 8.154
track_pt_sum_over_jet_pt	1	0.4907 ± 0.2191	0.392 ± 0.2634	0.4463 ± 0.2258	0.3723 ± 0.2675
avg_track_pt *	GeV	2.324 ± 1.466	1.658 ± 1.225	2.252 ± 1.238	2.163 ± 2.075
std_track_pt	GeV	0.7788 ± 0.8189	0.7049 ± 0.9009	0.6693 ± 0.7377	1.015 ± 1.615
n_tracks_below_avg_pt	count	1.438 ± 1.135	1.439 ± 1.344	1.154 ± 0.9806	1.385 ± 1.362
n_tracks_above_avg_pt	count	1.369 ± 0.6404	1.279 ± 0.836	1.256 ± 0.5131	1.238 ± 0.809
max_pt_ratio	1	0.294 ± 0.1642	0.218 ± 0.1574	0.2738 ± 0.1391	0.2102 ± 0.1553
min_pt_ratio	1	0.1449 ± 0.1583	0.088 ± 0.1015	0.1464 ± 0.126	0.08466 ± 0.09754
max_dr_ratio	1	0.1697 ± 0.1569	0.1094 ± 0.1074	0.1684 ± 0.1278	0.1035 ± 0.1068
min_dr_ratio	1	0.2555 ± 0.1822	0.1828 ± 0.1616	0.2408 ± 0.1502	0.1794 ± 0.1544
dr_max_pt	1	0.1381 ± 0.08403	0.1369 ± 0.09806	0.1456 ± 0.08267	0.1265 ± 0.09908
dr_min_pt	1	0.1997 ± 0.1016	0.1945 ± 0.1159	0.1996 ± 0.09463	0.1905 ± 0.1173
dr_max_dr	1	0.2374 ± 0.09857	0.2321 ± 0.118	0.2272 ± 0.08978	0.2214 ± 0.1207
dr_min_dr	1	0.1091 ± 0.07234	0.1075 ± 0.0842	0.1234 ± 0.07803	0.1037 ± 0.0882
pt_difference	GeV	1.82 ± 1.881	1.716 ± 2.391	1.535 ± 1.716	2.489 ± 4.078
r50	1	0.1448 ± 0.08161	0.1441 ± 0.09563	0.1515 ± 0.07961	0.1332 ± 0.09744
r95	1	0.235 ± 0.09818	0.2294 ± 0.1171	0.2249 ± 0.08993	0.2152 ± 0.1193
mean_abs_d0 *	mm	0.1609 ± 0.1414	2.084 ± 10.46	0.1315 ± 0.132	1.232 ± 7.836
max_abs_d0 *	mm	0.2711 ± 0.2344	5.665 ± 20.14	0.2163 ± 0.2025	3.263 ± 16.49
std_abs_d0 *	mm	0.08454 ± 0.09358	2.183 ± 7.796	0.06751 ± 0.08495	1.256 ± 6.626
mean_abs_dz *	mm	0.3899 ± 0.4376	3.113 ± 15.64	0.323 ± 0.4214	1.644 ± 9.612
max_abs_dz *	mm	0.6421 ± 0.67	8.078 ± 25.81	0.5109 ± 0.5825	4.304 ± 22.88
std_abs_dz *	mm	0.1949 ± 0.2486	3.067 ± 9.272	0.1532 ± 0.2216	1.668 ± 9.437
pt_weighted_abs_d0 *	mm	0.1676 ± 0.1432	1.96 ± 10.35	0.1372 ± 0.1328	1.174 ± 7.848
pt_weighted_abs_dz *	mm	0.408 ± 0.4529	2.677 ± 15.41	0.3377 ± 0.4259	1.351 ± 8.322

* Variable utilizada como entrada del modelo.

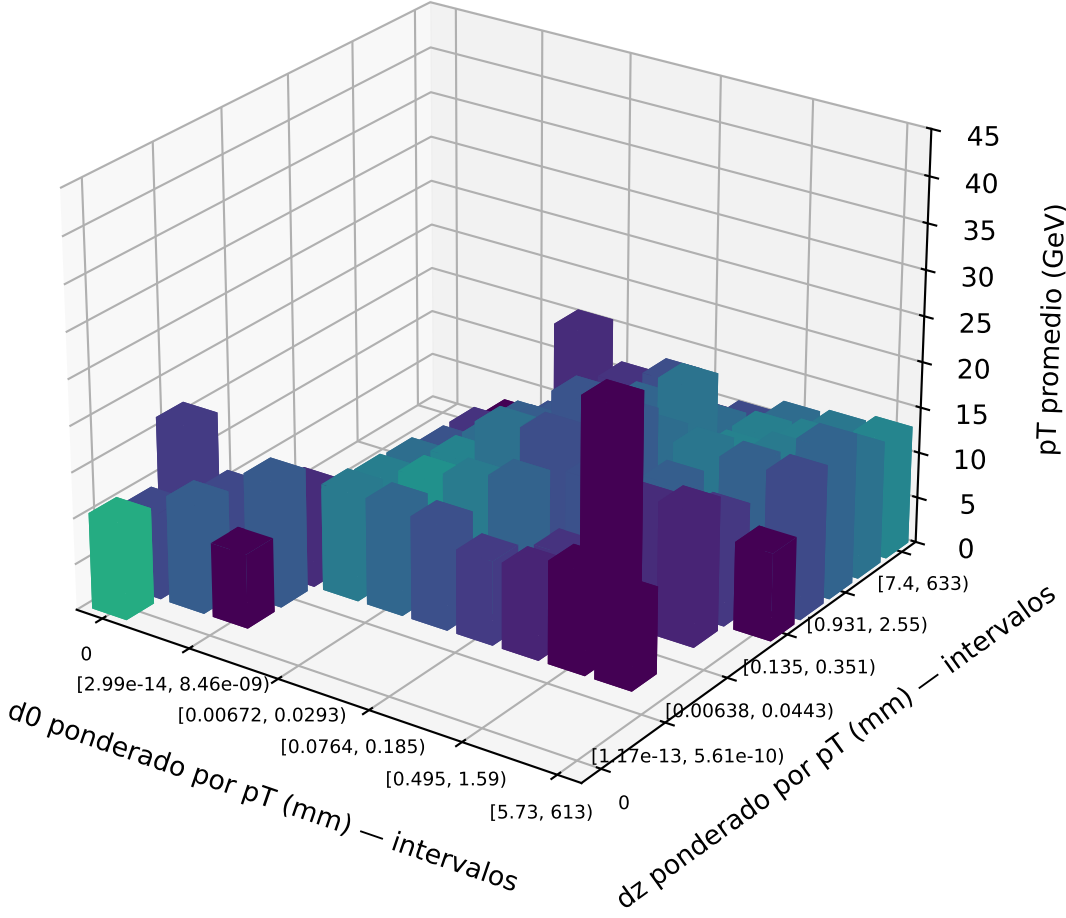
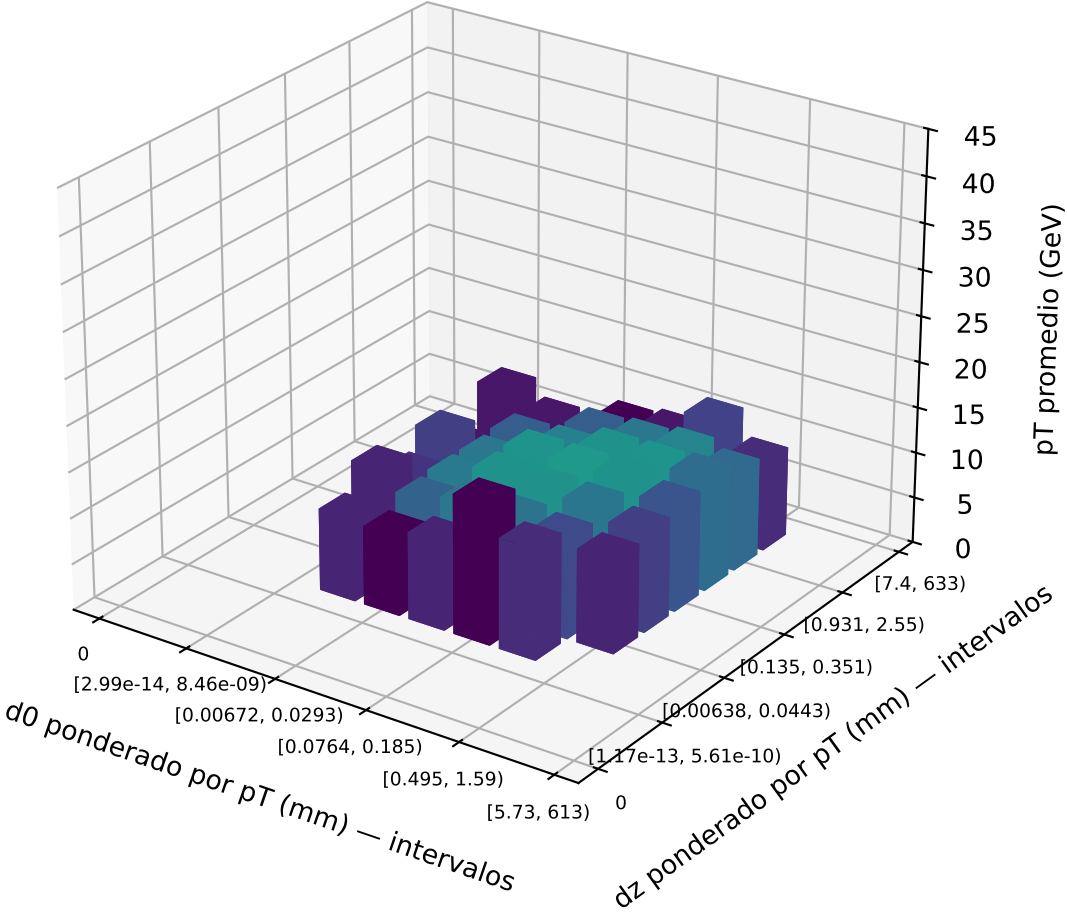
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RF32 · referencia existente

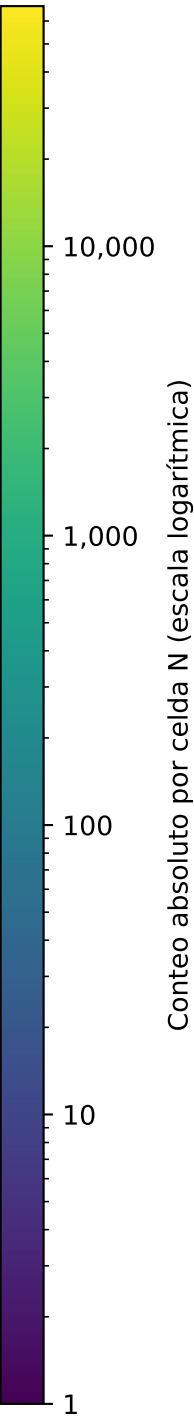
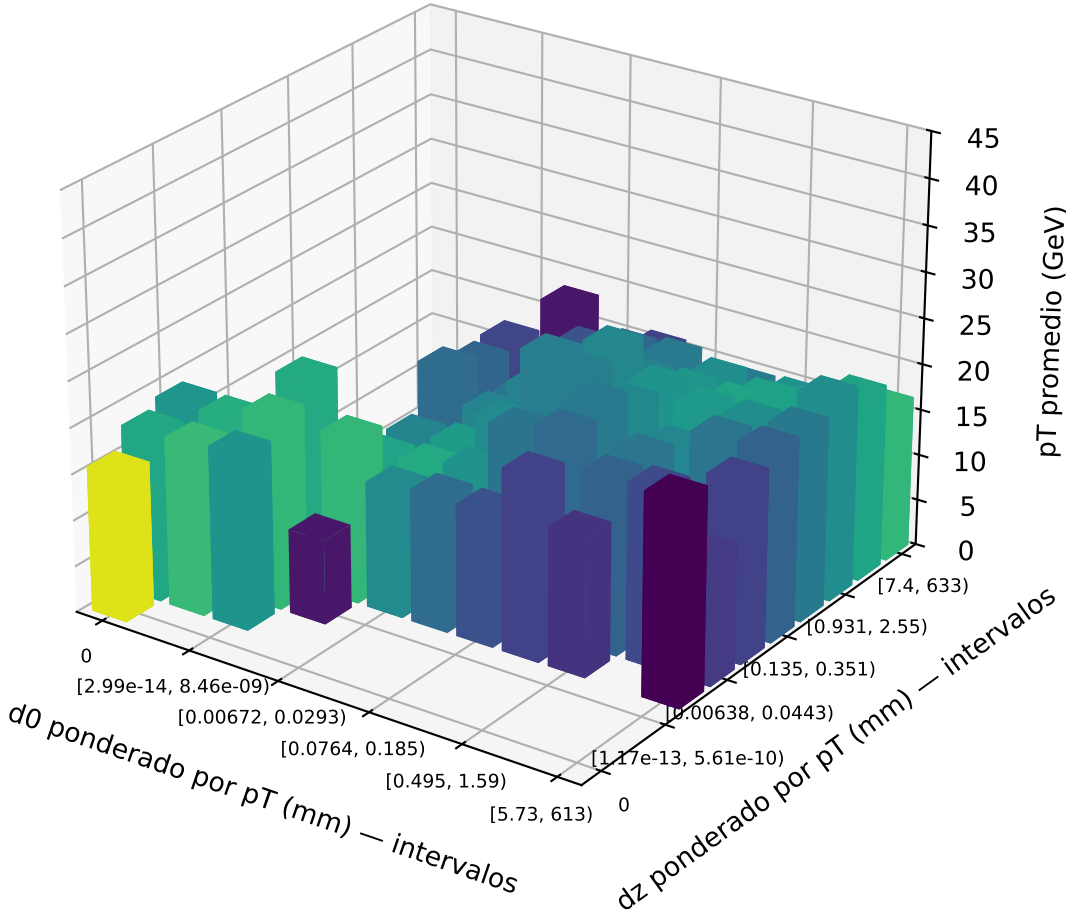
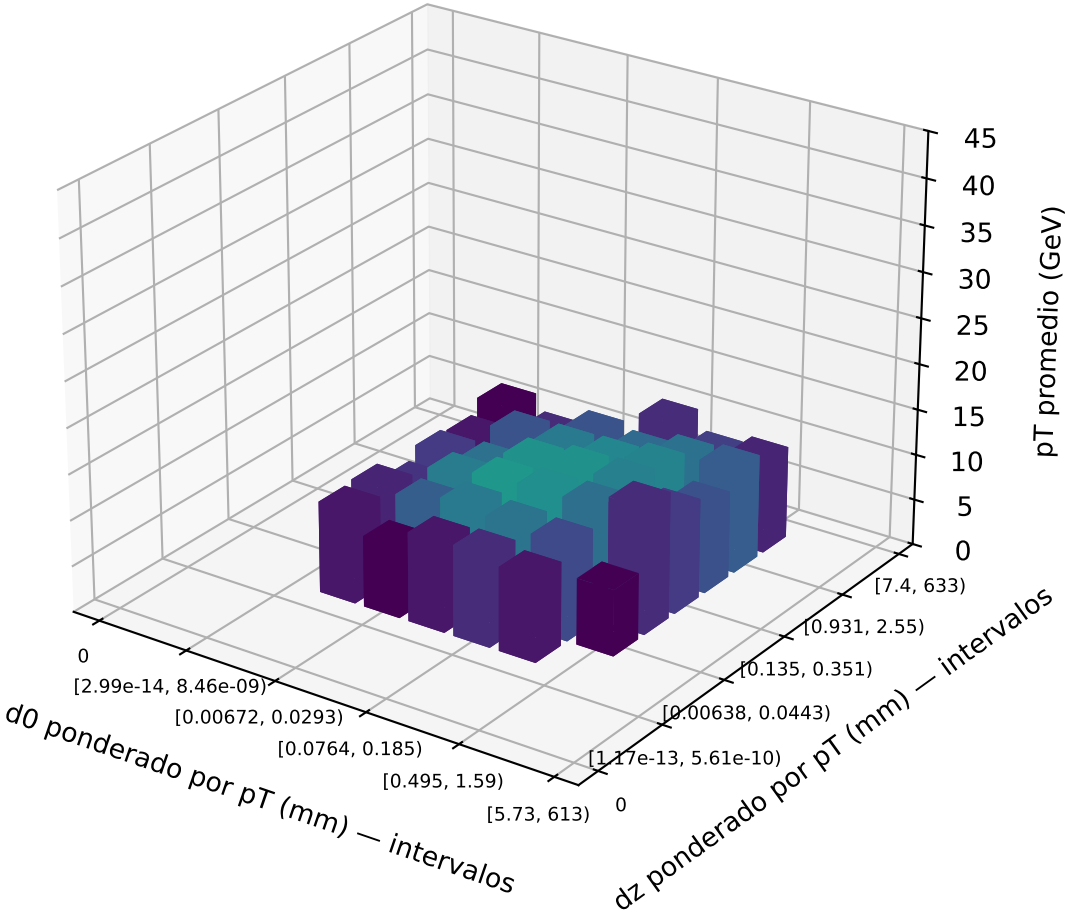
TP — N=3,204

FN — N=3,471



FP — N=2,300

TN — N=58,576



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF32 · referencia existente

Variable	Unidad	TP N=3,204	FN N=3,471	FP N=2,300	TN N=58,576
jet_pt *	GeV	11.97 ± 5.062	12.53 ± 7.803	11.67 ± 4.719	16.6 ± 11.8
jet_eta *	1	0.006546 ± 1.357	-0.001252 ± 1.466	0.01184 ± 1.384	0.002334 ± 1.5
jet_abs_eta *	1	1.166 ± 0.6941	1.261 ± 0.7484	1.189 ± 0.7083	1.279 ± 0.7839
jet_mass *	GeV	2.483 ± 1.325	2.573 ± 1.707	2.359 ± 1.163	2.871 ± 1.887
jet_ncharged *	count	2.8 ± 1.55	2.647 ± 1.967	2.483 ± 1.337	2.592 ± 1.949
jet_nneutral *	count	1.763 ± 1.534	2.117 ± 1.678	1.713 ± 1.351	2.395 ± 1.796
n_tracks_cone *	count	2.825 ± 1.542	2.693 ± 1.965	2.495 ± 1.322	2.621 ± 1.95
track_pt_sum *	GeV	5.806 ± 3.56	5.063 ± 5.821	5.169 ± 3.368	6.816 ± 8.18
track_pt_sum_over_jet_pt *	1	0.4899 ± 0.2206	0.383 ± 0.2645	0.4467 ± 0.2238	0.3717 ± 0.2678
avg_track_pt *	GeV	2.305 ± 1.465	1.611 ± 1.19	2.192 ± 1.232	2.165 ± 2.081
std_track_pt *	GeV	0.7866 ± 0.8247	0.6905 ± 0.9027	0.68 ± 0.7387	1.017 ± 1.619
n_tracks_below_avg_pt *	count	1.452 ± 1.154	1.426 ± 1.348	1.223 ± 1.027	1.384 ± 1.363
n_tracks_above_avg_pt *	count	1.373 ± 0.6537	1.267 ± 0.8424	1.273 ± 0.5333	1.237 ± 0.8104
max_pt_ratio *	1	0.2952 ± 0.1681	0.2095 ± 0.1502	0.2734 ± 0.146	0.2097 ± 0.1551
min_pt_ratio *	1	0.1431 ± 0.1597	0.08404 ± 0.09115	0.1415 ± 0.1301	0.08436 ± 0.09703
max_dr_ratio *	1	0.1673 ± 0.1578	0.1058 ± 0.09933	0.1633 ± 0.131	0.1032 ± 0.1064
min_dr_ratio *	1	0.2574 ± 0.1864	0.174 ± 0.1521	0.2394 ± 0.158	0.179 ± 0.1541
dr_max_pt *	1	0.1377 ± 0.08425	0.1371 ± 0.09916	0.1459 ± 0.08183	0.1264 ± 0.09921
dr_min_pt *	1	0.2002 ± 0.1011	0.1935 ± 0.1175	0.2033 ± 0.0946	0.1902 ± 0.1175
dr_max_dr *	1	0.2383 ± 0.09862	0.2308 ± 0.1196	0.2332 ± 0.09007	0.2211 ± 0.1209
dr_min_dr *	1	0.1086 ± 0.0717	0.1078 ± 0.08578	0.122 ± 0.07548	0.1036 ± 0.08836
pt_difference *	GeV	1.837 ± 1.882	1.69 ± 2.434	1.563 ± 1.707	2.496 ± 4.091
r50 *	1	0.1442 ± 0.0817	0.1445 ± 0.09683	0.1519 ± 0.07905	0.1331 ± 0.09757
r95 *	1	0.2358 ± 0.09815	0.2281 ± 0.1188	0.2312 ± 0.09013	0.2149 ± 0.1195
mean_abs_d0 *	mm	0.1534 ± 0.1427	2.279 ± 10.94	0.1218 ± 0.126	1.241 ± 7.866
max_abs_d0 *	mm	0.263 ± 0.2354	6.199 ± 21.02	0.2171 ± 0.208	3.287 ± 16.55
std_abs_d0 *	mm	0.08284 ± 0.09344	2.39 ± 8.138	0.07275 ± 0.09013	1.265 ± 6.651
mean_abs_dz *	mm	0.3764 ± 0.4479	3.391 ± 16.36	0.3123 ± 0.4376	1.655 ± 9.649
max_abs_dz *	mm	0.6311 ± 0.6939	8.814 ± 26.92	0.5213 ± 0.6099	4.333 ± 22.97
std_abs_dz *	mm	0.1934 ± 0.2555	3.349 ± 9.668	0.1647 ± 0.235	1.68 ± 9.474
pt_weighted_abs_d0 *	mm	0.1599 ± 0.1453	2.142 ± 10.83	0.127 ± 0.1282	1.182 ± 7.878
pt_weighted_abs_dz *	mm	0.3928 ± 0.461	2.913 ± 16.13	0.3244 ± 0.4428	1.359 ± 8.354

* Variable utilizada como entrada del modelo.

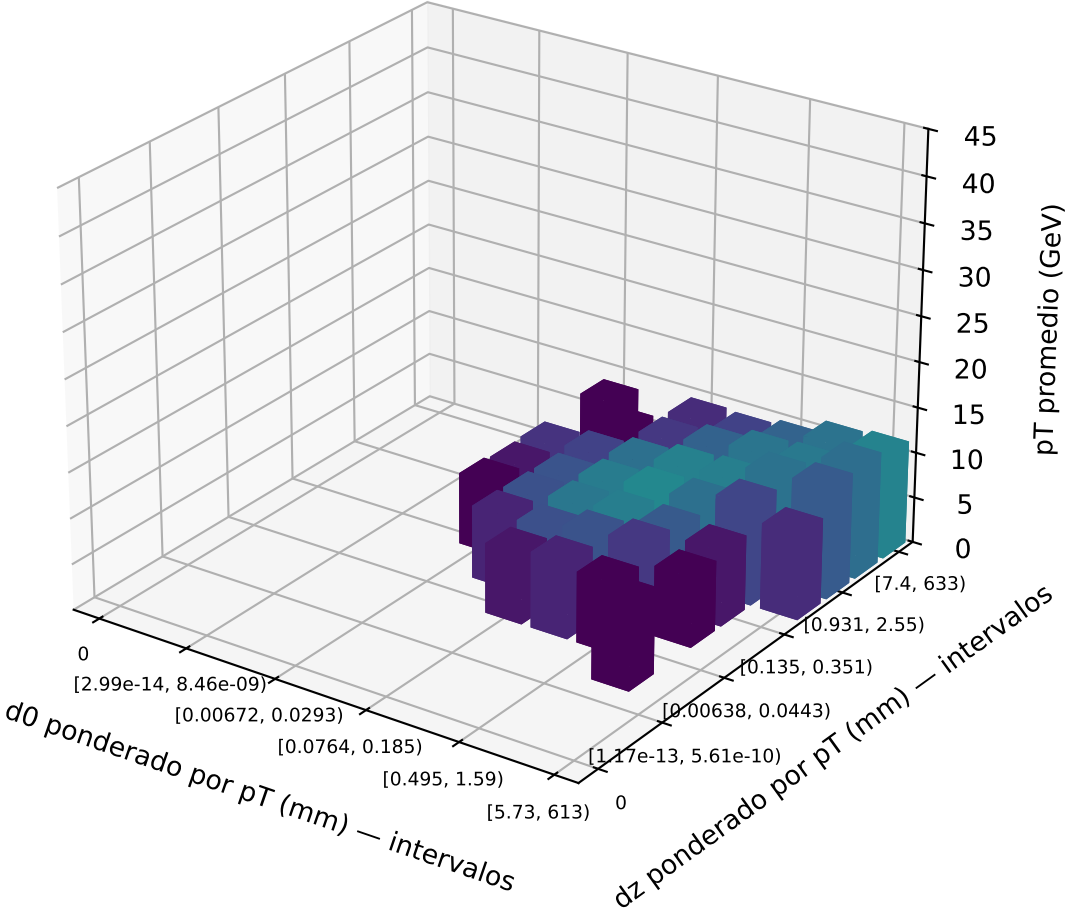
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

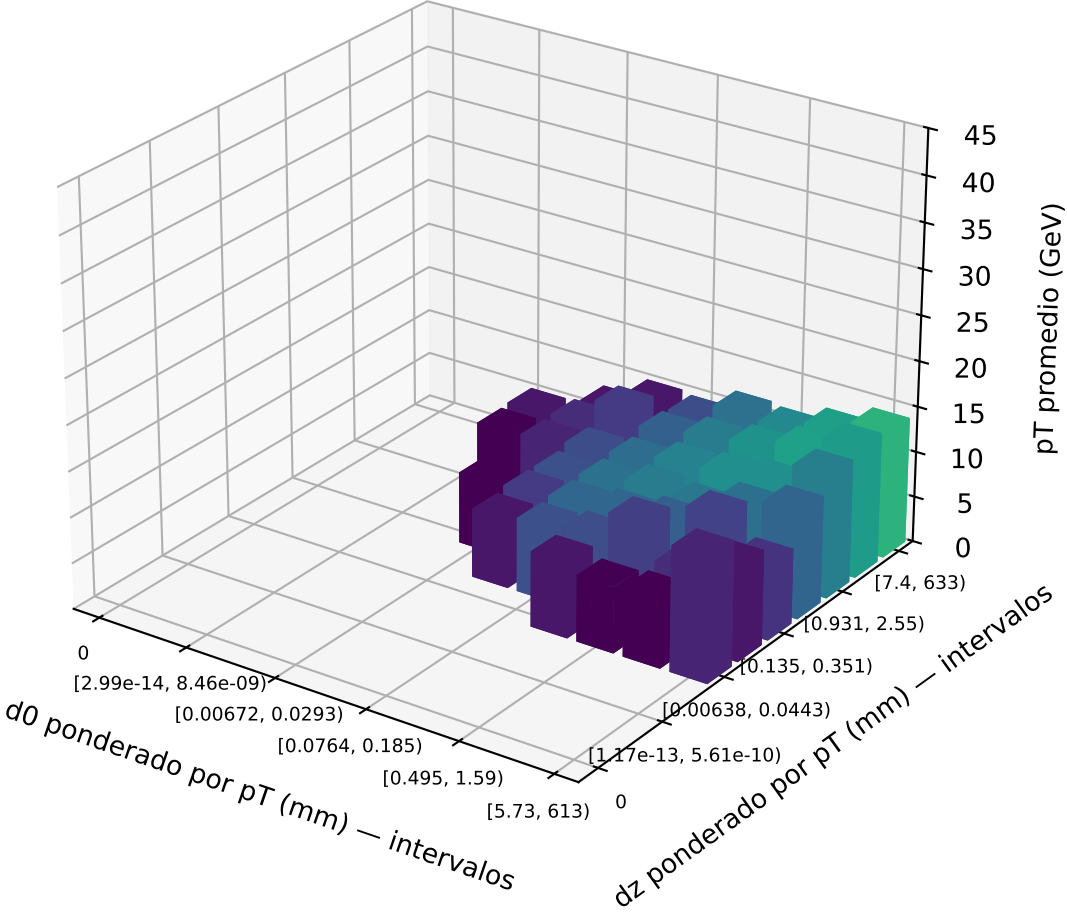
Perfil de pT medio en intervalos de d0/dz — LR12

TP — N=1,582

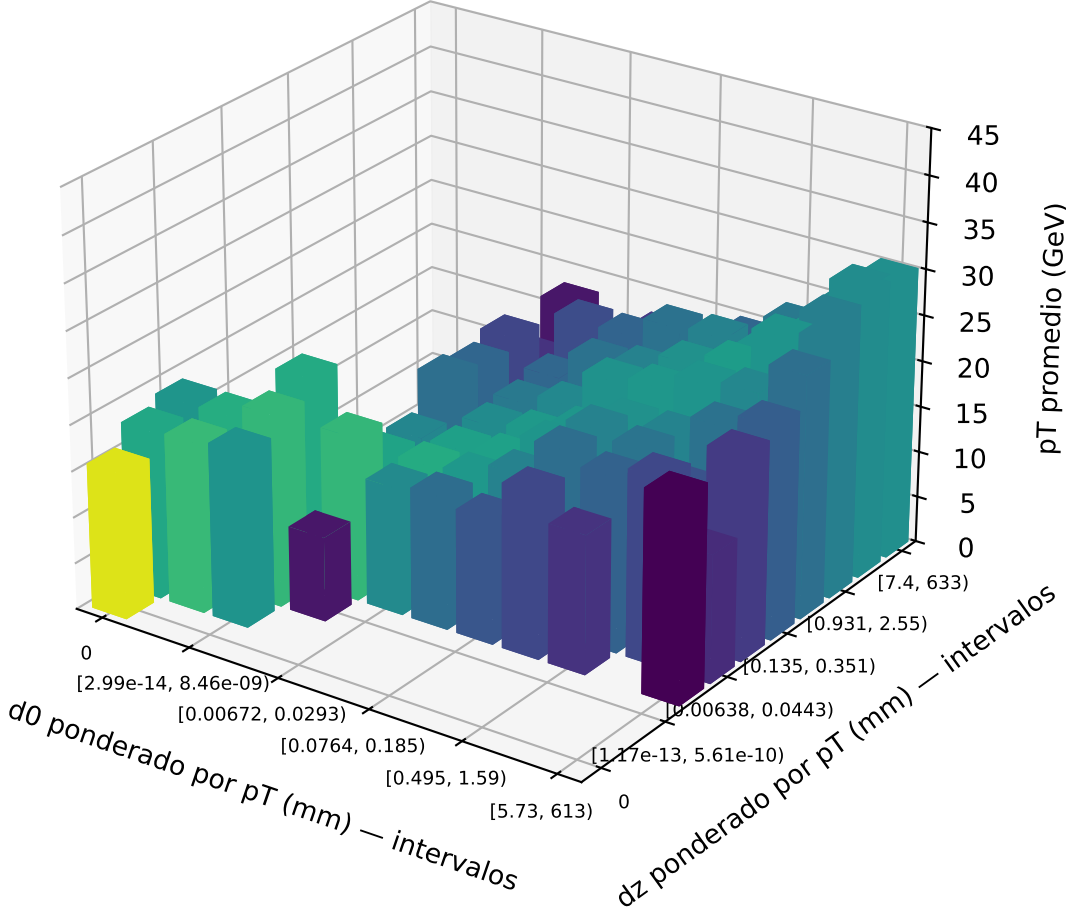
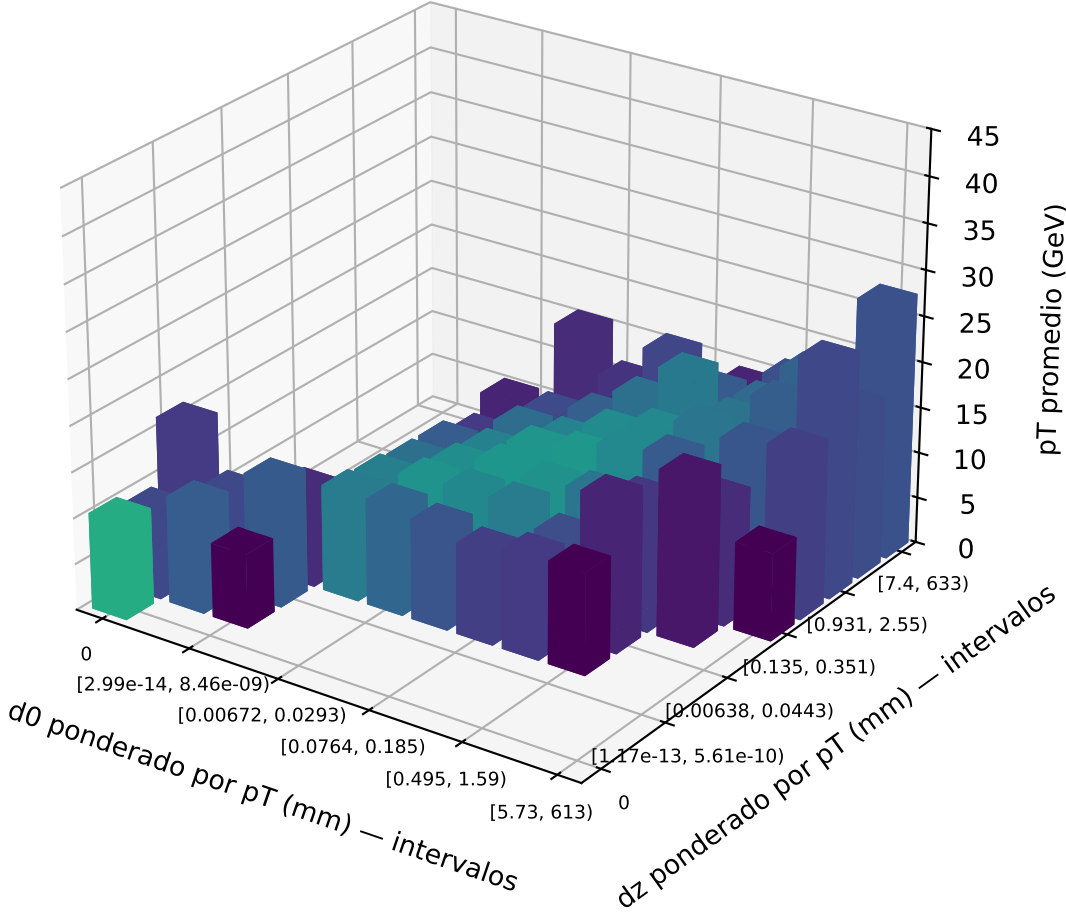
FN — N=5,093



FP — N=4,771



TN — N=56,105



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — LR12

Variable	Unidad	TP N=1,582	FN N=5,093	FP N=4,771	TN N=56,105
jet_pt *	GeV	9.825 ± 2.82	13.02 ± 7.269	12.07 ± 5.931	16.78 ± 11.94
jet_eta	1	0.004378 ± 1.432	0.001905 ± 1.409	0.01582 ± 1.437	0.001577 ± 1.501
jet_abs_eta *	1	1.242 ± 0.7133	1.207 ± 0.7276	1.23 ± 0.7441	1.279 ± 0.7843
jet_mass	GeV	2.149 ± 1.118	2.649 ± 1.626	2.653 ± 1.298	2.869 ± 1.907
jet_ncharged	count	2.566 ± 1.502	2.769 ± 1.857	2.764 ± 1.535	2.573 ± 1.959
jet_nneutral	count	1.547 ± 1.33	2.071 ± 1.681	1.861 ± 1.433	2.412 ± 1.806
n_tracks_cone	count	2.63 ± 1.516	2.795 ± 1.848	2.832 ± 1.522	2.598 ± 1.959
track_pt_sum *	GeV	4.556 ± 2.721	5.688 ± 5.352	4.988 ± 4.045	6.904 ± 8.292
track_pt_sum_over_jet_pt	1	0.465 ± 0.2346	0.4248 ± 0.2541	0.4138 ± 0.2388	0.3712 ± 0.2686
avg_track_pt *	GeV	1.968 ± 1.327	1.937 ± 1.388	1.759 ± 1.15	2.2 ± 2.11
std_track_pt	GeV	0.5328 ± 0.5883	0.7999 ± 0.9283	0.6467 ± 0.7882	1.035 ± 1.643
n_tracks_below_avg_pt	count	1.283 ± 1.137	1.487 ± 1.29	1.427 ± 1.121	1.374 ± 1.37
n_tracks_above_avg_pt	count	1.347 ± 0.6303	1.309 ± 0.7952	1.405 ± 0.659	1.224 ± 0.8111
max_pt_ratio	1	0.2787 ± 0.1628	0.2419 ± 0.1643	0.2233 ± 0.1246	0.2112 ± 0.1575
min_pt_ratio	1	0.1527 ± 0.1599	0.0999 ± 0.1194	0.1002 ± 0.08562	0.08536 ± 0.1001
max_dr_ratio	1	0.1769 ± 0.1575	0.1224 ± 0.1234	0.1245 ± 0.09518	0.1038 ± 0.1089
min_dr_ratio	1	0.2435 ± 0.1765	0.2048 ± 0.1728	0.1879 ± 0.1283	0.1807 ± 0.1567
dr_max_pt	1	0.1506 ± 0.08975	0.1333 ± 0.0927	0.1646 ± 0.09077	0.1239 ± 0.09867
dr_min_pt	1	0.1999 ± 0.1033	0.1957 ± 0.112	0.2204 ± 0.09484	0.1882 ± 0.1181
dr_max_dr	1	0.2355 ± 0.1011	0.2341 ± 0.1128	0.2594 ± 0.0859	0.2184 ± 0.1218
dr_min_dr	1	0.121 ± 0.08069	0.1042 ± 0.07848	0.1318 ± 0.08267	0.102 ± 0.08802
pt_difference	GeV	1.244 ± 1.344	1.921 ± 2.367	1.546 ± 1.907	2.538 ± 4.152
r50	1	0.1563 ± 0.08609	0.1407 ± 0.09071	0.1729 ± 0.08468	0.1304 ± 0.09725
r95	1	0.2343 ± 0.101	0.231 ± 0.1119	0.2569 ± 0.08618	0.212 ± 0.1202
mean_abs_d0 *	mm	4.045 ± 15.67	0.3931 ± 1.89	10.69 ± 24.17	0.3912 ± 2.577
max_abs_d0 *	mm	9.702 ± 27.87	1.377 ± 7.411	23.04 ± 42.78	1.482 ± 9.715
std_abs_d0 *	mm	3.661 ± 10.79	0.5439 ± 2.896	8.48 ± 17.43	0.6022 ± 3.944
mean_abs_dz *	mm	5.841 ± 23.32	0.7339 ± 3.239	13.59 ± 27.84	0.585 ± 4.257
max_abs_dz *	mm	12.87 ± 34.5	2.406 ± 10.98	28.92 ± 58.21	2.086 ± 14.38
std_abs_dz *	mm	4.66 ± 11.91	0.9569 ± 4.433	10.58 ± 24.19	0.8606 ± 6.051
pt_weighted_abs_d0 *	mm	4.132 ± 15.59	0.2772 ± 1.495	11.35 ± 24.4	0.2744 ± 2.146
pt_weighted_abs_dz *	mm	5.672 ± 23.37	0.4704 ± 1.869	13.11 ± 25.43	0.3176 ± 2.252

* Variable utilizada como entrada del modelo.

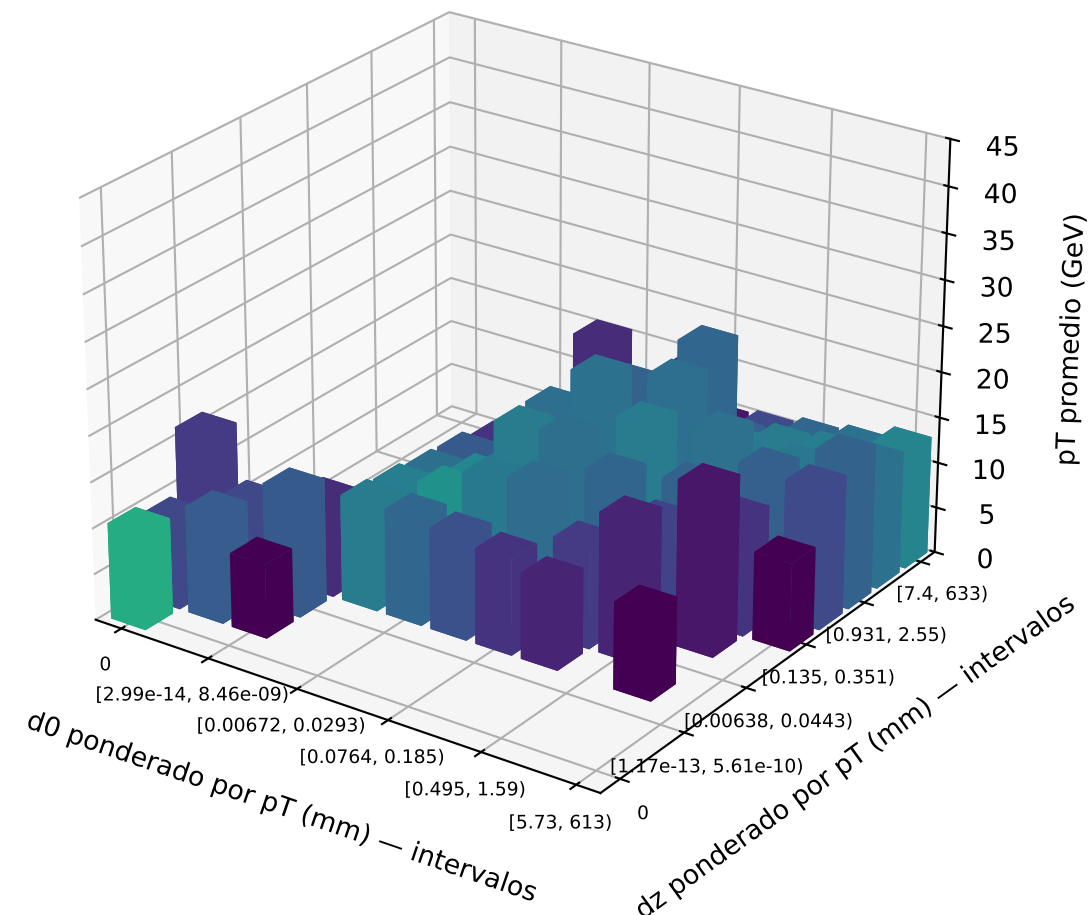
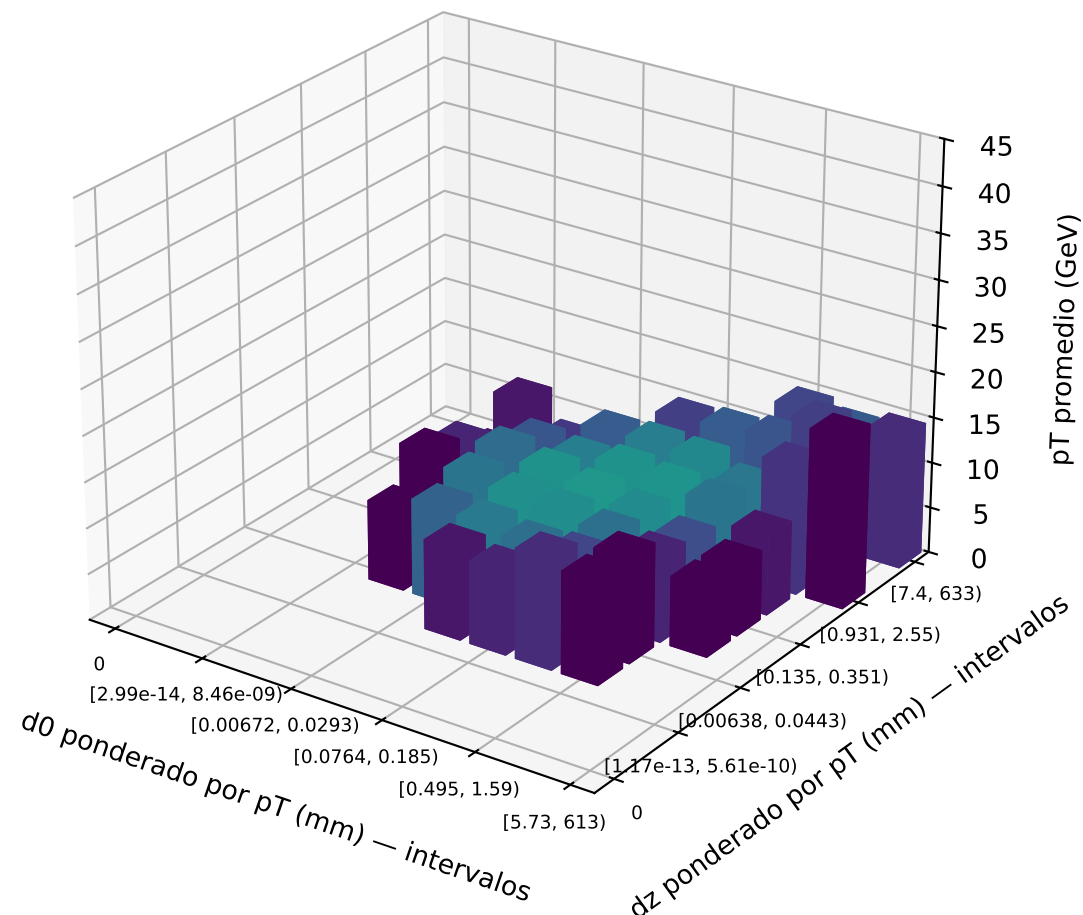
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF12 · seed 42

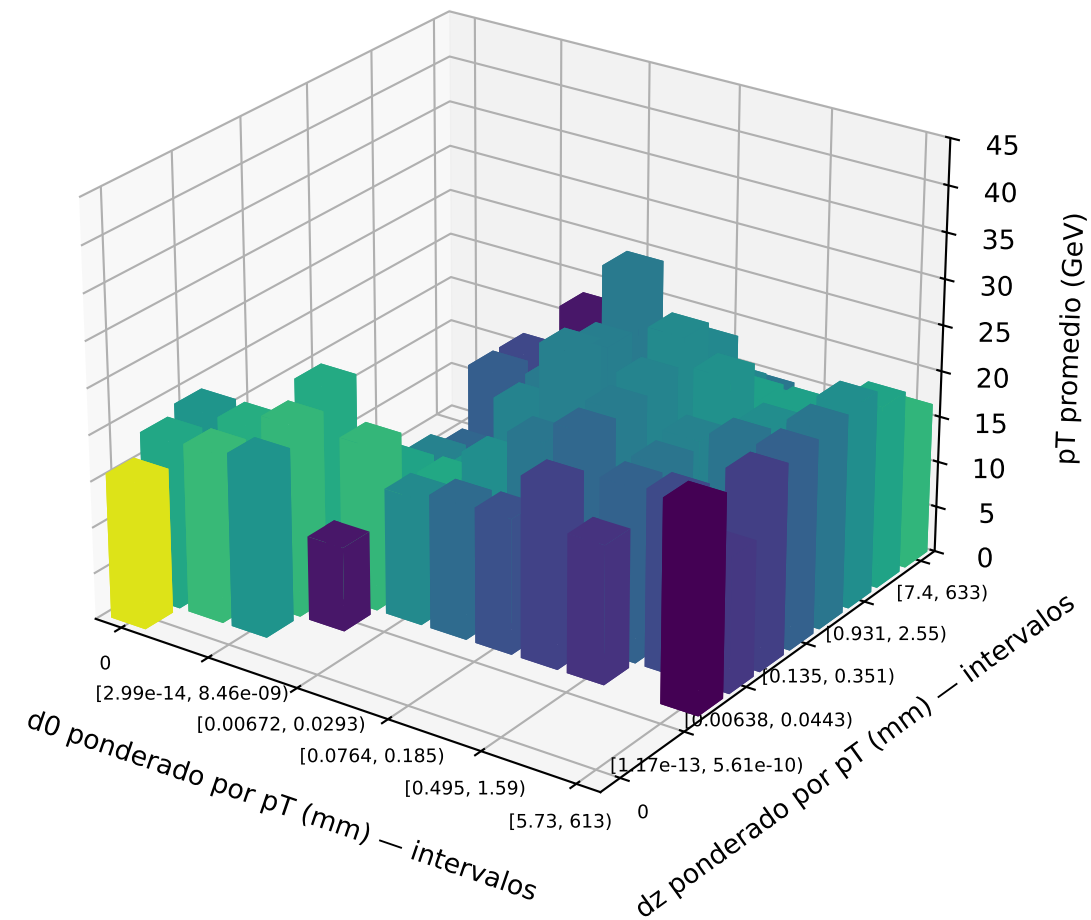
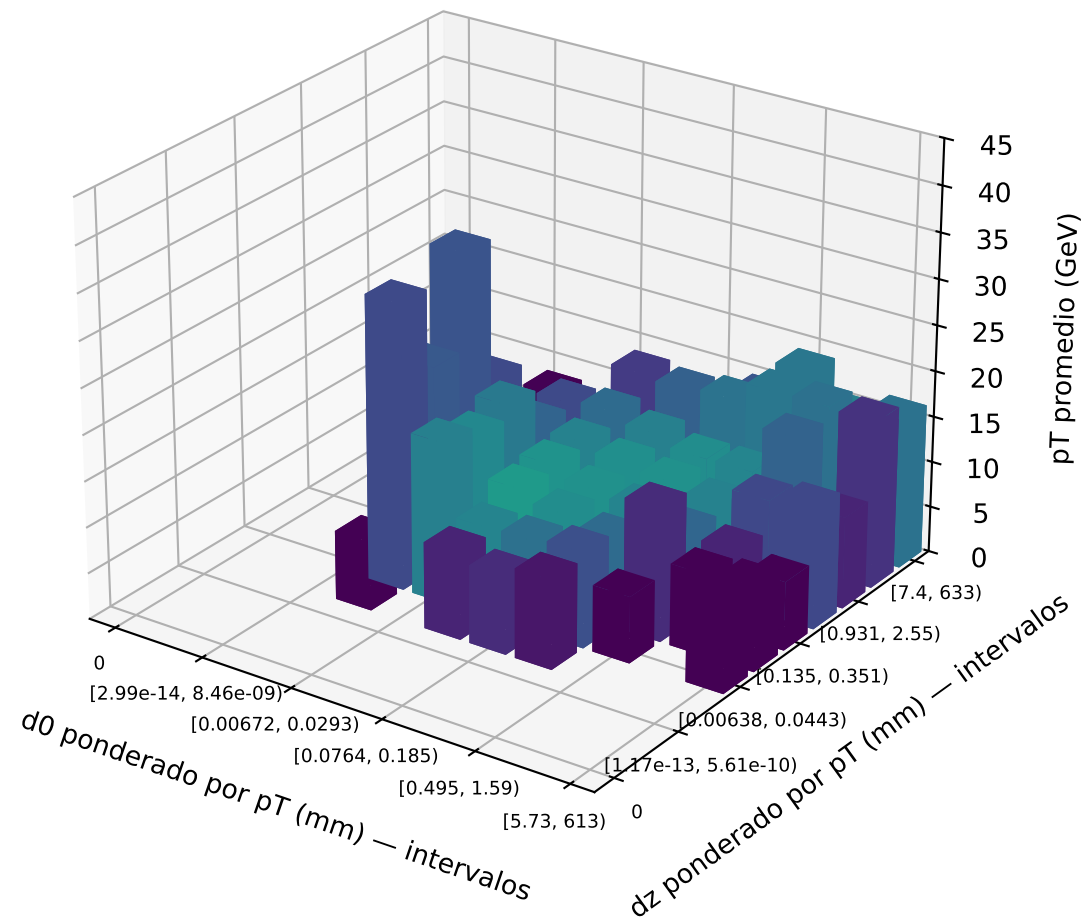
TP — N=3,035

FN — N=3,640



FP — N=4,123

TN — N=56,753



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF12 · seed 42

Variable	Unidad	TP N=3,035	FN N=3,640	FP N=4,123	TN N=56,753
jet_pt *	GeV	11.11 ± 4.023	13.22 ± 8.077	13.02 ± 7.766	16.66 ± 11.85
jet_eta	1	0.01076 ± 1.393	-0.004403 ± 1.432	-0.01076 ± 1.404	0.003671 ± 1.502
jet_abs_eta *	1	1.211 ± 0.6892	1.219 ± 0.7525	1.213 ± 0.7074	1.28 ± 0.7862
jet_mass	GeV	2.373 ± 1.163	2.661 ± 1.778	2.778 ± 1.495	2.857 ± 1.891
jet_ncharged	count	2.805 ± 1.471	2.65 ± 2	3.105 ± 1.654	2.55 ± 1.943
jet_nneutral	count	1.602 ± 1.416	2.234 ± 1.721	1.791 ± 1.542	2.411 ± 1.795
n_tracks_cone	count	2.849 ± 1.459	2.679 ± 1.999	3.157 ± 1.655	2.577 ± 1.942
track_pt_sum *	GeV	5.621 ± 3.272	5.252 ± 5.893	6.341 ± 5.221	6.784 ± 8.224
track_pt_sum_over_jet_pt	1	0.5059 ± 0.2183	0.3747 ± 0.2593	0.4843 ± 0.2214	0.3665 ± 0.2679
avg_track_pt *	GeV	2.162 ± 1.258	1.762 ± 1.438	2.024 ± 1.118	2.176 ± 2.106
std_track_pt	GeV	0.7965 ± 0.7867	0.6867 ± 0.9265	0.8812 ± 0.9902	1.013 ± 1.631
n_tracks_below_avg_pt	count	1.488 ± 1.116	1.397 ± 1.365	1.7 ± 1.235	1.355 ± 1.358
n_tracks_above_avg_pt	count	1.361 ± 0.6177	1.282 ± 0.8585	1.457 ± 0.7078	1.222 ± 0.8058
max_pt_ratio	1	0.2982 ± 0.157	0.2109 ± 0.1604	0.2618 ± 0.1286	0.2085 ± 0.1563
min_pt_ratio	1	0.1341 ± 0.1415	0.0943 ± 0.1206	0.1044 ± 0.09703	0.08522 ± 0.09911
max_dr_ratio	1	0.1601 ± 0.1415	0.1145 ± 0.1243	0.1304 ± 0.1043	0.1036 ± 0.1081
min_dr_ratio	1	0.257 ± 0.1764	0.1781 ± 0.1645	0.2196 ± 0.1418	0.1785 ± 0.1552
dr_max_pt	1	0.1391 ± 0.08452	0.136 ± 0.0983	0.1426 ± 0.08324	0.126 ± 0.09961
dr_min_pt	1	0.2047 ± 0.0993	0.1901 ± 0.1178	0.2184 ± 0.09657	0.1887 ± 0.1178
dr_max_dr	1	0.2433 ± 0.09416	0.227 ± 0.1213	0.2593 ± 0.08611	0.2189 ± 0.1215
dr_min_dr	1	0.1093 ± 0.07168	0.1072 ± 0.08518	0.1125 ± 0.07221	0.1037 ± 0.08898
pt_difference	GeV	1.853 ± 1.795	1.683 ± 2.465	2.151 ± 2.61	2.483 ± 4.113
r50	1	0.1459 ± 0.08211	0.1431 ± 0.09587	0.153 ± 0.07936	0.1324 ± 0.09801
r95	1	0.2409 ± 0.0938	0.2242 ± 0.1204	0.2544 ± 0.0861	0.2126 ± 0.1201
mean_abs_d0 *	mm	0.2798 ± 0.9106	2.075 ± 10.67	1.141 ± 5.633	1.203 ± 7.849
max_abs_d0 *	mm	0.6944 ± 3.196	5.564 ± 20.45	3.676 ± 18.78	3.134 ± 16.05
std_abs_d0 *	mm	0.2603 ± 1.326	2.135 ± 7.908	1.491 ± 8.042	1.2 ± 6.404
mean_abs_dz *	mm	0.9061 ± 3.778	2.81 ± 15.69	3.545 ± 13.97	1.463 ± 9.039
max_abs_dz *	mm	2.341 ± 12.22	7.008 ± 24.25	11.33 ± 43.57	3.67 ± 20.08
std_abs_dz *	mm	0.8844 ± 4.98	2.627 ± 8.467	4.673 ± 18.8	1.401 ± 8.144
pt_weighted_abs_d0 *	mm	0.2367 ± 0.5654	1.986 ± 10.58	0.6954 ± 3.101	1.175 ± 7.962
pt_weighted_abs_dz *	mm	0.7355 ± 2.408	2.51 ± 15.65	2.157 ± 7.303	1.26 ± 8.255

* Variable utilizada como entrada del modelo.

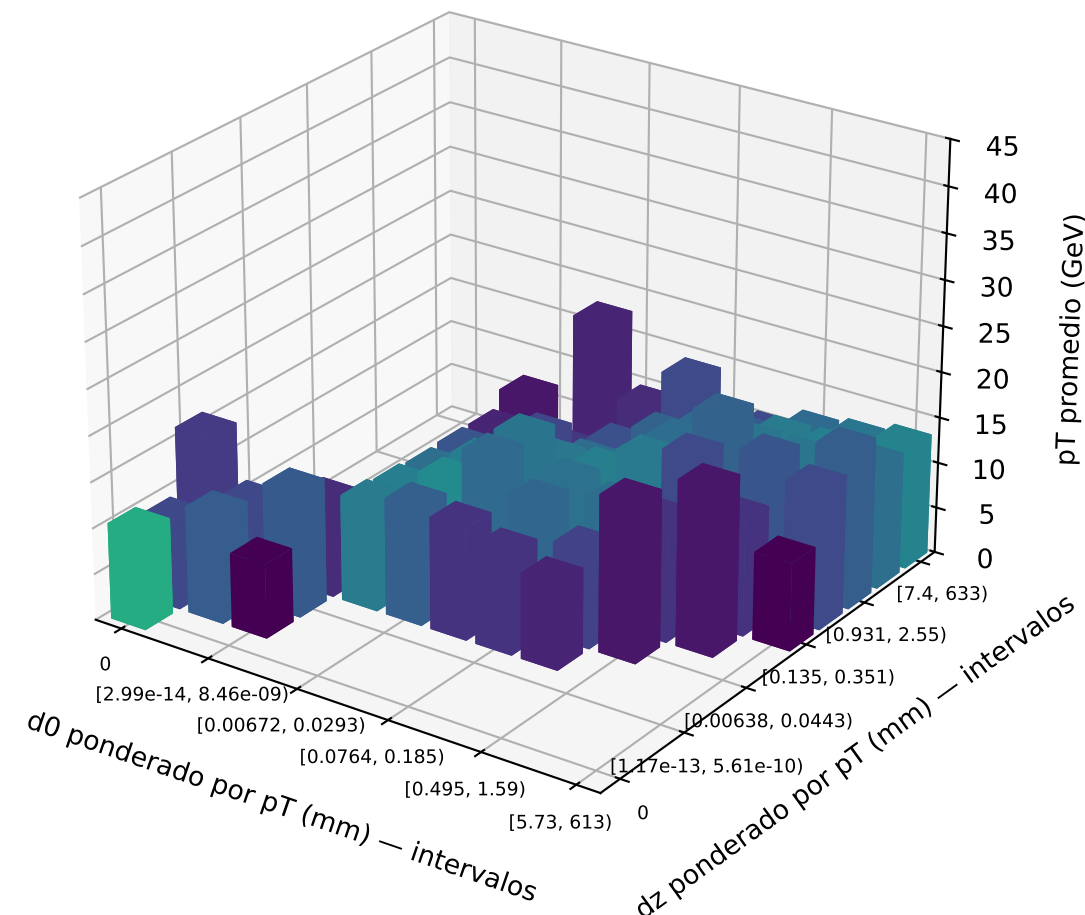
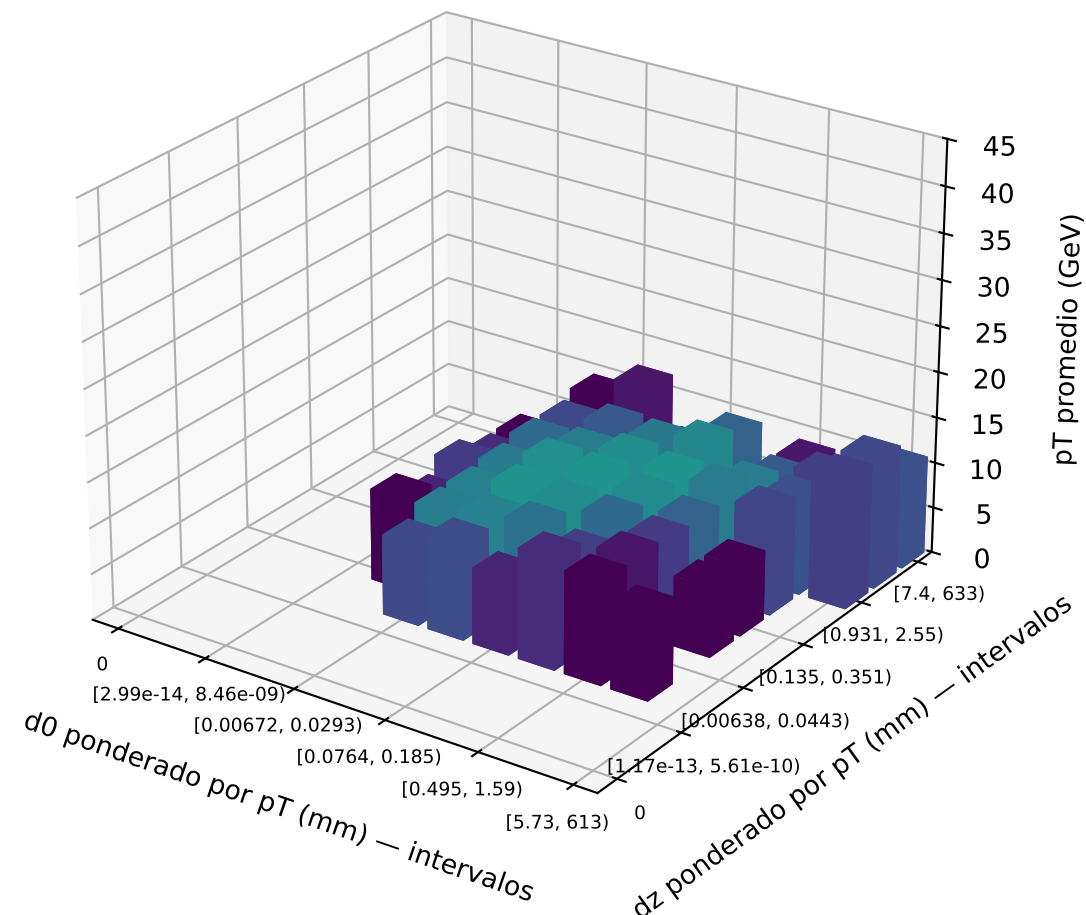
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF12 · seed 73

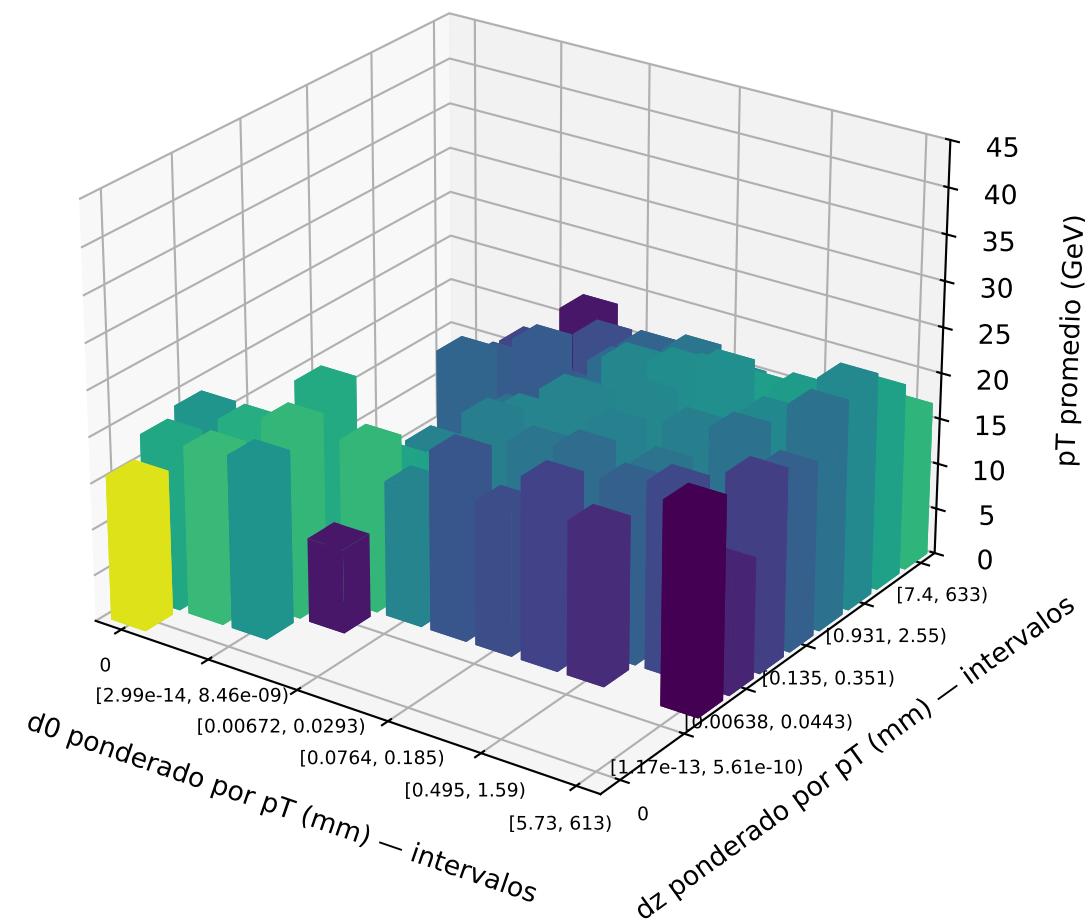
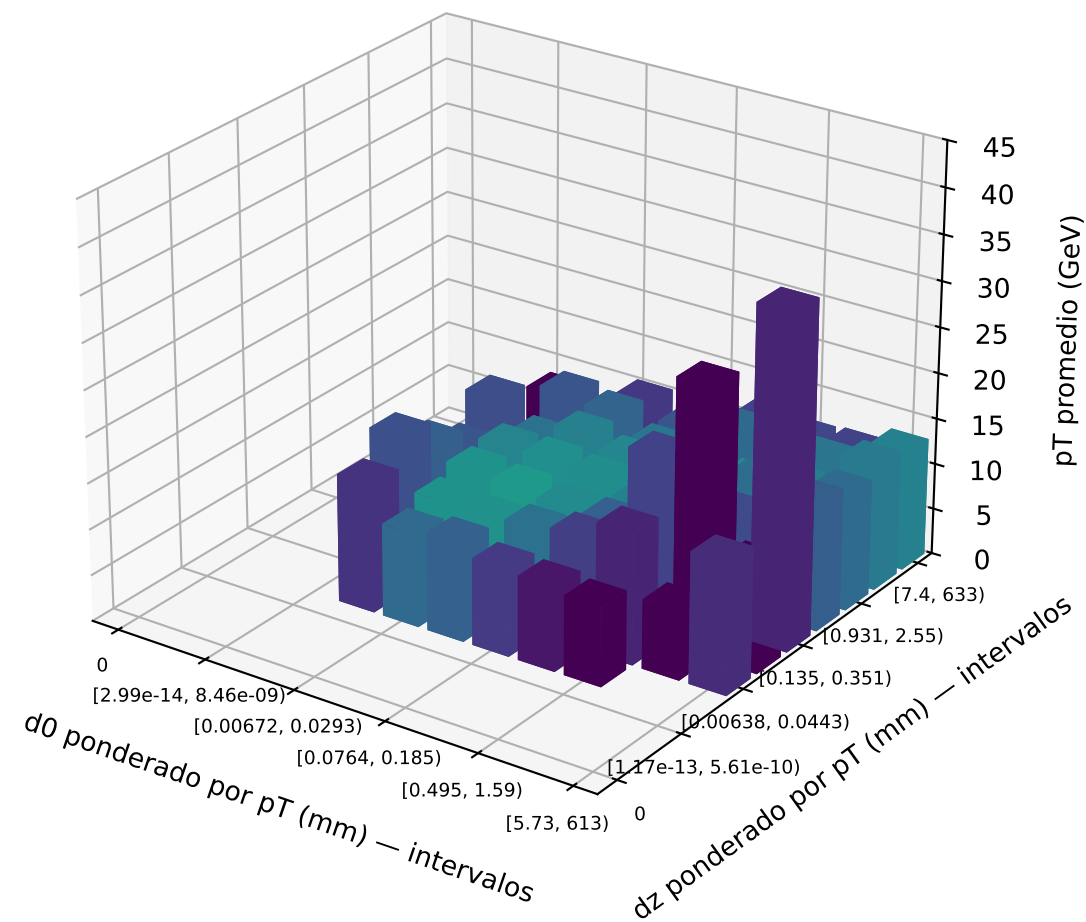
TP — N=3,269

FN — N=3,406



FP — N=4,472

TN — N=56,404



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF12 · seed 73

Variable	Unidad	TP N=3,269	FN N=3,406	FP N=4,472	TN N=56,404
jet_pt *	GeV	11.88 ± 5.937	12.63 ± 7.225	12.15 ± 6.694	16.75 ± 11.89
jet_eta	1	0.007487 ± 1.361	-0.002303 ± 1.464	-0.01248 ± 1.339	0.003896 ± 1.507
jet_abs_eta *	1	1.174 ± 0.6892	1.255 ± 0.7545	1.137 ± 0.7079	1.286 ± 0.7858
jet_mass	GeV	2.568 ± 1.356	2.494 ± 1.691	2.674 ± 1.38	2.866 ± 1.899
jet_ncharged	count	2.986 ± 1.535	2.466 ± 1.955	2.9 ± 1.499	2.563 ± 1.958
jet_nneutral	count	1.738 ± 1.594	2.147 ± 1.62	1.722 ± 1.424	2.421 ± 1.802
n_tracks_cone	count	3.028 ± 1.524	2.495 ± 1.953	2.945 ± 1.489	2.59 ± 1.958
track_pt_sum *	GeV	5.913 ± 4.277	4.946 ± 5.358	5.516 ± 4.333	6.852 ± 8.273
track_pt_sum_over_jet_pt	1	0.503 ± 0.2136	0.3684 ± 0.2646	0.4591 ± 0.2178	0.3678 ± 0.269
avg_track_pt *	GeV	2.119 ± 1.226	1.776 ± 1.483	1.924 ± 1.083	2.185 ± 2.112
std_track_pt	GeV	0.8132 ± 0.8023	0.6632 ± 0.9197	0.7272 ± 0.8559	1.026 ± 1.639
n_tracks_below_avg_pt	count	1.616 ± 1.139	1.268 ± 1.342	1.533 ± 1.103	1.366 ± 1.37
n_tracks_above_avg_pt	count	1.412 ± 0.6697	1.227 ± 0.8268	1.412 ± 0.6626	1.224 ± 0.8101
max_pt_ratio	1	0.2875 ± 0.1561	0.2152 ± 0.165	0.2508 ± 0.1269	0.2091 ± 0.1568
min_pt_ratio	1	0.1267 ± 0.141	0.09871 ± 0.1212	0.1121 ± 0.1022	0.08449 ± 0.09855
max_dr_ratio	1	0.1539 ± 0.1414	0.1174 ± 0.1245	0.1382 ± 0.1064	0.1028 ± 0.1078
min_dr_ratio	1	0.2445 ± 0.1756	0.1848 ± 0.1683	0.2112 ± 0.1397	0.1789 ± 0.1555
dr_max_pt	1	0.1404 ± 0.0842	0.1345 ± 0.09937	0.1514 ± 0.08548	0.1252 ± 0.09939
dr_min_pt	1	0.2072 ± 0.101	0.1867 ± 0.1172	0.2146 ± 0.09586	0.1888 ± 0.118
dr_max_dr	1	0.2494 ± 0.09482	0.22 ± 0.1213	0.254 ± 0.08711	0.219 ± 0.1217
dr_min_dr	1	0.1074 ± 0.06908	0.1089 ± 0.08805	0.119 ± 0.07431	0.1032 ± 0.08886
pt_difference	GeV	1.922 ± 1.987	1.605 ± 2.354	1.727 ± 2.126	2.519 ± 4.138
r50	1	0.148 ± 0.08147	0.1409 ± 0.09716	0.1608 ± 0.08136	0.1316 ± 0.09781
r95	1	0.2468 ± 0.09436	0.2174 ± 0.1204	0.2509 ± 0.08761	0.2127 ± 0.1202
mean_abs_d0 *	mm	0.6505 ± 6.687	1.842 ± 8.971	3.085 ± 14.61	1.049 ± 6.862
max_abs_d0 *	mm	1.53 ± 12.63	5.097 ± 17.55	7.646 ± 32.28	2.816 ± 14.16
std_abs_d0 *	mm	0.5355 ± 4.653	2 ± 6.949	2.866 ± 13.32	1.089 ± 5.63
mean_abs_dz *	mm	0.9004 ± 10.98	2.946 ± 12.64	2.364 ± 16	1.544 ± 8.742
max_abs_dz *	mm	1.68 ± 13.48	7.964 ± 24.04	5.898 ± 35.67	4.054 ± 21.15
std_abs_dz *	mm	0.5201 ± 3.474	3.096 ± 9.239	2.247 ± 14.87	1.573 ± 8.702
pt_weighted_abs_d0 *	mm	0.6409 ± 6.78	1.719 ± 8.763	2.795 ± 14.22	1.011 ± 6.945
pt_weighted_abs_dz *	mm	0.9091 ± 11.07	2.465 ± 12.23	1.885 ± 12.76	1.276 ± 7.719

* Variable utilizada como entrada del modelo.

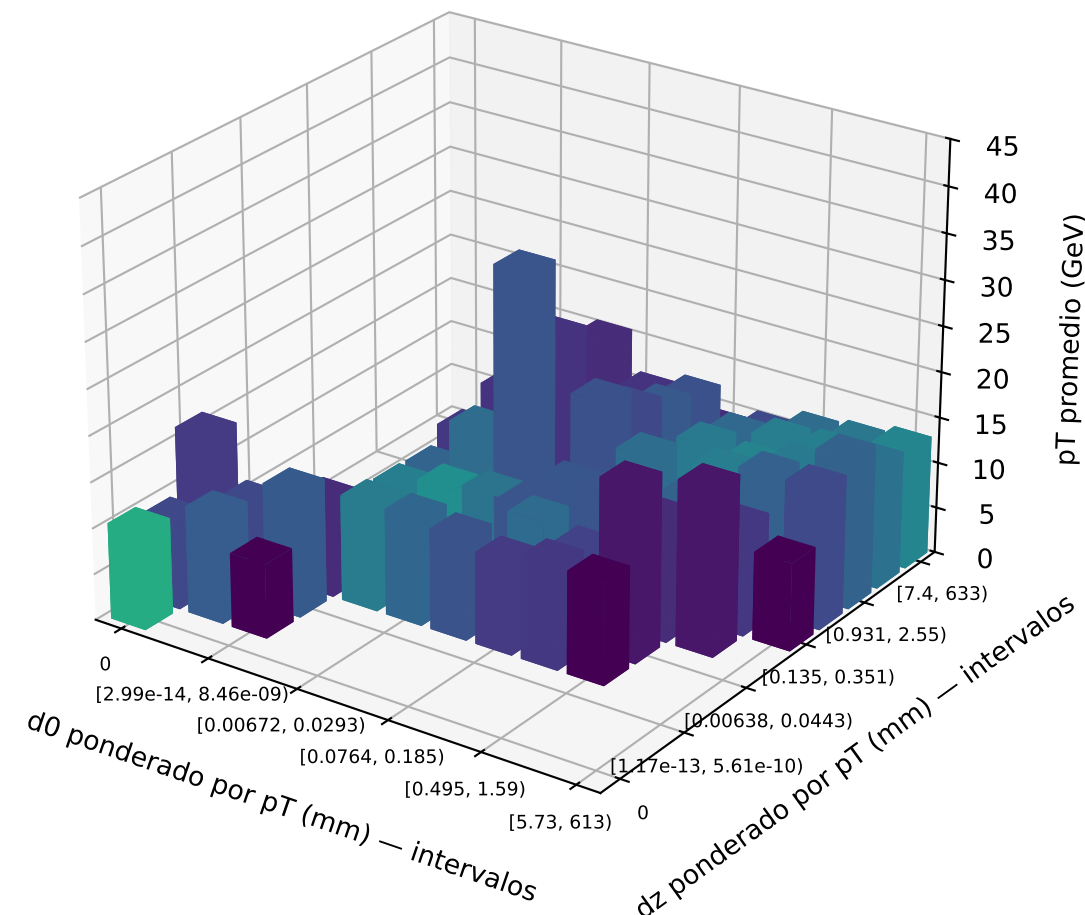
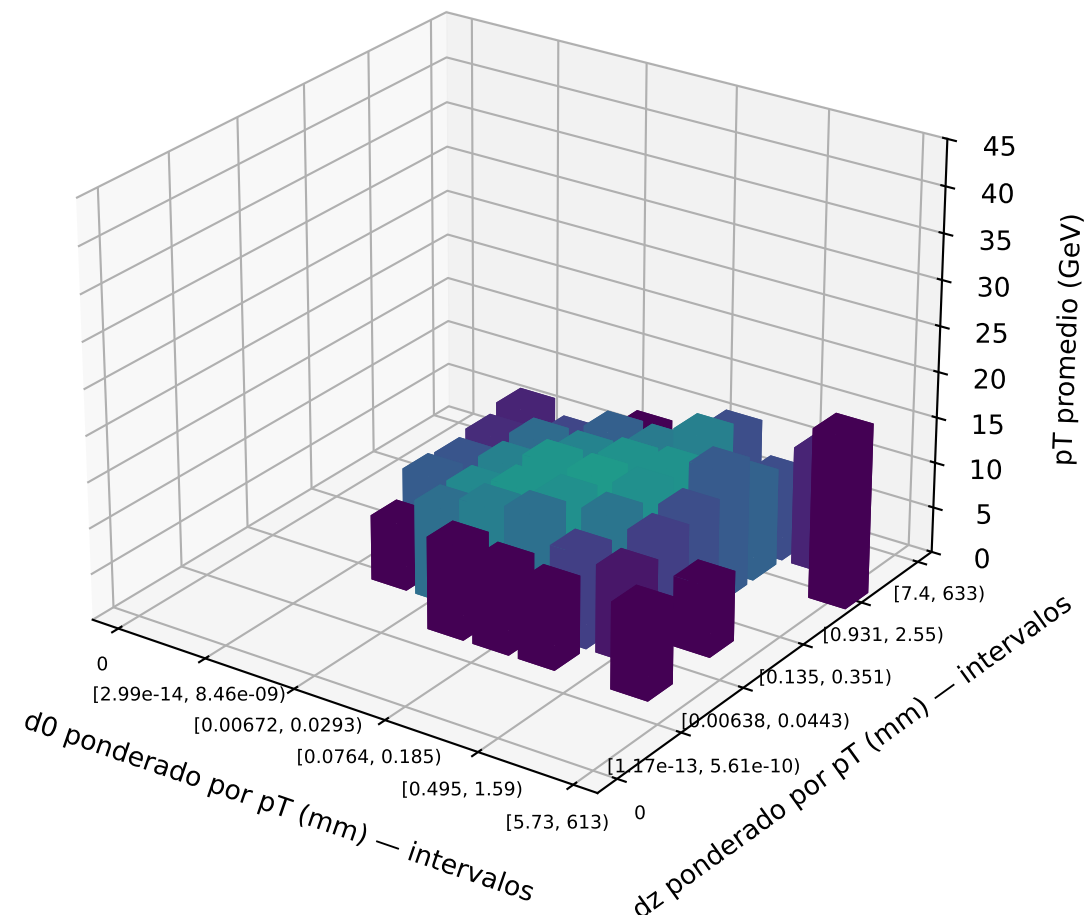
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF12 · seed 101

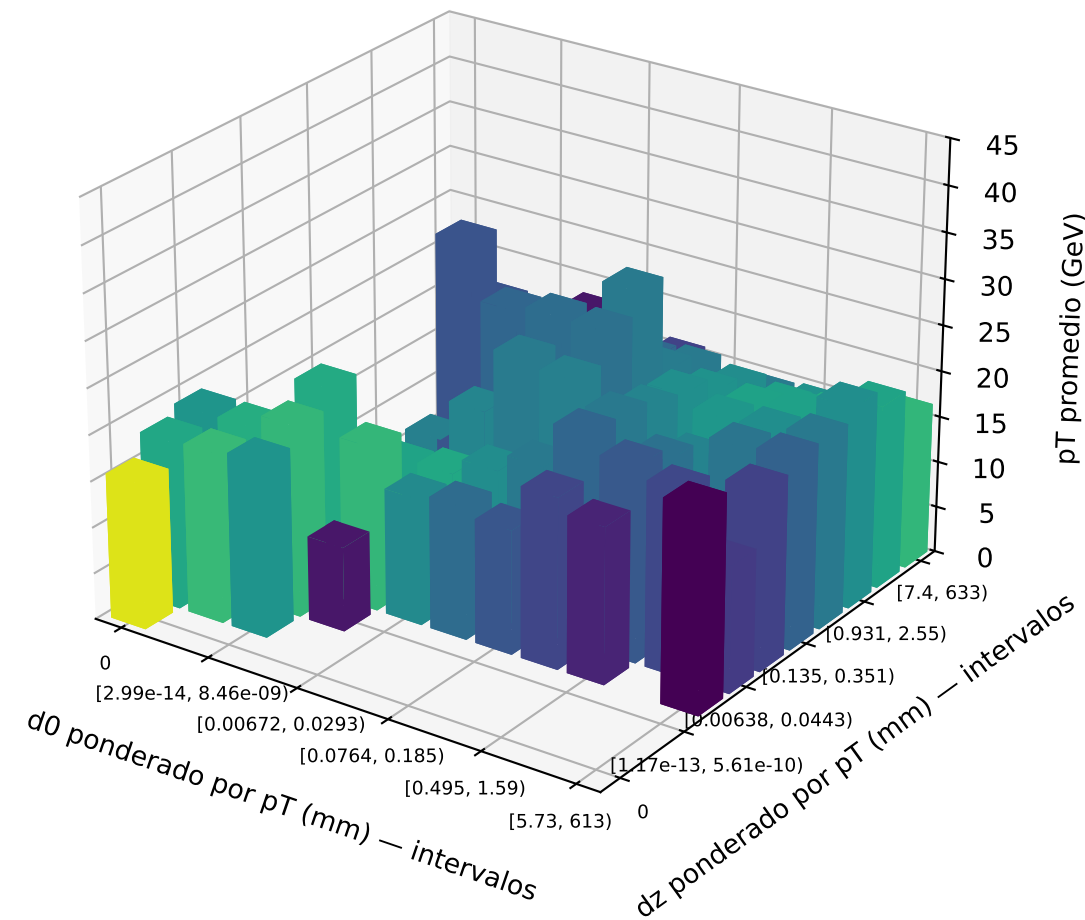
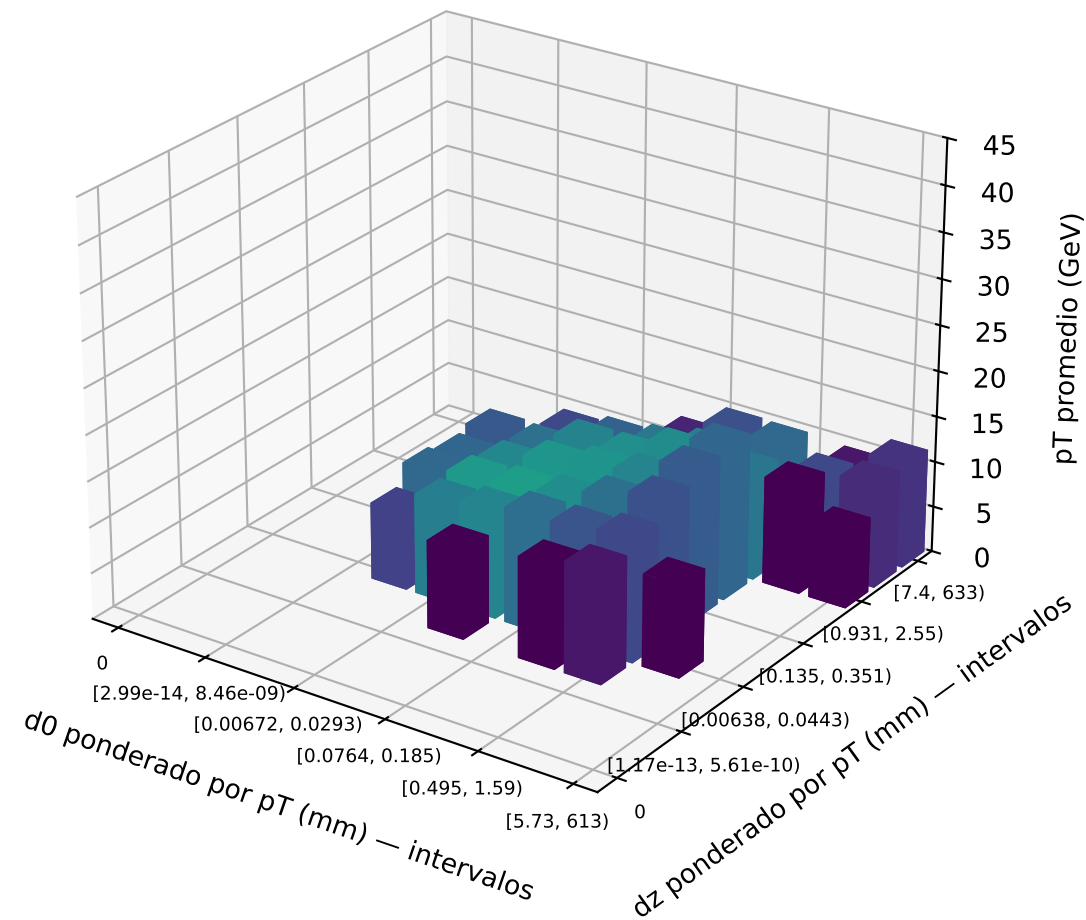
TP — N=3,399

FN — N=3,276



FP — N=3,725

TN — N=57,151



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF12 · seed 101

Variable	Unidad	TP N=3,399	FN N=3,276	FP N=3,725	TN N=57,151
jet_pt *	GeV	12.07 ± 5.784	12.46 ± 7.413	12.29 ± 5.437	16.68 ± 11.9
jet_eta	1	-0.006724 ± 1.406	0.01205 ± 1.424	0.006668 ± 1.477	0.002434 ± 1.497
jet_abs_eta *	1	1.216 ± 0.7048	1.214 ± 0.7442	1.292 ± 0.7176	1.274 ± 0.7853
jet_mass	GeV	2.553 ± 1.332	2.506 ± 1.723	2.606 ± 1.283	2.868 ± 1.898
jet_ncharged	count	2.899 ± 1.542	2.535 ± 1.982	2.86 ± 1.476	2.57 ± 1.954
jet_nneutral	count	1.818 ± 1.563	2.081 ± 1.666	1.836 ± 1.473	2.404 ± 1.799
n_tracks_cone	count	2.939 ± 1.531	2.566 ± 1.981	2.907 ± 1.475	2.597 ± 1.954
track_pt_sum *	GeV	5.868 ± 4.204	4.955 ± 5.461	5.743 ± 3.925	6.82 ± 8.25
track_pt_sum_over_jet_pt	1	0.4872 ± 0.2208	0.3795 ± 0.2666	0.4659 ± 0.2238	0.3686 ± 0.2681
avg_track_pt *	GeV	2.182 ± 1.381	1.698 ± 1.322	2.054 ± 1.202	2.173 ± 2.098
std_track_pt	GeV	0.8117 ± 0.8495	0.6588 ± 0.879	0.7752 ± 0.79	1.019 ± 1.634
n_tracks_below_avg_pt	count	1.537 ± 1.147	1.336 ± 1.358	1.51 ± 1.093	1.369 ± 1.367
n_tracks_above_avg_pt	count	1.402 ± 0.6664	1.23 ± 0.8364	1.396 ± 0.6436	1.228 ± 0.8099
max_pt_ratio	1	0.2825 ± 0.1591	0.2175 ± 0.1639	0.2582 ± 0.133	0.2091 ± 0.1561
min_pt_ratio	1	0.1267 ± 0.1398	0.09752 ± 0.1217	0.1133 ± 0.1026	0.08478 ± 0.0986
max_dr_ratio	1	0.1525 ± 0.1403	0.1174 ± 0.1253	0.1392 ± 0.1077	0.1032 ± 0.1077
min_dr_ratio	1	0.2419 ± 0.1764	0.1851 ± 0.1676	0.2195 ± 0.1451	0.1787 ± 0.1549
dr_max_pt	1	0.1412 ± 0.08429	0.1334 ± 0.09978	0.1465 ± 0.08406	0.1259 ± 0.09942
dr_min_pt	1	0.2068 ± 0.1005	0.1862 ± 0.1182	0.2135 ± 0.09592	0.1893 ± 0.1178
dr_max_dr	1	0.2468 ± 0.09465	0.2216 ± 0.1228	0.2508 ± 0.08775	0.2197 ± 0.1214
dr_min_dr	1	0.1098 ± 0.07166	0.1065 ± 0.08655	0.1167 ± 0.07382	0.1035 ± 0.08876
pt_difference	GeV	1.913 ± 2.069	1.602 ± 2.293	1.831 ± 1.88	2.501 ± 4.128
r50	1	0.1484 ± 0.08147	0.1403 ± 0.09768	0.1558 ± 0.08019	0.1323 ± 0.09782
r95	1	0.2441 ± 0.09421	0.219 ± 0.1219	0.2479 ± 0.08784	0.2134 ± 0.12
mean_abs_d0 *	mm	0.1423 ± 0.2624	2.417 ± 11.24	0.2964 ± 3.364	1.257 ± 7.917
max_abs_d0 *	mm	0.2914 ± 1.039	6.523 ± 21.57	0.6095 ± 4.482	3.338 ± 16.71
std_abs_d0 *	mm	0.1024 ± 0.3723	2.507 ± 8.355	0.1959 ± 1.198	1.286 ± 6.726
mean_abs_dz *	mm	0.3758 ± 0.5866	3.572 ± 16.82	0.4548 ± 0.9908	1.679 ± 9.765
max_abs_dz *	mm	0.7546 ± 2.318	9.173 ± 27.58	1.12 ± 2.751	4.389 ± 23.24
std_abs_dz *	mm	0.263 ± 0.8656	3.465 ± 9.908	0.4433 ± 1.122	1.699 ± 9.586
pt_weighted_abs_d0 *	mm	0.1434 ± 0.2066	2.278 ± 11.13	0.2754 ± 3.291	1.199 ± 7.931
pt_weighted_abs_dz *	mm	0.3803 ± 0.4808	3.075 ± 16.59	0.4286 ± 0.8669	1.379 ± 8.454

* Variable utilizada como entrada del modelo.

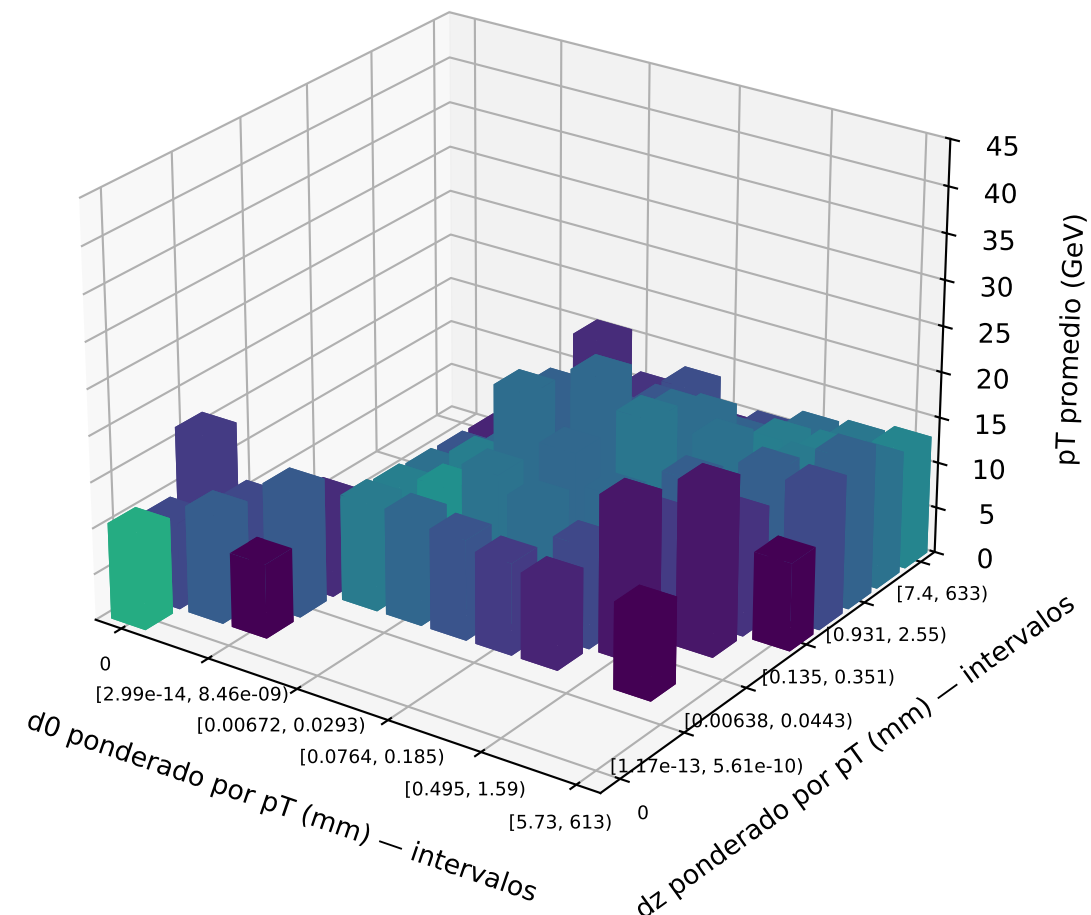
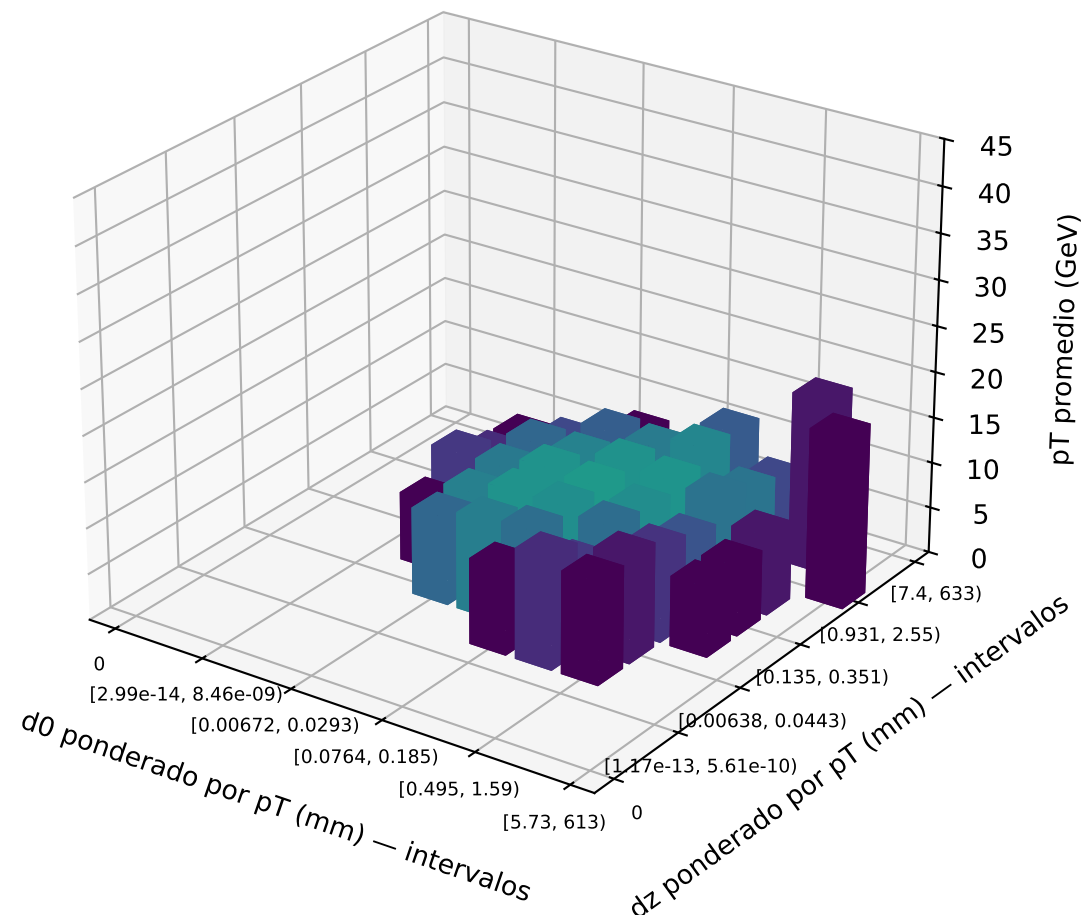
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF12 · ensemble

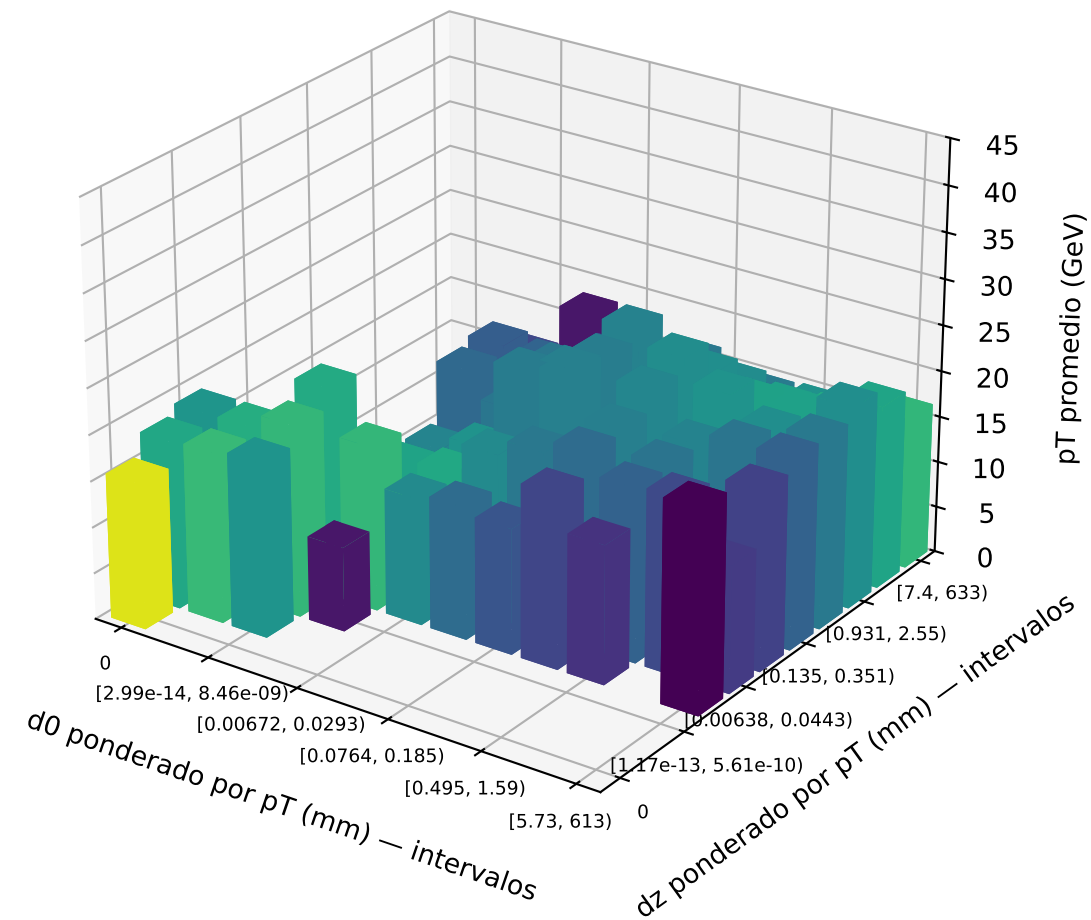
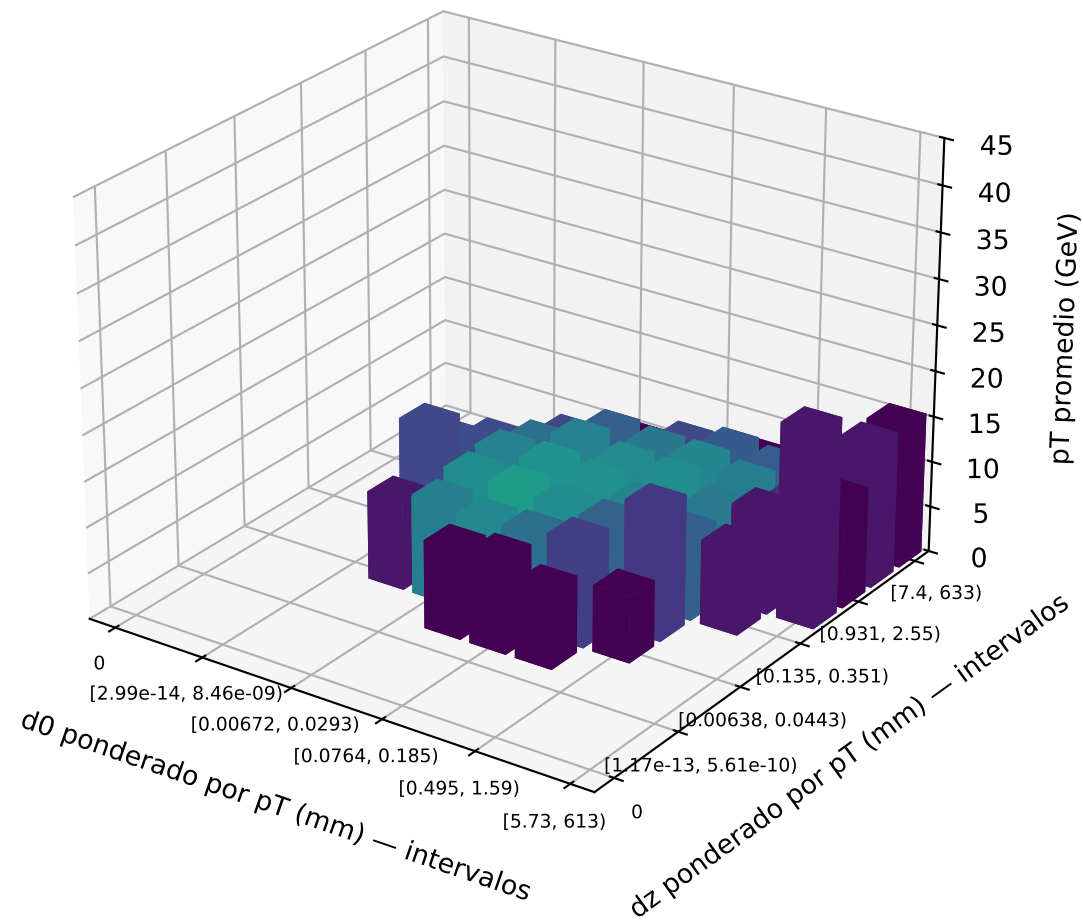
TP — N=3,239

FN — N=3,436



FP — N=3,444

TN — N=57,432



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF12 · ensemble

Variable	Unidad	TP N=3,239	FN N=3,436	FP N=3,444	TN N=57,432
jet_pt *	GeV	11.42 ± 5.022	13.06 ± 7.777	11.44 ± 4.355	16.71 ± 11.88
jet_eta	1	-0.00282 ± 1.392	0.007498 ± 1.436	0.000379 ± 1.402	0.002832 ± 1.501
jet_abs_eta *	1	1.208 ± 0.6918	1.222 ± 0.7538	1.209 ± 0.71	1.279 ± 0.7852
jet_mass	GeV	2.453 ± 1.256	2.603 ± 1.757	2.516 ± 1.184	2.872 ± 1.898
jet_ncharged	count	2.868 ± 1.49	2.582 ± 2.007	2.825 ± 1.389	2.573 ± 1.956
jet_nneutral	count	1.705 ± 1.511	2.175 ± 1.686	1.653 ± 1.347	2.412 ± 1.8
n_tracks_cone	count	2.907 ± 1.481	2.614 ± 2.005	2.875 ± 1.374	2.601 ± 1.957
track_pt_sum *	GeV	5.69 ± 3.889	5.165 ± 5.651	5.38 ± 3.118	6.836 ± 8.252
track_pt_sum_over_jet_pt	1	0.4987 ± 0.2151	0.3736 ± 0.2653	0.4746 ± 0.2167	0.3685 ± 0.2681
avg_track_pt *	GeV	2.141 ± 1.257	1.758 ± 1.451	1.966 ± 1.061	2.178 ± 2.099
std_track_pt	GeV	0.7947 ± 0.8047	0.6819 ± 0.9194	0.7133 ± 0.6595	1.022 ± 1.634
n_tracks_below_avg_pt	count	1.522 ± 1.112	1.359 ± 1.378	1.496 ± 1.033	1.371 ± 1.369
n_tracks_above_avg_pt	count	1.385 ± 0.6473	1.254 ± 0.847	1.378 ± 0.6247	1.23 ± 0.8103
max_pt_ratio	1	0.2913 ± 0.1575	0.2123 ± 0.1621	0.2618 ± 0.1283	0.2092 ± 0.1562
min_pt_ratio	1	0.1314 ± 0.1419	0.09449 ± 0.1192	0.1175 ± 0.1035	0.08467 ± 0.09851
max_dr_ratio	1	0.1577 ± 0.142	0.1141 ± 0.1228	0.1437 ± 0.1082	0.1031 ± 0.1076
min_dr_ratio	1	0.2501 ± 0.1762	0.18 ± 0.1658	0.2226 ± 0.1424	0.1788 ± 0.155
dr_max_pt	1	0.1399 ± 0.08411	0.135 ± 0.09934	0.1474 ± 0.08395	0.1259 ± 0.09936
dr_min_pt	1	0.2053 ± 0.09973	0.1886 ± 0.1183	0.2135 ± 0.09591	0.1894 ± 0.1177
dr_max_dr	1	0.2446 ± 0.09464	0.2248 ± 0.1221	0.2518 ± 0.08648	0.2198 ± 0.1214
dr_min_dr	1	0.1088 ± 0.07007	0.1077 ± 0.08716	0.1168 ± 0.07299	0.1036 ± 0.08874
pt_difference	GeV	1.86 ± 1.951	1.666 ± 2.385	1.668 ± 1.536	2.508 ± 4.127
r50	1	0.147 ± 0.0815	0.1419 ± 0.09707	0.1569 ± 0.07973	0.1324 ± 0.09776
r95	1	0.2421 ± 0.09421	0.2221 ± 0.1213	0.2494 ± 0.08685	0.2134 ± 0.1199
mean_abs_d0 *	mm	0.1766 ± 0.3386	2.278 ± 10.99	0.302 ± 2.346	1.252 ± 7.923
max_abs_d0 *	mm	0.3626 ± 1.455	6.166 ± 21.1	0.8547 ± 8.031	3.31 ± 16.6
std_abs_d0 *	mm	0.1264 ± 0.5382	2.372 ± 8.172	0.3513 ± 3.518	1.272 ± 6.662
mean_abs_dz *	mm	0.4398 ± 0.6466	3.362 ± 16.45	0.5498 ± 1.508	1.668 ± 9.738
max_abs_dz *	mm	0.8733 ± 2.509	8.67 ± 27.01	1.421 ± 5.39	4.355 ± 23.16
std_abs_dz *	mm	0.3028 ± 0.9361	3.278 ± 9.705	0.5798 ± 2.285	1.685 ± 9.552
pt_weighted_abs_d0 *	mm	0.1739 ± 0.2396	2.149 ± 10.88	0.2083 ± 0.8147	1.198 ± 7.953
pt_weighted_abs_dz *	mm	0.4434 ± 0.5356	2.891 ± 16.21	0.4727 ± 0.7435	1.371 ± 8.435

* Variable utilizada como entrada del modelo.

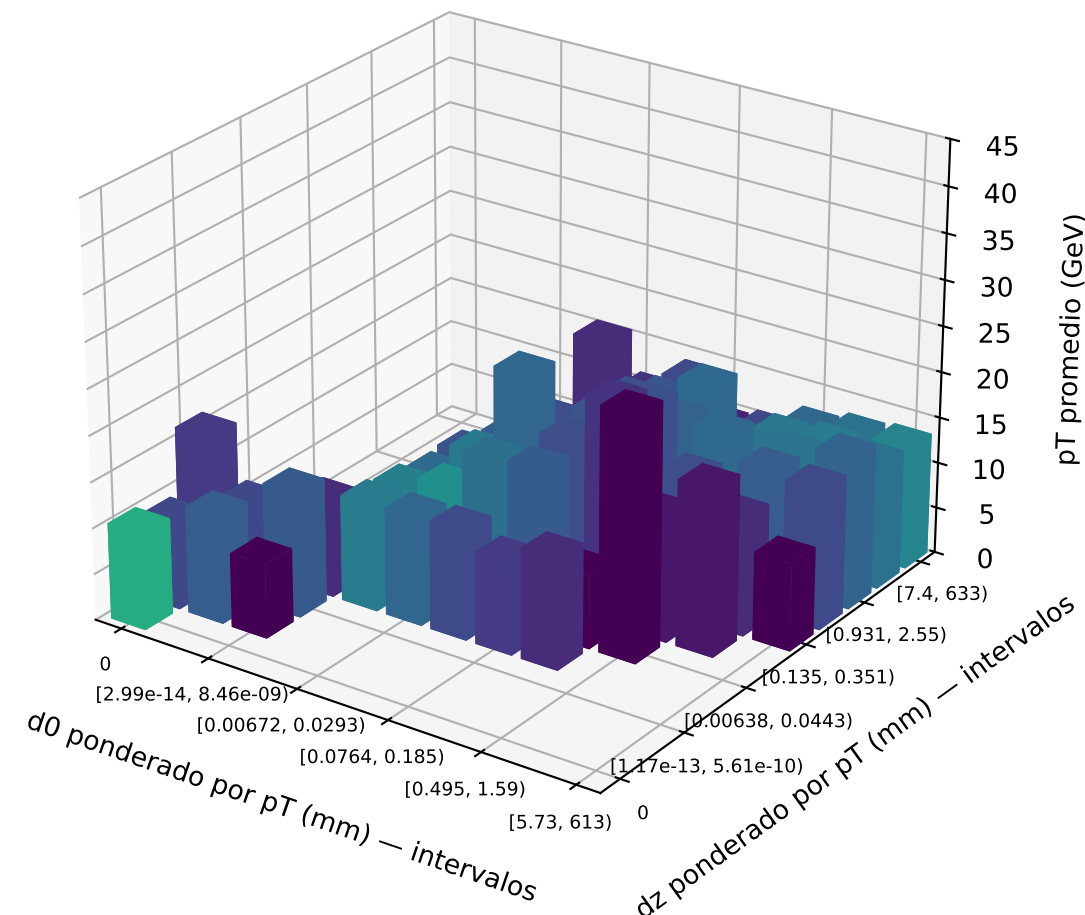
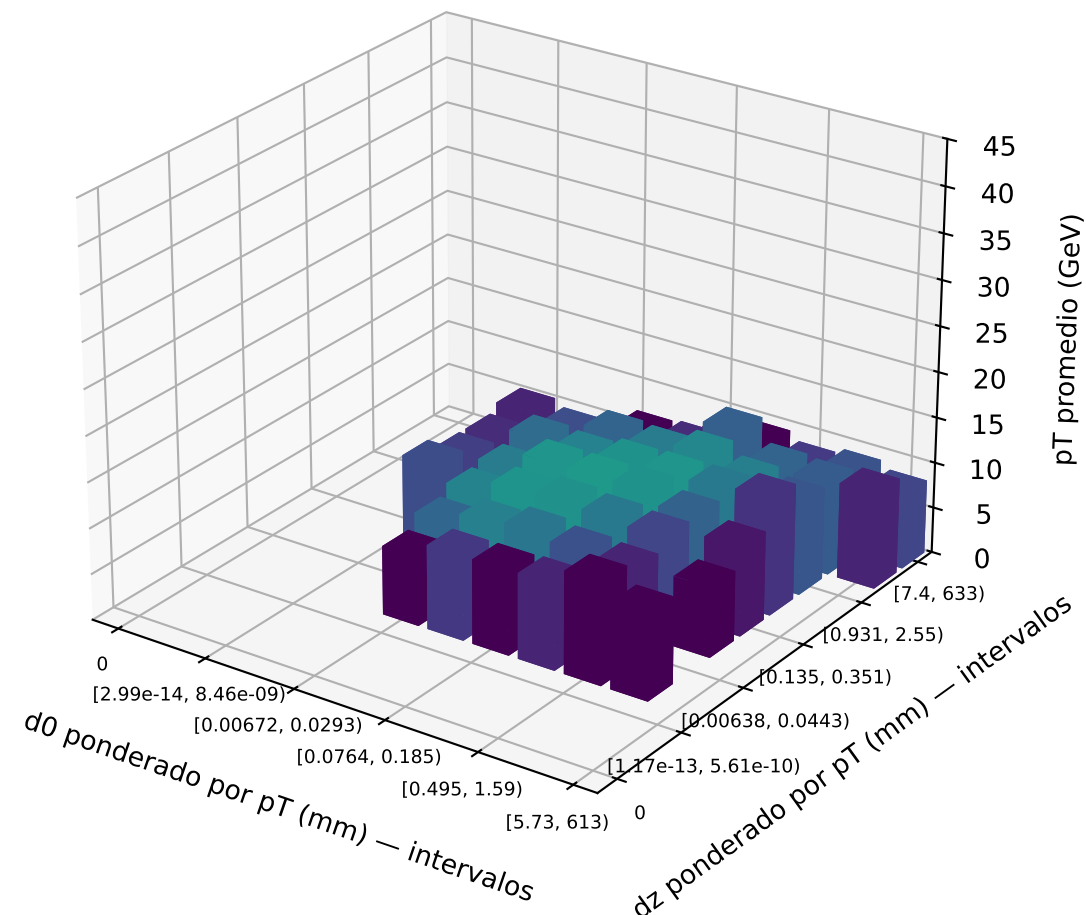
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF36 · seed 42

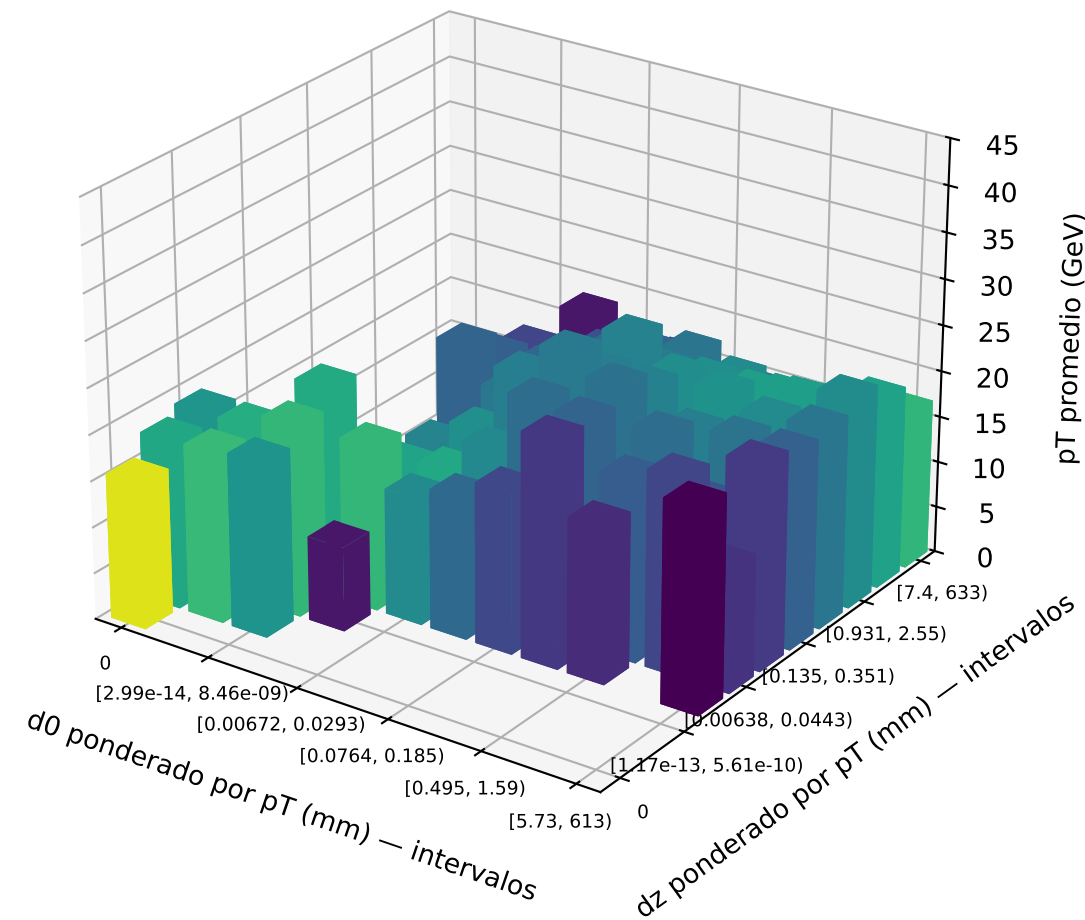
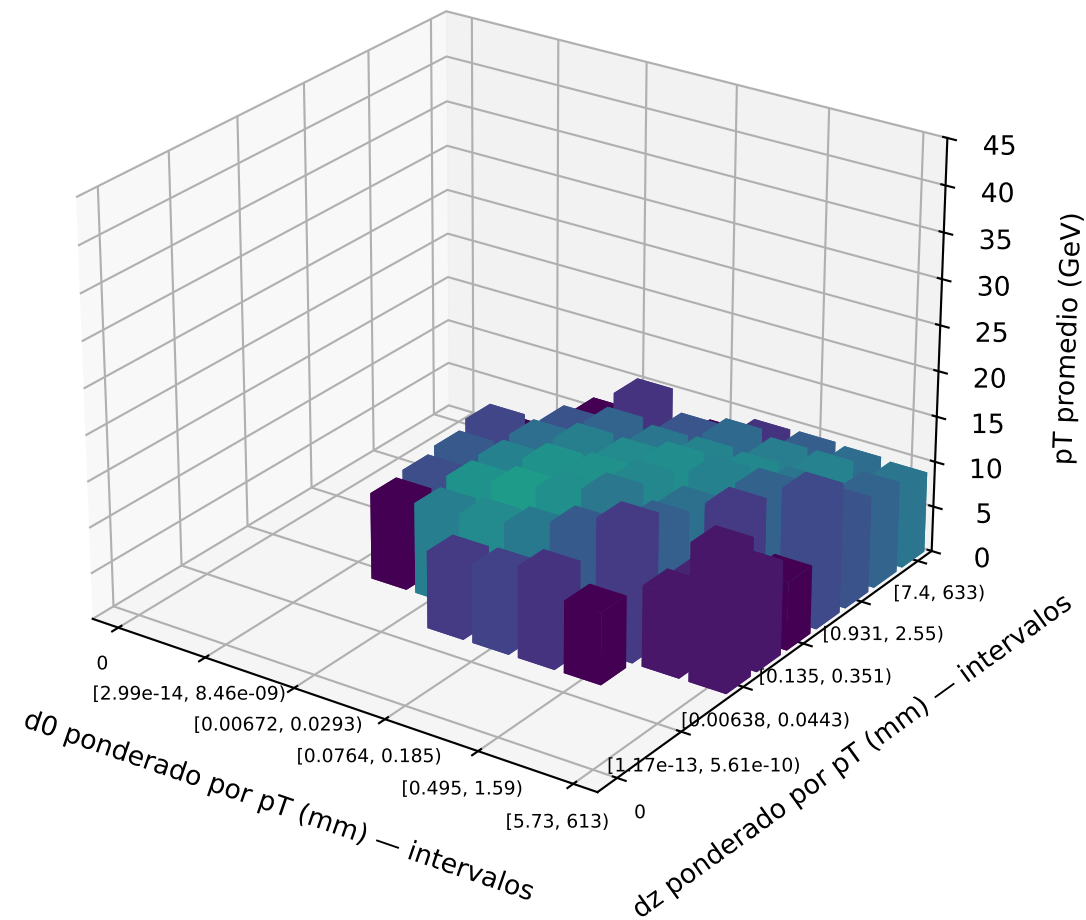
TP — N=3,779

FN — N=2,896



FP — N=4,335

TN — N=56,541



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF36 · seed 42

Variable	Unidad	TP N=3,779	FN N=2,896	FP N=4,335	TN N=56,541
jet_pt *	GeV	11.76 ± 4.92	12.91 ± 8.317	11.62 ± 5.054	16.78 ± 11.93
jet_eta	1	0.004102 ± 1.38	0.00039 ± 1.459	0.005368 ± 1.371	0.002488 ± 1.505
jet_abs_eta *	1	1.201 ± 0.679	1.234 ± 0.7793	1.202 ± 0.6594	1.281 ± 0.7896
jet_mass	GeV	2.478 ± 1.276	2.597 ± 1.818	2.508 ± 1.221	2.878 ± 1.905
jet_ncharged	count	2.816 ± 1.506	2.596 ± 2.08	2.723 ± 1.453	2.577 ± 1.961
jet_nneutral	count	1.769 ± 1.543	2.18 ± 1.687	1.692 ± 1.389	2.421 ± 1.802
n_tracks_cone	count	2.855 ± 1.499	2.627 ± 2.076	2.767 ± 1.452	2.605 ± 1.961
track_pt_sum *	GeV	5.59 ± 3.355	5.198 ± 6.338	5.027 ± 3.203	6.886 ± 8.298
track_pt_sum_over_jet_pt	1	0.4837 ± 0.2228	0.3699 ± 0.2687	0.4474 ± 0.2308	0.3689 ± 0.2683
avg_track_pt *	GeV	2.169 ± 1.356	1.651 ± 1.341	1.927 ± 1.055	2.184 ± 2.111
std_track_pt	GeV	0.7593 ± 0.7831	0.7071 ± 0.9658	0.6346 ± 0.6765	1.032 ± 1.642
n_tracks_below_avg_pt	count	1.476 ± 1.127	1.39 ± 1.41	1.39 ± 1.079	1.377 ± 1.371
n_tracks_above_avg_pt	count	1.379 ± 0.6445	1.237 ± 0.8811	1.376 ± 0.6342	1.228 ± 0.8122
max_pt_ratio	1	0.2845 ± 0.16	0.2064 ± 0.1603	0.2509 ± 0.1333	0.2092 ± 0.1563
min_pt_ratio	1	0.1338 ± 0.1458	0.08449 ± 0.1051	0.121 ± 0.1064	0.08388 ± 0.098
max_dr_ratio	1	0.1588 ± 0.1452	0.1046 ± 0.1113	0.1455 ± 0.1137	0.1024 ± 0.107
min_dr_ratio	1	0.2457 ± 0.1773	0.1727 ± 0.1617	0.2141 ± 0.1409	0.1787 ± 0.1554
dr_max_pt	1	0.1424 ± 0.08556	0.1309 ± 0.1	0.1534 ± 0.08618	0.1251 ± 0.09928
dr_min_pt	1	0.2045 ± 0.1006	0.1866 ± 0.1204	0.2113 ± 0.0961	0.1892 ± 0.118
dr_max_dr	1	0.2435 ± 0.09619	0.2225 ± 0.125	0.2474 ± 0.08969	0.2196 ± 0.1217
dr_min_dr	1	0.1117 ± 0.073	0.1036 ± 0.08669	0.1233 ± 0.07674	0.1029 ± 0.08861
pt_difference	GeV	1.775 ± 1.791	1.741 ± 2.616	1.488 ± 1.58	2.535 ± 4.15
r50	1	0.1494 ± 0.08255	0.1379 ± 0.09826	0.1614 ± 0.08155	0.1316 ± 0.09776
r95	1	0.2411 ± 0.09583	0.2197 ± 0.1239	0.2452 ± 0.08956	0.2132 ± 0.1202
mean_abs_d0 *	mm	0.2912 ± 1.053	2.521 ± 11.9	0.9216 ± 3.842	1.22 ± 7.938
max_abs_d0 *	mm	0.6256 ± 3.049	6.905 ± 22.7	2.053 ± 10.12	3.256 ± 16.62
std_abs_d0 *	mm	0.2279 ± 1.338	2.659 ± 8.762	0.7297 ± 4.131	1.257 ± 6.675
mean_abs_dz *	mm	0.6177 ± 1.668	3.675 ± 17.82	1.312 ± 3.953	1.627 ± 9.764
max_abs_dz *	mm	1.283 ± 4.99	9.589 ± 28.92	3.04 ± 10.49	4.277 ± 23.21
std_abs_dz *	mm	0.4551 ± 2.138	3.634 ± 10.3	1.156 ± 4.446	1.658 ± 9.567
pt_weighted_abs_d0 *	mm	0.2914 ± 1.046	2.364 ± 11.79	0.9631 ± 4.187	1.156 ± 7.937
pt_weighted_abs_dz *	mm	0.6237 ± 1.58	3.112 ± 17.58	1.298 ± 3.622	1.322 ± 8.447

* Variable utilizada como entrada del modelo.

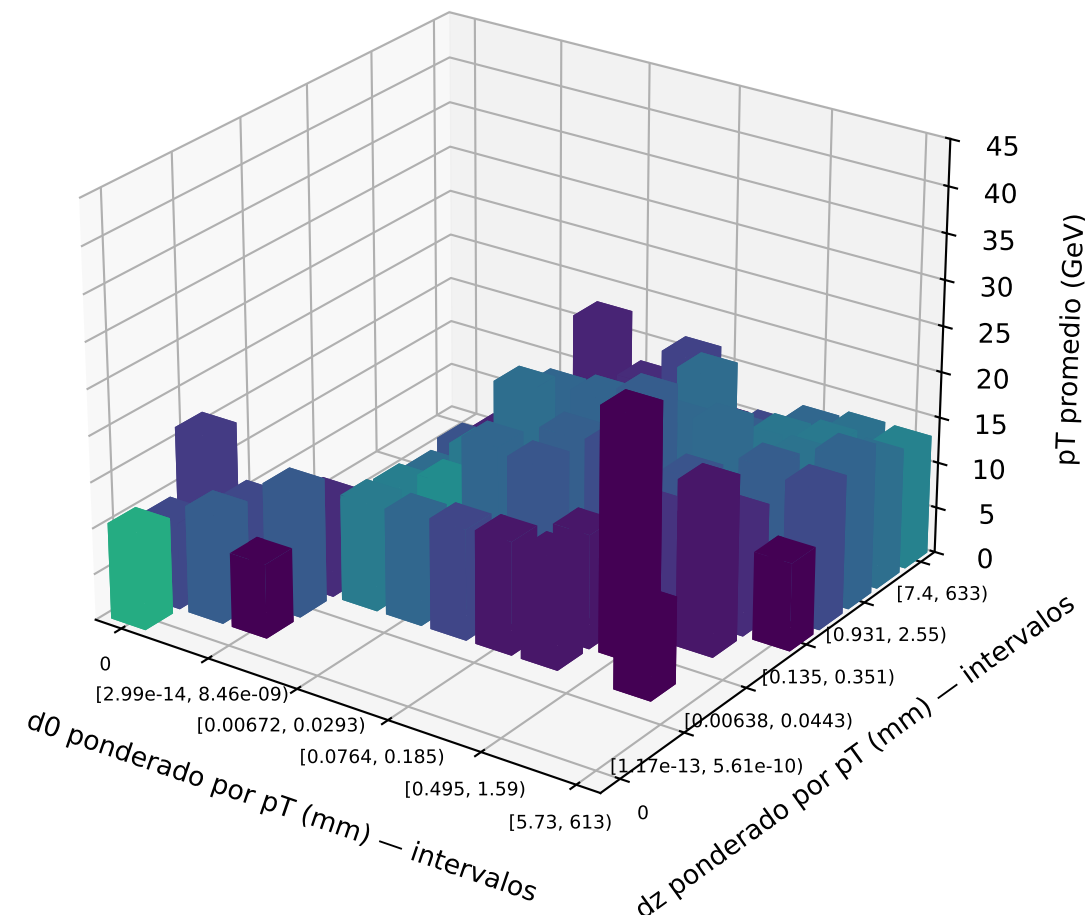
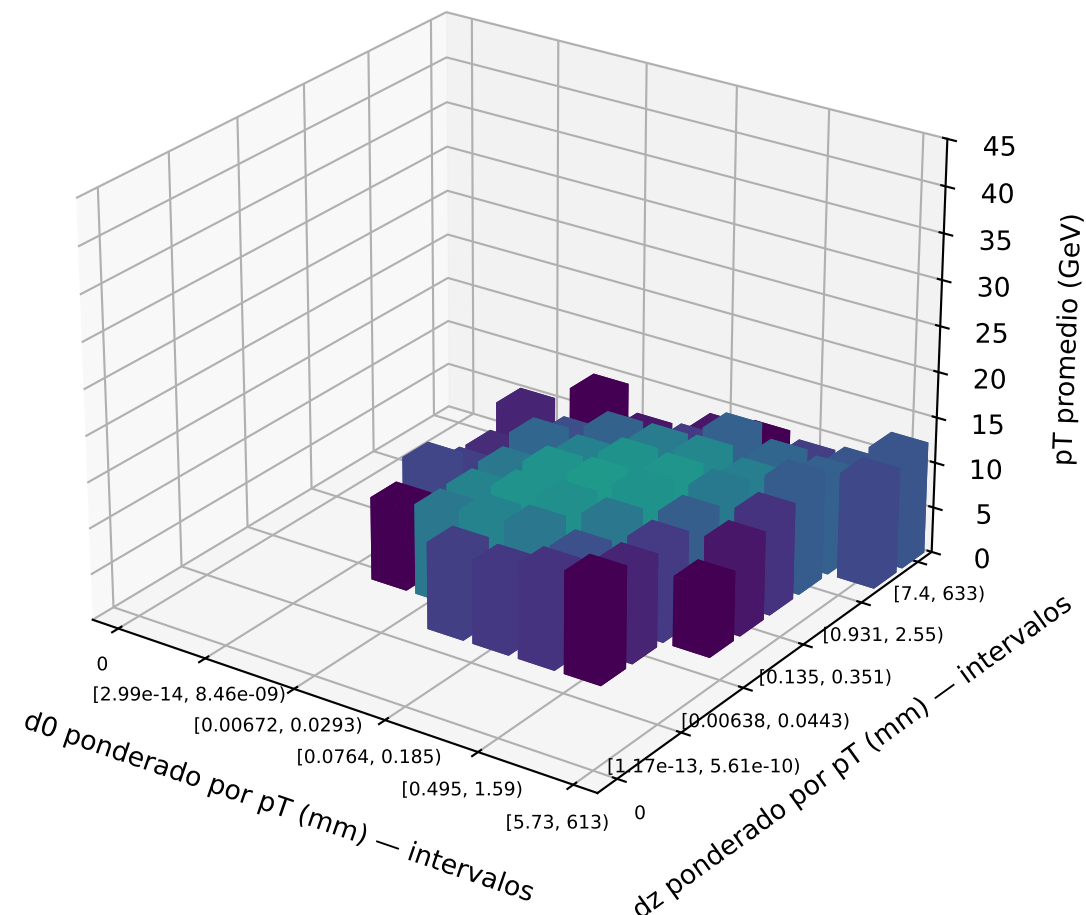
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF36 · seed 73

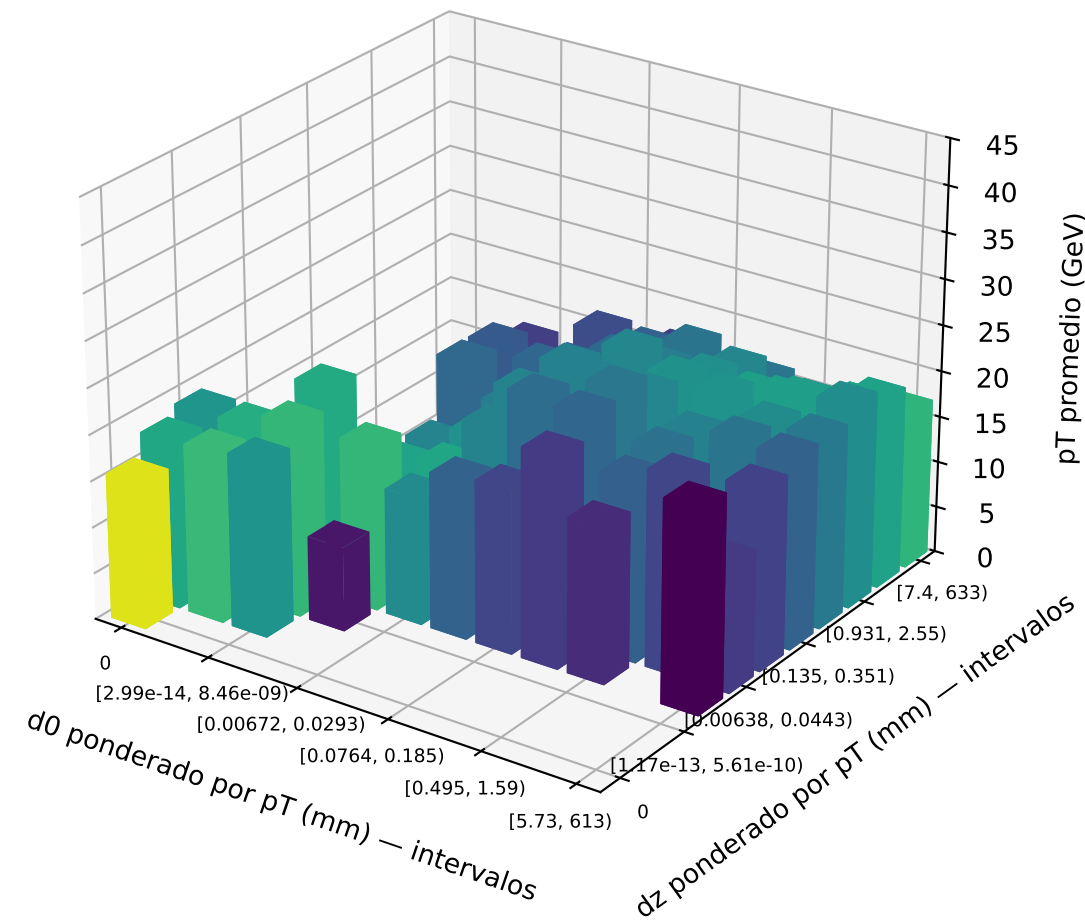
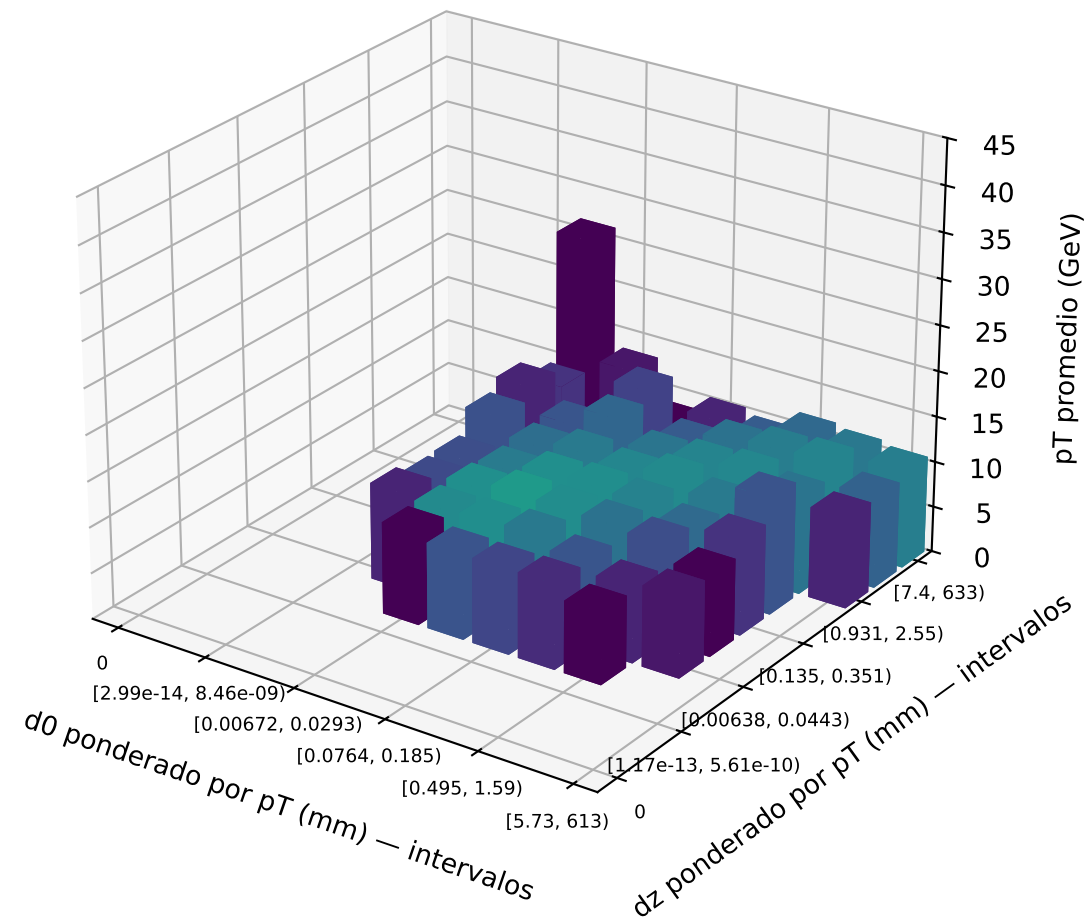
TP — N=3,727

FN — N=2,948



FP — N=4,162

TN — N=56,714



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF36 · seed 73

Variable	Unidad	TP N=3,727	FN N=2,948	FP N=4,162	TN N=56,714
jet_pt *	GeV	11.48 ± 4.471	13.24 ± 8.528	11.05 ± 4.243	16.81 ± 11.92
jet_eta	1	0.01262 ± 1.343	-0.01031 ± 1.5	0.01386 ± 1.323	0.001874 ± 1.508
jet_abs_eta *	1	1.157 ± 0.6816	1.288 ± 0.769	1.138 ± 0.6751	1.286 ± 0.7876
jet_mass	GeV	2.463 ± 1.265	2.615 ± 1.819	2.411 ± 1.134	2.884 ± 1.906
jet_ncharged	count	2.874 ± 1.547	2.527 ± 2.022	2.747 ± 1.458	2.576 ± 1.959
jet_nneutral	count	1.668 ± 1.452	2.299 ± 1.748	1.509 ± 1.222	2.432 ± 1.804
n_tracks_cone	count	2.91 ± 1.534	2.561 ± 2.024	2.785 ± 1.45	2.604 ± 1.96
track_pt_sum *	GeV	5.666 ± 3.358	5.109 ± 6.289	5.12 ± 3.239	6.874 ± 8.288
track_pt_sum_over_jet_pt	1	0.4959 ± 0.2196	0.3565 ± 0.2644	0.466 ± 0.2297	0.3678 ± 0.2679
avg_track_pt *	GeV	2.157 ± 1.342	1.675 ± 1.366	1.929 ± 1.092	2.183 ± 2.107
std_track_pt	GeV	0.7637 ± 0.7573	0.7024 ± 0.9882	0.6407 ± 0.6398	1.031 ± 1.641
n_tracks_below_avg_pt	count	1.517 ± 1.148	1.339 ± 1.38	1.41 ± 1.078	1.376 ± 1.371
n_tracks_above_avg_pt	count	1.393 ± 0.6624	1.222 ± 0.8575	1.375 ± 0.636	1.228 ± 0.8116
max_pt_ratio	1	0.2887 ± 0.1578	0.2024 ± 0.1606	0.2584 ± 0.129	0.2088 ± 0.1564
min_pt_ratio	1	0.1337 ± 0.1453	0.08548 ± 0.1071	0.1228 ± 0.1066	0.08386 ± 0.09798
max_dr_ratio	1	0.1596 ± 0.1449	0.1046 ± 0.1123	0.1476 ± 0.1103	0.1023 ± 0.1072
min_dr_ratio	1	0.2485 ± 0.1762	0.1703 ± 0.1621	0.2208 ± 0.141	0.1783 ± 0.1552
dr_max_pt	1	0.141 ± 0.08447	0.1329 ± 0.1012	0.1501 ± 0.08409	0.1254 ± 0.09945
dr_min_pt	1	0.2045 ± 0.1005	0.1868 ± 0.1202	0.2107 ± 0.09533	0.1893 ± 0.118
dr_max_dr	1	0.2446 ± 0.09553	0.2216 ± 0.125	0.2465 ± 0.08729	0.2198 ± 0.1217
dr_min_dr	1	0.1102 ± 0.07144	0.1057 ± 0.08824	0.1209 ± 0.07556	0.1031 ± 0.0887
pt_difference	GeV	1.793 ± 1.747	1.719 ± 2.641	1.508 ± 1.511	2.53 ± 4.147
r50	1	0.148 ± 0.0814	0.1399 ± 0.0994	0.1587 ± 0.07928	0.1319 ± 0.09792
r95	1	0.2422 ± 0.09518	0.2187 ± 0.1239	0.2444 ± 0.08739	0.2133 ± 0.1202
mean_abs_d0 *	mm	0.4081 ± 2.103	2.334 ± 11.65	1.306 ± 6.18	1.191 ± 7.82
max_abs_d0 *	mm	1.103 ± 8.29	6.19 ± 20.95	3.527 ± 18.96	3.145 ± 16.03
std_abs_d0 *	mm	0.4211 ± 3.468	2.372 ± 7.98	1.37 ± 7.872	1.209 ± 6.419
mean_abs_dz *	mm	0.7642 ± 2.745	3.436 ± 17.52	2.752 ± 12.42	1.52 ± 9.21
max_abs_dz *	mm	1.708 ± 9.391	8.904 ± 27.41	7.441 ± 48.31	3.951 ± 19.32
std_abs_dz *	mm	0.6146 ± 3.818	3.377 ± 9.647	2.959 ± 20.02	1.524 ± 7.952
pt_weighted_abs_d0 *	mm	0.4034 ± 2.16	2.186 ± 11.52	1.145 ± 5.953	1.142 ± 7.845
pt_weighted_abs_dz *	mm	0.7242 ± 2.136	2.941 ± 17.37	2.157 ± 7.951	1.259 ± 8.212

* Variable utilizada como entrada del modelo.

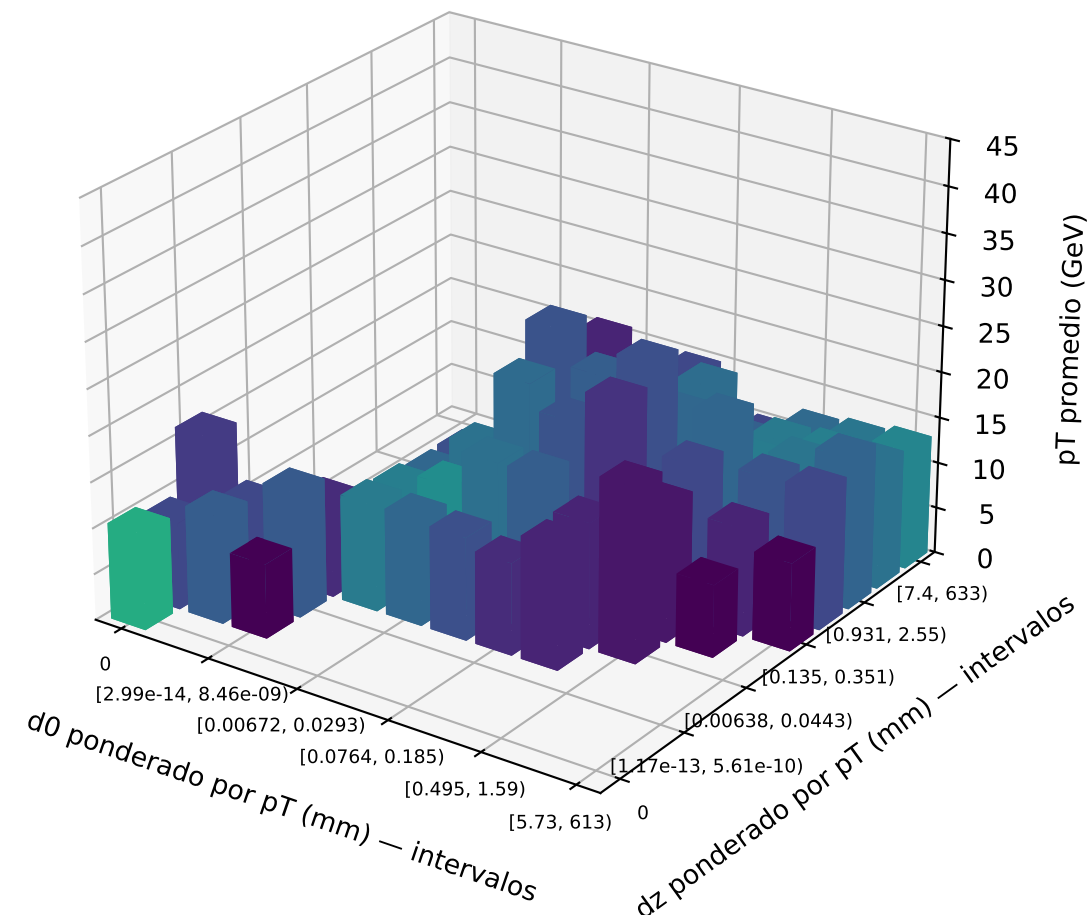
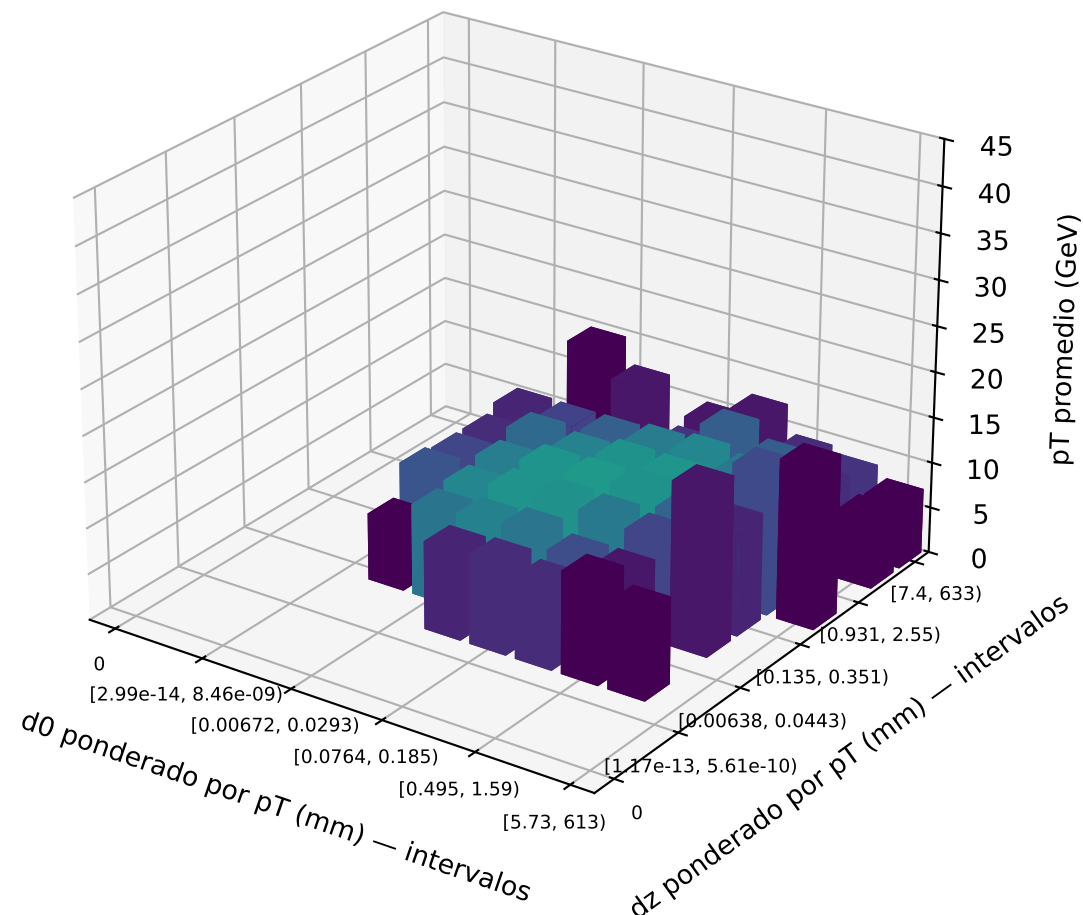
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF36 · seed 101

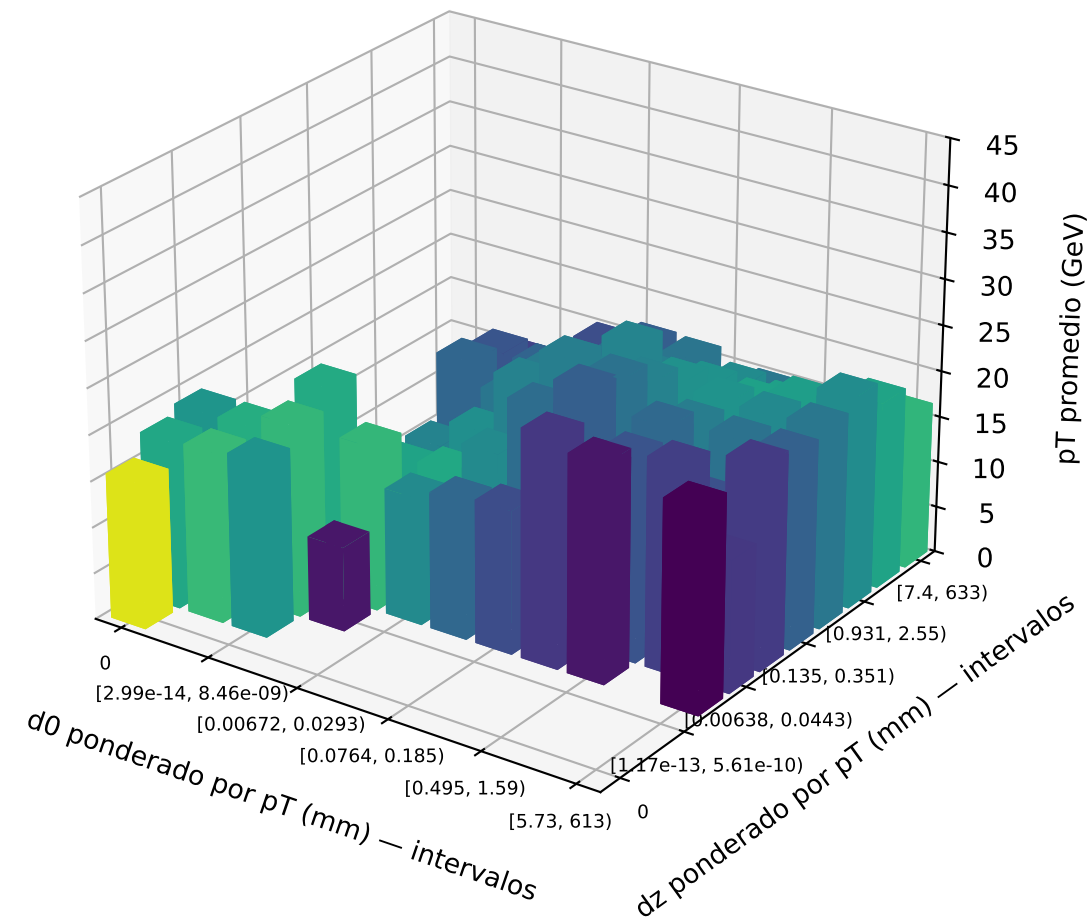
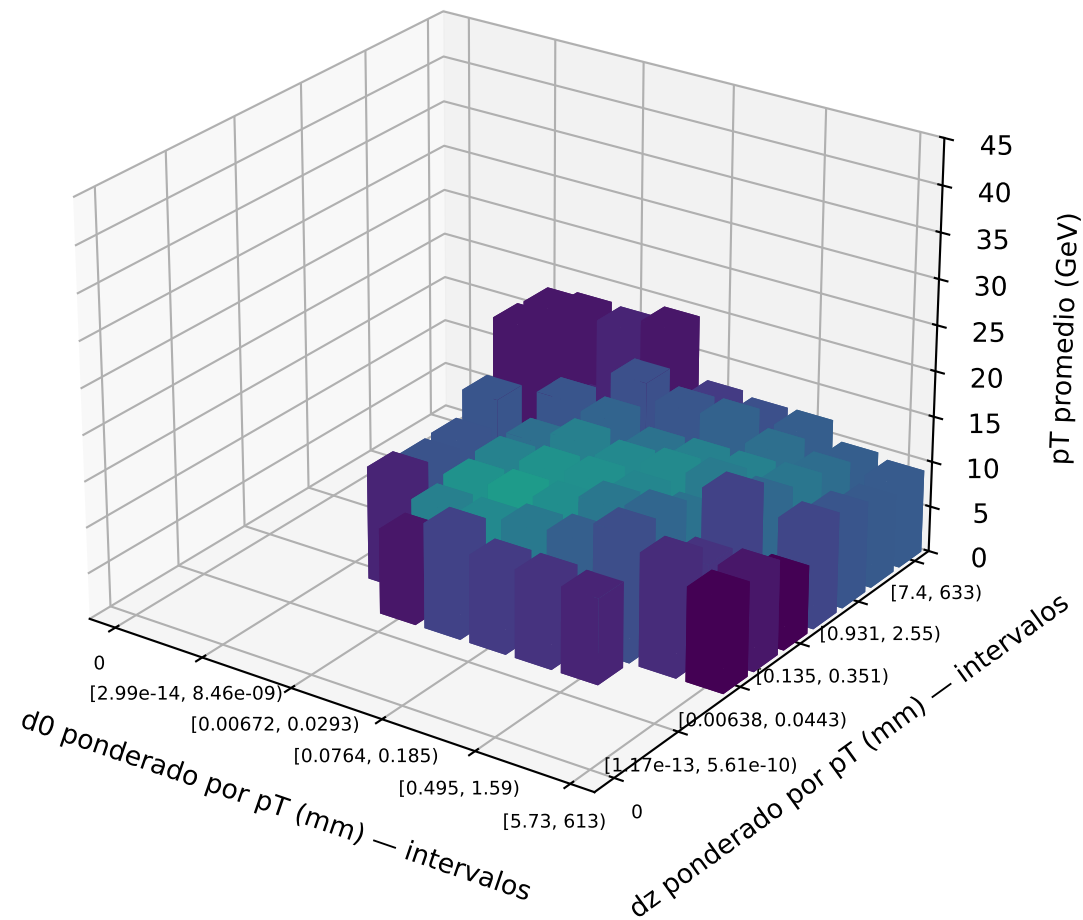
TP — N=3,714

FN — N=2,961



FP — N=4,016

TN — N=56,860



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF36 · seed 101

Variable	Unidad	TP N=3,714	FN N=2,961	FP N=4,016	TN N=56,860
jet_pt *	GeV	11.65 ± 4.438	13.03 ± 8.575	11.64 ± 4.558	16.75 ± 11.92
jet_eta	1	-0.000472 ± 1.369	0.006208 ± 1.471	0.04361 ± 1.373	-0.000197 ± 1.504
jet_abs_eta *	1	1.177 ± 0.6978	1.263 ± 0.7538	1.17 ± 0.7189	1.283 ± 0.785
jet_mass	GeV	2.477 ± 1.269	2.597 ± 1.815	2.514 ± 1.221	2.876 ± 1.902
jet_ncharged	count	2.839 ± 1.519	2.573 ± 2.053	2.76 ± 1.482	2.576 ± 1.957
jet_nneutral	count	1.751 ± 1.504	2.193 ± 1.724	1.742 ± 1.373	2.414 ± 1.803
n_tracks_cone	count	2.877 ± 1.512	2.605 ± 2.05	2.797 ± 1.47	2.604 ± 1.957
track_pt_sum *	GeV	5.552 ± 3.093	5.254 ± 6.457	5.033 ± 2.891	6.876 ± 8.287
track_pt_sum_over_jet_pt	1	0.4849 ± 0.2191	0.3709 ± 0.2714	0.4442 ± 0.2203	0.3696 ± 0.2689
avg_track_pt *	GeV	2.174 ± 1.386	1.656 ± 1.302	1.961 ± 1.193	2.18 ± 2.102
std_track_pt	GeV	0.7506 ± 0.7593	0.7191 ± 0.9862	0.6378 ± 0.6862	1.03 ± 1.638
n_tracks_below_avg_pt	count	1.495 ± 1.144	1.367 ± 1.386	1.404 ± 1.093	1.376 ± 1.369
n_tracks_above_avg_pt	count	1.382 ± 0.6436	1.238 ± 0.8772	1.393 ± 0.6463	1.227 ± 0.8104
max_pt_ratio	1	0.2861 ± 0.1624	0.2061 ± 0.1566	0.2515 ± 0.1368	0.2094 ± 0.156
min_pt_ratio	1	0.1357 ± 0.151	0.08318 ± 0.09571	0.1235 ± 0.1171	0.08391 ± 0.09716
max_dr_ratio	1	0.1603 ± 0.1499	0.1039 ± 0.1033	0.1476 ± 0.1196	0.1024 ± 0.1066
min_dr_ratio	1	0.2476 ± 0.1802	0.1718 ± 0.1572	0.2152 ± 0.1481	0.1788 ± 0.1548
dr_max_pt	1	0.1414 ± 0.08527	0.1323 ± 0.1002	0.1529 ± 0.08544	0.1253 ± 0.09929
dr_min_pt	1	0.2042 ± 0.1016	0.1873 ± 0.119	0.2109 ± 0.0976	0.1893 ± 0.1178
dr_max_dr	1	0.2434 ± 0.09756	0.2231 ± 0.1231	0.2476 ± 0.09113	0.2198 ± 0.1214
dr_min_dr	1	0.1108 ± 0.07254	0.1049 ± 0.08699	0.1216 ± 0.07579	0.1031 ± 0.08864
pt_difference	GeV	1.752 ± 1.714	1.77 ± 2.665	1.486 ± 1.536	2.529 ± 4.141
r50	1	0.1486 ± 0.08248	0.1391 ± 0.09814	0.1616 ± 0.08086	0.1318 ± 0.09774
r95	1	0.2411 ± 0.09723	0.2202 ± 0.122	0.2453 ± 0.09146	0.2134 ± 0.1199
mean_abs_d0 *	mm	0.2243 ± 0.545	2.556 ± 11.8	0.833 ± 6.288	1.224 ± 7.81
max_abs_d0 *	mm	0.5025 ± 1.918	6.921 ± 22.59	1.431 ± 7.645	3.294 ± 16.68
std_abs_d0 *	mm	0.1752 ± 0.7747	2.672 ± 8.742	0.3944 ± 2.102	1.278 ± 6.728
mean_abs_dz *	mm	0.5197 ± 1.853	3.731 ± 17.58	1.025 ± 4.186	1.645 ± 9.733
max_abs_dz *	mm	1.078 ± 6.207	9.663 ± 28.26	2.278 ± 10.5	4.324 ± 23.15
std_abs_dz *	mm	0.3724 ± 2.577	3.668 ± 10.04	0.8277 ± 4.473	1.679 ± 9.544
pt_weighted_abs_d0 *	mm	0.225 ± 0.5834	2.402 ± 11.69	0.8445 ± 6.495	1.163 ± 7.81
pt_weighted_abs_dz *	mm	0.522 ± 1.51	3.184 ± 17.38	1.046 ± 4	1.34 ± 8.415

* Variable utilizada como entrada del modelo.

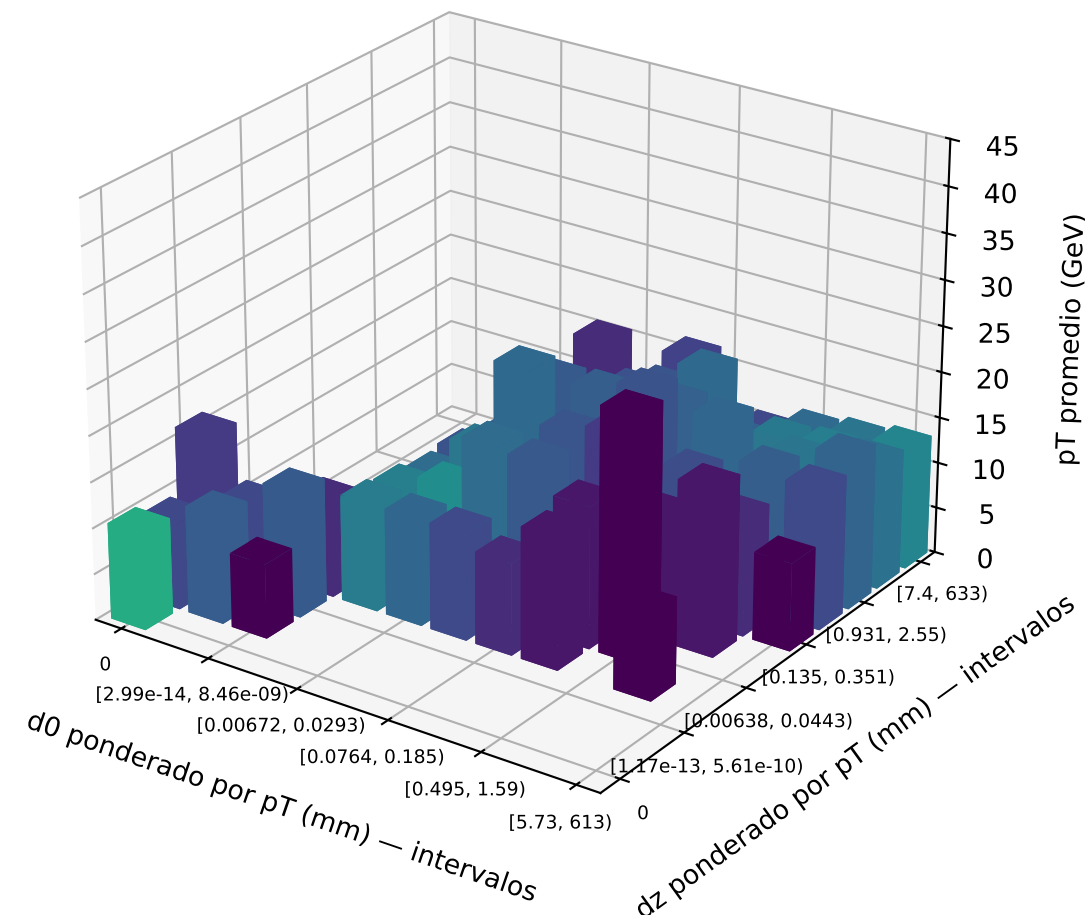
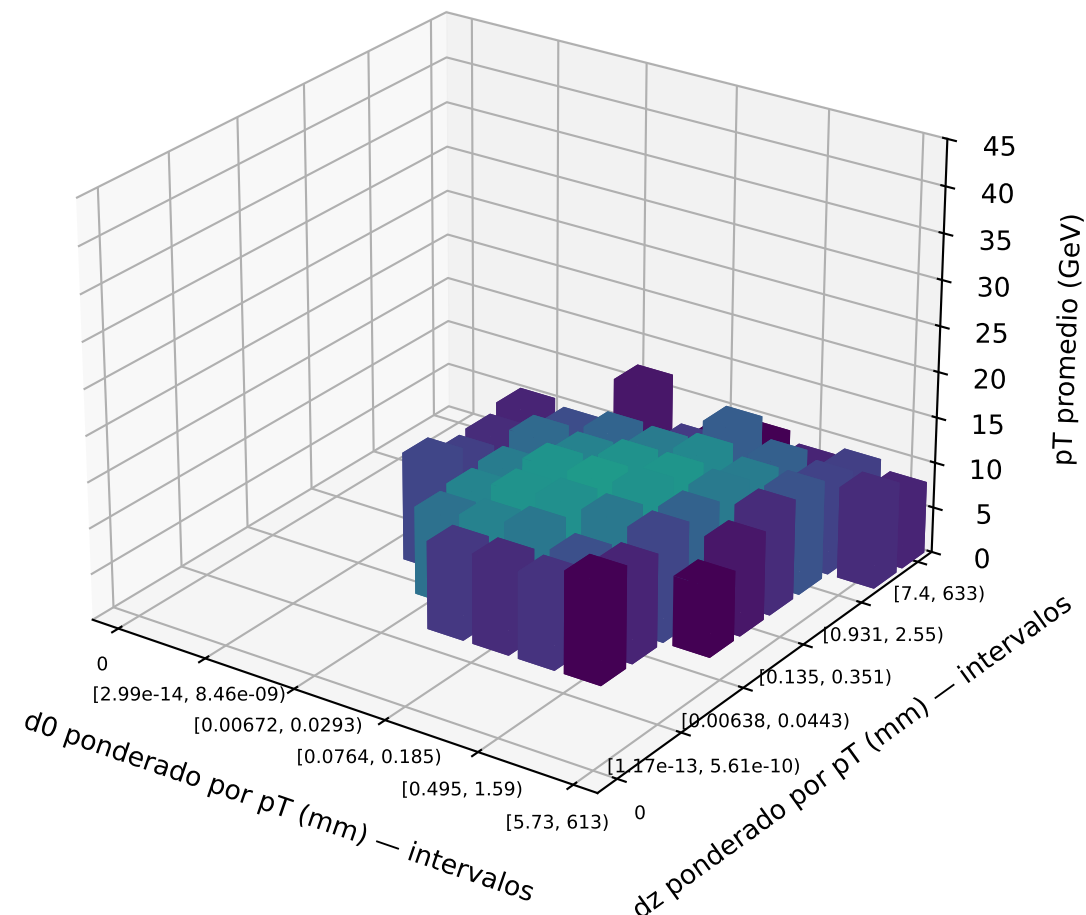
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF36 · ensemble

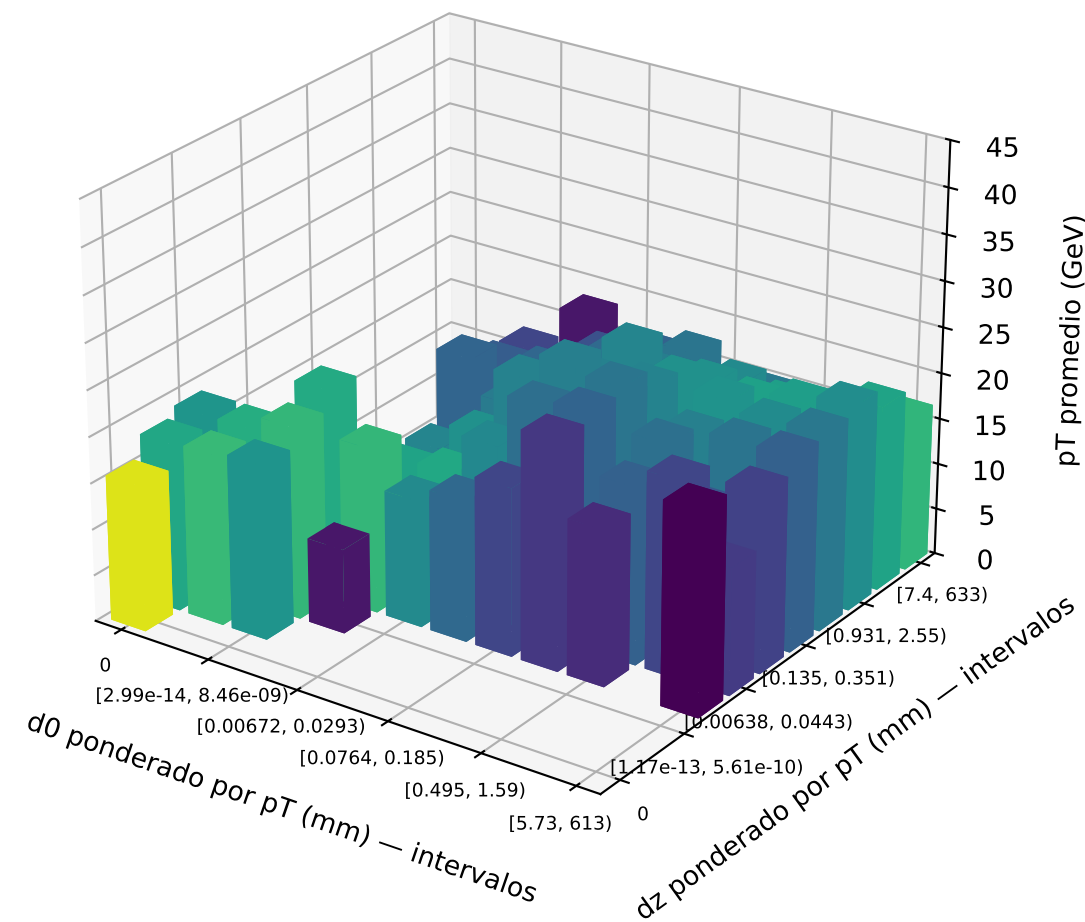
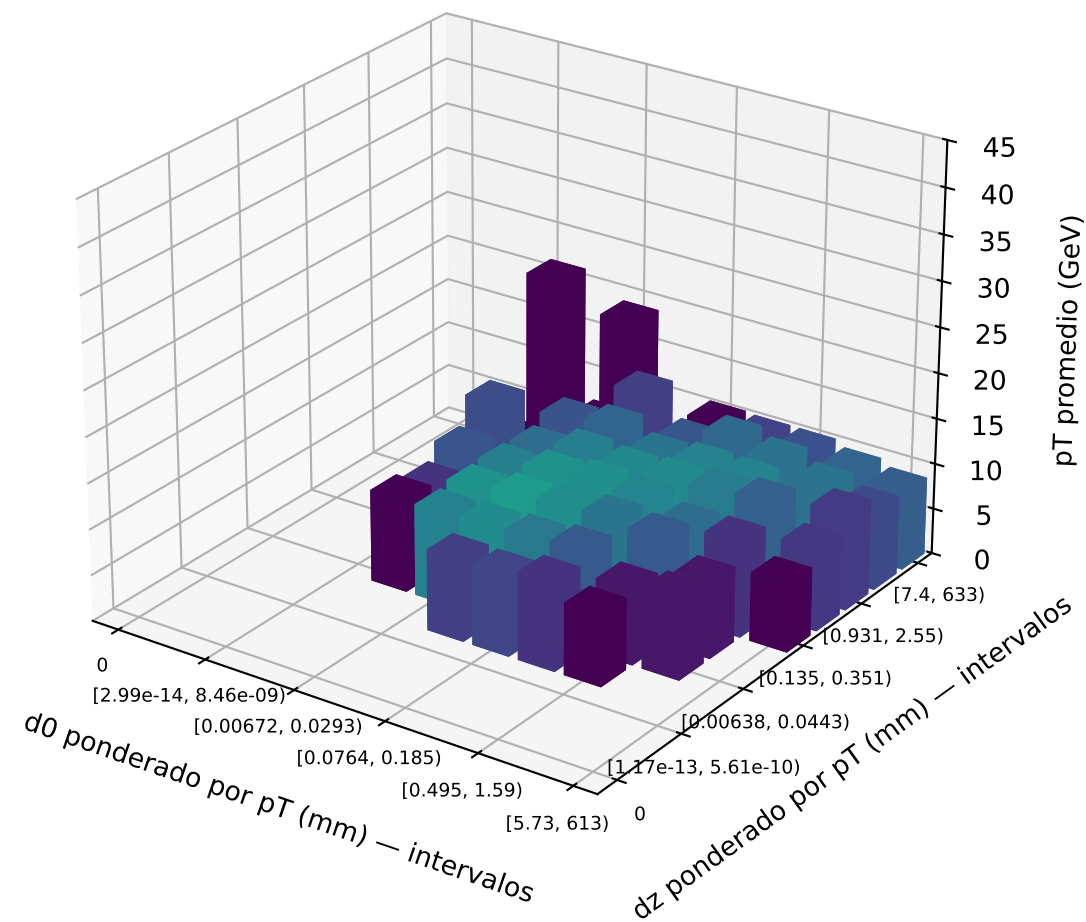
TP — N=3,736

FN — N=2,939



FP — N=3,858

TN — N=57,018



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF36 · ensemble

Variable	Unidad	TP N=3,736	FN N=2,939	FP N=3,858	TN N=57,018
jet_pt *	GeV	11.6 ± 4.61	13.1 ± 8.471	11.1 ± 3.87	16.77 ± 11.91
jet_eta	1	-0.000813 ± 1.363	0.006691 ± 1.478	0.0212 ± 1.345	0.001441 ± 1.505
jet_abs_eta *	1	1.177 ± 0.6875	1.264 ± 0.766	1.159 ± 0.682	1.283 ± 0.787
jet_mass	GeV	2.462 ± 1.254	2.616 ± 1.829	2.399 ± 1.119	2.882 ± 1.903
jet_ncharged	count	2.822 ± 1.508	2.592 ± 2.07	2.668 ± 1.398	2.582 ± 1.96
jet_nneutral	count	1.727 ± 1.485	2.227 ± 1.737	1.601 ± 1.285	2.421 ± 1.803
n_tracks_cone	count	2.859 ± 1.5	2.625 ± 2.067	2.702 ± 1.388	2.611 ± 1.961
track_pt_sum *	GeV	5.56 ± 3.233	5.241 ± 6.388	4.928 ± 2.872	6.878 ± 8.277
track_pt_sum_over_jet_pt	1	0.4858 ± 0.2199	0.3689 ± 0.2703	0.4529 ± 0.2259	0.3692 ± 0.2683
avg_track_pt *	GeV	2.166 ± 1.364	1.662 ± 1.333	1.948 ± 1.068	2.18 ± 2.104
std_track_pt	GeV	0.7549 ± 0.7689	0.7134 ± 0.978	0.6189 ± 0.625	1.03 ± 1.638
n_tracks_below_avg_pt	count	1.485 ± 1.132	1.379 ± 1.401	1.344 ± 1.034	1.38 ± 1.371
n_tracks_above_avg_pt	count	1.374 ± 0.6414	1.246 ± 0.8822	1.358 ± 0.6196	1.23 ± 0.8119
max_pt_ratio	1	0.2863 ± 0.1599	0.2053 ± 0.1595	0.2572 ± 0.1331	0.2091 ± 0.1561
min_pt_ratio	1	0.1351 ± 0.1478	0.08353 ± 0.1017	0.1273 ± 0.1131	0.08376 ± 0.09745
max_dr_ratio	1	0.1602 ± 0.147	0.1035 ± 0.1082	0.152 ± 0.1161	0.1023 ± 0.1068
min_dr_ratio	1	0.2474 ± 0.1778	0.1715 ± 0.1604	0.2211 ± 0.1445	0.1785 ± 0.155
dr_max_pt	1	0.1418 ± 0.0853	0.1317 ± 0.1002	0.1513 ± 0.0852	0.1255 ± 0.09931
dr_min_pt	1	0.2038 ± 0.101	0.1877 ± 0.1199	0.2087 ± 0.09581	0.1895 ± 0.1179
dr_max_dr	1	0.2431 ± 0.09666	0.2233 ± 0.1243	0.2443 ± 0.08924	0.2201 ± 0.1215
dr_min_dr	1	0.1114 ± 0.07272	0.1042 ± 0.08685	0.1226 ± 0.0762	0.1031 ± 0.08858
pt_difference	GeV	1.763 ± 1.749	1.758 ± 2.642	1.439 ± 1.437	2.53 ± 4.138
r50	1	0.1486 ± 0.08235	0.1391 ± 0.09838	0.1596 ± 0.081	0.132 ± 0.09774
r95	1	0.2408 ± 0.09637	0.2203 ± 0.1231	0.2425 ± 0.08933	0.2136 ± 0.12
mean_abs_d0 *	mm	0.2387 ± 0.7426	2.555 ± 11.84	0.5559 ± 2.703	1.242 ± 7.943
max_abs_d0 *	mm	0.4931 ± 2.284	6.981 ± 22.62	1.169 ± 7.166	3.306 ± 16.67
std_abs_d0 *	mm	0.1709 ± 0.9623	2.696 ± 8.747	0.4118 ± 3.135	1.274 ± 6.693
mean_abs_dz *	mm	0.5155 ± 1.216	3.76 ± 17.71	1.035 ± 4.502	1.643 ± 9.713
max_abs_dz *	mm	0.9685 ± 3.382	9.867 ± 28.91	2.27 ± 16.52	4.319 ± 22.89
std_abs_dz *	mm	0.3238 ± 1.465	3.755 ± 10.33	0.8328 ± 6.475	1.676 ± 9.456
pt_weighted_abs_d0 *	mm	0.2437 ± 0.8084	2.395 ± 11.72	0.5614 ± 2.948	1.182 ± 7.949
pt_weighted_abs_dz *	mm	0.5333 ± 1.273	3.19 ± 17.47	1.017 ± 3.849	1.341 ± 8.41

* Variable utilizada como entrada del modelo.

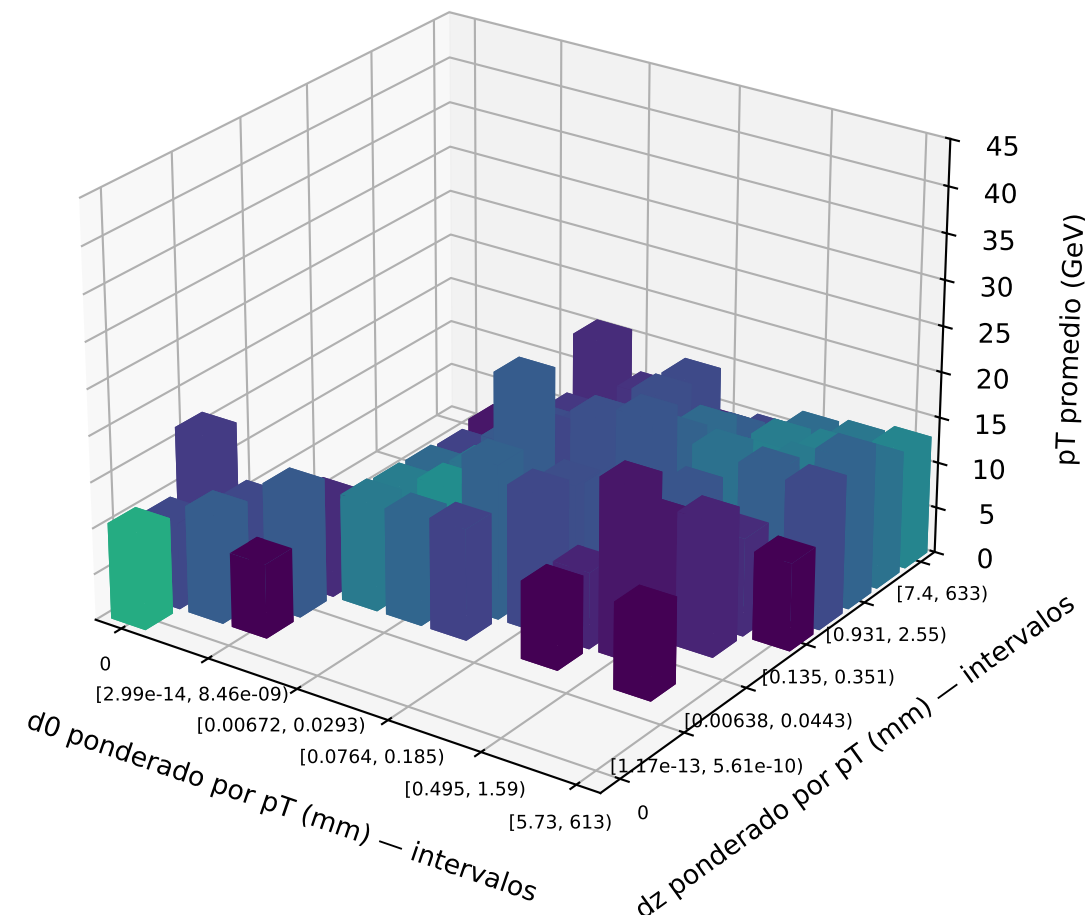
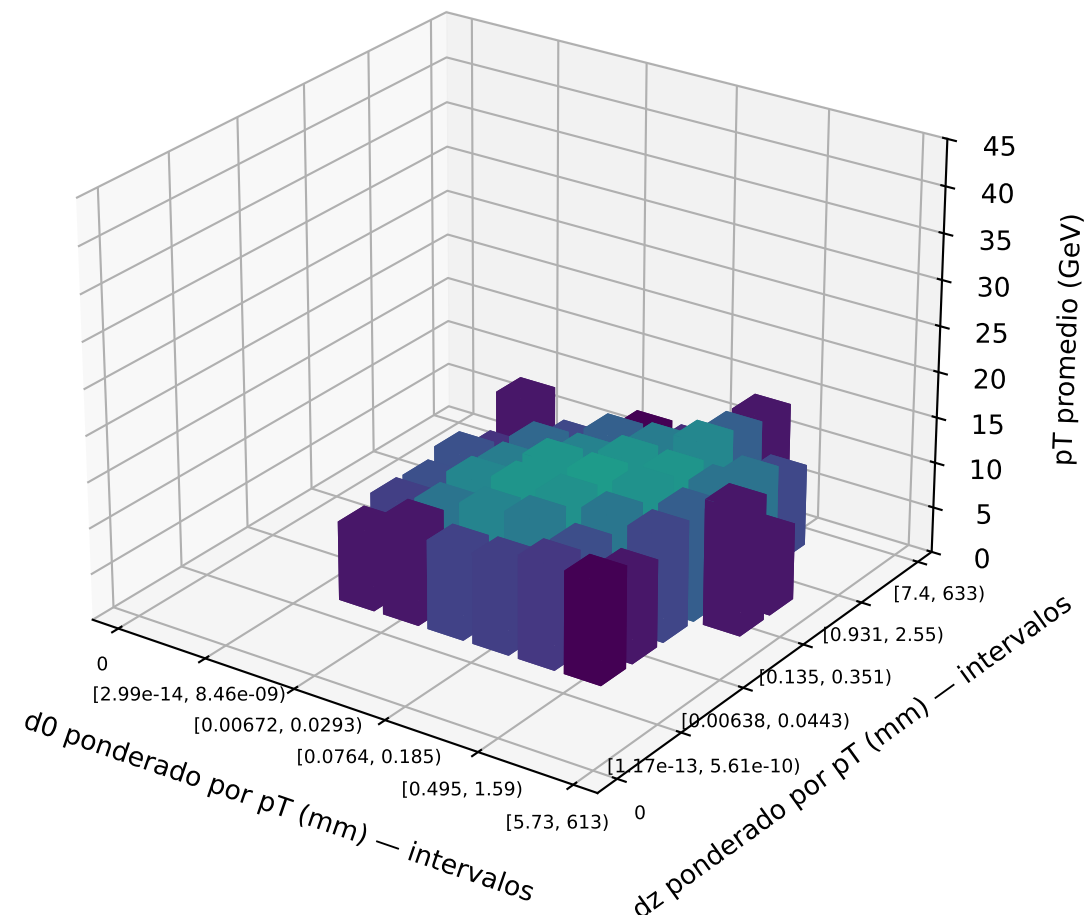
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — HGB12

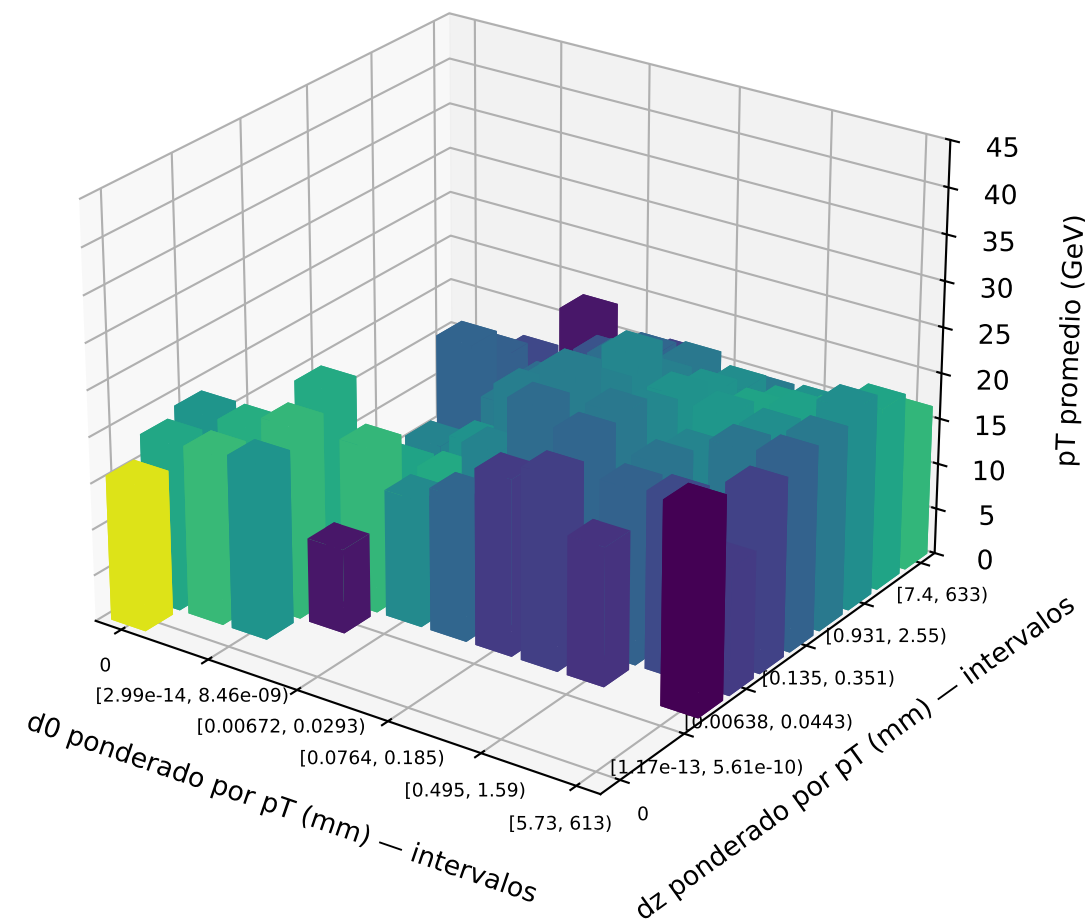
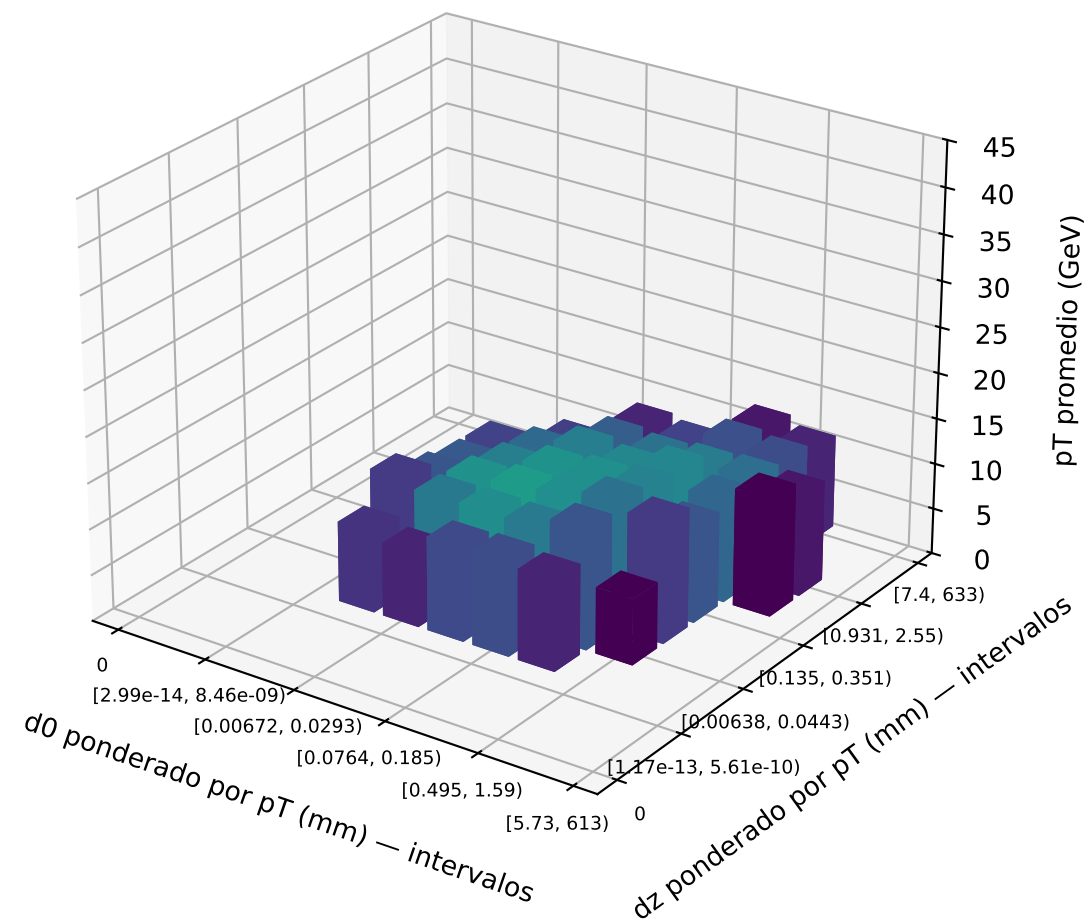
TP — N=3,723

FN — N=2,952



FP — N=3,179

TN — N=57,697



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — HGB12

Variable	Unidad	TP N=3,723	FN N=2,952	FP N=3,179	TN N=57,697
jet_pt *	GeV	12.12 ± 6.024	12.44 ± 7.332	11.88 ± 5.354	16.66 ± 11.85
jet_eta	1	0.005398 ± 1.369	-0.001175 ± 1.47	-0.02016 ± 1.389	0.003952 ± 1.501
jet_abs_eta *	1	1.177 ± 0.699	1.263 ± 0.7525	1.192 ± 0.7132	1.28 ± 0.7847
jet_mass	GeV	2.504 ± 1.373	2.563 ± 1.719	2.371 ± 1.171	2.878 ± 1.894
jet_ncharged	count	2.807 ± 1.586	2.612 ± 1.995	2.489 ± 1.378	2.593 ± 1.955
jet_nneutral	count	1.784 ± 1.549	2.152 ± 1.684	1.751 ± 1.376	2.403 ± 1.8
n_tracks_cone	count	2.839 ± 1.578	2.651 ± 1.993	2.508 ± 1.363	2.622 ± 1.956
track_pt_sum *	GeV	5.857 ± 4.338	4.868 ± 5.444	5.176 ± 3.686	6.841 ± 8.221
track_pt_sum_over_jet_pt	1	0.4851 ± 0.225	0.3703 ± 0.2652	0.4384 ± 0.2279	0.371 ± 0.2681
avg_track_pt *	GeV	2.275 ± 1.457	1.527 ± 1.131	2.171 ± 1.286	2.165 ± 2.089
std_track_pt	GeV	0.7932 ± 0.8725	0.6653 ± 0.8557	0.6663 ± 0.7649	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.464 ± 1.182	1.407 ± 1.348	1.216 ± 1.038	1.387 ± 1.367
n_tracks_above_avg_pt	count	1.376 ± 0.6574	1.245 ± 0.866	1.292 ± 0.563	1.235 ± 0.8127
max_pt_ratio	1	0.2904 ± 0.1658	0.2004 ± 0.1489	0.2648 ± 0.143	0.2092 ± 0.1553
min_pt_ratio	1	0.1397 ± 0.1554	0.07796 ± 0.08271	0.1393 ± 0.1273	0.08362 ± 0.09646
max_dr_ratio	1	0.1639 ± 0.1541	0.09922 ± 0.09227	0.1614 ± 0.1291	0.1023 ± 0.1059
min_dr_ratio	1	0.2516 ± 0.1833	0.1666 ± 0.1498	0.2311 ± 0.1539	0.1785 ± 0.1542
dr_max_pt	1	0.1398 ± 0.08568	0.1343 ± 0.09994	0.1481 ± 0.08535	0.126 ± 0.09923
dr_min_pt	1	0.2019 ± 0.1017	0.1901 ± 0.1193	0.2042 ± 0.09712	0.19 ± 0.1177
dr_max_dr	1	0.2406 ± 0.09836	0.2266 ± 0.1229	0.2341 ± 0.0921	0.2209 ± 0.1212
dr_min_dr	1	0.1104 ± 0.07392	0.1054 ± 0.08559	0.124 ± 0.07923	0.1032 ± 0.0883
pt_difference	GeV	1.868 ± 2.123	1.625 ± 2.259	1.538 ± 1.786	2.511 ± 4.112
r50	1	0.1463 ± 0.083	0.1419 ± 0.09782	0.1546 ± 0.0825	0.1326 ± 0.0976
r95	1	0.238 ± 0.09782	0.224 ± 0.1221	0.2319 ± 0.09211	0.2146 ± 0.1198
mean_abs_d0 *	mm	0.1577 ± 0.1779	2.647 ± 11.82	0.1259 ± 0.1784	1.258 ± 7.925
max_abs_d0 *	mm	0.2829 ± 0.4032	7.218 ± 22.64	0.2267 ± 0.348	3.333 ± 16.67
std_abs_d0 *	mm	0.09122 ± 0.1532	2.785 ± 8.764	0.07705 ± 0.1482	1.283 ± 6.7
mean_abs_dz *	mm	0.4074 ± 0.6211	3.882 ± 17.69	0.3534 ± 0.7305	1.673 ± 9.72
max_abs_dz *	mm	0.7271 ± 2.002	10.13 ± 28.92	0.6097 ± 1.574	4.386 ± 23.14
std_abs_dz *	mm	0.2298 ± 0.7468	3.858 ± 10.37	0.1977 ± 0.6527	1.701 ± 9.543
pt_weighted_abs_d0 *	mm	0.1628 ± 0.1759	2.487 ± 11.71	0.1298 ± 0.1736	1.198 ± 7.937
pt_weighted_abs_dz *	mm	0.4214 ± 0.6224	3.319 ± 17.45	0.3617 ± 0.7124	1.373 ± 8.415

* Variable utilizada como entrada del modelo.

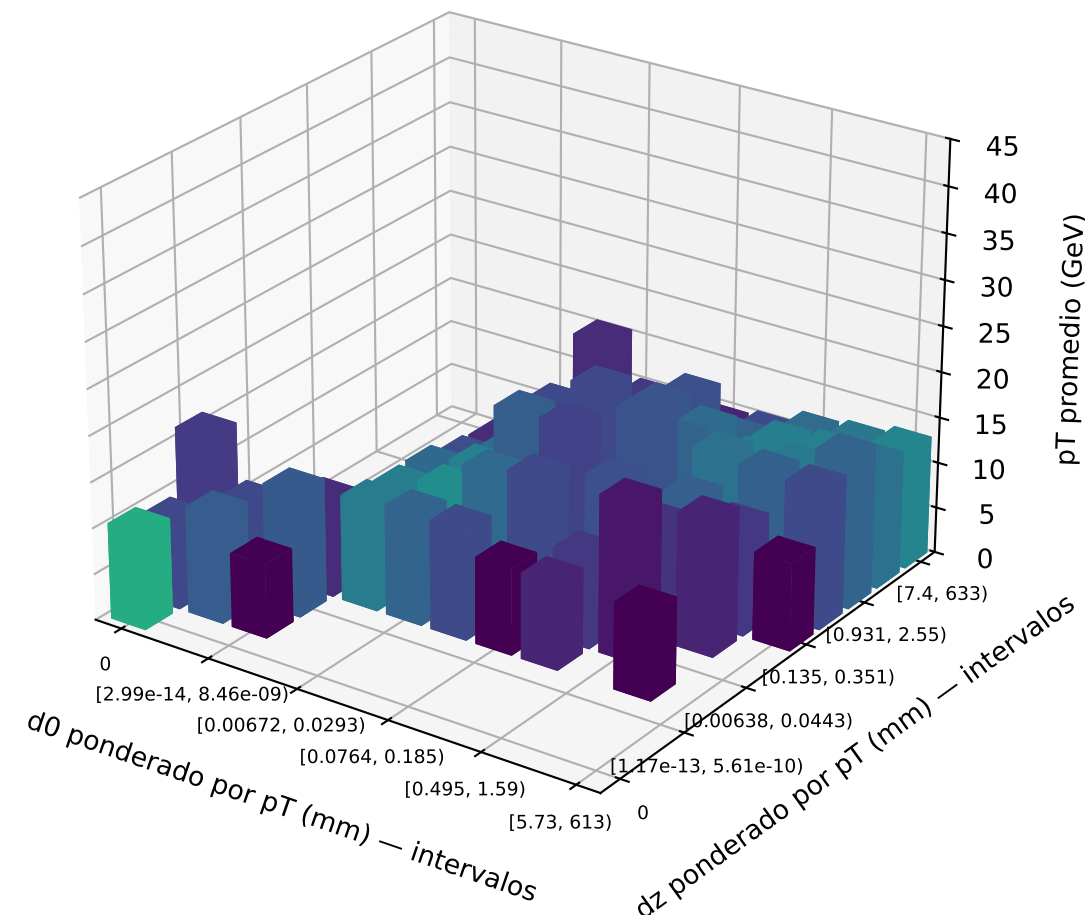
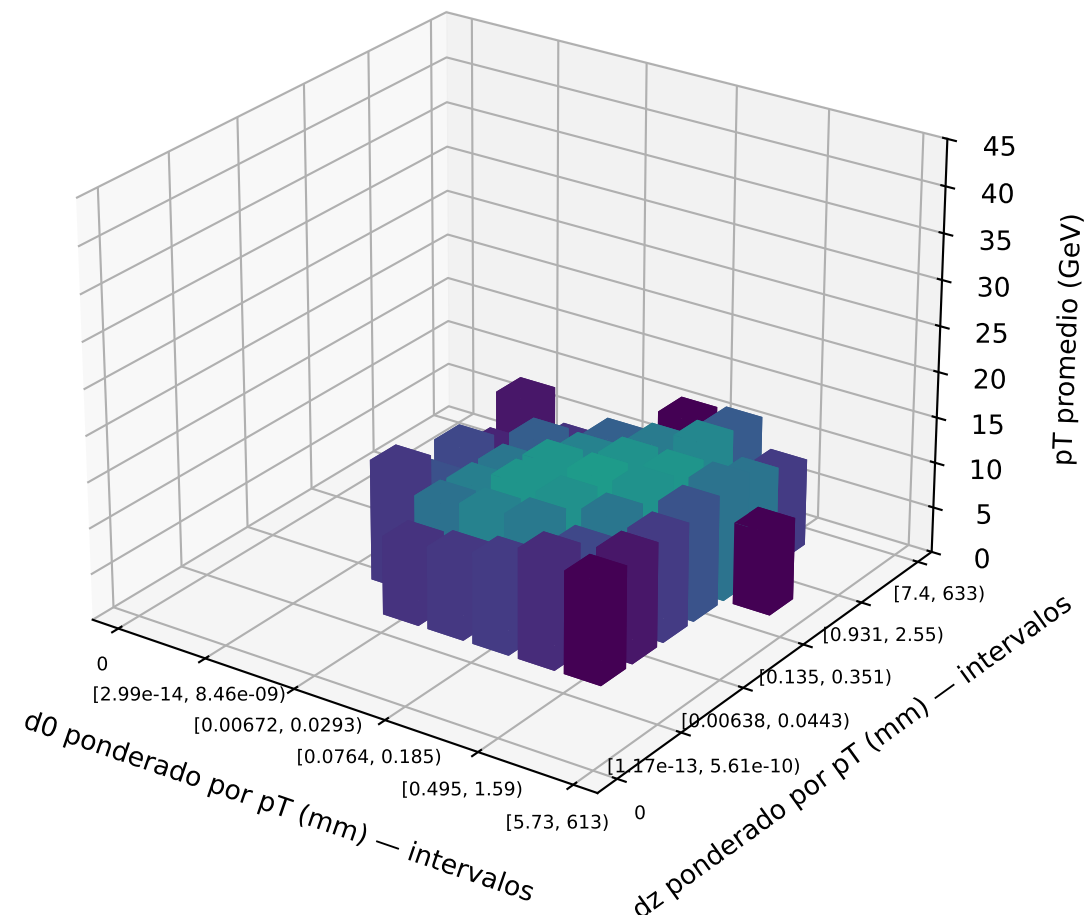
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF36-HGB · seed 42

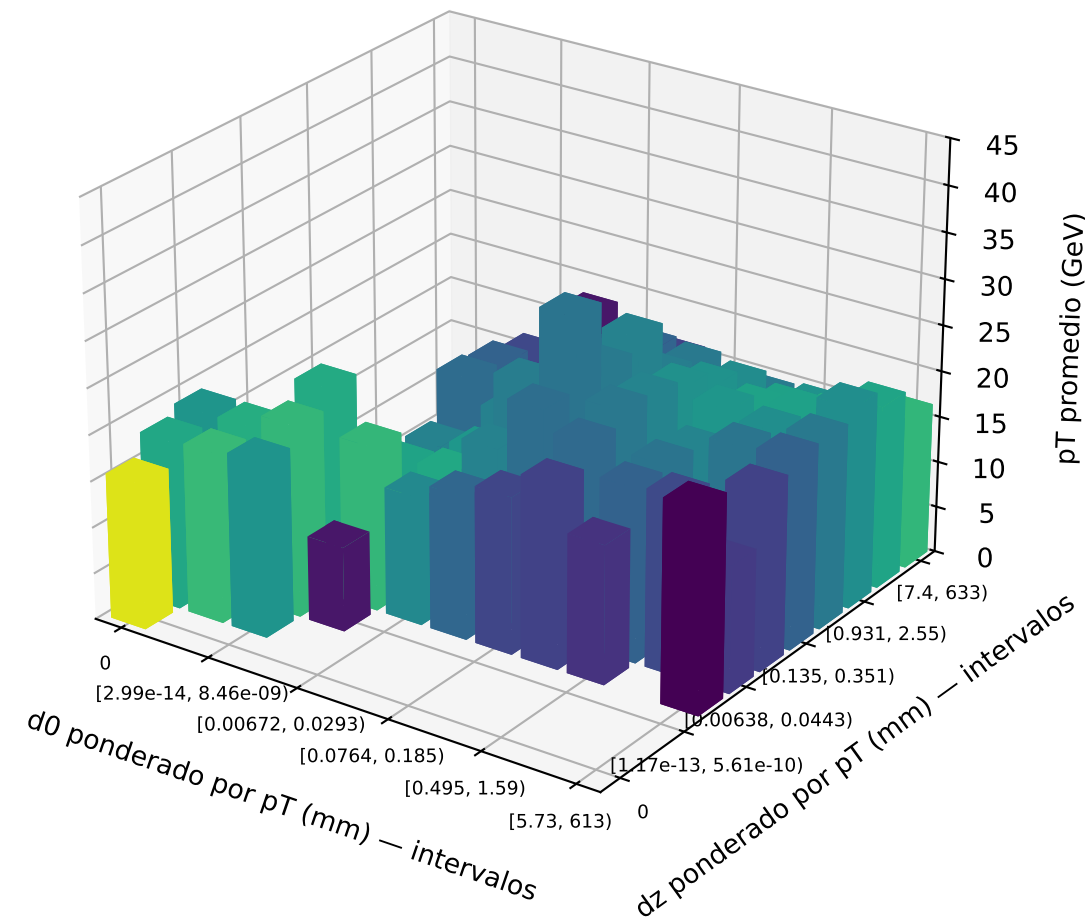
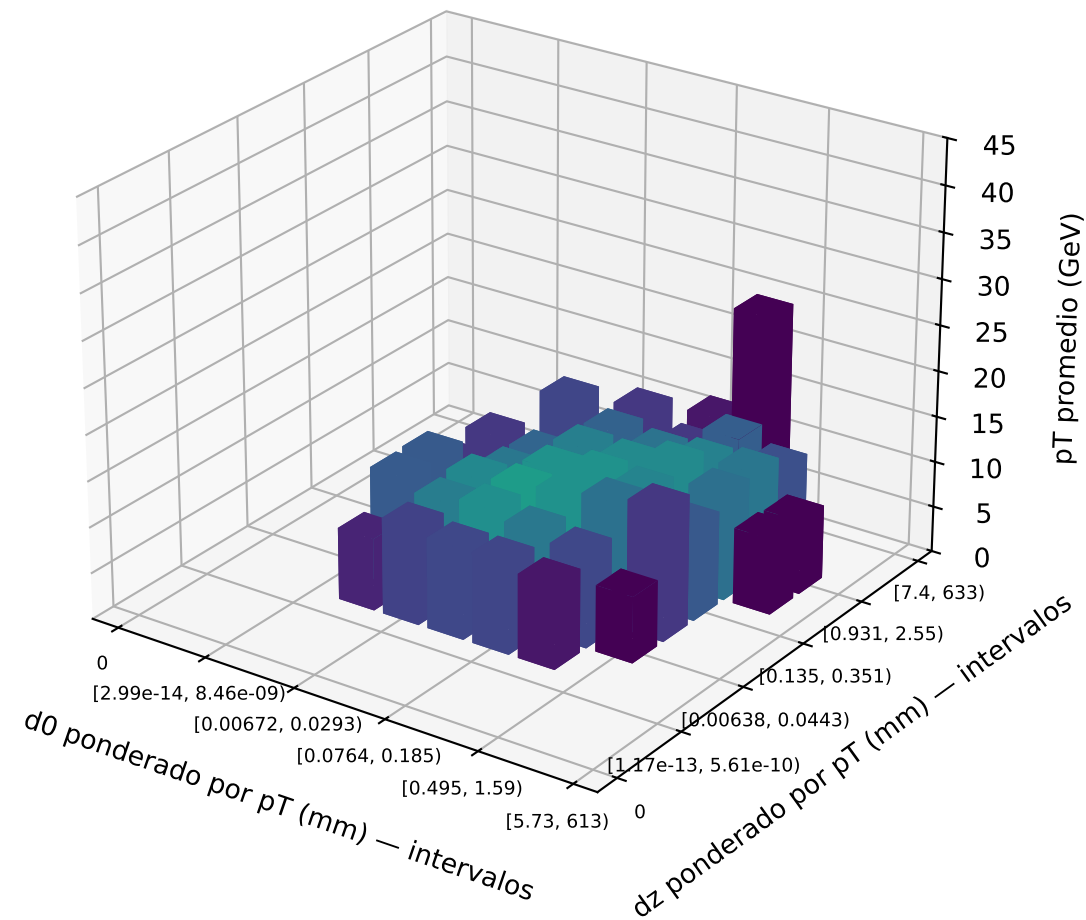
TP — N=3,560

FN — N=3,115



FP — N=3,316

TN — N=57,560



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF36-HGB · seed 42

Variable	Unidad	TP N=3,560	FN N=3,115	FP N=3,316	TN N=57,560
jet_pt *	GeV	12.13 ± 6.052	12.41 ± 7.244	11.94 ± 5.346	16.67 ± 11.86
jet_eta	1	0.00405 ± 1.372	0.00071 ± 1.462	0.0181 ± 1.4	0.001806 ± 1.501
jet_abs_eta *	1	1.176 ± 0.7064	1.26 ± 0.742	1.193 ± 0.7328	1.28 ± 0.7838
jet_mass	GeV	2.524 ± 1.376	2.538 ± 1.701	2.472 ± 1.219	2.874 ± 1.895
jet_ncharged	count	2.831 ± 1.565	2.594 ± 1.992	2.63 ± 1.398	2.585 ± 1.956
jet_nneutral	count	1.798 ± 1.57	2.117 ± 1.66	1.748 ± 1.405	2.405 ± 1.799
n_tracks_cone	count	2.862 ± 1.554	2.635 ± 1.993	2.657 ± 1.385	2.614 ± 1.956
track_pt_sum *	GeV	5.864 ± 4.271	4.912 ± 5.455	5.322 ± 3.548	6.837 ± 8.234
track_pt_sum_over_jet_pt	1	0.4862 ± 0.2212	0.3751 ± 0.2677	0.4506 ± 0.2231	0.3701 ± 0.2683
avg_track_pt *	GeV	2.266 ± 1.452	1.576 ± 1.175	2.121 ± 1.259	2.168 ± 2.091
std_track_pt	GeV	0.7946 ± 0.8548	0.6703 ± 0.877	0.6881 ± 0.7329	1.022 ± 1.63
n_tracks_below_avg_pt	count	1.488 ± 1.17	1.382 ± 1.351	1.326 ± 1.04	1.381 ± 1.368
n_tracks_above_avg_pt	count	1.374 ± 0.6498	1.254 ± 0.8637	1.331 ± 0.5944	1.233 ± 0.8117
max_pt_ratio	1	0.2898 ± 0.1649	0.2058 ± 0.1526	0.2627 ± 0.1384	0.2092 ± 0.1556
min_pt_ratio	1	0.1382 ± 0.1547	0.08289 ± 0.09146	0.1316 ± 0.1214	0.08393 ± 0.09701
max_dr_ratio	1	0.163 ± 0.1537	0.1036 ± 0.09886	0.155 ± 0.123	0.1026 ± 0.1064
min_dr_ratio	1	0.2501 ± 0.1827	0.1728 ± 0.1547	0.2281 ± 0.1507	0.1785 ± 0.1545
dr_max_pt	1	0.1398 ± 0.0854	0.1346 ± 0.09953	0.1484 ± 0.08598	0.1259 ± 0.09922
dr_min_pt	1	0.2021 ± 0.1016	0.1906 ± 0.1186	0.2066 ± 0.09703	0.1898 ± 0.1177
dr_max_dr	1	0.241 ± 0.0979	0.2268 ± 0.1221	0.2398 ± 0.09145	0.2205 ± 0.1212
dr_min_dr	1	0.1099 ± 0.07299	0.1063 ± 0.08597	0.1213 ± 0.0777	0.1033 ± 0.08843
pt_difference	GeV	1.871 ± 2.084	1.634 ± 2.294	1.599 ± 1.707	2.51 ± 4.119
r50	1	0.1465 ± 0.08291	0.142 ± 0.09719	0.1562 ± 0.08275	0.1325 ± 0.0976
r95	1	0.2383 ± 0.09742	0.2243 ± 0.1213	0.2376 ± 0.09154	0.2142 ± 0.1198
mean_abs_d0 *	mm	0.1582 ± 0.1651	2.516 ± 11.52	0.1421 ± 0.1925	1.26 ± 7.934
max_abs_d0 *	mm	0.2892 ± 0.3643	6.848 ± 22.09	0.2933 ± 0.451	3.337 ± 16.69
std_abs_d0 *	mm	0.09638 ± 0.1492	2.638 ± 8.555	0.1093 ± 0.1933	1.284 ± 6.708
mean_abs_dz *	mm	0.3985 ± 0.5404	3.711 ± 17.24	0.3975 ± 0.7704	1.674 ± 9.732
max_abs_dz *	mm	0.7003 ± 0.9069	9.671 ± 28.29	0.8466 ± 2.56	4.382 ± 23.16
std_abs_dz *	mm	0.2245 ± 0.3412	3.674 ± 10.15	0.3106 ± 1.044	1.698 ± 9.553
pt_weighted_abs_d0 *	mm	0.1638 ± 0.1629	2.365 ± 11.41	0.1445 ± 0.184	1.2 ± 7.946
pt_weighted_abs_dz *	mm	0.414 ± 0.5526	3.176 ± 17	0.4013 ± 0.7311	1.373 ± 8.425

* Variable utilizada como entrada del modelo.

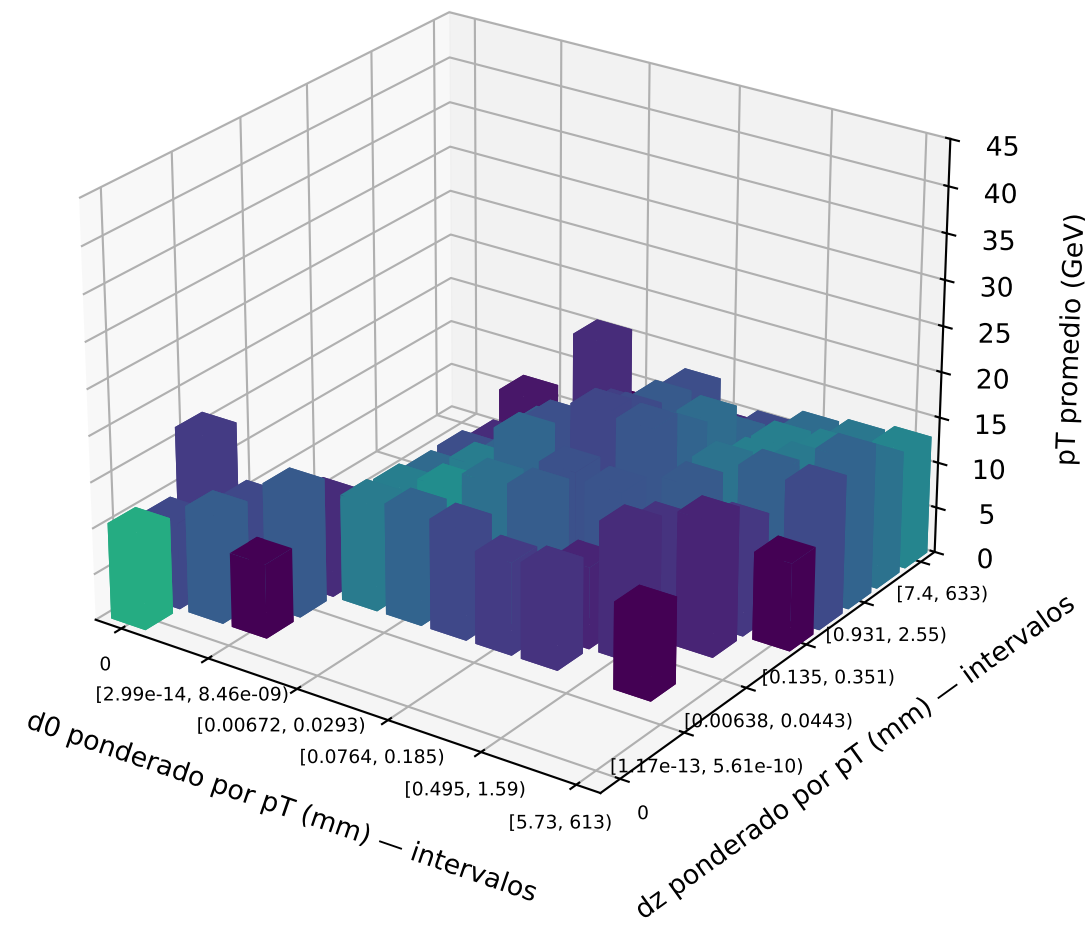
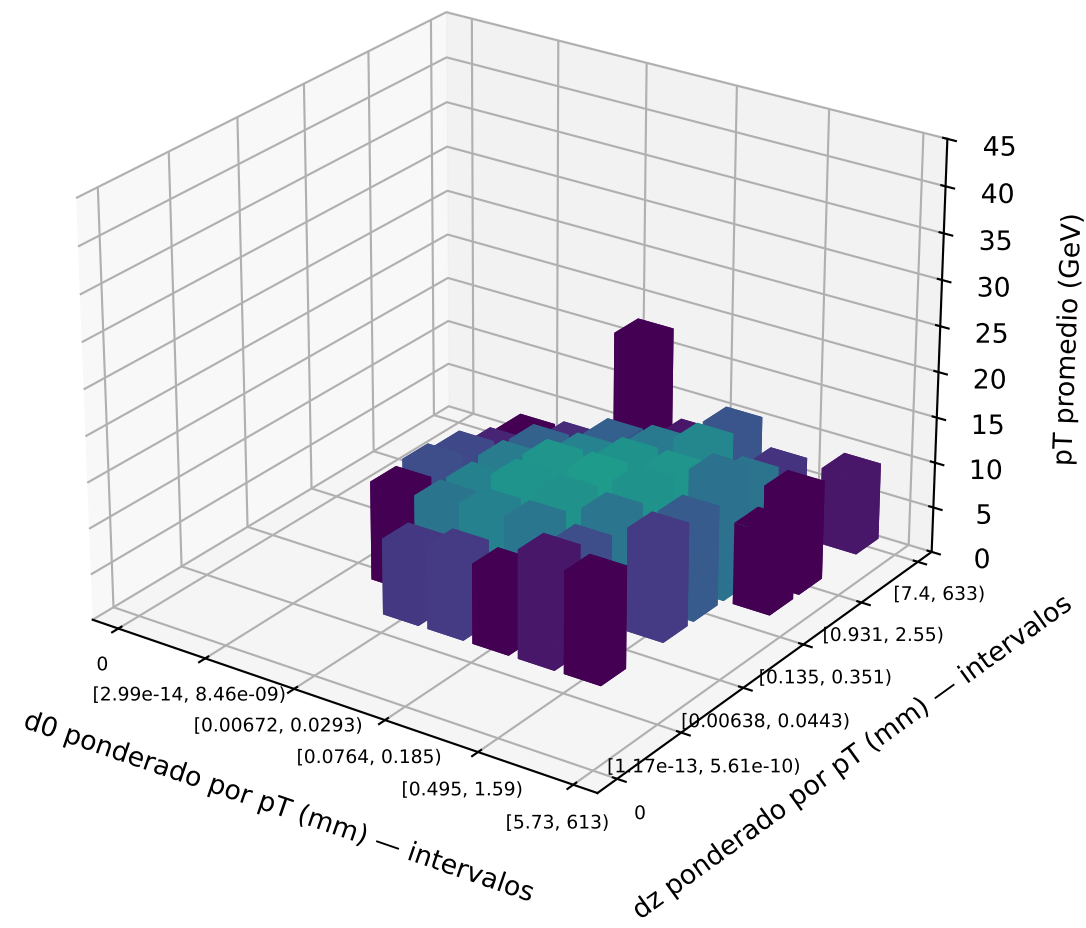
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF36-HGB · seed 73

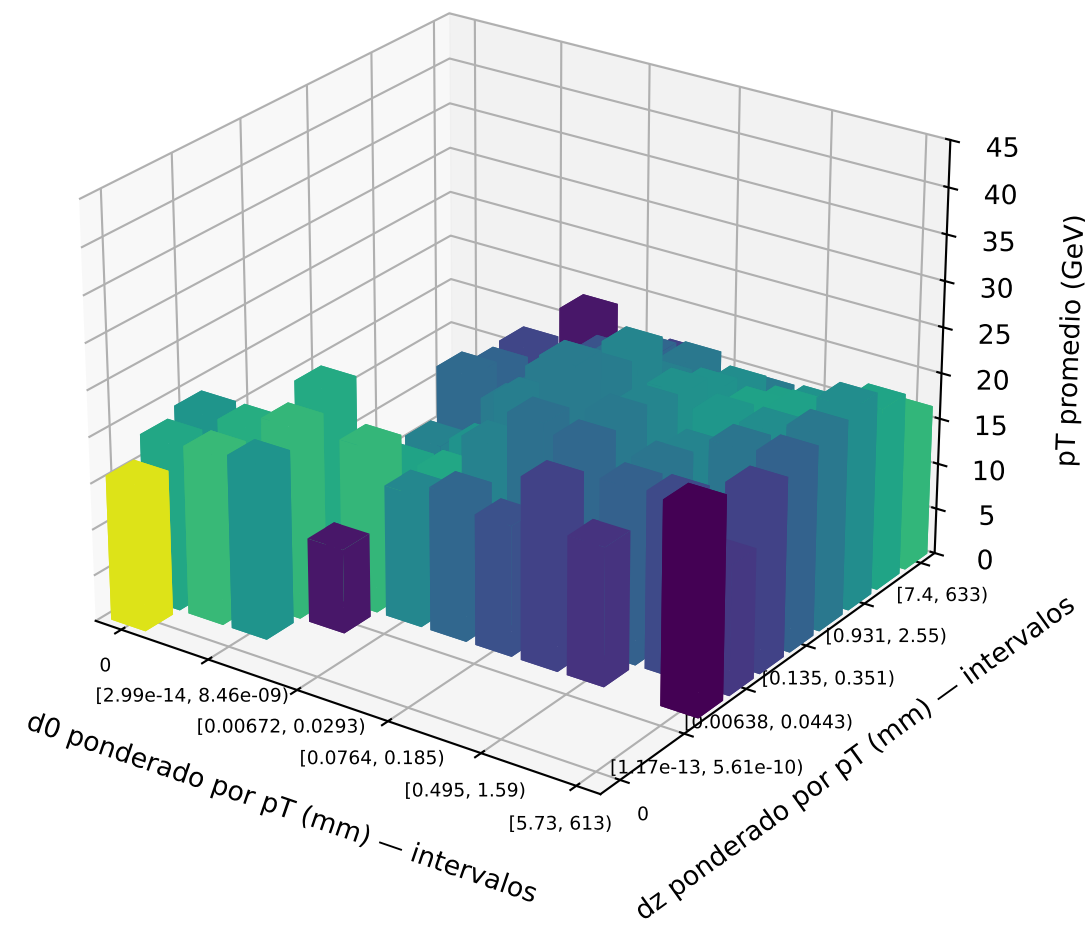
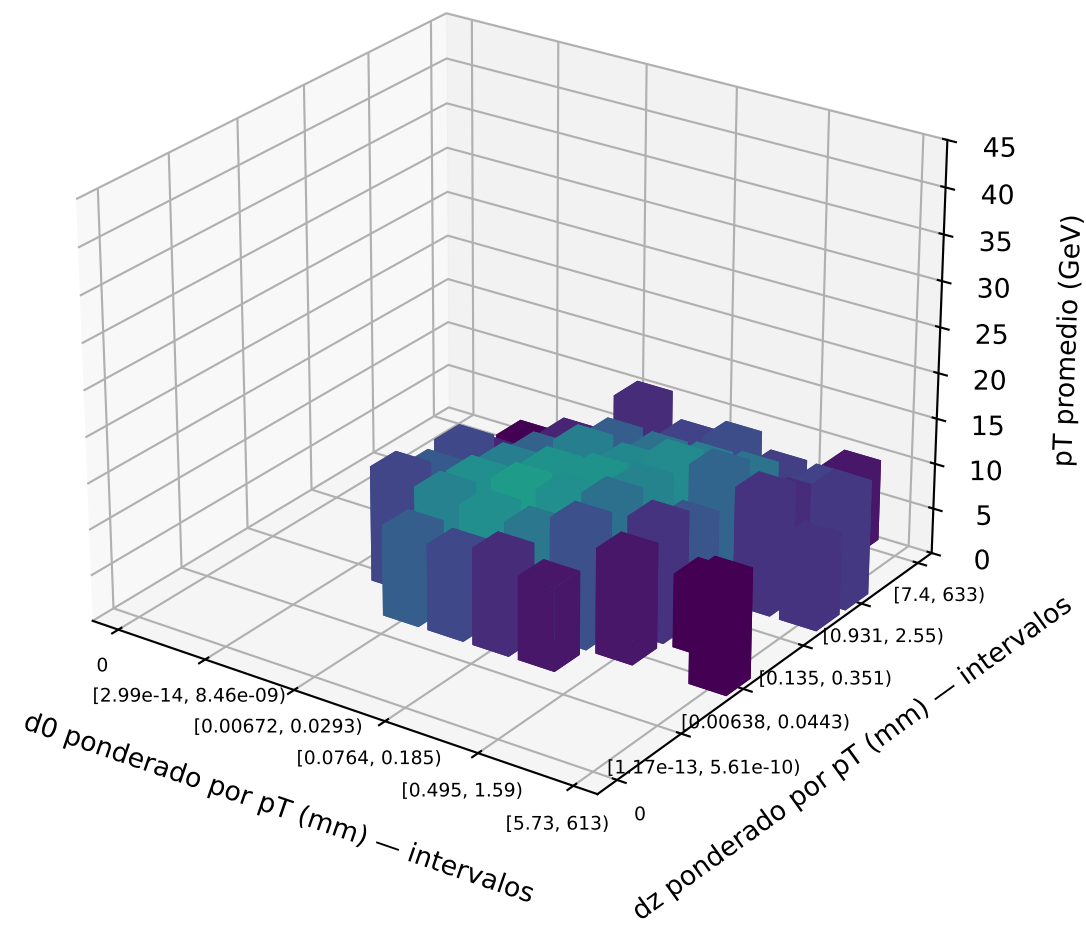
TP — N=3,525

FN — N=3,150



FP — N=3,301

TN — N=57,575



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF36-HGB · seed 73

Variable	Unidad	TP N=3,525	FN N=3,150	FP N=3,301	TN N=57,575
jet_pt *	GeV	12.11 ± 6.045	12.43 ± 7.237	12.03 ± 5.741	16.67 ± 11.85
jet_eta	1	0.009296 ± 1.374	-0.005123 ± 1.459	0.01161 ± 1.396	0.002182 ± 1.501
jet_abs_eta *	1	1.179 ± 0.705	1.256 ± 0.7434	1.189 ± 0.7319	1.28 ± 0.7838
jet_mass	GeV	2.516 ± 1.383	2.546 ± 1.691	2.459 ± 1.234	2.874 ± 1.894
jet_ncharged	count	2.819 ± 1.579	2.611 ± 1.977	2.596 ± 1.406	2.587 ± 1.955
jet_nneutral	count	1.801 ± 1.575	2.11 ± 1.654	1.755 ± 1.429	2.405 ± 1.798
n_tracks_cone	count	2.85 ± 1.567	2.651 ± 1.978	2.628 ± 1.403	2.616 ± 1.956
track_pt_sum *	GeV	5.88 ± 4.387	4.904 ± 5.337	5.364 ± 3.771	6.834 ± 8.228
track_pt_sum_over_jet_pt	1	0.4875 ± 0.2215	0.3748 ± 0.2666	0.4508 ± 0.2261	0.3701 ± 0.2681
avg_track_pt *	GeV	2.273 ± 1.438	1.576 ± 1.195	2.144 ± 1.236	2.167 ± 2.092
std_track_pt	GeV	0.7966 ± 0.8579	0.6695 ± 0.8732	0.7048 ± 0.7951	1.021 ± 1.629
n_tracks_below_avg_pt	count	1.479 ± 1.173	1.393 ± 1.347	1.314 ± 1.072	1.382 ± 1.367
n_tracks_above_avg_pt	count	1.371 ± 0.6581	1.258 ± 0.8551	1.314 ± 0.5763	1.234 ± 0.8125
max_pt_ratio	1	0.291 ± 0.163	0.2055 ± 0.1546	0.2656 ± 0.139	0.2091 ± 0.1555
min_pt_ratio	1	0.1392 ± 0.1535	0.08246 ± 0.09416	0.1325 ± 0.1191	0.08389 ± 0.09716
max_dr_ratio	1	0.1641 ± 0.1523	0.103 ± 0.1014	0.1557 ± 0.1215	0.1025 ± 0.1065
min_dr_ratio	1	0.2512 ± 0.1812	0.1724 ± 0.1565	0.2303 ± 0.151	0.1784 ± 0.1544
dr_max_pt	1	0.1399 ± 0.08526	0.1346 ± 0.09951	0.1475 ± 0.0847	0.1259 ± 0.09929
dr_min_pt	1	0.202 ± 0.1015	0.1908 ± 0.1186	0.2066 ± 0.09725	0.1898 ± 0.1177
dr_max_dr	1	0.2402 ± 0.0978	0.2279 ± 0.1221	0.2389 ± 0.09139	0.2206 ± 0.1212
dr_min_dr	1	0.1104 ± 0.07318	0.1057 ± 0.08562	0.1221 ± 0.07812	0.1033 ± 0.0884
pt_difference	GeV	1.873 ± 2.09	1.634 ± 2.286	1.636 ± 1.848	2.508 ± 4.115
r50	1	0.1463 ± 0.08259	0.1423 ± 0.09736	0.155 ± 0.08191	0.1325 ± 0.09765
r95	1	0.2375 ± 0.09731	0.2254 ± 0.1213	0.2362 ± 0.09153	0.2143 ± 0.1198
mean_abs_d0 *	mm	0.1574 ± 0.1768	2.491 ± 11.46	0.1744 ± 0.6253	1.257 ± 7.932
max_abs_d0 *	mm	0.2819 ± 0.3531	6.783 ± 21.98	0.3191 ± 0.8693	3.334 ± 16.69
std_abs_d0 *	mm	0.09237 ± 0.1483	2.614 ± 8.51	0.1036 ± 0.2999	1.284 ± 6.707
mean_abs_dz *	mm	0.3915 ± 0.5033	3.682 ± 17.15	0.3351 ± 0.55	1.677 ± 9.731
max_abs_dz *	mm	0.6895 ± 0.954	9.583 ± 28.14	0.6438 ± 1.17	4.392 ± 23.16
std_abs_dz *	mm	0.2201 ± 0.3875	3.641 ± 10.1	0.2207 ± 0.4867	1.703 ± 9.553
pt_weighted_abs_d0 *	mm	0.165 ± 0.1929	2.339 ± 11.35	0.1881 ± 0.681	1.197 ± 7.944
pt_weighted_abs_dz *	mm	0.4097 ± 0.5486	3.15 ± 16.91	0.3587 ± 0.6051	1.376 ± 8.425

* Variable utilizada como entrada del modelo.

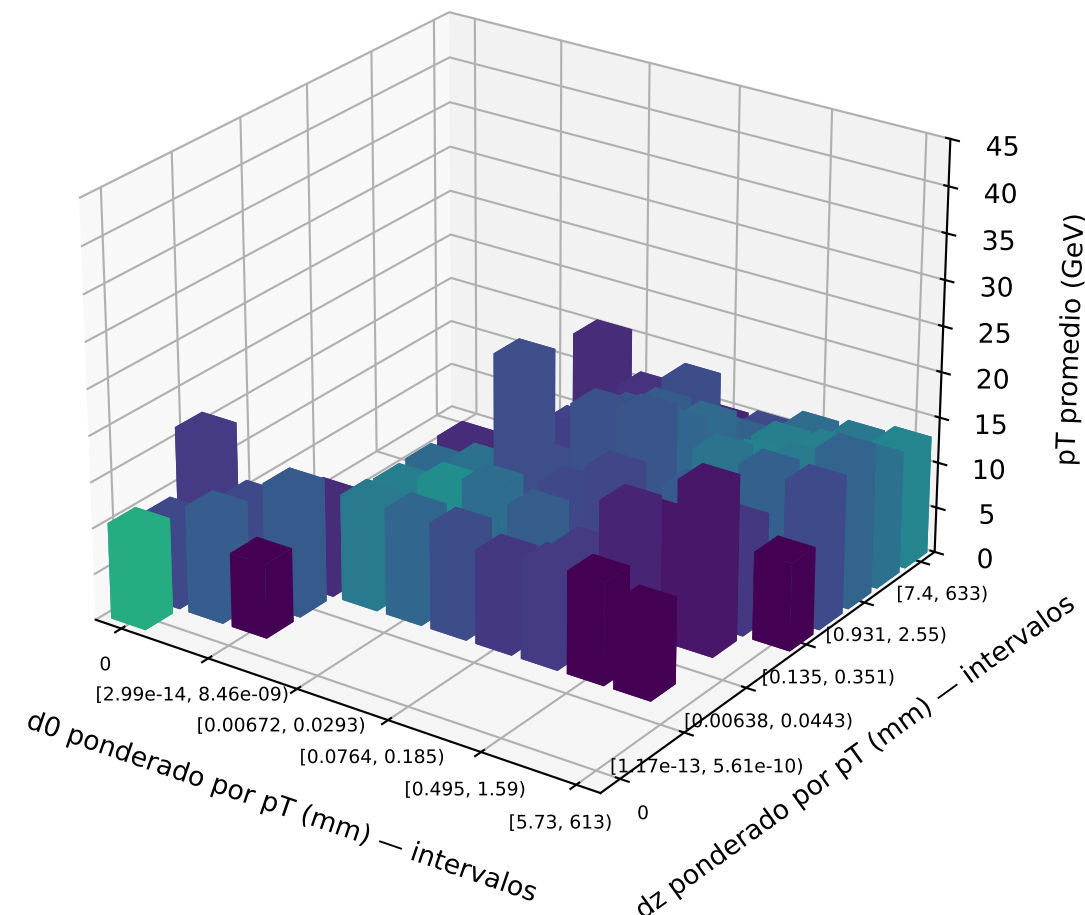
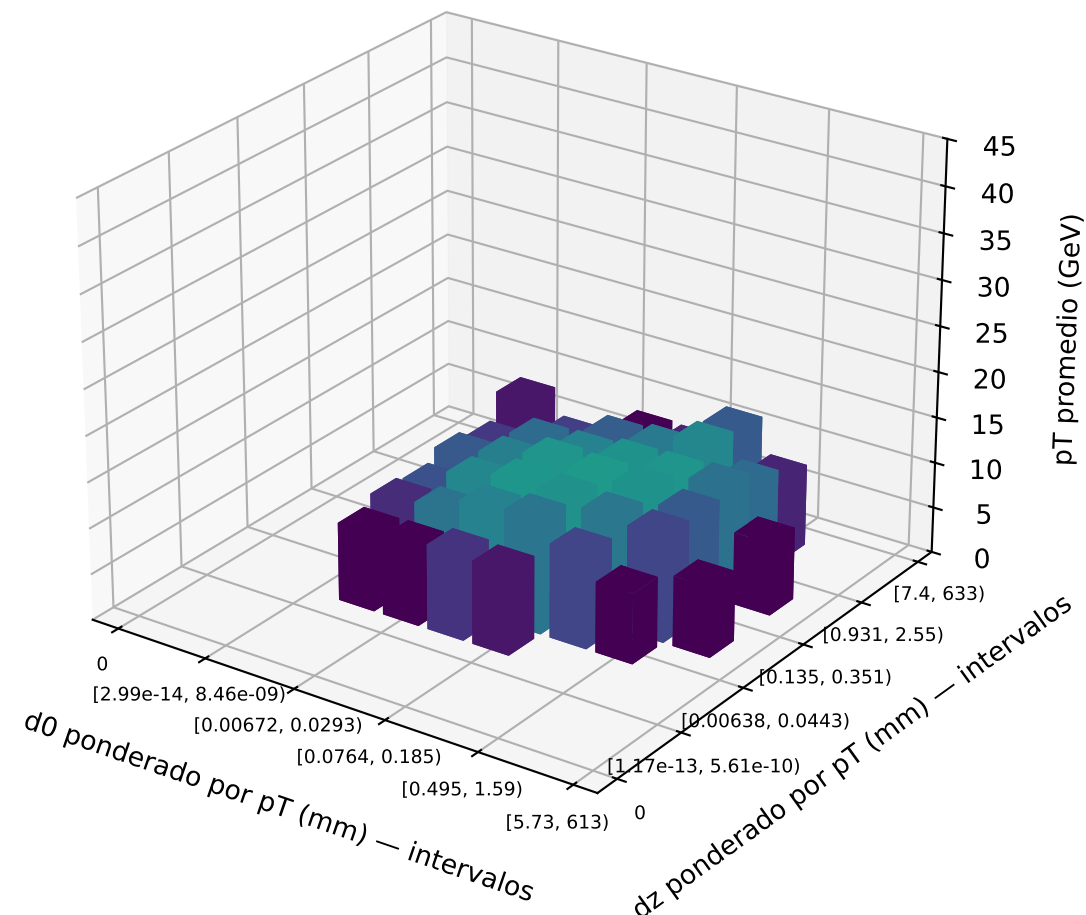
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF36-HGB · seed 101

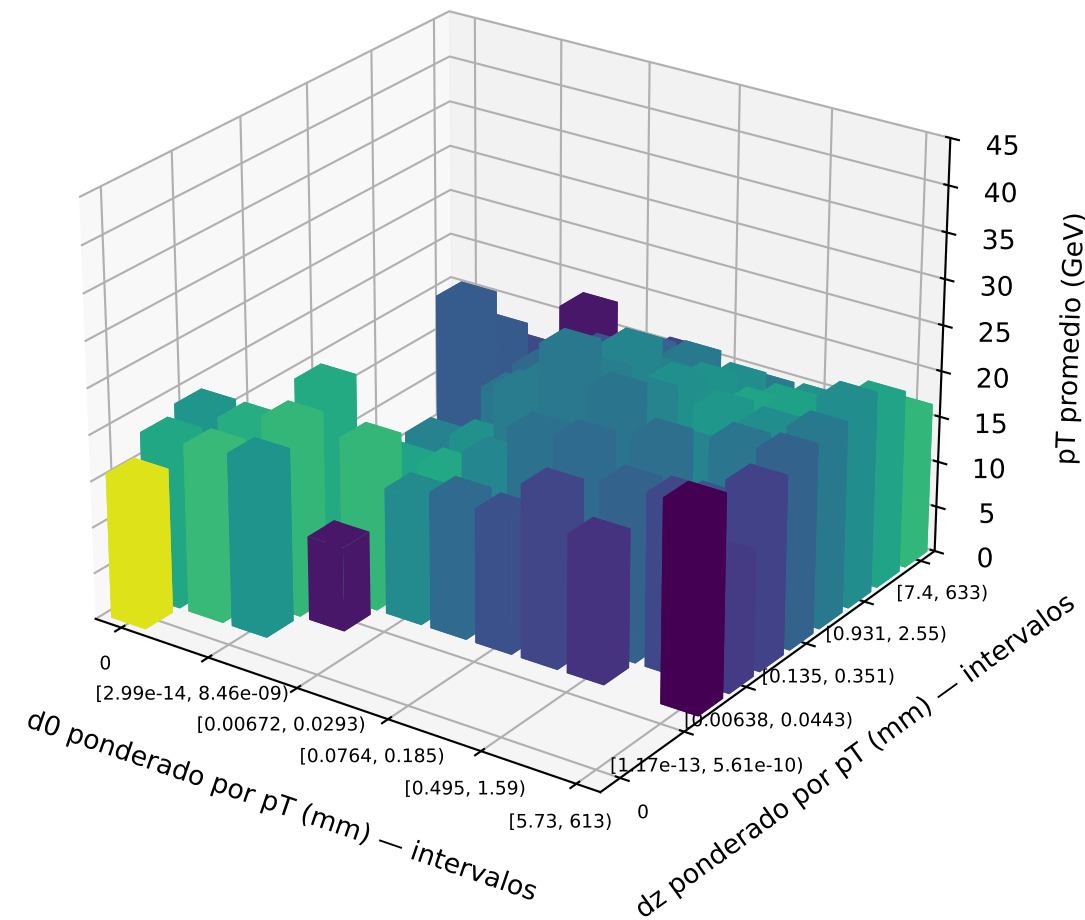
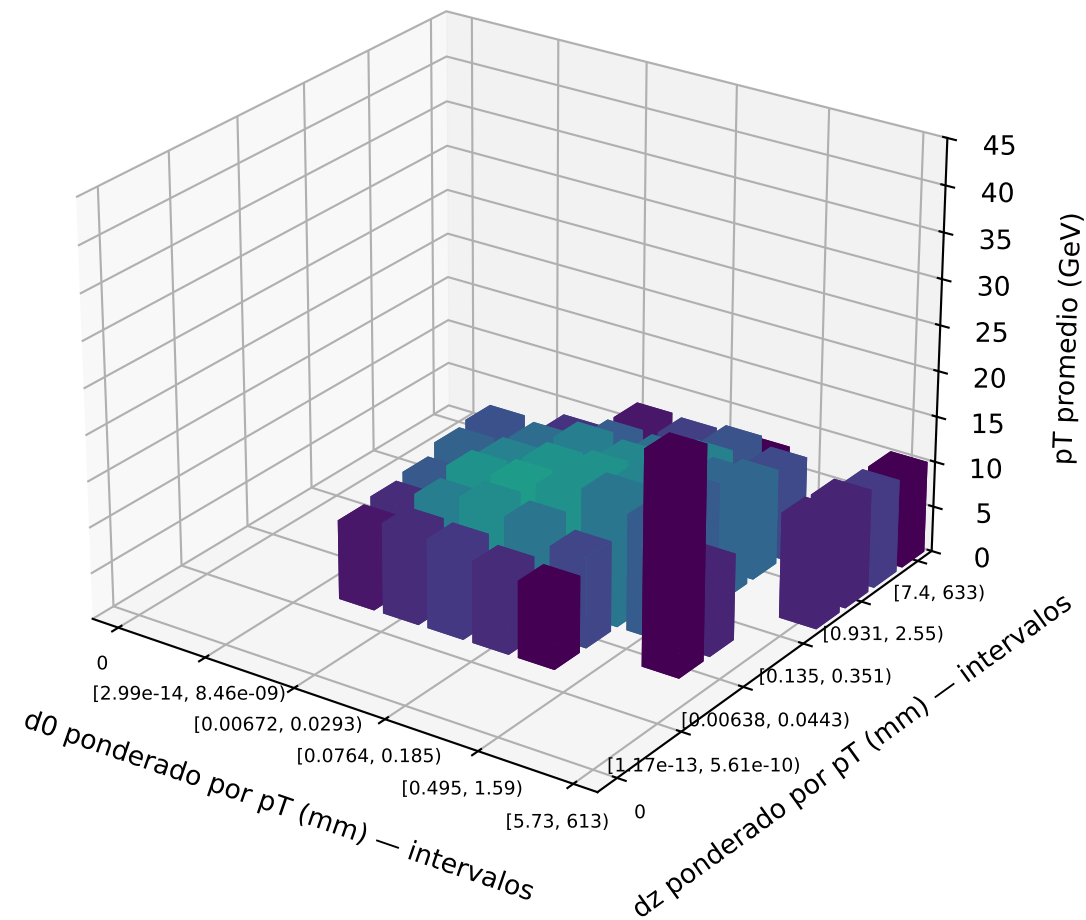
TP — N=3,620

FN — N=3,055



FP — N=3,359

TN — N=57,517



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF36-HGB · seed 101

Variable	Unidad	TP N=3,620	FN N=3,055	FP N=3,359	TN N=57,517
jet_pt *	GeV	12.15 ± 6.126	12.39 ± 7.193	12.11 ± 5.666	16.67 ± 11.86
jet_eta	1	0.003714 ± 1.38	0.001042 ± 1.455	0.006647 ± 1.415	0.002462 ± 1.5
jet_abs_eta *	1	1.186 ± 0.705	1.25 ± 0.7453	1.213 ± 0.7288	1.279 ± 0.7841
jet_mass	GeV	2.516 ± 1.381	2.546 ± 1.702	2.481 ± 1.254	2.873 ± 1.894
jet_ncharged	count	2.819 ± 1.575	2.605 ± 1.991	2.621 ± 1.434	2.586 ± 1.955
jet_nneutral	count	1.812 ± 1.586	2.107 ± 1.645	1.78 ± 1.399	2.404 ± 1.8
n_tracks_cone	count	2.849 ± 1.561	2.646 ± 1.995	2.647 ± 1.425	2.615 ± 1.955
track_pt_sum *	GeV	5.851 ± 4.275	4.909 ± 5.474	5.388 ± 3.872	6.834 ± 8.229
track_pt_sum_over_jet_pt	1	0.4849 ± 0.2221	0.3744 ± 0.2678	0.4467 ± 0.2258	0.3703 ± 0.2682
avg_track_pt *	GeV	2.271 ± 1.446	1.557 ± 1.17	2.135 ± 1.242	2.168 ± 2.093
std_track_pt	GeV	0.7914 ± 0.8635	0.6717 ± 0.8676	0.688 ± 0.7633	1.023 ± 1.63
n_tracks_below_avg_pt	count	1.473 ± 1.169	1.397 ± 1.356	1.315 ± 1.071	1.382 ± 1.367
n_tracks_above_avg_pt	count	1.376 ± 0.6554	1.249 ± 0.8619	1.332 ± 0.601	1.233 ± 0.8116
max_pt_ratio	1	0.2895 ± 0.1645	0.2045 ± 0.1526	0.2613 ± 0.1404	0.2093 ± 0.1555
min_pt_ratio	1	0.1392 ± 0.1543	0.08067 ± 0.08963	0.1325 ± 0.1226	0.08384 ± 0.09687
max_dr_ratio	1	0.1639 ± 0.1531	0.1014 ± 0.09758	0.1557 ± 0.1246	0.1025 ± 0.1063
min_dr_ratio	1	0.2502 ± 0.1822	0.1711 ± 0.1542	0.2261 ± 0.1518	0.1786 ± 0.1544
dr_max_pt	1	0.1397 ± 0.08521	0.1346 ± 0.09999	0.1491 ± 0.08562	0.1258 ± 0.09923
dr_min_pt	1	0.2018 ± 0.1017	0.1907 ± 0.1188	0.208 ± 0.09715	0.1897 ± 0.1177
dr_max_dr	1	0.2406 ± 0.09851	0.2271 ± 0.122	0.24 ± 0.09136	0.2205 ± 0.1213
dr_min_dr	1	0.1101 ± 0.07276	0.106 ± 0.08642	0.1229 ± 0.0788	0.1032 ± 0.08836
pt_difference	GeV	1.862 ± 2.094	1.64 ± 2.288	1.607 ± 1.806	2.51 ± 4.118
r50	1	0.1464 ± 0.08268	0.142 ± 0.09768	0.157 ± 0.08272	0.1324 ± 0.0976
r95	1	0.2381 ± 0.09797	0.2243 ± 0.1212	0.2376 ± 0.09134	0.2142 ± 0.1198
mean_abs_d0 *	mm	0.1481 ± 0.1551	2.574 ± 11.62	0.4003 ± 4.713	1.245 ± 7.857
max_abs_d0 *	mm	0.2648 ± 0.3016	7.005 ± 22.28	0.5384 ± 4.721	3.325 ± 16.66
std_abs_d0 *	mm	0.08603 ± 0.1178	2.7 ± 8.628	0.09778 ± 0.1832	1.285 ± 6.711
mean_abs_dz *	mm	0.3807 ± 0.5158	3.797 ± 17.4	0.346 ± 0.6053	1.678 ± 9.735
max_abs_dz *	mm	0.6575 ± 0.8259	9.898 ± 28.52	0.6254 ± 0.924	4.397 ± 23.17
std_abs_dz *	mm	0.2071 ± 0.3061	3.763 ± 10.23	0.2106 ± 0.372	1.705 ± 9.558
pt_weighted_abs_d0 *	mm	0.1535 ± 0.154	2.42 ± 11.52	0.4063 ± 4.713	1.185 ± 7.869
pt_weighted_abs_dz *	mm	0.3957 ± 0.5249	3.252 ± 17.16	0.3606 ± 0.6233	1.376 ± 8.429

* Variable utilizada como entrada del modelo.

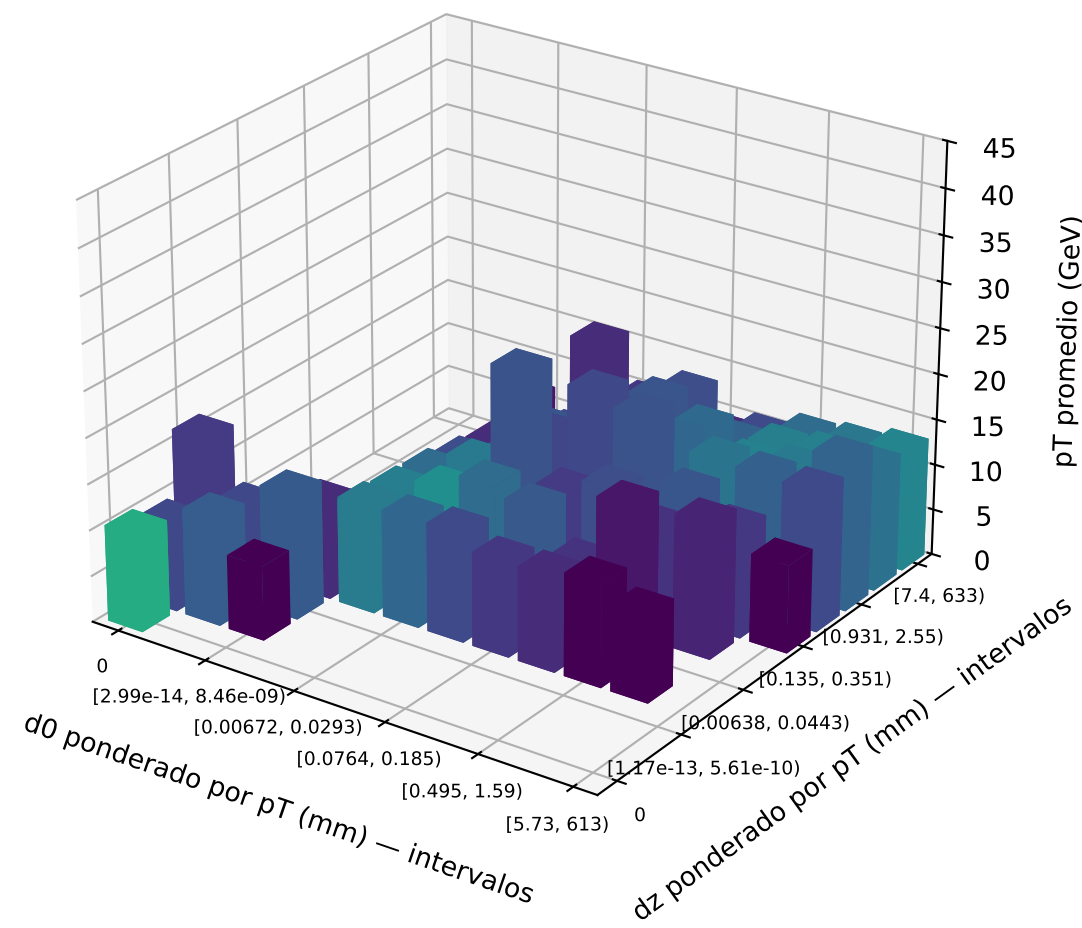
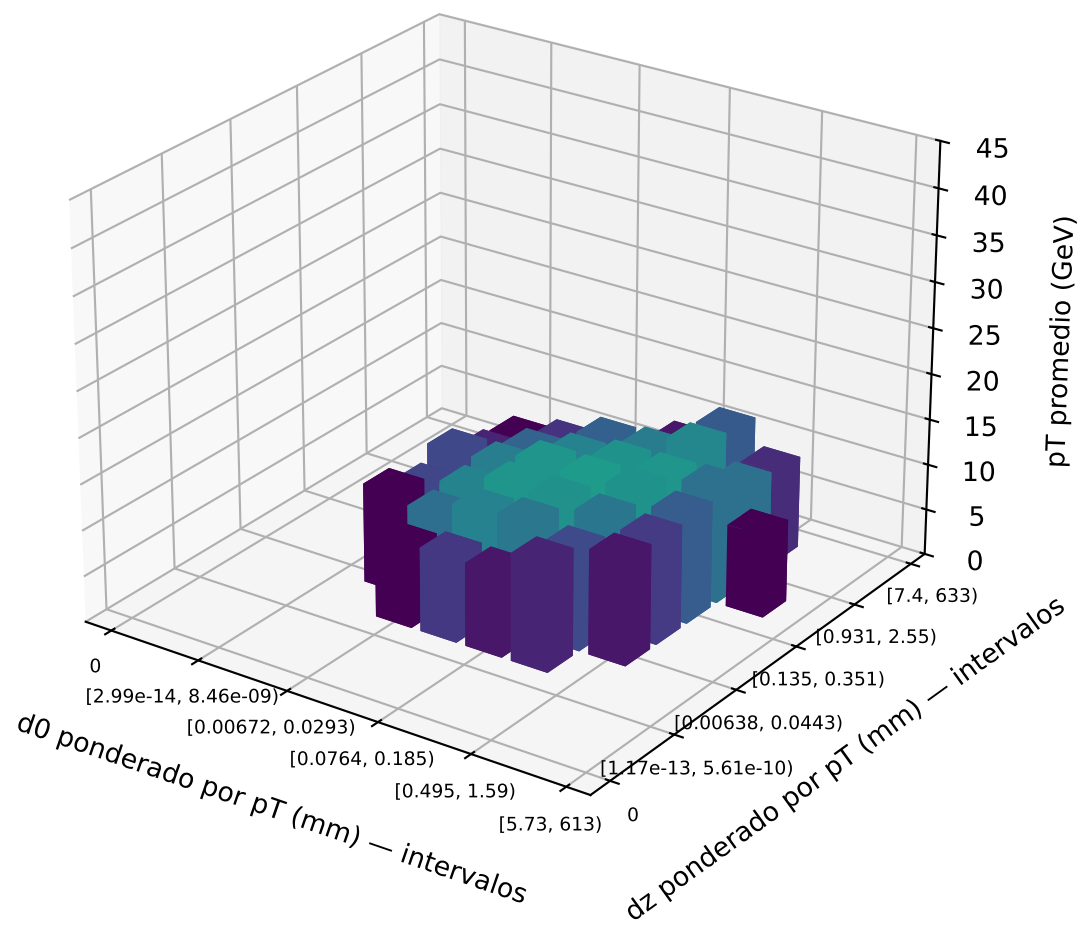
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF36-HGB · ensemble

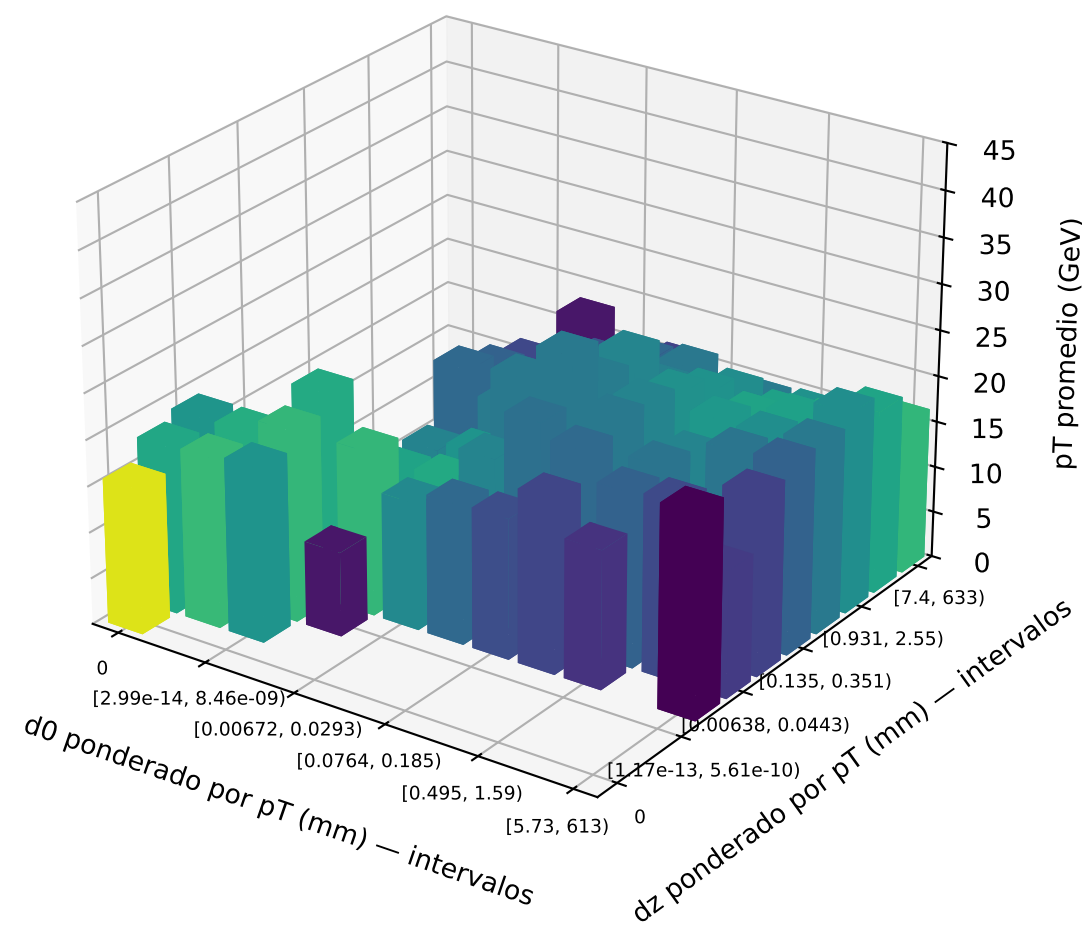
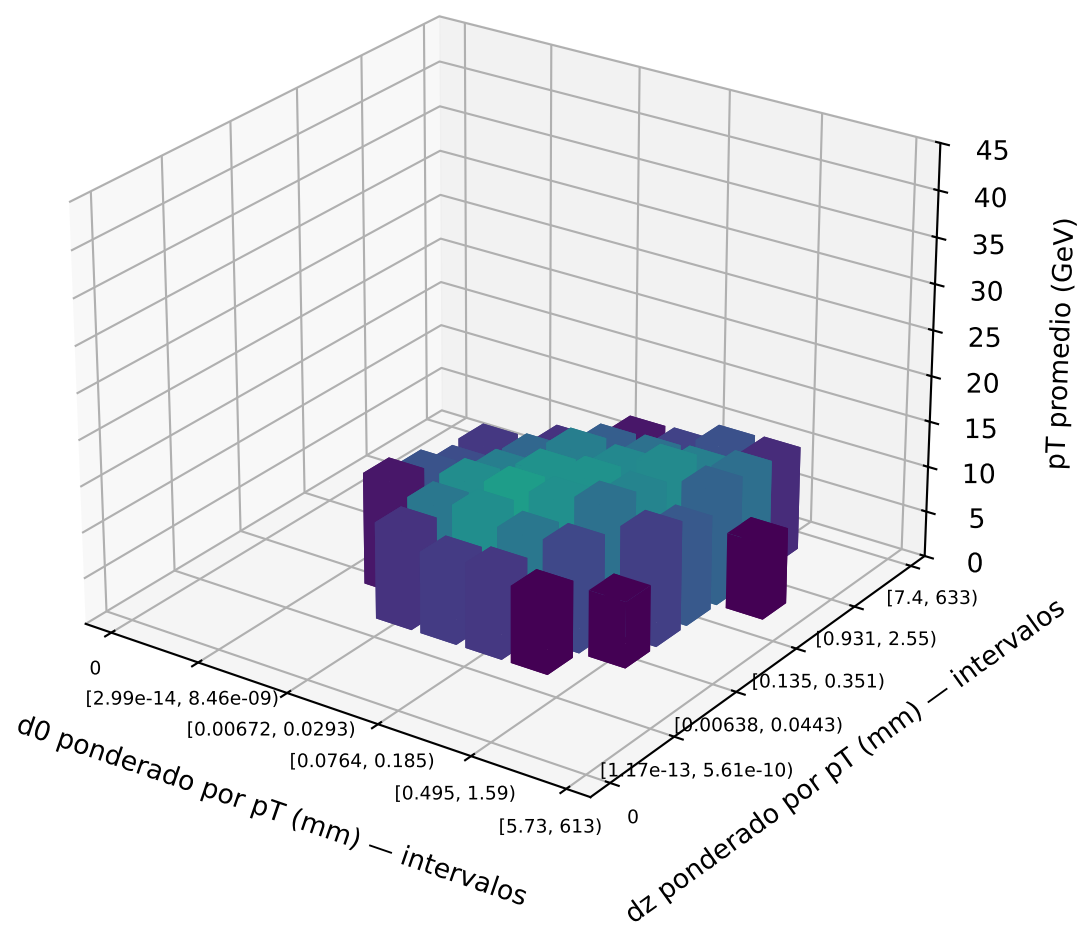
TP — N=3,571

FN — N=3,104



FP — N=3,184

TN — N=57,692



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF36-HGB · ensemble

Variable	Unidad	TP N=3,571	FN N=3,104	FP N=3,184	TN N=57,692
jet_pt *	GeV	12.13 ± 5.994	12.41 ± 7.303	11.96 ± 5.465	16.66 ± 11.85
jet_eta	1	0.002584 ± 1.373	0.002385 ± 1.461	0.002712 ± 1.406	0.002692 ± 1.501
jet_abs_eta *	1	1.179 ± 0.7039	1.257 ± 0.7451	1.203 ± 0.729	1.28 ± 0.7839
jet_mass	GeV	2.512 ± 1.368	2.551 ± 1.71	2.443 ± 1.21	2.874 ± 1.894
jet_ncharged	count	2.804 ± 1.561	2.625 ± 2	2.578 ± 1.384	2.588 ± 1.955
jet_nneutral	count	1.809 ± 1.574	2.106 ± 1.657	1.753 ± 1.384	2.403 ± 1.799
n_tracks_cone	count	2.834 ± 1.55	2.667 ± 2.001	2.606 ± 1.372	2.617 ± 1.956
track_pt_sum *	GeV	5.83 ± 4.232	4.947 ± 5.5	5.318 ± 3.694	6.833 ± 8.223
track_pt_sum_over_jet_pt	1	0.4842 ± 0.2217	0.377 ± 0.2682	0.4483 ± 0.2252	0.3704 ± 0.2681
avg_track_pt *	GeV	2.277 ± 1.451	1.561 ± 1.168	2.145 ± 1.235	2.167 ± 2.091
std_track_pt	GeV	0.7873 ± 0.8532	0.6784 ± 0.8799	0.6838 ± 0.7561	1.022 ± 1.628
n_tracks_below_avg_pt	count	1.464 ± 1.166	1.409 ± 1.357	1.289 ± 1.039	1.383 ± 1.368
n_tracks_above_avg_pt	count	1.369 ± 0.6481	1.258 ± 0.8665	1.317 ± 0.5832	1.234 ± 0.8118
max_pt_ratio	1	0.2899 ± 0.1646	0.2054 ± 0.1528	0.2636 ± 0.1392	0.2093 ± 0.1555
min_pt_ratio	1	0.14 ± 0.155	0.08064 ± 0.0892	0.1341 ± 0.1213	0.0839 ± 0.09704
max_dr_ratio	1	0.1646 ± 0.1537	0.1015 ± 0.09723	0.1572 ± 0.1236	0.1026 ± 0.1064
min_dr_ratio	1	0.2504 ± 0.1824	0.1721 ± 0.1547	0.2287 ± 0.1505	0.1786 ± 0.1545
dr_max_pt	1	0.1404 ± 0.08539	0.1339 ± 0.09955	0.1486 ± 0.08511	0.1259 ± 0.09924
dr_min_pt	1	0.2017 ± 0.1014	0.191 ± 0.1189	0.2065 ± 0.09686	0.1899 ± 0.1177
dr_max_dr	1	0.2402 ± 0.09792	0.2277 ± 0.1223	0.2384 ± 0.09127	0.2207 ± 0.1212
dr_min_dr	1	0.1106 ± 0.07332	0.1054 ± 0.08565	0.1226 ± 0.07796	0.1033 ± 0.08838
pt_difference	GeV	1.85 ± 2.071	1.658 ± 2.31	1.587 ± 1.774	2.509 ± 4.114
r50	1	0.1468 ± 0.08287	0.1416 ± 0.09725	0.1559 ± 0.08215	0.1325 ± 0.09761
r95	1	0.2377 ± 0.09751	0.2249 ± 0.1214	0.2362 ± 0.0915	0.2143 ± 0.1197
mean_abs_d0 *	mm	0.153 ± 0.155	2.53 ± 11.54	0.1252 ± 0.1553	1.258 ± 7.925
max_abs_d0 *	mm	0.2704 ± 0.2761	6.893 ± 22.12	0.2426 ± 0.2903	3.332 ± 16.67
std_abs_d0 *	mm	0.08769 ± 0.1112	2.657 ± 8.567	0.08594 ± 0.1255	1.282 ± 6.701
mean_abs_dz *	mm	0.3851 ± 0.477	3.738 ± 17.27	0.3261 ± 0.4832	1.675 ± 9.721
max_abs_dz *	mm	0.6657 ± 0.7932	9.742 ± 28.32	0.5916 ± 0.7617	4.388 ± 23.14
std_abs_dz *	mm	0.2106 ± 0.3044	3.703 ± 10.16	0.2005 ± 0.3149	1.701 ± 9.544
pt_weighted_abs_d0 *	mm	0.1596 ± 0.1579	2.377 ± 11.43	0.1315 ± 0.1602	1.198 ± 7.937
pt_weighted_abs_dz *	mm	0.4023 ± 0.4948	3.2 ± 17.03	0.3442 ± 0.5083	1.374 ± 8.416

* Variable utilizada como entrada del modelo.

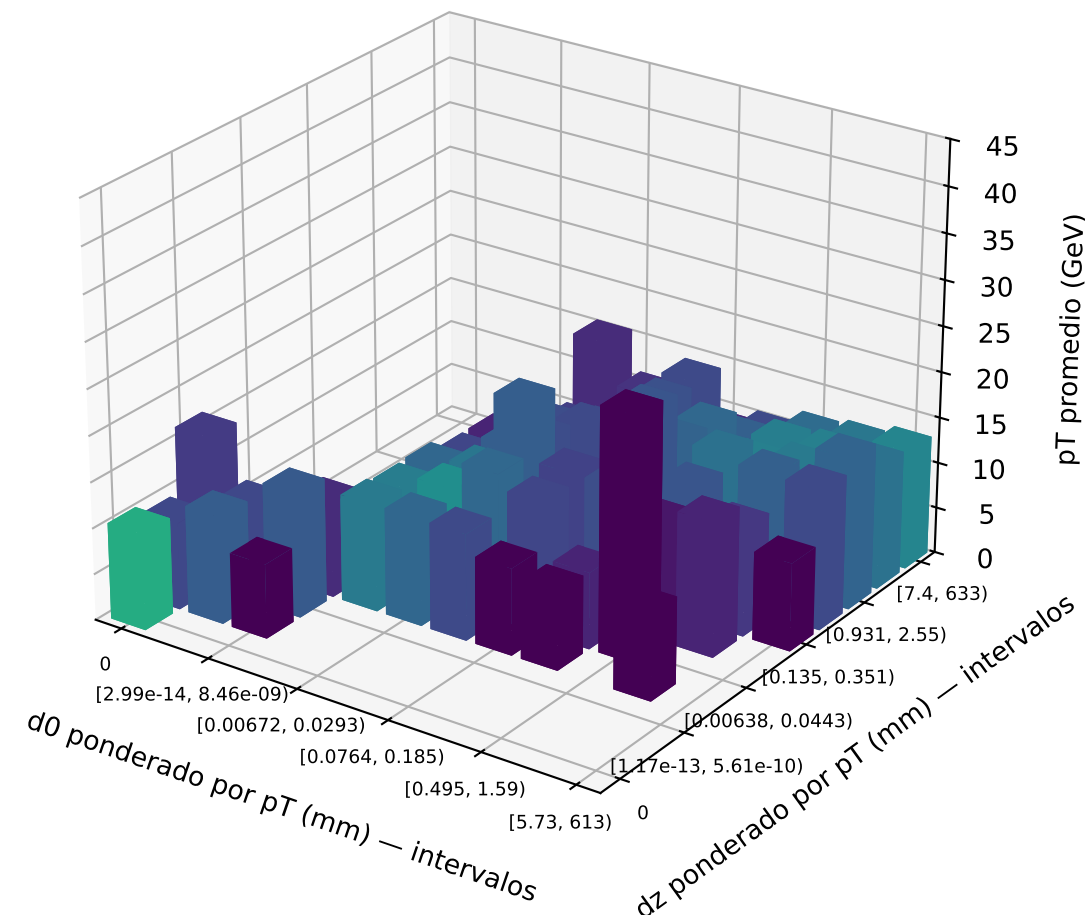
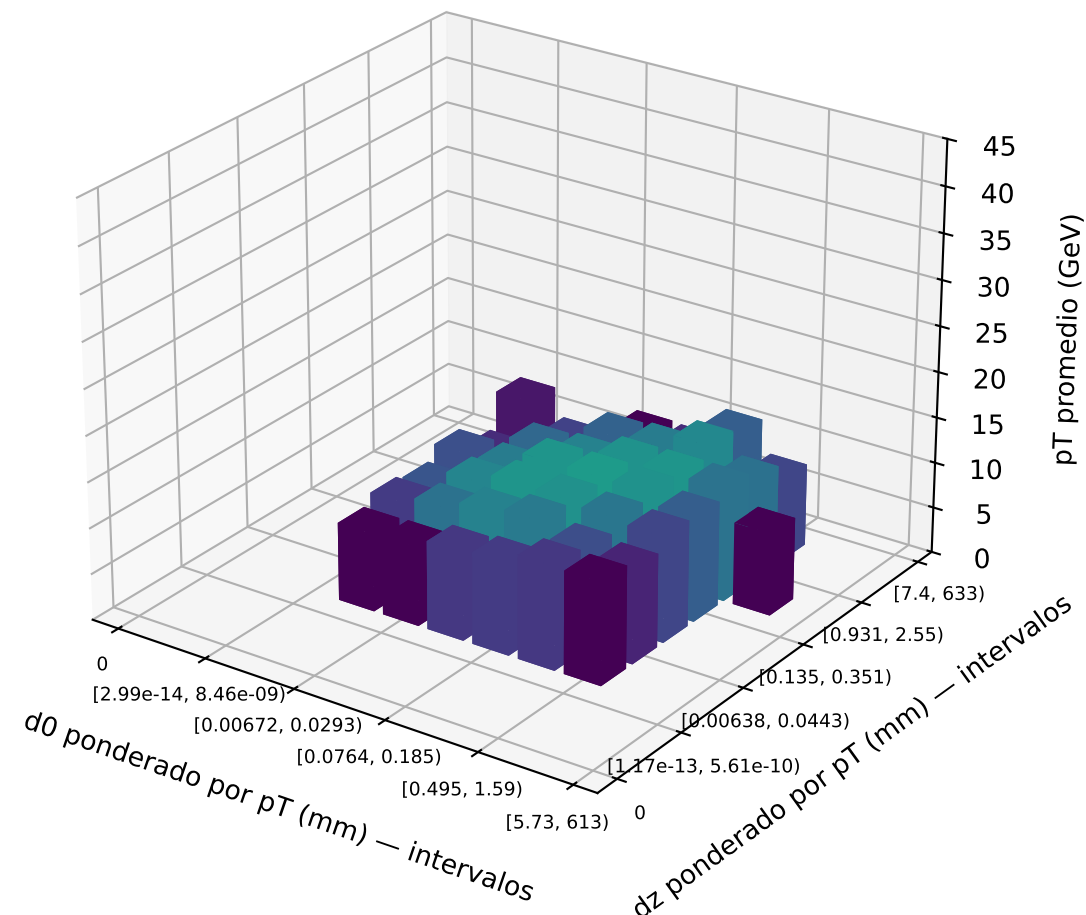
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF48-HGB · seed 42

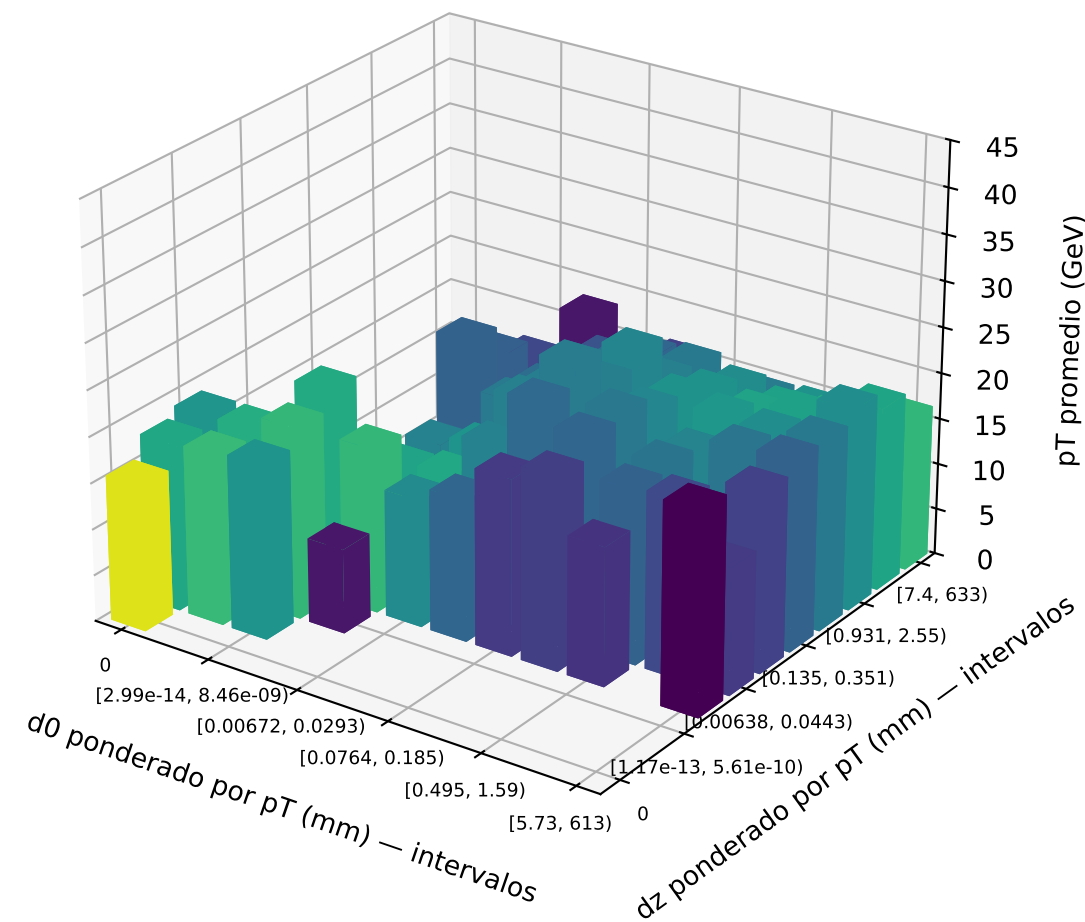
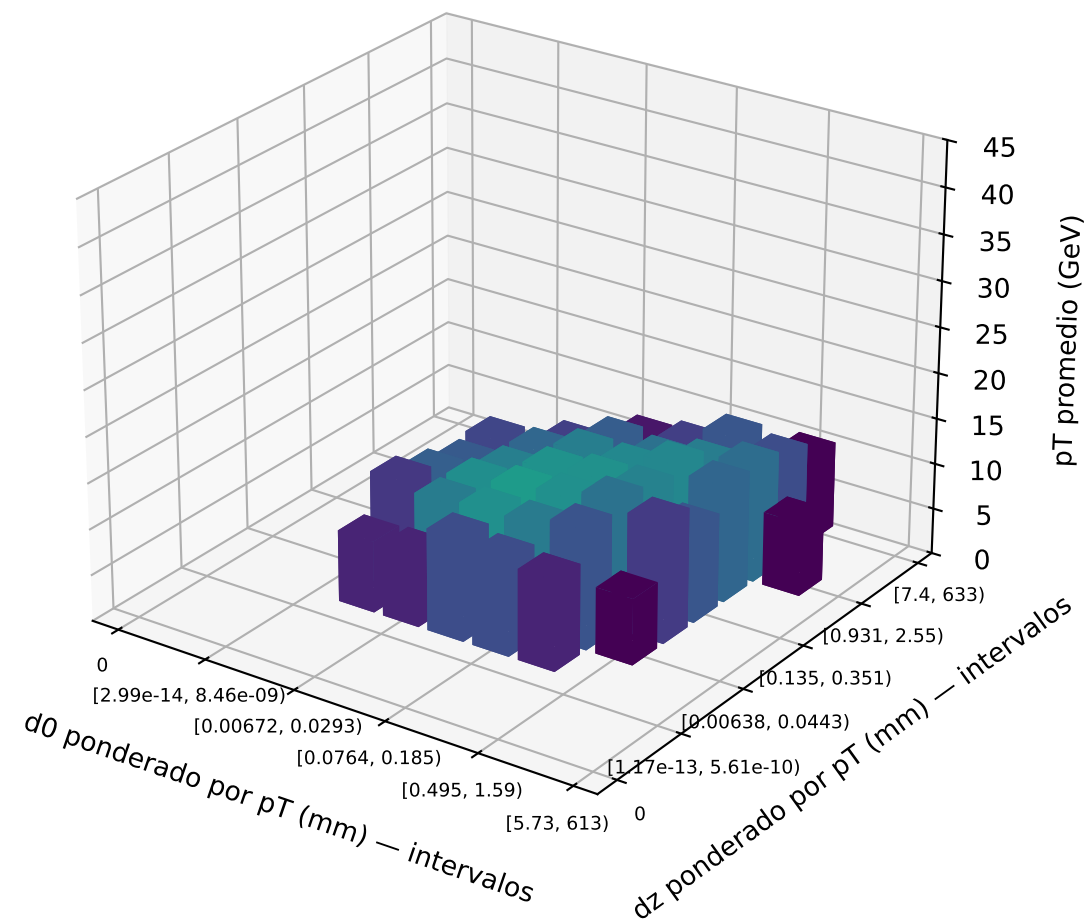
TP — N=3,684

FN — N=2,991



FP — N=3,205

TN — N=57,671



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF48-HGB · seed 42

Variable	Unidad	TP N=3,684	FN N=2,991	FP N=3,205	TN N=57,671
jet_pt *	GeV	12.12 ± 6.065	12.43 ± 7.275	11.93 ± 5.457	16.66 ± 11.85
jet_eta	1	0.002407 ± 1.371	0.002595 ± 1.466	0.004946 ± 1.377	0.002568 ± 1.502
jet_abs_eta *	1	1.178 ± 0.7024	1.261 ± 0.748	1.173 ± 0.7199	1.281 ± 0.7842
jet_mass	GeV	2.506 ± 1.377	2.56 ± 1.711	2.391 ± 1.187	2.877 ± 1.894
jet_ncharged	count	2.797 ± 1.576	2.627 ± 2.001	2.51 ± 1.373	2.592 ± 1.956
jet_nneutral	count	1.799 ± 1.559	2.129 ± 1.674	1.735 ± 1.368	2.405 ± 1.8
n_tracks_cone	count	2.827 ± 1.566	2.668 ± 2.001	2.529 ± 1.366	2.621 ± 1.956
track_pt_sum *	GeV	5.834 ± 4.321	4.909 ± 5.454	5.24 ± 3.778	6.838 ± 8.221
track_pt_sum_over_jet_pt	1	0.4833 ± 0.2233	0.3741 ± 0.2678	0.441 ± 0.2265	0.3708 ± 0.2682
avg_track_pt *	GeV	2.278 ± 1.462	1.533 ± 1.129	2.18 ± 1.304	2.165 ± 2.089
std_track_pt	GeV	0.7888 ± 0.8675	0.6724 ± 0.8631	0.6755 ± 0.7996	1.022 ± 1.627
n_tracks_below_avg_pt	count	1.458 ± 1.175	1.415 ± 1.354	1.232 ± 1.039	1.386 ± 1.367
n_tracks_above_avg_pt	count	1.37 ± 0.6504	1.254 ± 0.8712	1.297 ± 0.5672	1.235 ± 0.8126
max_pt_ratio	1	0.2899 ± 0.1653	0.2022 ± 0.1504	0.2651 ± 0.1421	0.2092 ± 0.1554
min_pt_ratio	1	0.1402 ± 0.1556	0.07816 ± 0.08331	0.1391 ± 0.1265	0.0836 ± 0.0965
max_dr_ratio	1	0.1645 ± 0.1542	0.09924 ± 0.09268	0.1614 ± 0.1281	0.1023 ± 0.106
min_dr_ratio	1	0.2515 ± 0.1829	0.1678 ± 0.1513	0.231 ± 0.1534	0.1785 ± 0.1543
dr_max_pt	1	0.1399 ± 0.08566	0.1343 ± 0.09979	0.1472 ± 0.08454	0.126 ± 0.09928
dr_min_pt	1	0.2019 ± 0.1019	0.1904 ± 0.119	0.2033 ± 0.09693	0.19 ± 0.1177
dr_max_dr	1	0.2401 ± 0.09852	0.2274 ± 0.1225	0.2336 ± 0.092	0.2209 ± 0.1212
dr_min_dr	1	0.1108 ± 0.07389	0.105 ± 0.08546	0.123 ± 0.07861	0.1033 ± 0.08835
pt_difference	GeV	1.856 ± 2.115	1.643 ± 2.268	1.561 ± 1.861	2.51 ± 4.112
r50	1	0.1463 ± 0.08297	0.1421 ± 0.09768	0.1541 ± 0.08188	0.1326 ± 0.09765
r95	1	0.2375 ± 0.09801	0.2247 ± 0.1217	0.2313 ± 0.0921	0.2146 ± 0.1198
mean_abs_d0 *	mm	0.1563 ± 0.1684	2.616 ± 11.74	0.1256 ± 0.1698	1.258 ± 7.927
max_abs_d0 *	mm	0.2769 ± 0.3152	7.135 ± 22.5	0.2346 ± 0.379	3.334 ± 16.68
std_abs_d0 *	mm	0.08958 ± 0.1261	2.752 ± 8.712	0.0815 ± 0.1578	1.283 ± 6.702
mean_abs_dz *	mm	0.3992 ± 0.5314	3.847 ± 17.58	0.3382 ± 0.5924	1.675 ± 9.723
max_abs_dz *	mm	0.6876 ± 0.8887	10.06 ± 28.8	0.5935 ± 1.106	4.389 ± 23.14
std_abs_dz *	mm	0.216 ± 0.3344	3.828 ± 10.33	0.195 ± 0.4368	1.702 ± 9.545
pt_weighted_abs_d0 *	mm	0.1618 ± 0.1667	2.458 ± 11.64	0.1298 ± 0.1634	1.199 ± 7.938
pt_weighted_abs_dz *	mm	0.4139 ± 0.539	3.291 ± 17.34	0.3524 ± 0.6342	1.374 ± 8.418

* Variable utilizada como entrada del modelo.

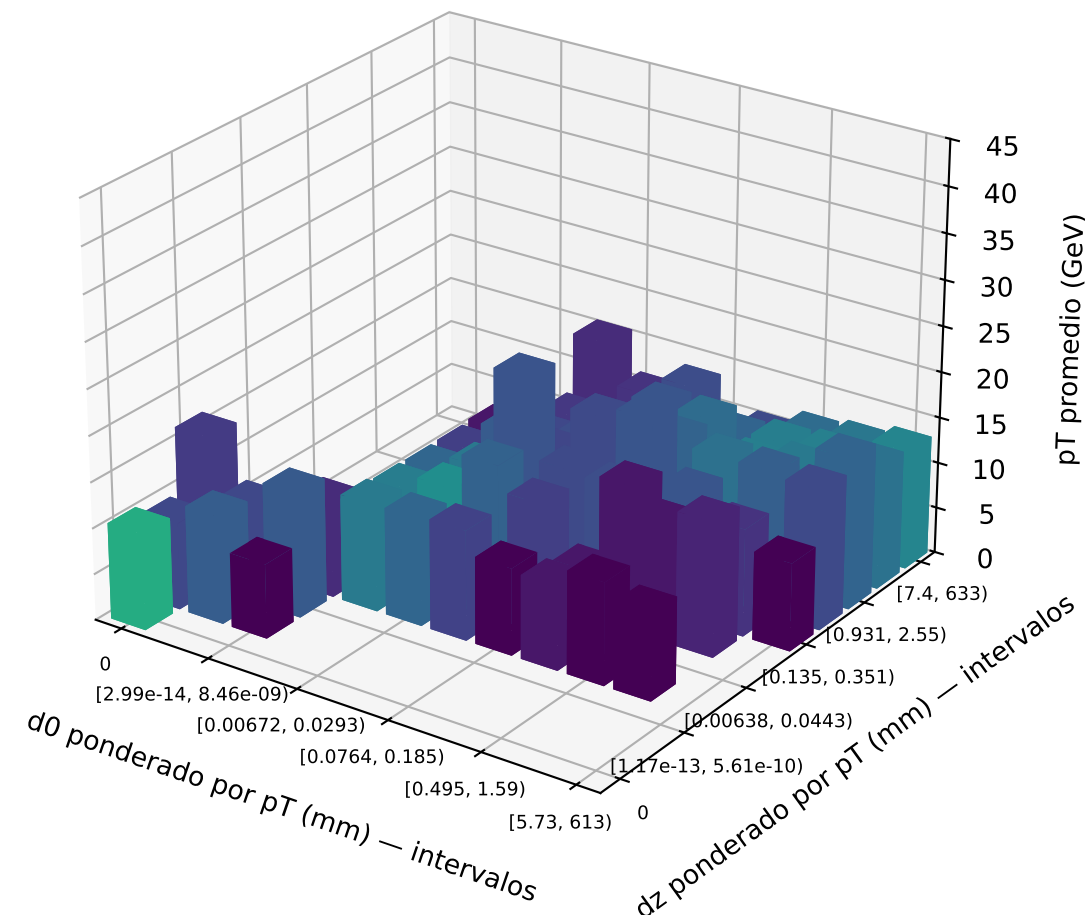
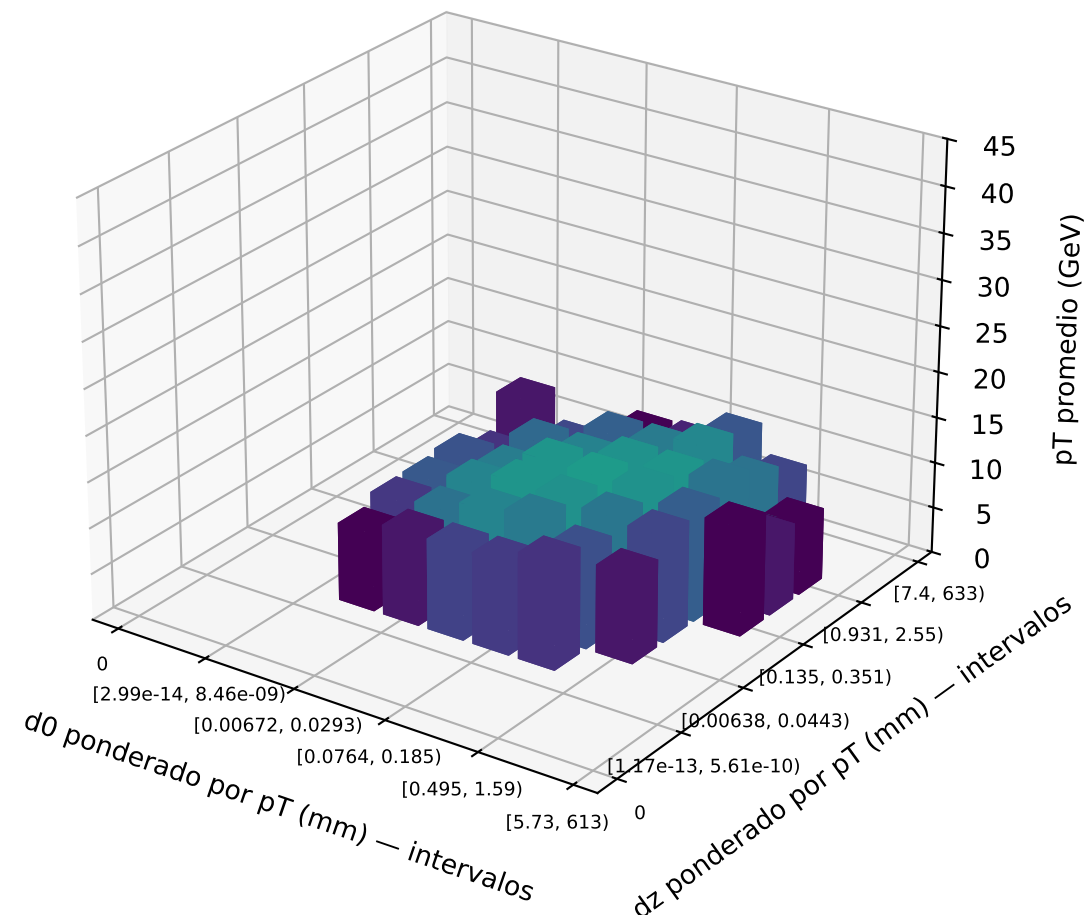
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF48-HGB · seed 73

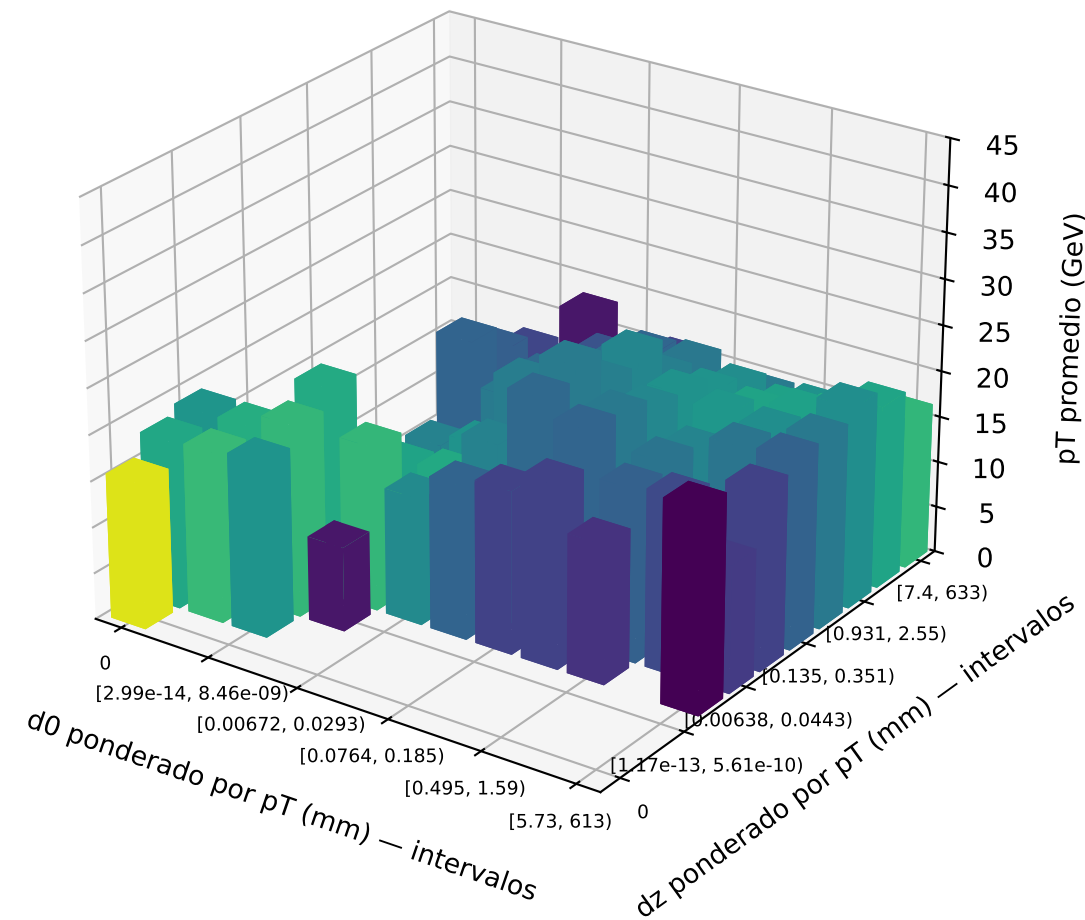
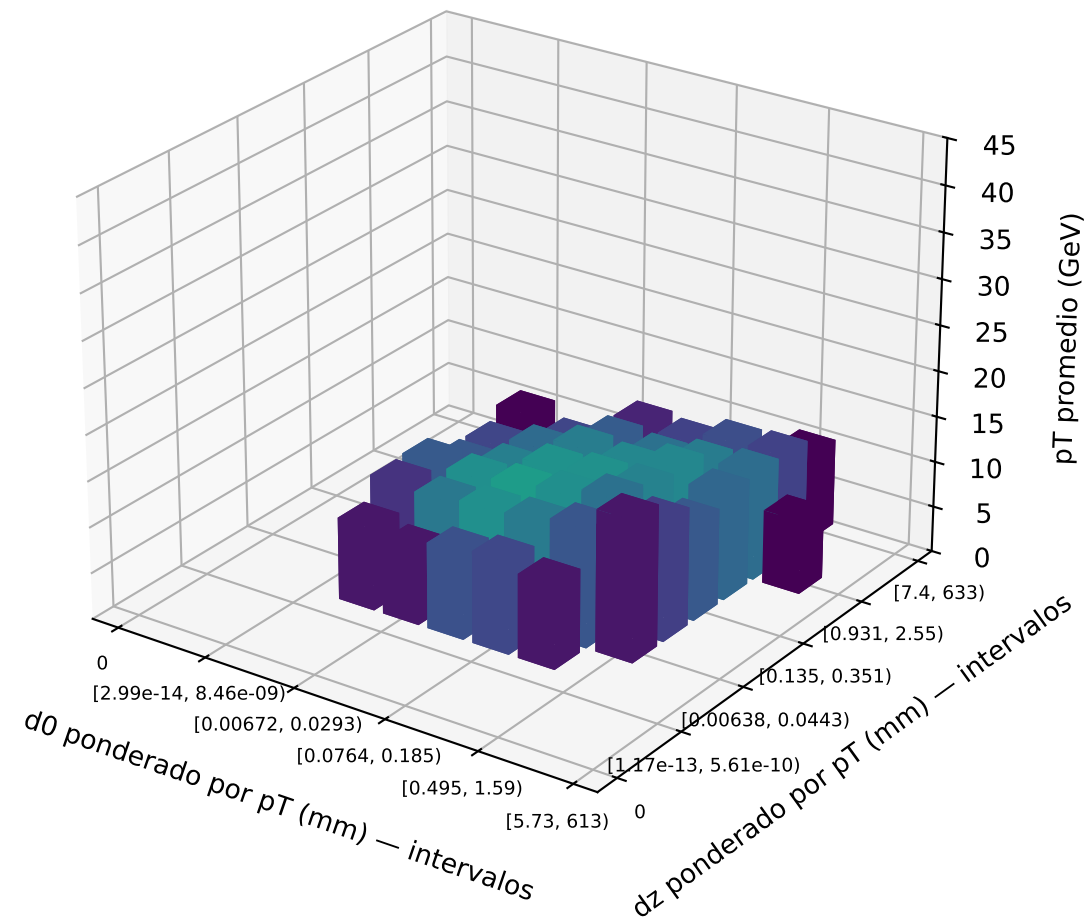
TP — N=3,696

FN — N=2,979



FP — N=3,197

TN — N=57,679



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF48-HGB · seed 73

Variable	Unidad	TP N=3,696	FN N=2,979	FP N=3,197	TN N=57,679
jet_pt *	GeV	12.11 ± 6.078	12.45 ± 7.265	11.87 ± 5.302	16.67 ± 11.85
jet_eta	1	0.00727 ± 1.37	-0.003438 ± 1.468	-0.006223 ± 1.388	0.003187 ± 1.502
jet_abs_eta *	1	1.178 ± 0.6998	1.262 ± 0.7512	1.19 ± 0.7149	1.28 ± 0.7846
jet_mass	GeV	2.505 ± 1.376	2.561 ± 1.714	2.388 ± 1.184	2.877 ± 1.894
jet_ncharged	count	2.802 ± 1.569	2.619 ± 2.008	2.504 ± 1.364	2.592 ± 1.956
jet_nneutral	count	1.785 ± 1.554	2.148 ± 1.677	1.745 ± 1.367	2.404 ± 1.8
n_tracks_cone	count	2.836 ± 1.564	2.657 ± 2.004	2.523 ± 1.349	2.622 ± 1.957
track_pt_sum *	GeV	5.858 ± 4.422	4.876 ± 5.351	5.178 ± 3.619	6.841 ± 8.224
track_pt_sum_over_jet_pt	1	0.4853 ± 0.224	0.3712 ± 0.2661	0.4393 ± 0.2262	0.3709 ± 0.2682
avg_track_pt *	GeV	2.276 ± 1.456	1.533 ± 1.136	2.161 ± 1.279	2.166 ± 2.089
std_track_pt	GeV	0.7943 ± 0.8718	0.665 ± 0.8566	0.6667 ± 0.7625	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.464 ± 1.174	1.406 ± 1.356	1.232 ± 1.035	1.386 ± 1.368
n_tracks_above_avg_pt	count	1.371 ± 0.6503	1.251 ± 0.8718	1.291 ± 0.5586	1.235 ± 0.813
max_pt_ratio	1	0.2908 ± 0.1653	0.2008 ± 0.1496	0.2638 ± 0.1413	0.2093 ± 0.1554
min_pt_ratio	1	0.1399 ± 0.1555	0.07832 ± 0.08339	0.1382 ± 0.1256	0.08366 ± 0.0966
max_dr_ratio	1	0.1643 ± 0.1541	0.09928 ± 0.0927	0.1603 ± 0.1273	0.1024 ± 0.1061
min_dr_ratio	1	0.2519 ± 0.183	0.167 ± 0.1506	0.2304 ± 0.1522	0.1785 ± 0.1543
dr_max_pt	1	0.1396 ± 0.08575	0.1346 ± 0.09976	0.1484 ± 0.08553	0.1259 ± 0.09922
dr_min_pt	1	0.2017 ± 0.1019	0.1905 ± 0.119	0.2043 ± 0.09701	0.19 ± 0.1177
dr_max_dr	1	0.2403 ± 0.09854	0.2271 ± 0.1226	0.2351 ± 0.09208	0.2208 ± 0.1212
dr_min_dr	1	0.1105 ± 0.07392	0.1053 ± 0.08549	0.1235 ± 0.07892	0.1033 ± 0.08833
pt_difference	GeV	1.869 ± 2.127	1.625 ± 2.253	1.538 ± 1.778	2.512 ± 4.113
r50	1	0.1463 ± 0.08315	0.1421 ± 0.09755	0.155 ± 0.08282	0.1326 ± 0.09759
r95	1	0.2377 ± 0.09803	0.2245 ± 0.1217	0.2329 ± 0.0921	0.2145 ± 0.1198
mean_abs_d0 *	mm	0.1576 ± 0.1783	2.624 ± 11.77	0.124 ± 0.1722	1.258 ± 7.926
max_abs_d0 *	mm	0.2802 ± 0.3634	7.158 ± 22.54	0.2261 ± 0.3146	3.334 ± 16.68
std_abs_d0 *	mm	0.09055 ± 0.1412	2.761 ± 8.728	0.07779 ± 0.1372	1.283 ± 6.701
mean_abs_dz *	mm	0.3888 ± 0.5014	3.874 ± 17.61	0.329 ± 0.5992	1.675 ± 9.722
max_abs_dz *	mm	0.6712 ± 0.849	10.12 ± 28.85	0.5746 ± 1.343	4.39 ± 23.14
std_abs_dz *	mm	0.2107 ± 0.3218	3.849 ± 10.35	0.1886 ± 0.5623	1.702 ± 9.544
pt_weighted_abs_d0 *	mm	0.163 ± 0.176	2.466 ± 11.66	0.1289 ± 0.1678	1.199 ± 7.938
pt_weighted_abs_dz *	mm	0.4039 ± 0.5135	3.315 ± 17.37	0.3387 ± 0.5494	1.375 ± 8.417

* Variable utilizada como entrada del modelo.

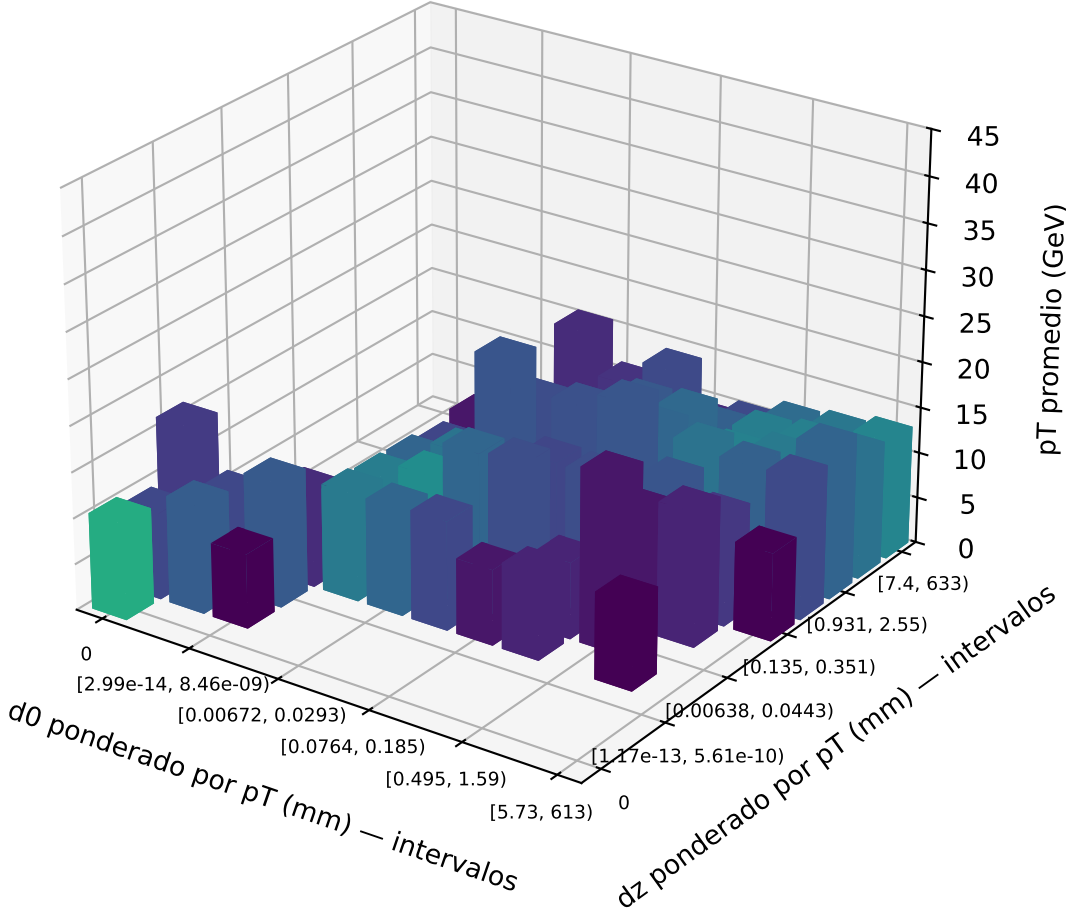
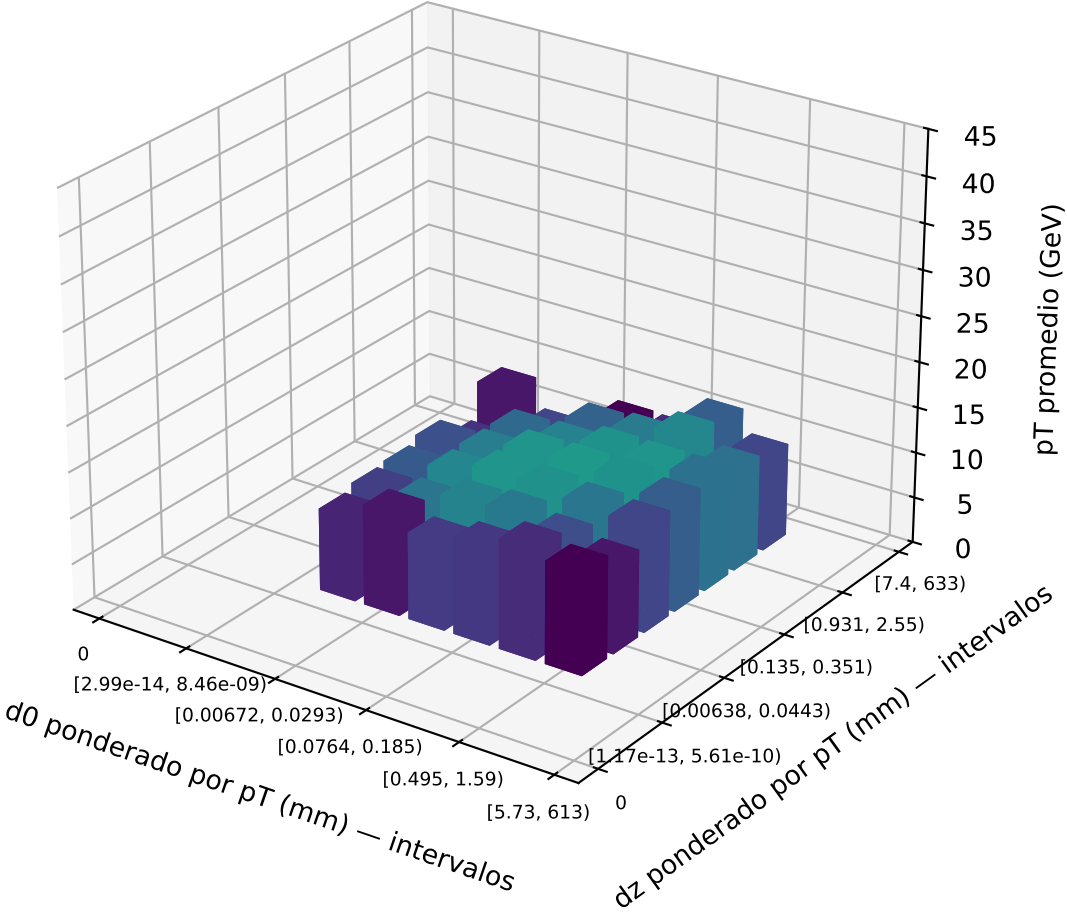
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF48-HGB · seed 101

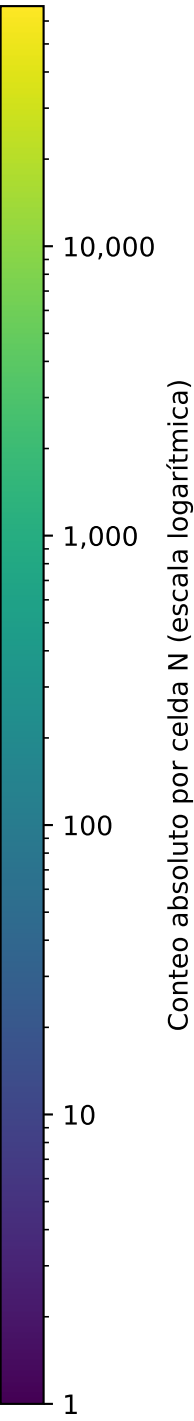
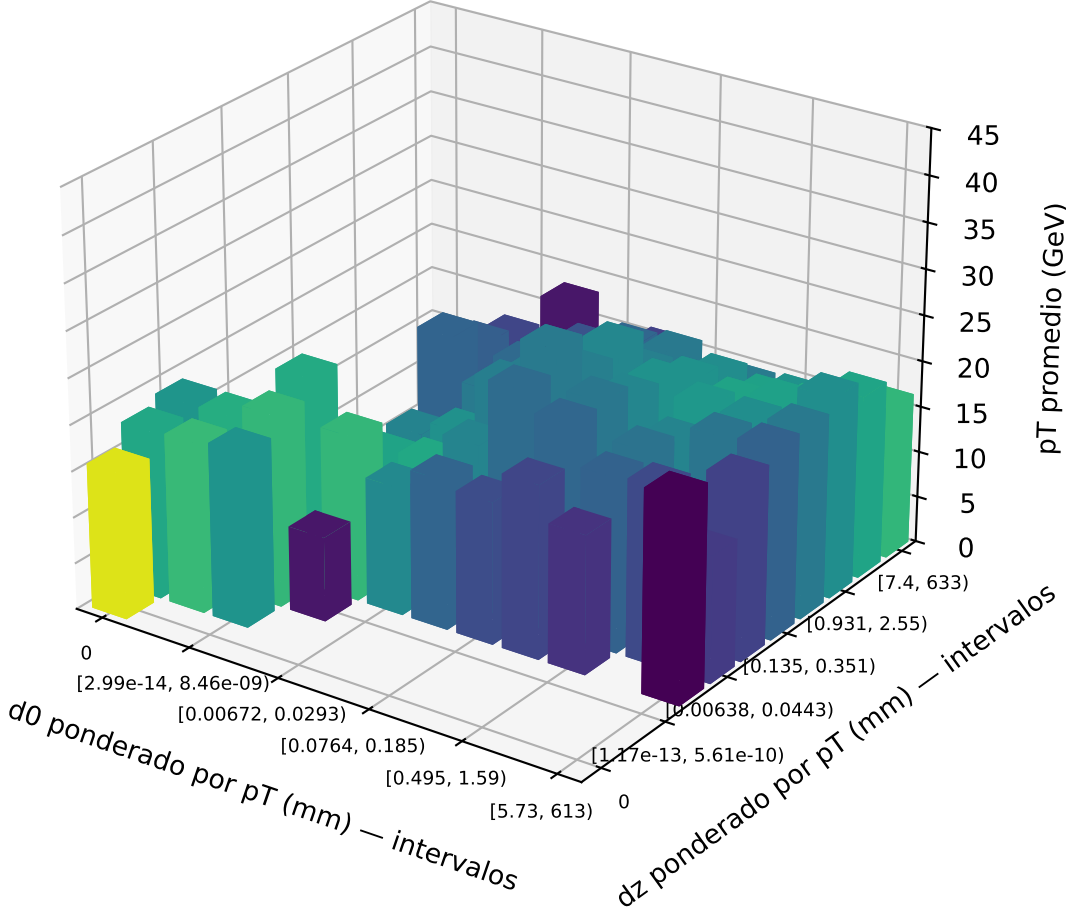
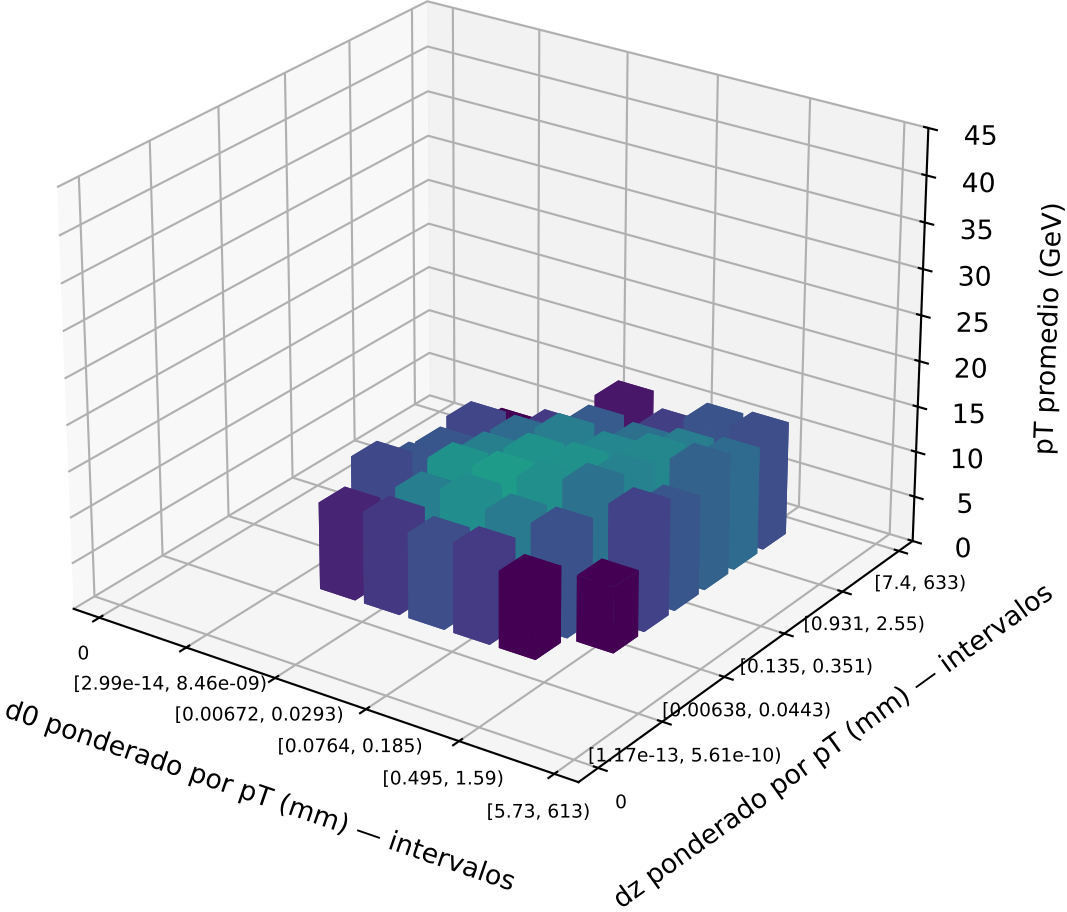
TP — N=3,691

FN — N=2,984



FP — N=3,221

TN — N=57,655



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF48-HGB · seed 101

Variable	Unidad	TP N=3,691	FN N=2,984	FP N=3,221	TN N=57,655
jet_pt *	GeV	12.07 ± 5.977	12.5 ± 7.364	11.94 ± 5.408	16.66 ± 11.86
jet_eta	1	0.01016 ± 1.372	-0.007001 ± 1.466	0.002669 ± 1.392	0.002695 ± 1.501
jet_abs_eta *	1	1.18 ± 0.7005	1.259 ± 0.7506	1.192 ± 0.7195	1.28 ± 0.7844
jet_mass	GeV	2.496 ± 1.366	2.573 ± 1.723	2.405 ± 1.187	2.877 ± 1.895
jet_ncharged	count	2.789 ± 1.56	2.636 ± 2.018	2.524 ± 1.369	2.591 ± 1.956
jet_nneutral	count	1.788 ± 1.56	2.143 ± 1.671	1.759 ± 1.38	2.403 ± 1.8
n_tracks_cone	count	2.821 ± 1.55	2.676 ± 2.018	2.543 ± 1.355	2.62 ± 1.957
track_pt_sum *	GeV	5.814 ± 4.276	4.932 ± 5.503	5.246 ± 3.699	6.838 ± 8.224
track_pt_sum_over_jet_pt	1	0.4845 ± 0.2238	0.3722 ± 0.2667	0.442 ± 0.2262	0.3707 ± 0.2682
avg_track_pt *	GeV	2.28 ± 1.466	1.529 ± 1.118	2.172 ± 1.287	2.165 ± 2.089
std_track_pt	GeV	0.7867 ± 0.8603	0.6747 ± 0.8723	0.6777 ± 0.7769	1.022 ± 1.628
n_tracks_below_avg_pt	count	1.453 ± 1.165	1.42 ± 1.365	1.24 ± 1.03	1.386 ± 1.368
n_tracks_above_avg_pt	count	1.368 ± 0.6472	1.256 ± 0.875	1.303 ± 0.5693	1.235 ± 0.8126
max_pt_ratio	1	0.291 ± 0.1659	0.2006 ± 0.1487	0.2644 ± 0.1423	0.2092 ± 0.1554
min_pt_ratio	1	0.1407 ± 0.1562	0.07738 ± 0.08119	0.1373 ± 0.1258	0.08368 ± 0.09659
max_dr_ratio	1	0.1649 ± 0.155	0.09865 ± 0.09052	0.16 ± 0.1276	0.1024 ± 0.106
min_dr_ratio	1	0.2523 ± 0.1834	0.1666 ± 0.1498	0.2303 ± 0.1535	0.1785 ± 0.1543
dr_max_pt	1	0.14 ± 0.08579	0.1342 ± 0.09967	0.1485 ± 0.08541	0.1259 ± 0.09923
dr_min_pt	1	0.2022 ± 0.1018	0.19 ± 0.1191	0.2048 ± 0.09688	0.1899 ± 0.1177
dr_max_dr	1	0.2405 ± 0.09855	0.2269 ± 0.1225	0.2358 ± 0.09188	0.2208 ± 0.1212
dr_min_dr	1	0.1108 ± 0.07372	0.105 ± 0.08567	0.1232 ± 0.07871	0.1033 ± 0.08834
pt_difference	GeV	1.847 ± 2.086	1.653 ± 2.302	1.564 ± 1.794	2.511 ± 4.114
r50	1	0.1465 ± 0.08313	0.1417 ± 0.09753	0.1553 ± 0.08269	0.1326 ± 0.0976
r95	1	0.2379 ± 0.09801	0.2242 ± 0.1217	0.2334 ± 0.09199	0.2145 ± 0.1198
mean_abs_d0 *	mm	0.1509 ± 0.1561	2.629 ± 11.76	0.12 ± 0.1507	1.259 ± 7.928
max_abs_d0 *	mm	0.2622 ± 0.2657	7.169 ± 22.52	0.2185 ± 0.2584	3.336 ± 16.68
std_abs_d0 *	mm	0.0838 ± 0.106	2.765 ± 8.72	0.07499 ± 0.111	1.284 ± 6.703
mean_abs_dz *	mm	0.3922 ± 0.5272	3.864 ± 17.6	0.3215 ± 0.5142	1.676 ± 9.724
max_abs_dz *	mm	0.6688 ± 0.8518	10.1 ± 28.83	0.5469 ± 0.7544	4.393 ± 23.15
std_abs_dz *	mm	0.2086 ± 0.3195	3.846 ± 10.34	0.1771 ± 0.2999	1.703 ± 9.547
pt_weighted_abs_d0 *	mm	0.157 ± 0.1586	2.47 ± 11.65	0.1259 ± 0.156	1.199 ± 7.94
pt_weighted_abs_dz *	mm	0.4085 ± 0.5379	3.304 ± 17.36	0.3386 ± 0.5381	1.375 ± 8.419

* Variable utilizada como entrada del modelo.

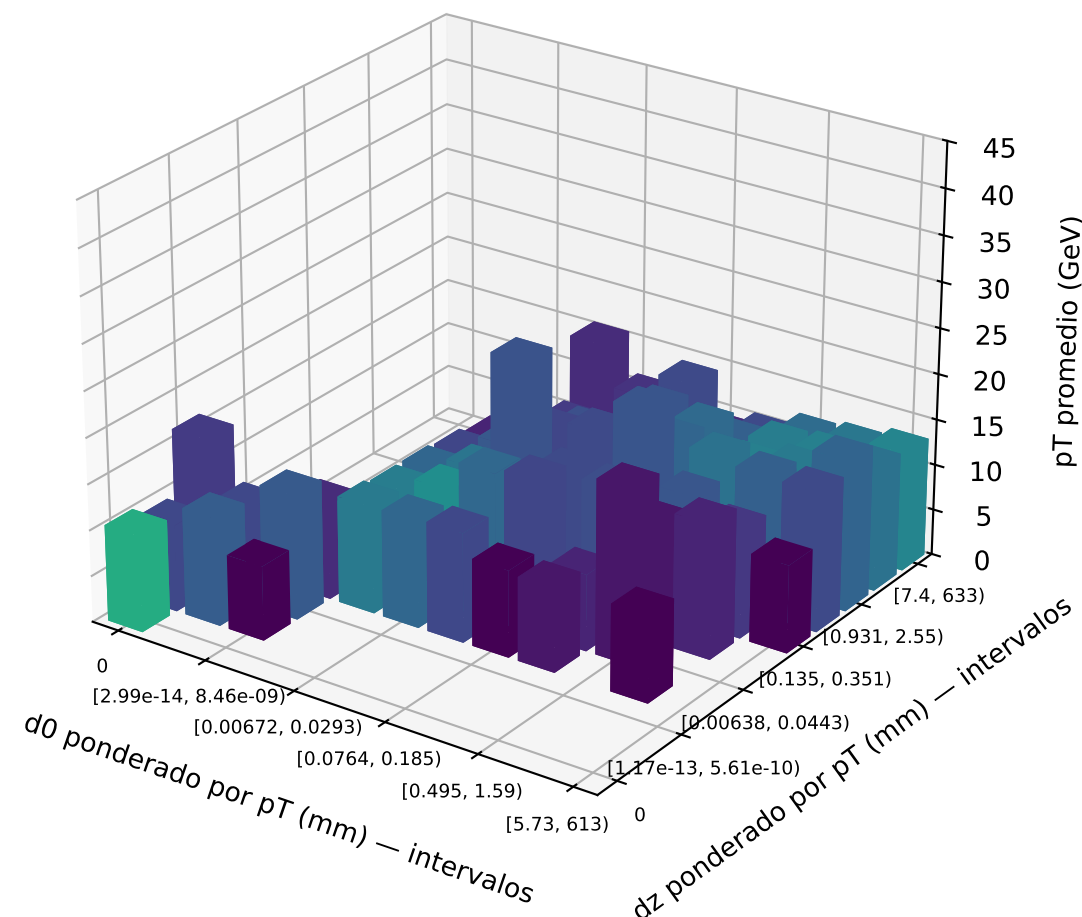
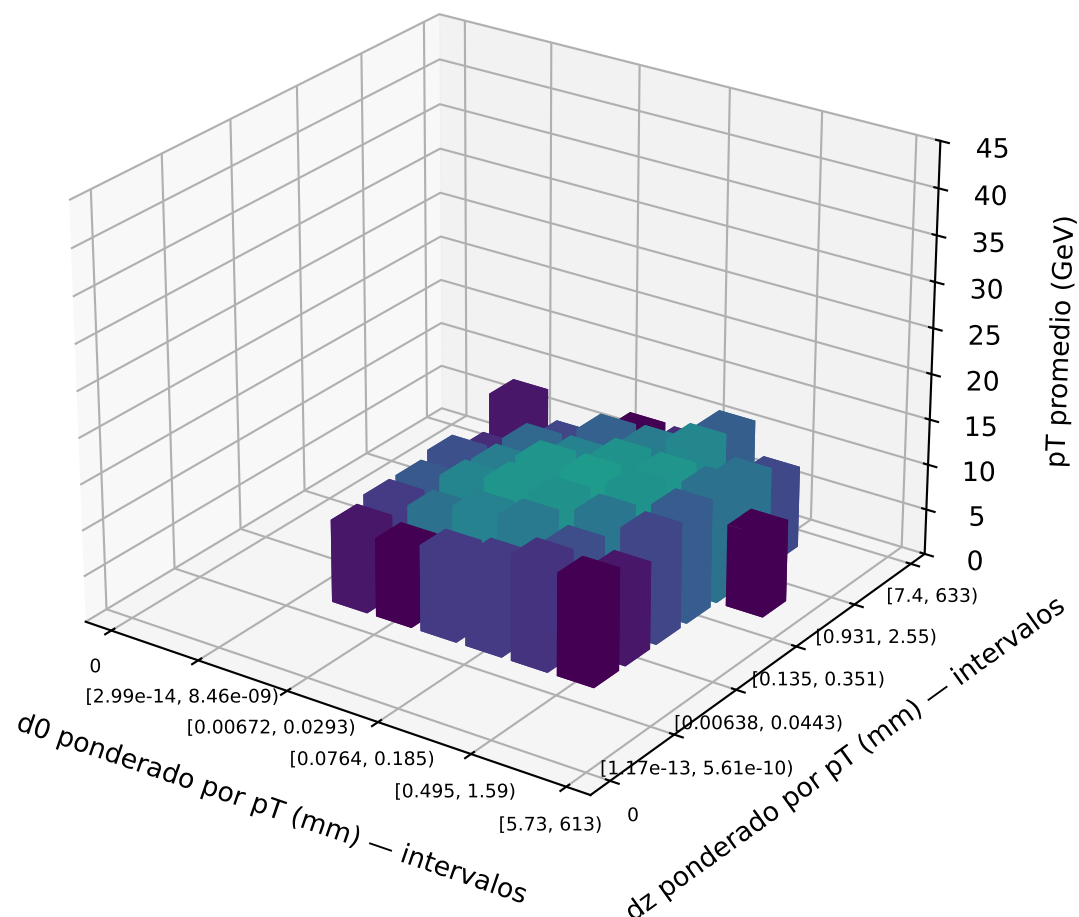
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — QRF48-HGB · ensemble

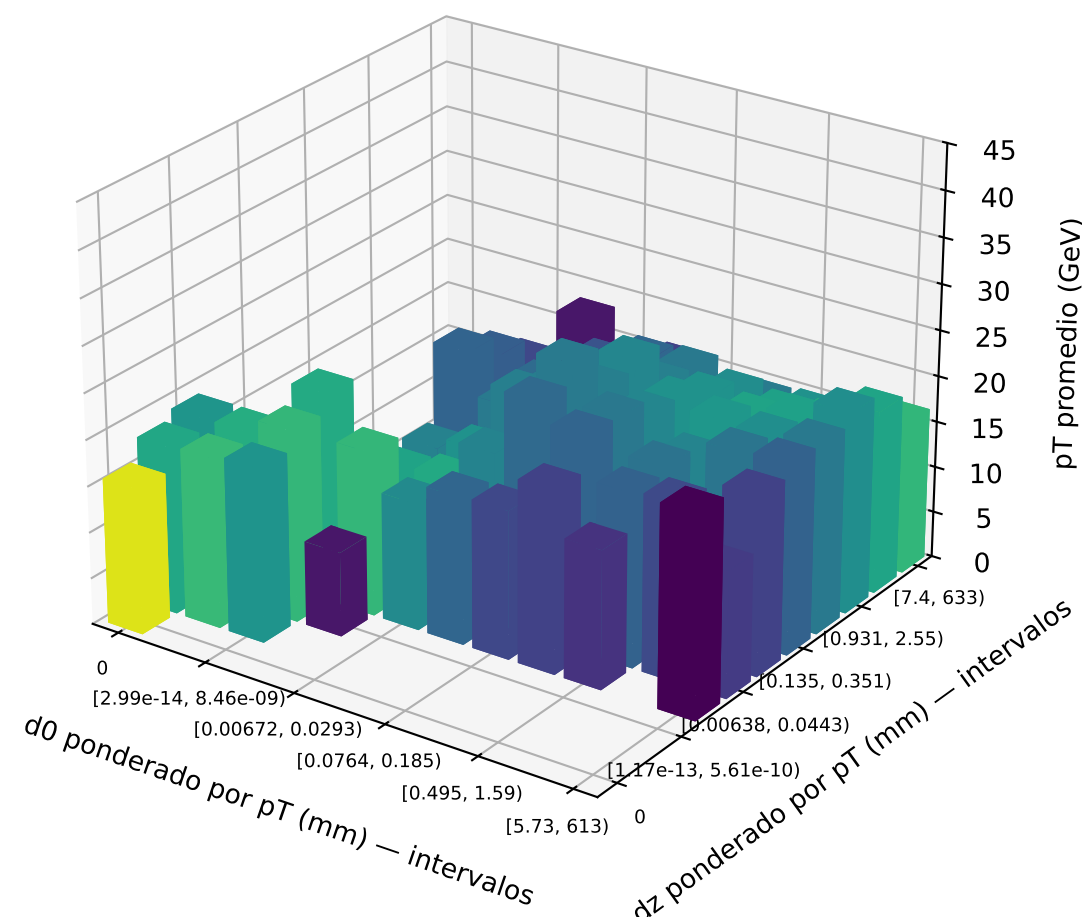
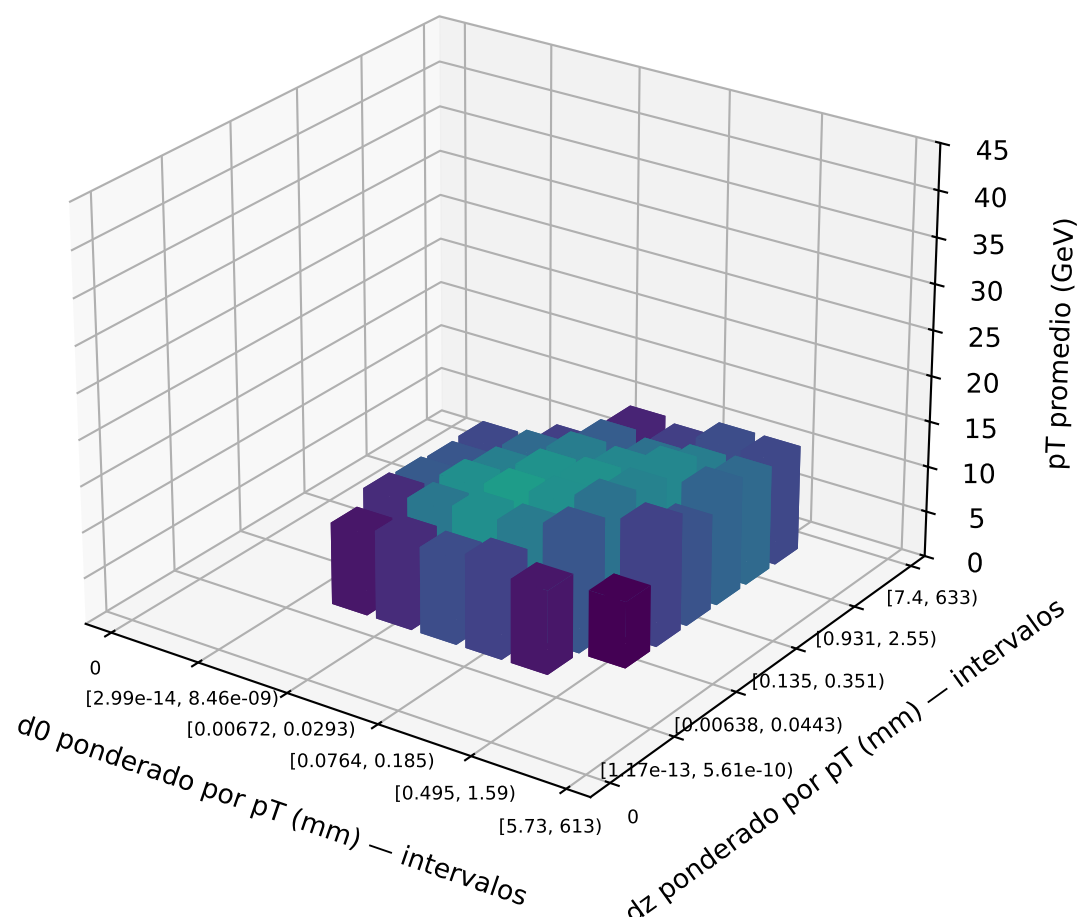
TP — N=3,684

FN — N=2,991



FP — N=3,163

TN — N=57,713



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — QRF48-HGB · ensemble

Variable	Unidad	TP N=3,684	FN N=2,991	FP N=3,163	TN N=57,713
jet_pt *	GeV	12.09 ± 6.025	12.48 ± 7.314	11.88 ± 5.289	16.66 ± 11.85
jet_eta	1	0.005145 ± 1.373	-0.000778 ± 1.464	-0.004789 ± 1.385	0.003103 ± 1.502
jet_abs_eta *	1	1.18 ± 0.7016	1.258 ± 0.7494	1.184 ± 0.7184	1.281 ± 0.7843
jet_mass	GeV	2.498 ± 1.371	2.57 ± 1.717	2.386 ± 1.175	2.877 ± 1.894
jet_ncharged	count	2.792 ± 1.569	2.634 ± 2.008	2.497 ± 1.357	2.593 ± 1.956
jet_nneutral	count	1.795 ± 1.562	2.135 ± 1.671	1.751 ± 1.373	2.403 ± 1.8
n_tracks_cone	count	2.824 ± 1.56	2.672 ± 2.007	2.517 ± 1.345	2.622 ± 1.957
track_pt_sum *	GeV	5.827 ± 4.315	4.918 ± 5.46	5.186 ± 3.661	6.84 ± 8.221
track_pt_sum_over_jet_pt	1	0.4842 ± 0.2241	0.3729 ± 0.2665	0.4387 ± 0.2261	0.371 ± 0.2682
avg_track_pt *	GeV	2.278 ± 1.464	1.533 ± 1.124	2.169 ± 1.287	2.166 ± 2.089
std_track_pt	GeV	0.788 ± 0.8659	0.6733 ± 0.8652	0.664 ± 0.7696	1.023 ± 1.627
n_tracks_below_avg_pt	count	1.454 ± 1.17	1.419 ± 1.359	1.225 ± 1.028	1.387 ± 1.368
n_tracks_above_avg_pt	count	1.37 ± 0.6492	1.253 ± 0.8723	1.292 ± 0.5605	1.235 ± 0.8128
max_pt_ratio	1	0.2906 ± 0.1657	0.2013 ± 0.1494	0.2636 ± 0.1412	0.2093 ± 0.1554
min_pt_ratio	1	0.1405 ± 0.1559	0.07783 ± 0.08239	0.1388 ± 0.1262	0.08365 ± 0.09657
max_dr_ratio	1	0.1645 ± 0.1546	0.09923 ± 0.09194	0.1614 ± 0.1279	0.1024 ± 0.106
min_dr_ratio	1	0.2521 ± 0.1832	0.167 ± 0.1503	0.2294 ± 0.1521	0.1786 ± 0.1544
dr_max_pt	1	0.1397 ± 0.08563	0.1345 ± 0.09983	0.1487 ± 0.08534	0.1259 ± 0.09922
dr_min_pt	1	0.2019 ± 0.1017	0.1903 ± 0.1192	0.2038 ± 0.09703	0.19 ± 0.1177
dr_max_dr	1	0.24 ± 0.09844	0.2275 ± 0.1226	0.2344 ± 0.09201	0.2209 ± 0.1212
dr_min_dr	1	0.1107 ± 0.07392	0.1051 ± 0.08543	0.1238 ± 0.0788	0.1033 ± 0.08833
pt_difference	GeV	1.853 ± 2.108	1.646 ± 2.276	1.533 ± 1.791	2.511 ± 4.112
r50	1	0.1462 ± 0.08301	0.1422 ± 0.09764	0.1552 ± 0.08265	0.1326 ± 0.09759
r95	1	0.2375 ± 0.09794	0.2247 ± 0.1217	0.2322 ± 0.09201	0.2146 ± 0.1197
mean_abs_d0 *	mm	0.1543 ± 0.1661	2.619 ± 11.74	0.1201 ± 0.1487	1.258 ± 7.924
max_abs_d0 *	mm	0.2701 ± 0.2945	7.143 ± 22.5	0.2178 ± 0.2591	3.333 ± 16.67
std_abs_d0 *	mm	0.08658 ± 0.1174	2.756 ± 8.711	0.0748 ± 0.1115	1.282 ± 6.699
mean_abs_dz *	mm	0.395 ± 0.5271	3.852 ± 17.58	0.3194 ± 0.5004	1.675 ± 9.719
max_abs_dz *	mm	0.6737 ± 0.8609	10.08 ± 28.8	0.5401 ± 0.7164	4.389 ± 23.13
std_abs_dz *	mm	0.2095 ± 0.321	3.836 ± 10.33	0.1731 ± 0.2784	1.702 ± 9.542
pt_weighted_abs_d0 *	mm	0.1598 ± 0.1643	2.461 ± 11.64	0.1259 ± 0.1547	1.198 ± 7.936
pt_weighted_abs_dz *	mm	0.4108 ± 0.538	3.295 ± 17.34	0.3346 ± 0.5138	1.374 ± 8.415

* Variable utilizada como entrada del modelo.

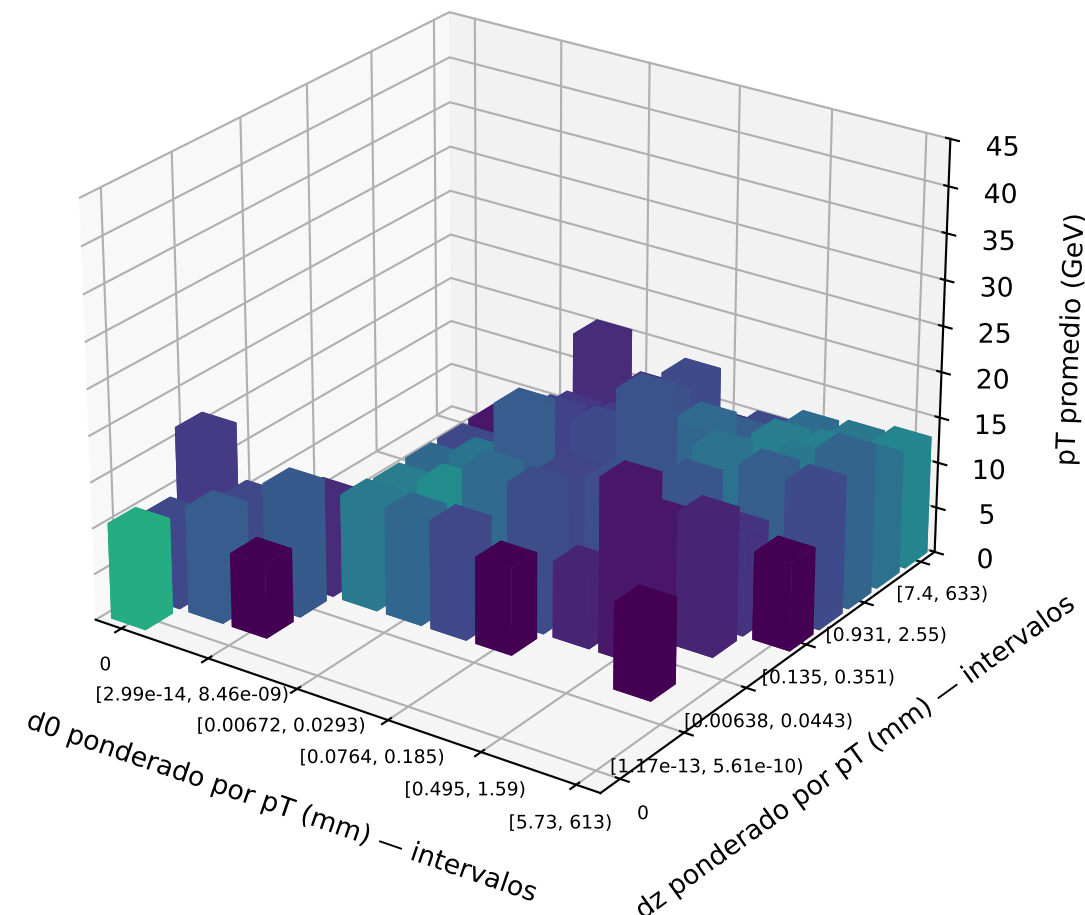
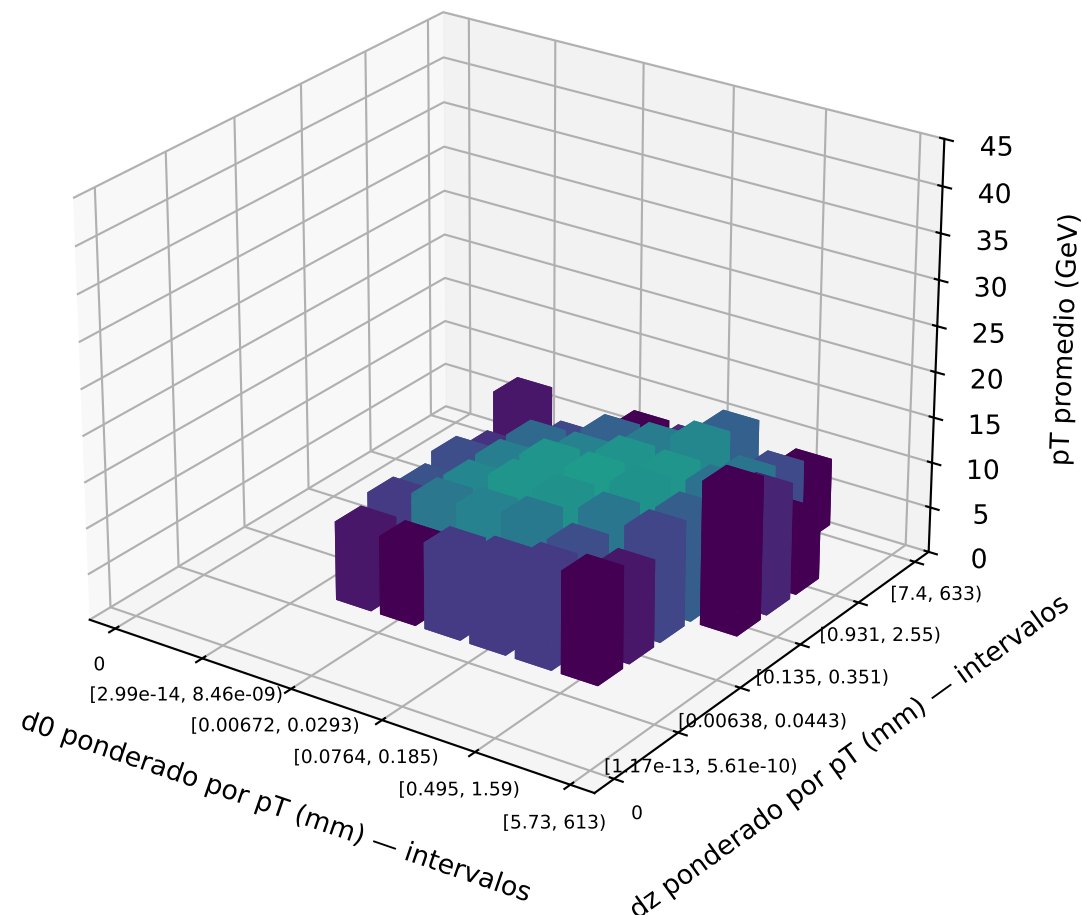
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF48-HGB · seed 42

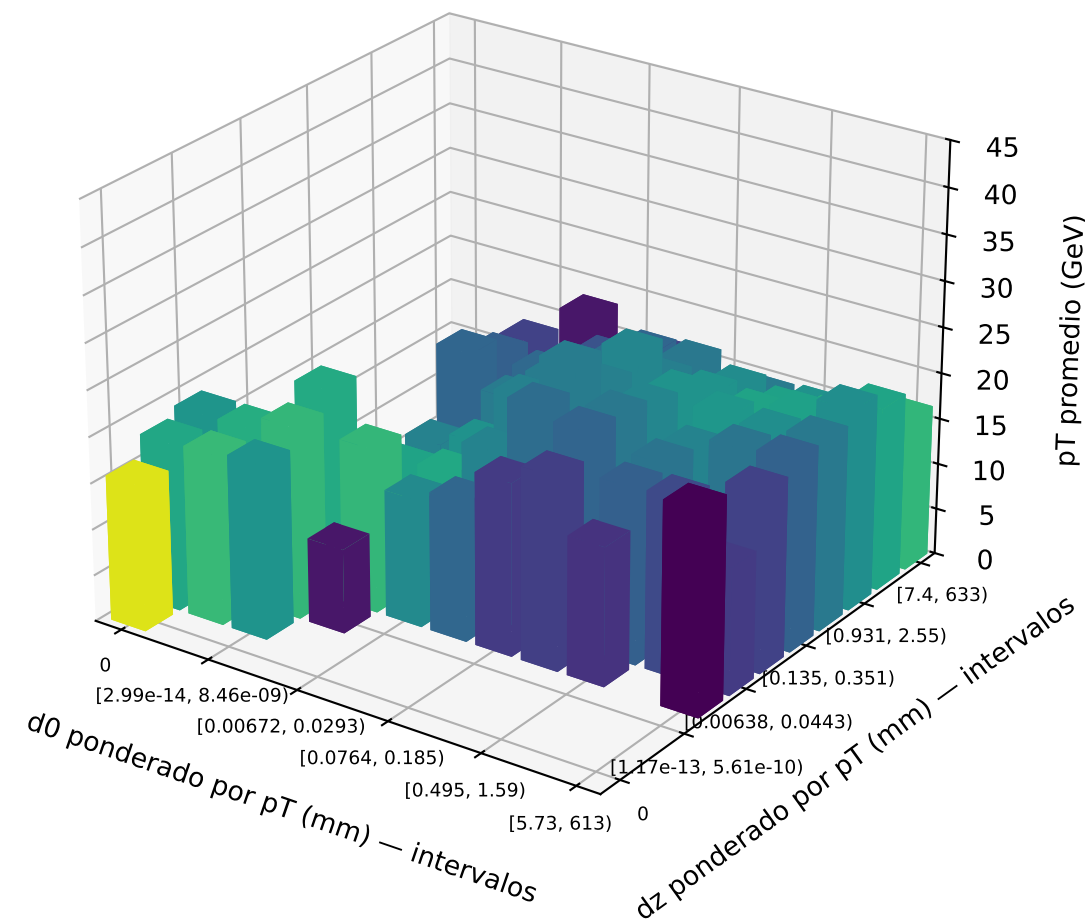
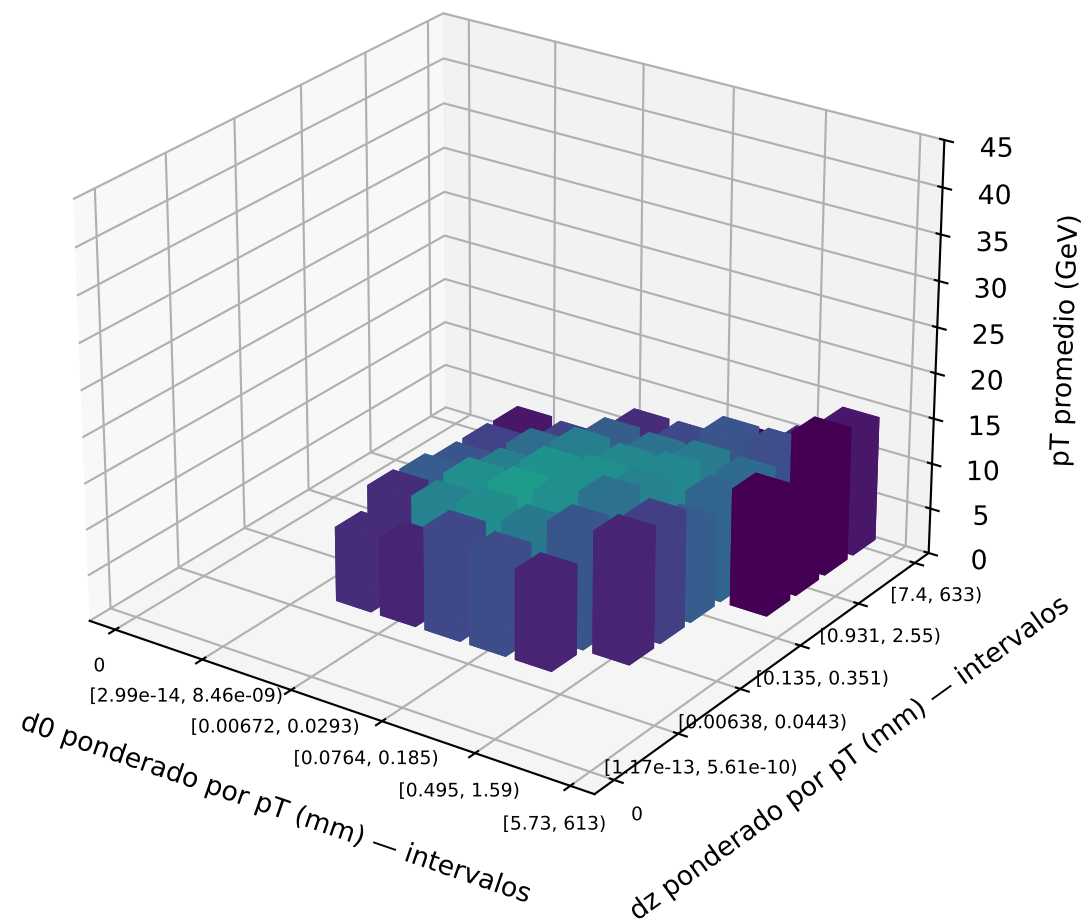
TP — N=3,732

FN — N=2,943



FP — N=3,238

TN — N=57,638



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF48-HGB · seed 42

Variable	Unidad	TP N=3,732	FN N=2,943	FP N=3,238	TN N=57,638
jet_pt *	GeV	12.12 ± 6.058	12.44 ± 7.3	11.71 ± 5.093	16.68 ± 11.86
jet_eta	1	0.008267 ± 1.369	-0.004833 ± 1.47	-0.00375 ± 1.39	0.003055 ± 1.502
jet_abs_eta *	1	1.175 ± 0.7028	1.266 ± 0.7479	1.19 ± 0.7179	1.28 ± 0.7845
jet_mass	GeV	2.511 ± 1.383	2.555 ± 1.711	2.371 ± 1.167	2.879 ± 1.895
jet_ncharged	count	2.82 ± 1.589	2.595 ± 1.991	2.547 ± 1.402	2.59 ± 1.955
jet_nneutral	count	1.785 ± 1.562	2.152 ± 1.668	1.714 ± 1.371	2.406 ± 1.799
n_tracks_cone	count	2.852 ± 1.582	2.634 ± 1.988	2.572 ± 1.395	2.619 ± 1.955
track_pt_sum *	GeV	5.857 ± 4.332	4.865 ± 5.452	5.17 ± 3.545	6.843 ± 8.228
track_pt_sum_over_jet_pt	1	0.4853 ± 0.2244	0.3697 ± 0.2658	0.445 ± 0.2307	0.3706 ± 0.2679
avg_track_pt *	GeV	2.268 ± 1.458	1.533 ± 1.133	2.13 ± 1.268	2.168 ± 2.09
std_track_pt	GeV	0.7886 ± 0.8656	0.6707 ± 0.8652	0.6591 ± 0.7471	1.024 ± 1.629
n_tracks_below_avg_pt	count	1.474 ± 1.185	1.393 ± 1.345	1.258 ± 1.061	1.385 ± 1.367
n_tracks_above_avg_pt	count	1.378 ± 0.6584	1.241 ± 0.865	1.314 ± 0.5807	1.234 ± 0.8121
max_pt_ratio	1	0.2898 ± 0.1658	0.2009 ± 0.1491	0.2643 ± 0.1432	0.2092 ± 0.1553
min_pt_ratio	1	0.1395 ± 0.1555	0.07803 ± 0.08231	0.1371 ± 0.1259	0.08368 ± 0.09658
max_dr_ratio	1	0.1636 ± 0.154	0.09937 ± 0.09233	0.1594 ± 0.1277	0.1024 ± 0.106
min_dr_ratio	1	0.2515 ± 0.1832	0.1665 ± 0.1498	0.2298 ± 0.1544	0.1785 ± 0.1542
dr_max_pt	1	0.1397 ± 0.08551	0.1345 ± 0.1002	0.149 ± 0.08525	0.1259 ± 0.09923
dr_min_pt	1	0.2019 ± 0.1016	0.1901 ± 0.1195	0.2058 ± 0.09738	0.1899 ± 0.1177
dr_max_dr	1	0.2408 ± 0.09833	0.2263 ± 0.123	0.2366 ± 0.09233	0.2208 ± 0.1212
dr_min_dr	1	0.1104 ± 0.07391	0.1054 ± 0.08564	0.1237 ± 0.07894	0.1032 ± 0.08833
pt_difference	GeV	1.859 ± 2.109	1.635 ± 2.277	1.528 ± 1.73	2.513 ± 4.115
r50	1	0.1465 ± 0.08288	0.1417 ± 0.09799	0.156 ± 0.08249	0.1325 ± 0.0976
r95	1	0.2382 ± 0.09775	0.2237 ± 0.1222	0.2343 ± 0.09248	0.2144 ± 0.1197
mean_abs_d0 *	mm	0.1583 ± 0.1837	2.654 ± 11.84	0.1254 ± 0.1889	1.259 ± 7.929
max_abs_d0 *	mm	0.2843 ± 0.4252	7.237 ± 22.67	0.2276 ± 0.3088	3.336 ± 16.68
std_abs_d0 *	mm	0.09128 ± 0.1606	2.793 ± 8.776	0.07622 ± 0.1222	1.284 ± 6.704
mean_abs_dz *	mm	0.4062 ± 0.5673	3.895 ± 17.72	0.3579 ± 0.8142	1.674 ± 9.725
max_abs_dz *	mm	0.7098 ± 1.043	10.18 ± 29.01	0.637 ± 1.714	4.389 ± 23.15
std_abs_dz *	mm	0.2221 ± 0.3925	3.879 ± 10.41	0.2078 ± 0.68	1.702 ± 9.547
pt_weighted_abs_d0 *	mm	0.164 ± 0.1821	2.493 ± 11.73	0.1319 ± 0.1959	1.199 ± 7.941
pt_weighted_abs_dz *	mm	0.421 ± 0.5788	3.329 ± 17.48	0.3719 ± 0.7933	1.374 ± 8.419

* Variable utilizada como entrada del modelo.

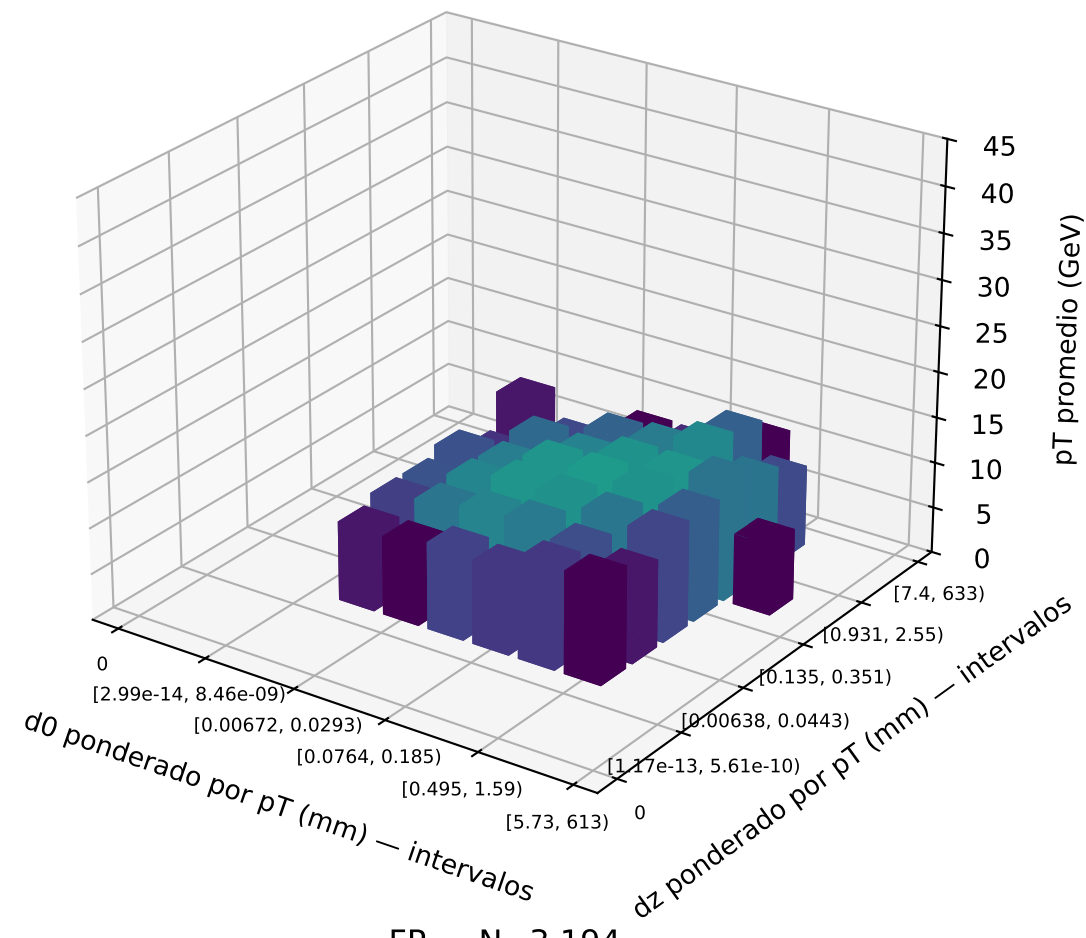
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

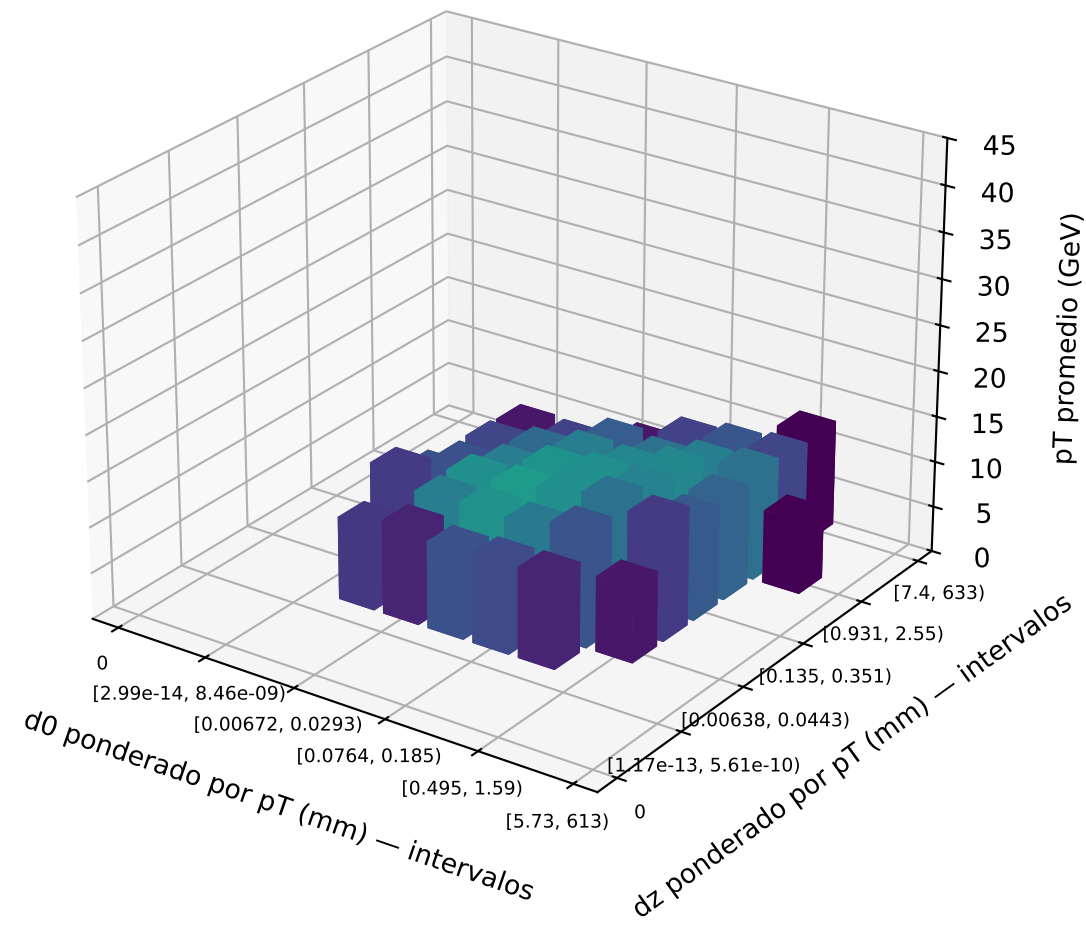
Perfil de pT medio en intervalos de d0/dz — RFF48-HGB · seed 73

TP — N=3,726

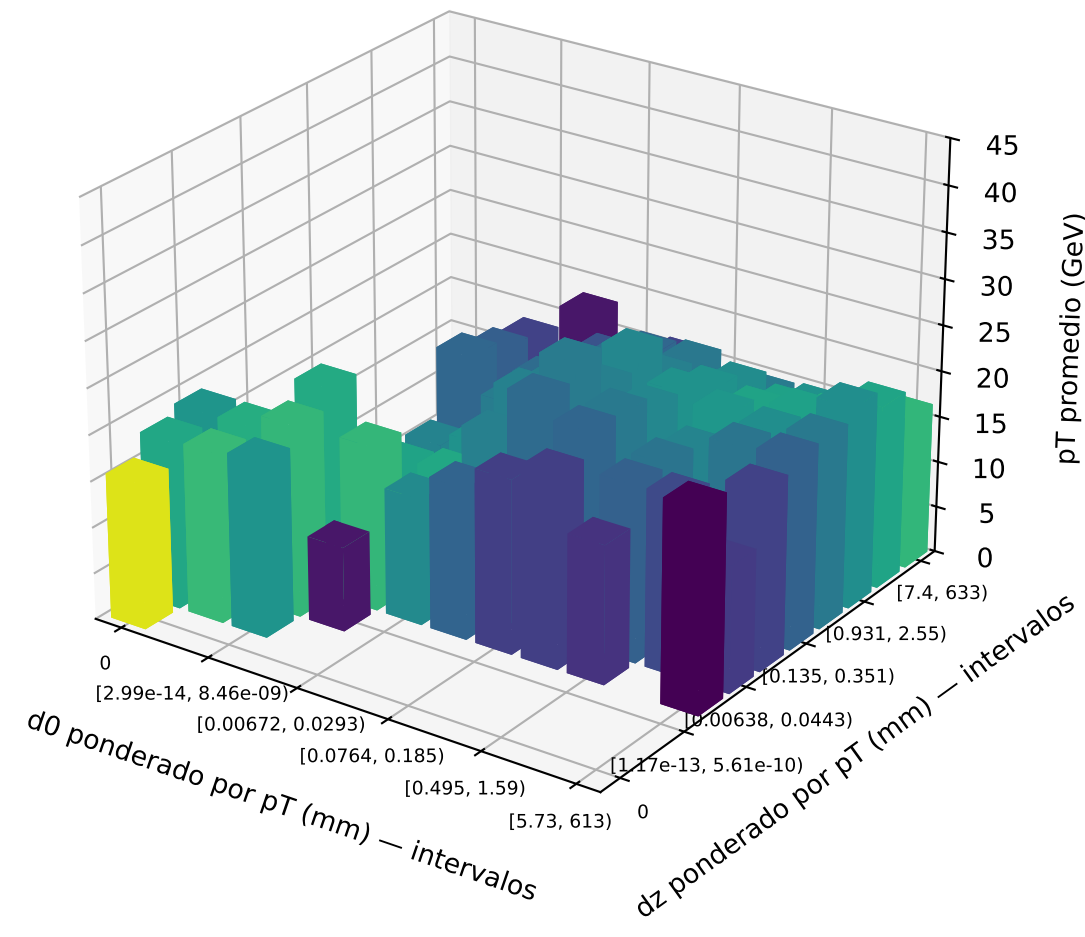
FN — N=2,949



FP — N=3,194



TN — N=57,682



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF48-HGB · seed 73

Variable	Unidad	TP N=3,726	FN N=2,949	FP N=3,194	TN N=57,682
jet_pt *	GeV	12.07 ± 6.023	12.5 ± 7.331	11.7 ± 5.146	16.68 ± 11.85
jet_eta	1	0.009389 ± 1.375	-0.006224 ± 1.463	0.00021 ± 1.383	0.002831 ± 1.502
jet_abs_eta *	1	1.183 ± 0.701	1.255 ± 0.751	1.186 ± 0.7121	1.28 ± 0.7847
jet_mass	GeV	2.505 ± 1.375	2.561 ± 1.718	2.358 ± 1.155	2.879 ± 1.895
jet_ncharged	count	2.811 ± 1.58	2.607 ± 2	2.536 ± 1.399	2.591 ± 1.955
jet_nneutral	count	1.785 ± 1.564	2.152 ± 1.666	1.688 ± 1.345	2.407 ± 1.8
n_tracks_cone	count	2.845 ± 1.573	2.644 ± 1.998	2.559 ± 1.39	2.62 ± 1.955
track_pt_sum *	GeV	5.839 ± 4.368	4.89 ± 5.418	5.243 ± 3.712	6.838 ± 8.222
track_pt_sum_over_jet_pt	1	0.4856 ± 0.2243	0.3695 ± 0.2657	0.4491 ± 0.2306	0.3704 ± 0.2679
avg_track_pt *	GeV	2.266 ± 1.455	1.537 ± 1.14	2.154 ± 1.261	2.166 ± 2.09
std_track_pt	GeV	0.785 ± 0.856	0.6754 ± 0.8778	0.6659 ± 0.7541	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.468 ± 1.179	1.401 ± 1.351	1.245 ± 1.051	1.385 ± 1.367
n_tracks_above_avg_pt	count	1.377 ± 0.6556	1.242 ± 0.8676	1.313 ± 0.5808	1.234 ± 0.812
max_pt_ratio	1	0.29 ± 0.1651	0.2008 ± 0.1501	0.2669 ± 0.141	0.2091 ± 0.1554
min_pt_ratio	1	0.1397 ± 0.1553	0.07793 ± 0.08282	0.139 ± 0.1267	0.08362 ± 0.0965
max_dr_ratio	1	0.164 ± 0.154	0.09892 ± 0.09214	0.1613 ± 0.1279	0.1023 ± 0.106
min_dr_ratio	1	0.2513 ± 0.1828	0.1669 ± 0.1507	0.2326 ± 0.1524	0.1784 ± 0.1543
dr_max_pt	1	0.14 ± 0.08579	0.1341 ± 0.09983	0.1468 ± 0.08489	0.126 ± 0.09927
dr_min_pt	1	0.2019 ± 0.1015	0.1902 ± 0.1196	0.2041 ± 0.09677	0.19 ± 0.1177
dr_max_dr	1	0.2409 ± 0.09828	0.2262 ± 0.123	0.2348 ± 0.09242	0.2209 ± 0.1212
dr_min_dr	1	0.1104 ± 0.07366	0.1054 ± 0.08589	0.1227 ± 0.07887	0.1033 ± 0.08834
pt_difference	GeV	1.85 ± 2.084	1.648 ± 2.307	1.549 ± 1.781	2.511 ± 4.113
r50	1	0.1467 ± 0.08305	0.1415 ± 0.09776	0.1541 ± 0.08229	0.1326 ± 0.09762
r95	1	0.2383 ± 0.09779	0.2236 ± 0.1221	0.2323 ± 0.0925	0.2145 ± 0.1197
mean_abs_d0 *	mm	0.1569 ± 0.1729	2.65 ± 11.82	0.1245 ± 0.1693	1.258 ± 7.926
max_abs_d0 *	mm	0.2808 ± 0.3719	7.228 ± 22.65	0.2311 ± 0.3439	3.334 ± 16.67
std_abs_d0 *	mm	0.09076 ± 0.1448	2.788 ± 8.768	0.07932 ± 0.1472	1.283 ± 6.701
mean_abs_dz *	mm	0.4029 ± 0.5534	3.892 ± 17.7	0.3307 ± 0.5488	1.675 ± 9.722
max_abs_dz *	mm	0.6956 ± 0.9478	10.18 ± 28.99	0.5774 ± 1.047	4.389 ± 23.14
std_abs_dz *	mm	0.2173 ± 0.3574	3.878 ± 10.4	0.1871 ± 0.4261	1.702 ± 9.545
pt_weighted_abs_d0 *	mm	0.163 ± 0.1733	2.489 ± 11.72	0.1308 ± 0.1679	1.198 ± 7.938
pt_weighted_abs_dz *	mm	0.4214 ± 0.5761	3.323 ± 17.46	0.3511 ± 0.5665	1.374 ± 8.417

* Variable utilizada como entrada del modelo.

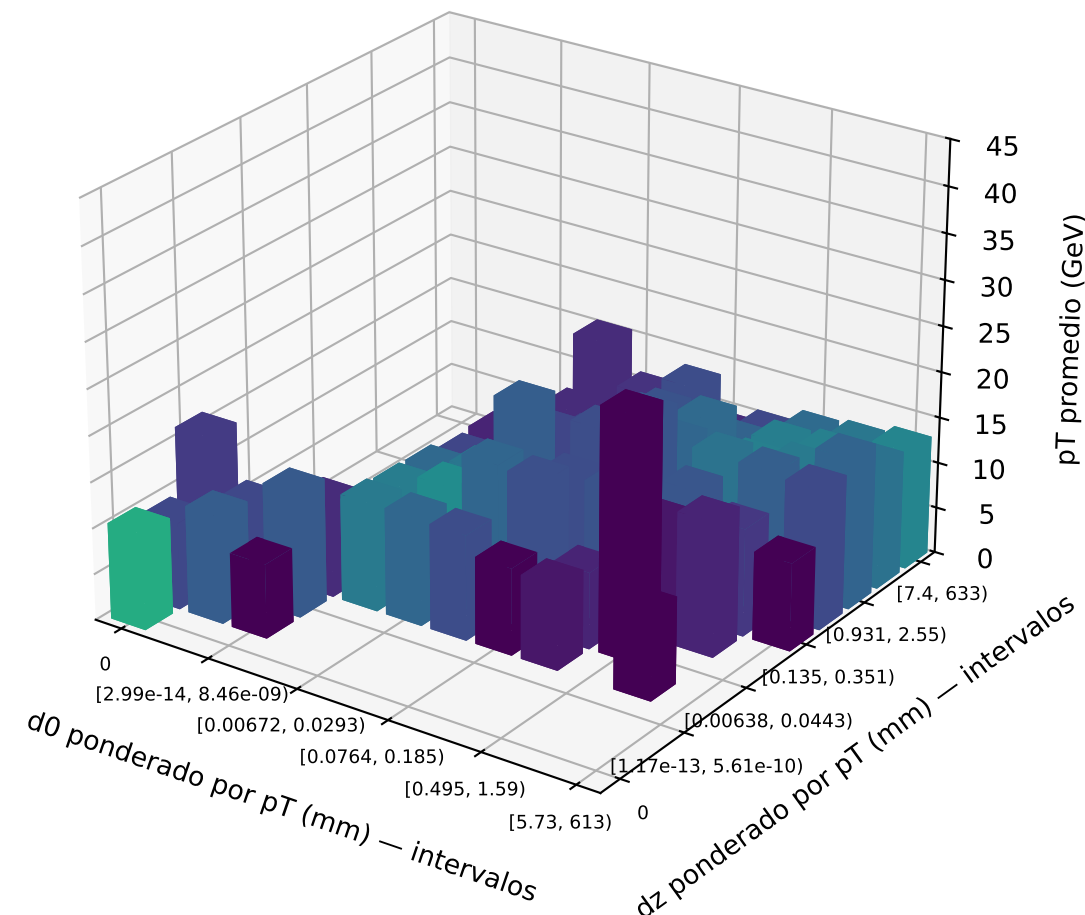
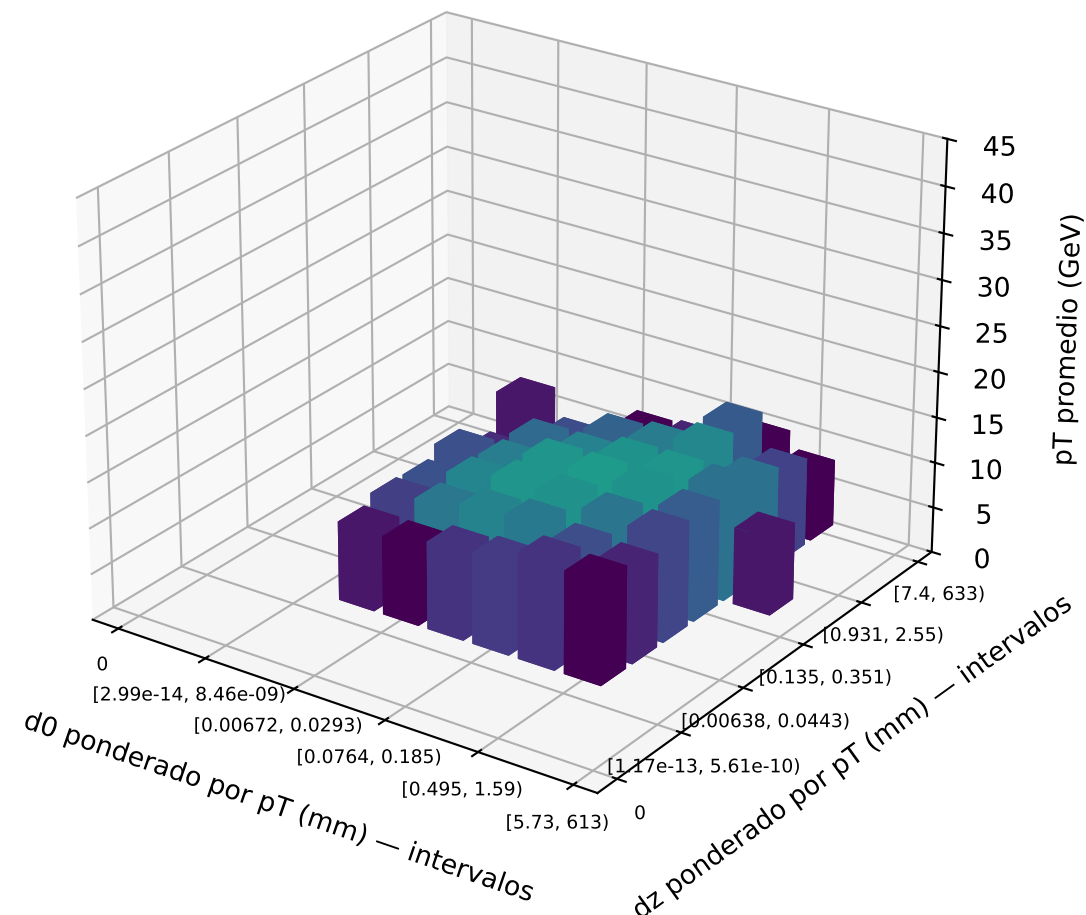
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF48-HGB · seed 101

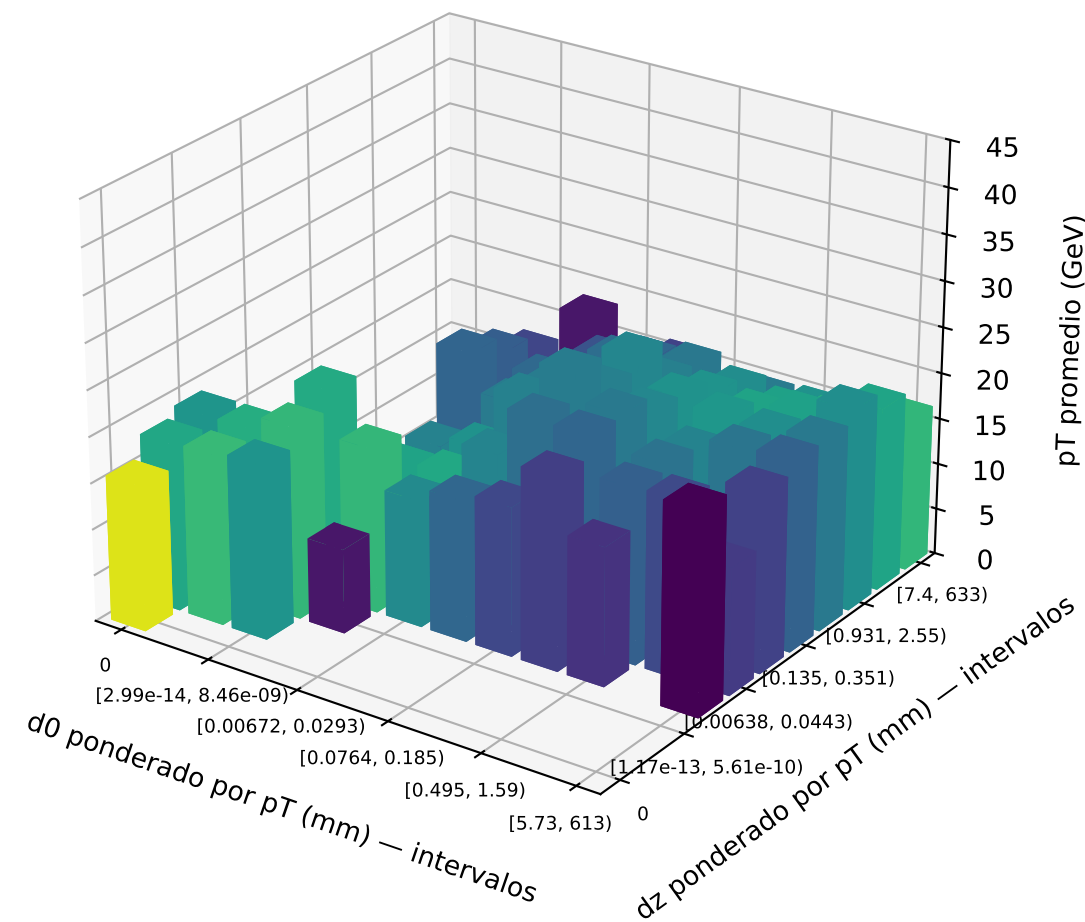
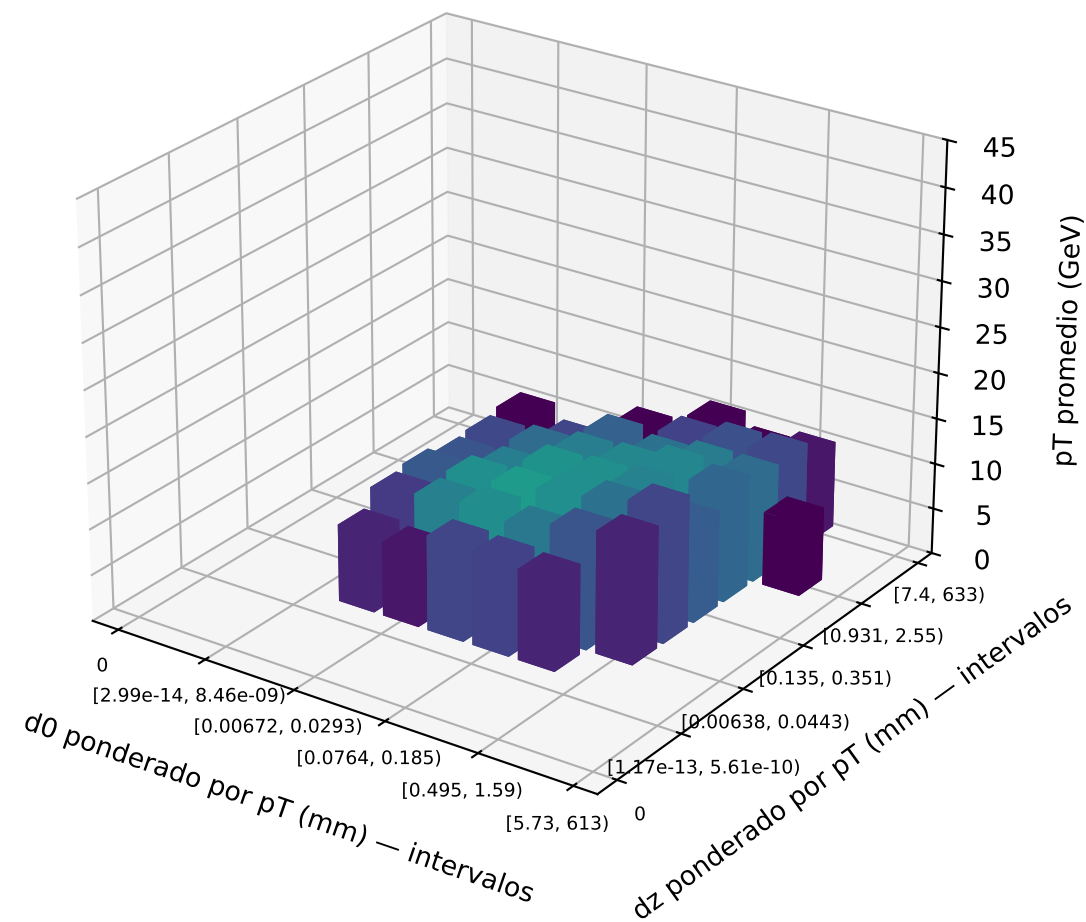
TP — N=3,684

FN — N=2,991



FP — N=3,176

TN — N=57,700



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF48-HGB · seed 101

Variable	Unidad	TP N=3,684	FN N=2,991	FP N=3,176	TN N=57,700
jet_pt *	GeV	12.11 ± 6.027	12.45 ± 7.313	11.83 ± 5.315	16.67 ± 11.85
jet_eta	1	0.003925 ± 1.368	0.000725 ± 1.47	-0.000569 ± 1.382	0.002873 ± 1.502
jet_abs_eta *	1	1.175 ± 0.7013	1.265 ± 0.7489	1.18 ± 0.7192	1.281 ± 0.7843
jet_mass	GeV	2.502 ± 1.366	2.565 ± 1.722	2.371 ± 1.166	2.878 ± 1.895
jet_ncharged	count	2.799 ± 1.57	2.625 ± 2.007	2.503 ± 1.365	2.592 ± 1.956
jet_nneutral	count	1.789 ± 1.553	2.141 ± 1.679	1.735 ± 1.375	2.404 ± 1.799
n_tracks_cone	count	2.83 ± 1.561	2.665 ± 2.006	2.523 ± 1.355	2.622 ± 1.956
track_pt_sum *	GeV	5.831 ± 4.315	4.913 ± 5.46	5.2 ± 3.663	6.84 ± 8.222
track_pt_sum_over_jet_pt	1	0.4842 ± 0.2233	0.3729 ± 0.2673	0.4428 ± 0.2276	0.3708 ± 0.2681
avg_track_pt *	GeV	2.277 ± 1.459	1.534 ± 1.134	2.176 ± 1.344	2.165 ± 2.087
std_track_pt	GeV	0.7859 ± 0.8646	0.676 ± 0.8671	0.6643 ± 0.7786	1.023 ± 1.627
n_tracks_below_avg_pt	count	1.46 ± 1.175	1.411 ± 1.354	1.225 ± 1.028	1.387 ± 1.368
n_tracks_above_avg_pt	count	1.369 ± 0.6491	1.254 ± 0.8725	1.298 ± 0.5681	1.235 ± 0.8125
max_pt_ratio	1	0.2903 ± 0.1654	0.2017 ± 0.1501	0.2656 ± 0.1418	0.2092 ± 0.1554
min_pt_ratio	1	0.1406 ± 0.1559	0.07771 ± 0.08243	0.1397 ± 0.1268	0.0836 ± 0.09649
max_dr_ratio	1	0.1646 ± 0.1546	0.09917 ± 0.09193	0.1619 ± 0.1282	0.1023 ± 0.106
min_dr_ratio	1	0.2522 ± 0.1827	0.167 ± 0.1511	0.2317 ± 0.1529	0.1785 ± 0.1543
dr_max_pt	1	0.1397 ± 0.08561	0.1346 ± 0.09986	0.1477 ± 0.08491	0.126 ± 0.09925
dr_min_pt	1	0.2021 ± 0.1016	0.1901 ± 0.1192	0.2042 ± 0.09661	0.19 ± 0.1177
dr_max_dr	1	0.2403 ± 0.09848	0.2272 ± 0.1226	0.234 ± 0.09181	0.2209 ± 0.1212
dr_min_dr	1	0.1105 ± 0.07361	0.1054 ± 0.08578	0.1235 ± 0.07876	0.1033 ± 0.08833
pt_difference	GeV	1.851 ± 2.11	1.649 ± 2.275	1.536 ± 1.804	2.511 ± 4.112
r50	1	0.1463 ± 0.083	0.1421 ± 0.09765	0.1548 ± 0.08229	0.1326 ± 0.09761
r95	1	0.2378 ± 0.09792	0.2244 ± 0.1217	0.2317 ± 0.09176	0.2146 ± 0.1198
mean_abs_d0 *	mm	0.1544 ± 0.1709	2.618 ± 11.74	0.1249 ± 0.1746	1.258 ± 7.925
max_abs_d0 *	mm	0.2714 ± 0.3141	7.142 ± 22.5	0.224 ± 0.3069	3.333 ± 16.67
std_abs_d0 *	mm	0.08692 ± 0.1236	2.755 ± 8.711	0.07577 ± 0.1327	1.283 ± 6.7
mean_abs_dz *	mm	0.3994 ± 0.5576	3.847 ± 17.58	0.3737 ± 1.977	1.672 ± 9.71
max_abs_dz *	mm	0.695 ± 1.026	10.05 ± 28.8	0.6378 ± 2.421	4.385 ± 23.13
std_abs_dz *	mm	0.2199 ± 0.3996	3.823 ± 10.33	0.1999 ± 0.6298	1.701 ± 9.543
pt_weighted_abs_d0 *	mm	0.1595 ± 0.1673	2.461 ± 11.64	0.1301 ± 0.172	1.198 ± 7.936
pt_weighted_abs_dz *	mm	0.4129 ± 0.5636	3.292 ± 17.34	0.3839 ± 1.955	1.372 ± 8.405

* Variable utilizada como entrada del modelo.

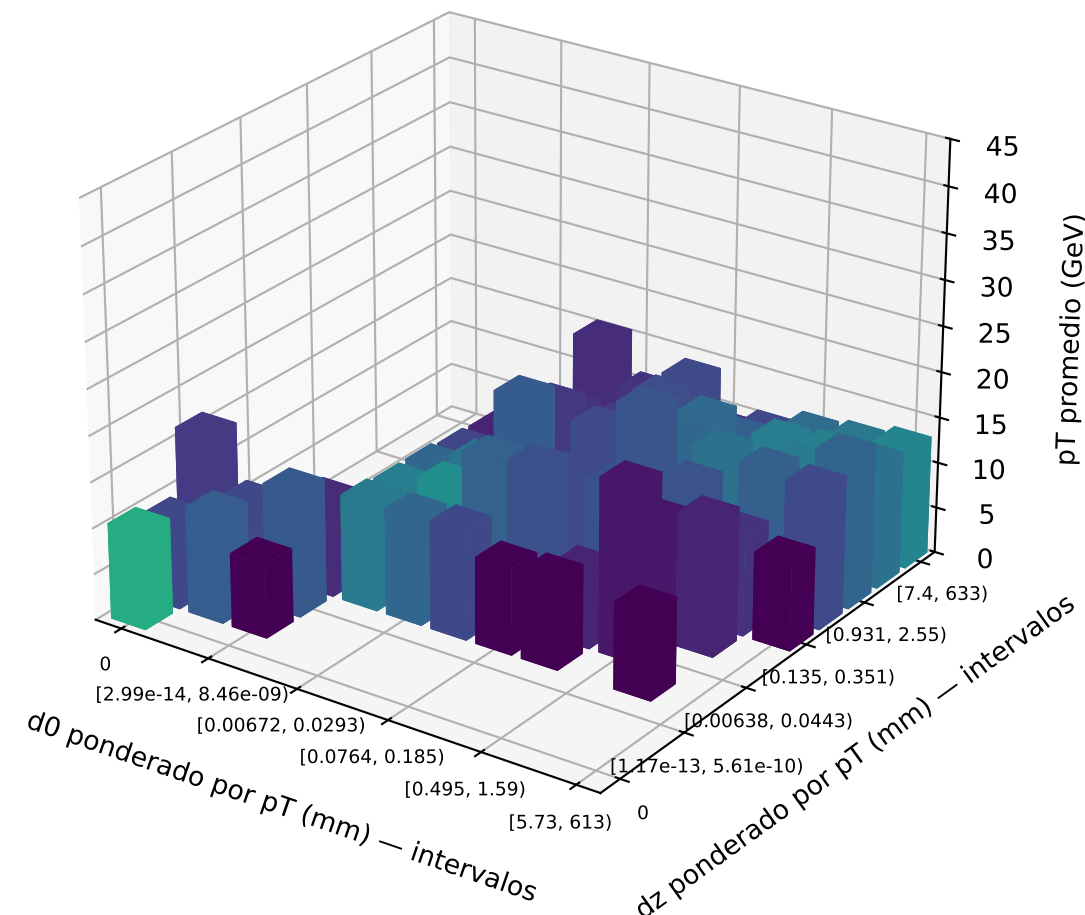
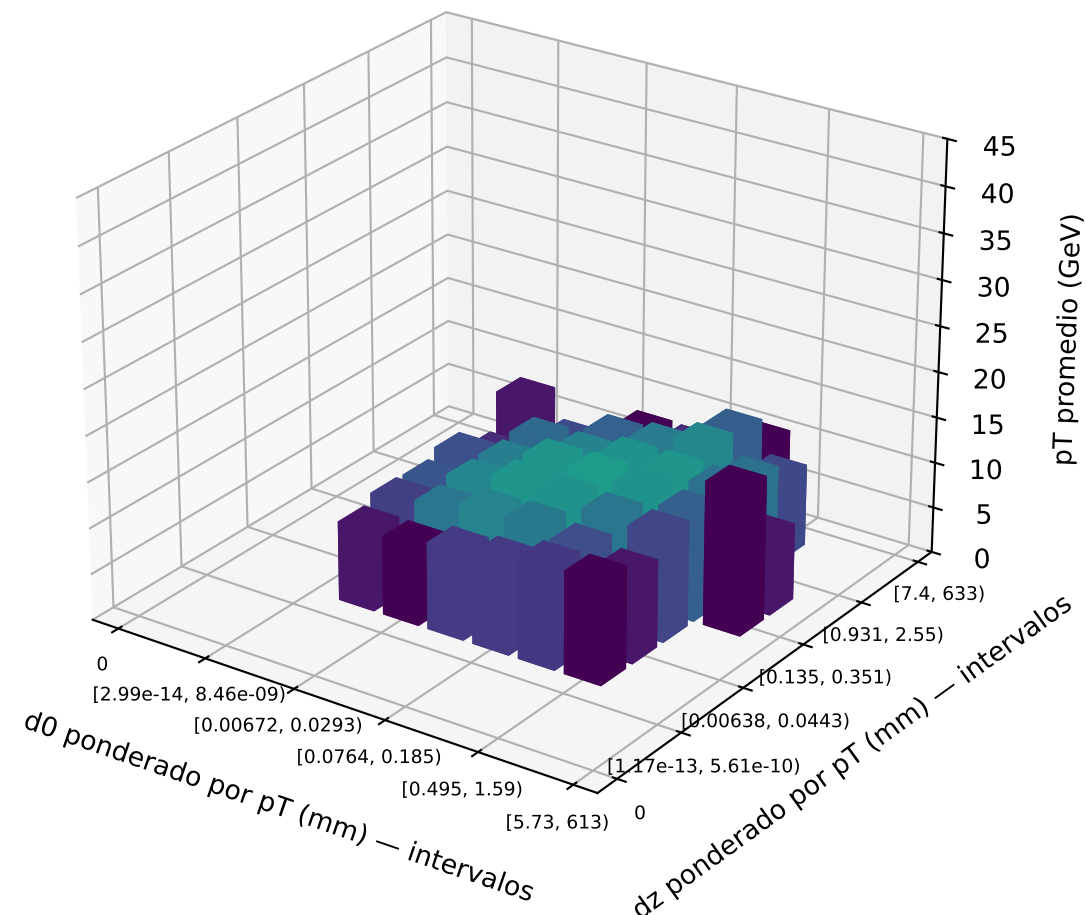
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RFF48-HGB · ensemble

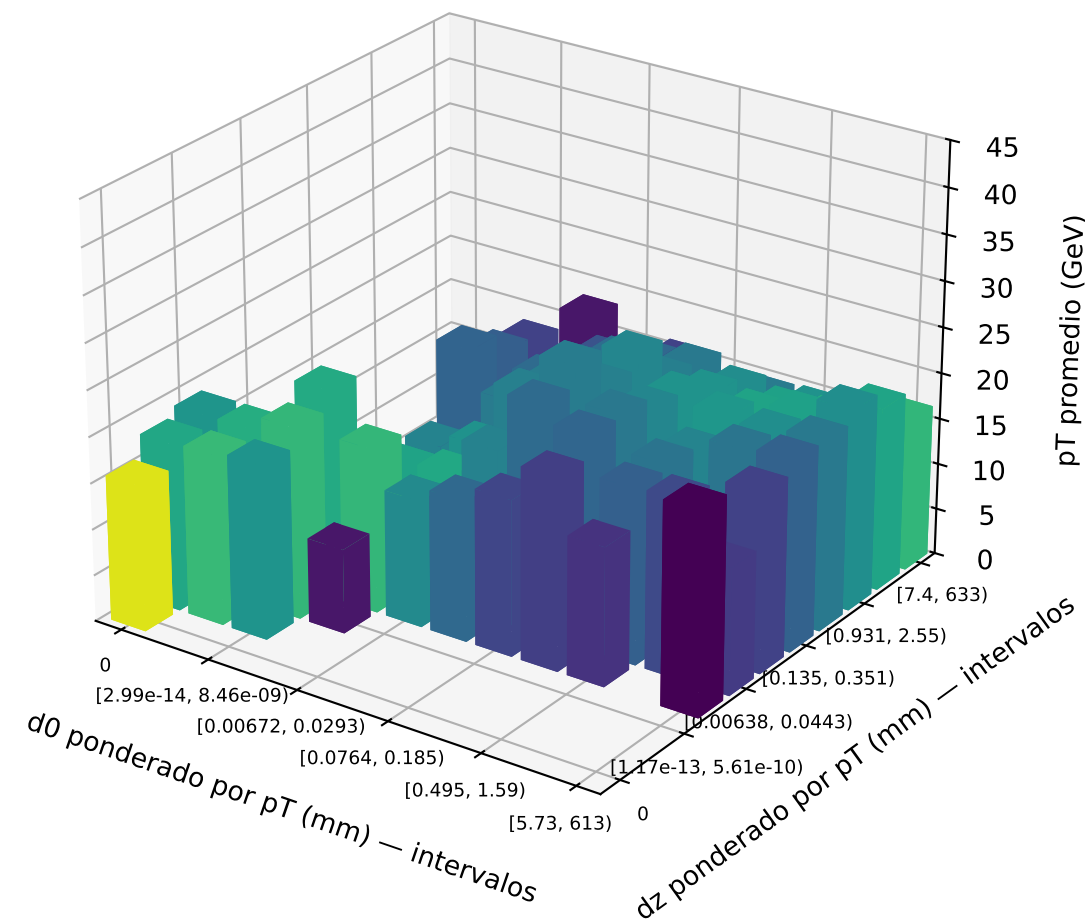
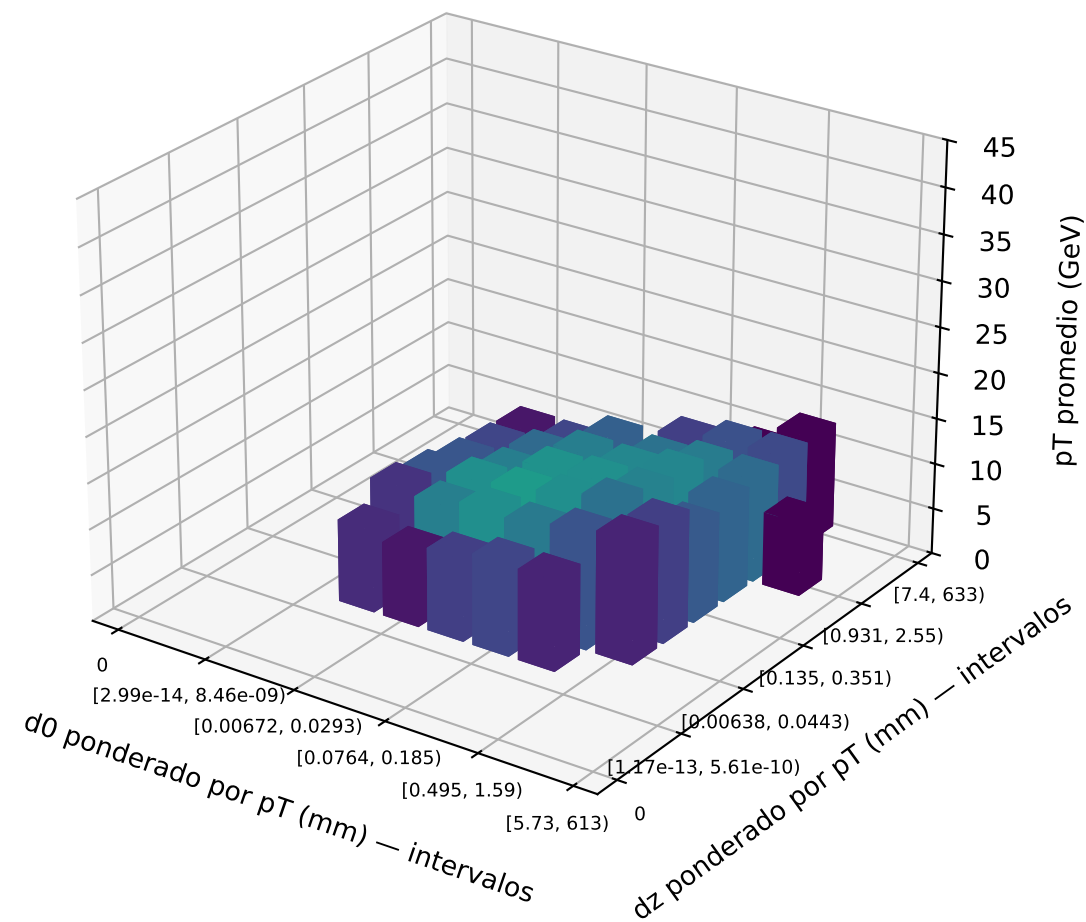
TP — N=3,724

FN — N=2,951



FP — N=3,188

TN — N=57,688



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RFF48-HGB · ensemble

Variable	Unidad	TP N=3,724	FN N=2,951	FP N=3,188	TN N=57,688
jet_pt *	GeV	12.07 ± 5.992	12.5 ± 7.363	11.72 ± 5.137	16.67 ± 11.85
jet_eta	1	0.005793 ± 1.371	-0.001675 ± 1.468	0.00489 ± 1.387	0.002572 ± 1.502
jet_abs_eta *	1	1.178 ± 0.7019	1.262 ± 0.7492	1.187 ± 0.7169	1.28 ± 0.7844
jet_mass	GeV	2.5 ± 1.364	2.569 ± 1.728	2.361 ± 1.152	2.879 ± 1.895
jet_ncharged	count	2.802 ± 1.575	2.618 ± 2.007	2.511 ± 1.371	2.592 ± 1.956
jet_nneutral	count	1.787 ± 1.56	2.148 ± 1.671	1.721 ± 1.374	2.405 ± 1.799
n_tracks_cone	count	2.835 ± 1.568	2.657 ± 2.003	2.534 ± 1.363	2.621 ± 1.956
track_pt_sum *	GeV	5.826 ± 4.309	4.907 ± 5.48	5.173 ± 3.618	6.841 ± 8.224
track_pt_sum_over_jet_pt	1	0.4844 ± 0.2241	0.3712 ± 0.2665	0.444 ± 0.2293	0.3707 ± 0.268
avg_track_pt *	GeV	2.269 ± 1.457	1.534 ± 1.134	2.153 ± 1.269	2.166 ± 2.09
std_track_pt	GeV	0.7856 ± 0.8622	0.6749 ± 0.8701	0.66 ± 0.7583	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.462 ± 1.178	1.409 ± 1.353	1.231 ± 1.035	1.386 ± 1.368
n_tracks_above_avg_pt	count	1.373 ± 0.6524	1.248 ± 0.8713	1.304 ± 0.5695	1.235 ± 0.8125
max_pt_ratio	1	0.29 ± 0.1652	0.2009 ± 0.15	0.2653 ± 0.1415	0.2092 ± 0.1554
min_pt_ratio	1	0.14 ± 0.1554	0.0776 ± 0.0824	0.139 ± 0.1261	0.08362 ± 0.09654
max_dr_ratio	1	0.1642 ± 0.1541	0.09871 ± 0.09181	0.161 ± 0.1276	0.1024 ± 0.106
min_dr_ratio	1	0.2515 ± 0.1825	0.1666 ± 0.1511	0.2312 ± 0.1526	0.1785 ± 0.1543
dr_max_pt	1	0.1399 ± 0.08576	0.1342 ± 0.09986	0.1476 ± 0.08483	0.126 ± 0.09926
dr_min_pt	1	0.202 ± 0.1015	0.1901 ± 0.1196	0.2049 ± 0.09702	0.19 ± 0.1177
dr_max_dr	1	0.2405 ± 0.0983	0.2267 ± 0.123	0.2351 ± 0.09212	0.2208 ± 0.1212
dr_min_dr	1	0.1107 ± 0.07394	0.105 ± 0.08555	0.1235 ± 0.07871	0.1033 ± 0.08834
pt_difference	GeV	1.851 ± 2.103	1.646 ± 2.284	1.528 ± 1.772	2.512 ± 4.113
r50	1	0.1467 ± 0.08321	0.1415 ± 0.09758	0.1549 ± 0.08225	0.1326 ± 0.09761
r95	1	0.238 ± 0.09778	0.224 ± 0.1221	0.2327 ± 0.09214	0.2145 ± 0.1197
mean_abs_d0 *	mm	0.1558 ± 0.1738	2.65 ± 11.82	0.1217 ± 0.1649	1.258 ± 7.925
max_abs_d0 *	mm	0.2765 ± 0.346	7.228 ± 22.64	0.219 ± 0.2875	3.334 ± 16.67
std_abs_d0 *	mm	0.08899 ± 0.1361	2.789 ± 8.765	0.07414 ± 0.1257	1.283 ± 6.701
mean_abs_dz *	mm	0.3993 ± 0.5448	3.894 ± 17.69	0.3303 ± 0.5684	1.675 ± 9.721
max_abs_dz *	mm	0.6857 ± 0.9101	10.19 ± 28.97	0.5747 ± 1.059	4.389 ± 23.14
std_abs_dz *	mm	0.2133 ± 0.3394	3.88 ± 10.39	0.1864 ± 0.4261	1.702 ± 9.544
pt_weighted_abs_d0 *	mm	0.1616 ± 0.1739	2.49 ± 11.71	0.1271 ± 0.1622	1.198 ± 7.937
pt_weighted_abs_dz *	mm	0.4153 ± 0.5609	3.328 ± 17.45	0.3452 ± 0.5723	1.374 ± 8.417

* Variable utilizada como entrada del modelo.

Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.